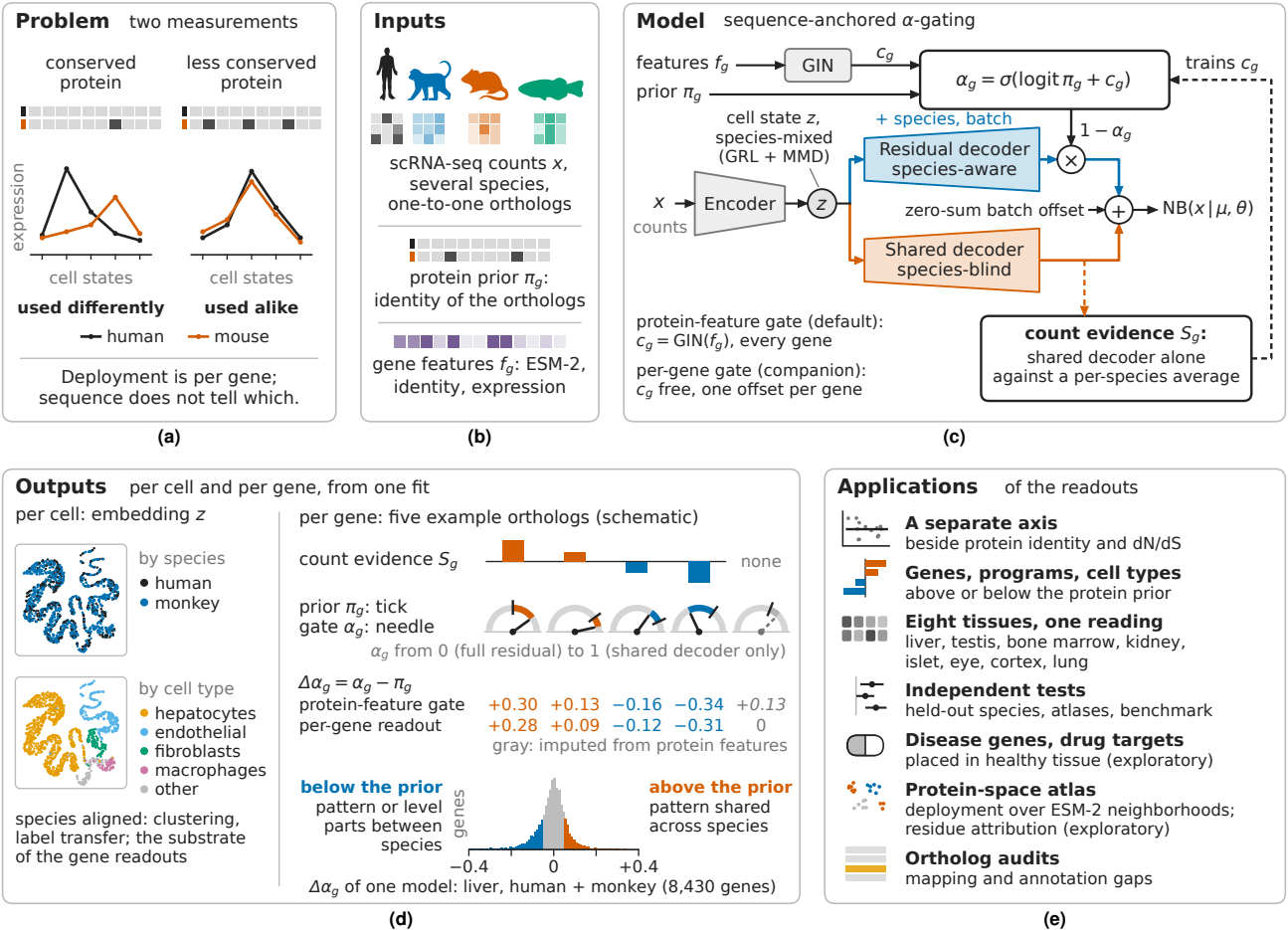
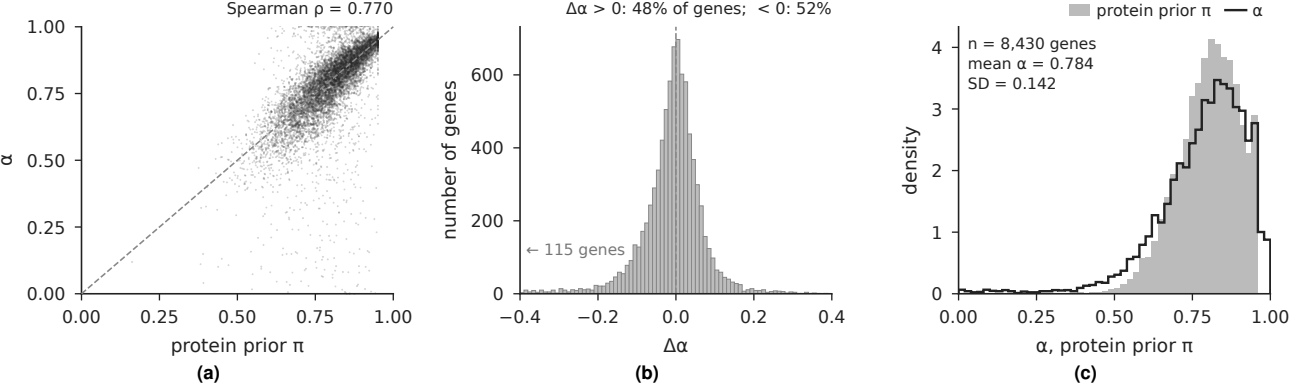
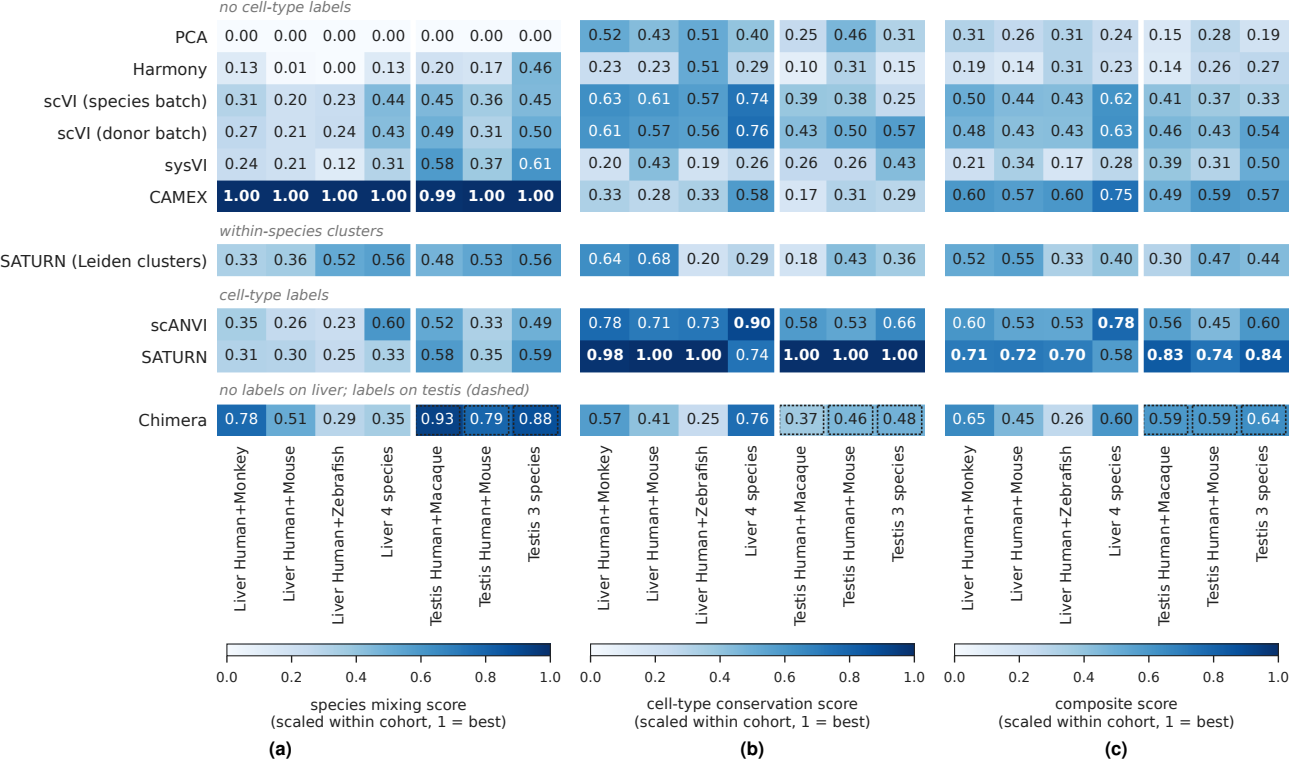
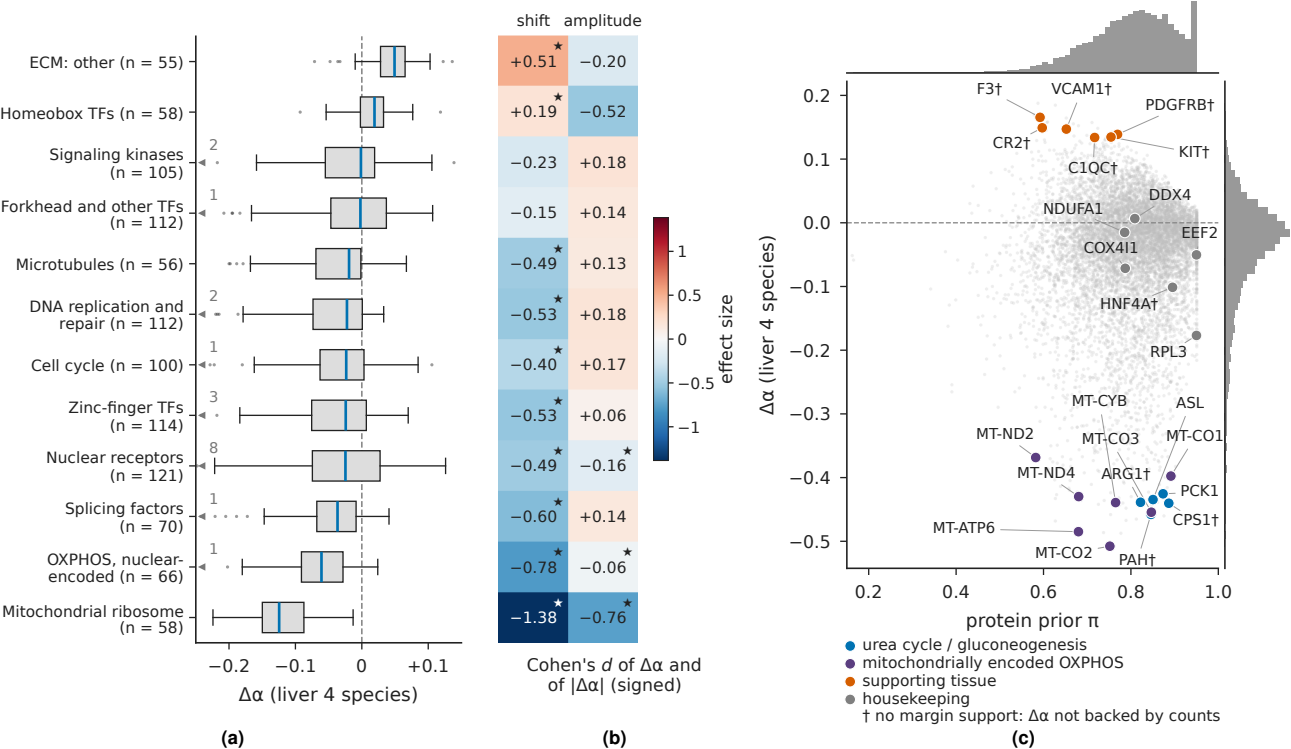
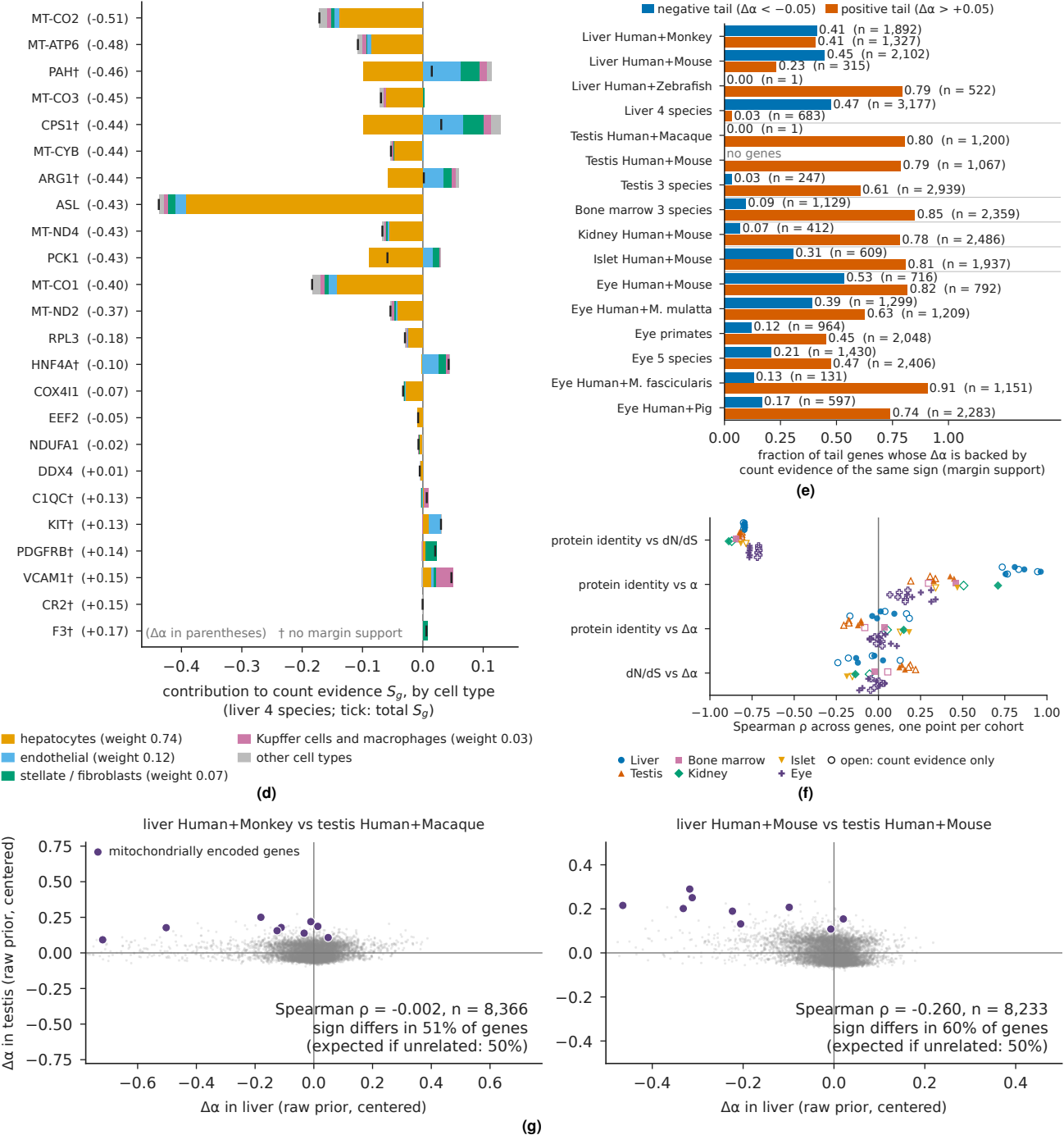
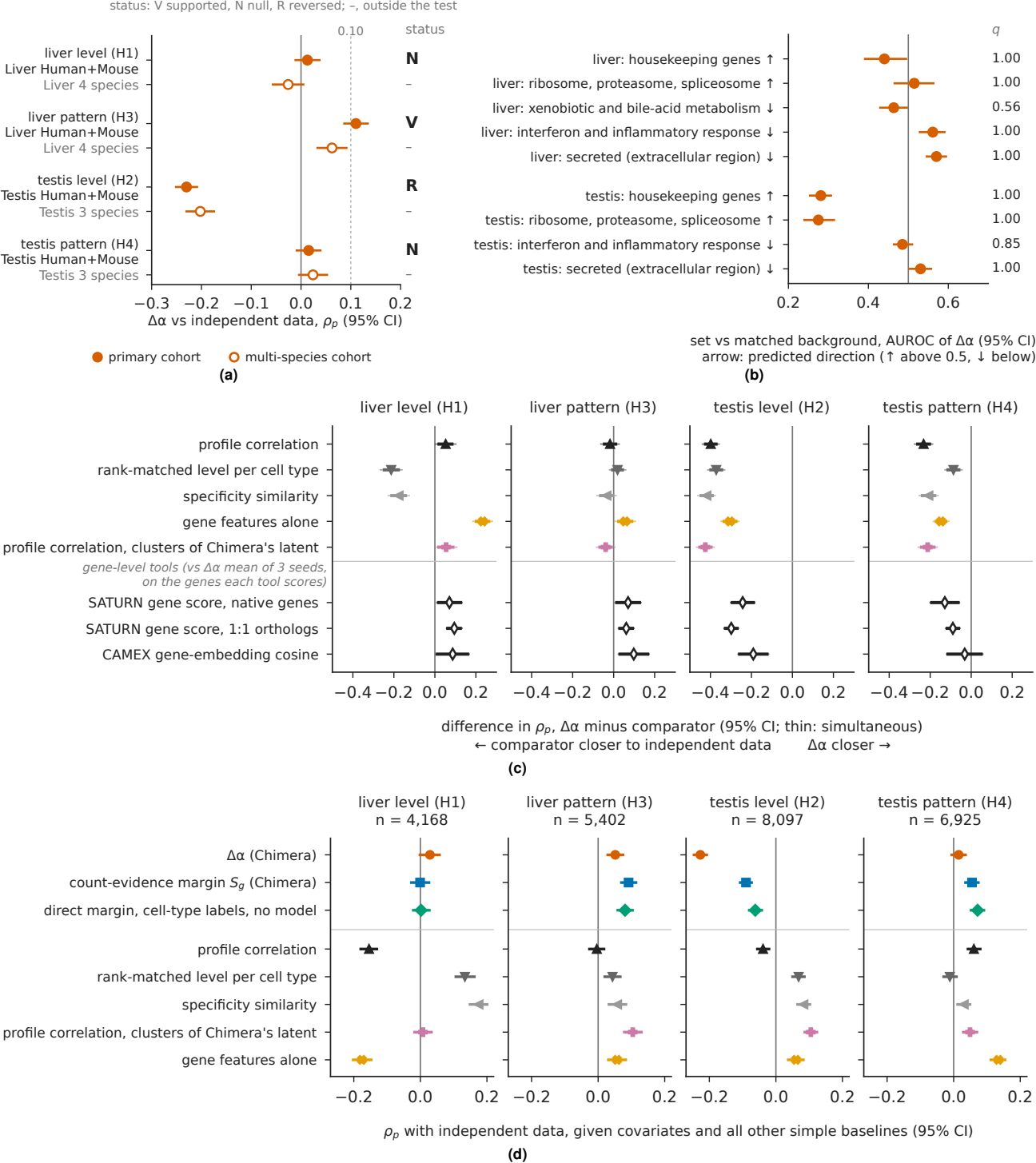

Problem two measurements
conserved protein
less conserved protein
expression
cell states
cell states
used differently
used alike
human
mouse
Deployment is per gene; sequence does not tell which.
(a)
Inputs
scRNA-seq counts x, several species, one-to-one orthologs
protein prior π_g: identity of the orthologs
gene features f_g: ESM-2, identity, expression
(b)
Model sequence-anchored α-gating
features f_g
prior π_g
GIN
c_g
$\alpha_g = \sigma(\text{logit}\,\pi_g + c_g)$
trains c_g
cell state z, species-mixed (GRL + MMD)
+ species, batch
$1 - \alpha_g$
Residual decoder species-aware
x counts
Encoder
z
zero-sum batch offset
NB(x | μ, θ)
Shared decoder species-blind
protein-feature gate (default): $c_g = \text{GIN}(f_g)$, every gene
per-gene gate (companion): c_g free, one offset per gene
count evidence S_g: shared decoder alone against a per-species average
(c)
Outputs per cell and per gene, from one fit
per cell: embedding z
by species
human
monkey
by cell type
hepatocytes
endothelial
fibroblasts
macrophages
other
species aligned: clustering, label transfer; the substrate of the gene readouts
per gene: five example orthologs (schematic)
count evidence S_g
none
prior π_g: tick
gate α_g: needle
α_g from 0 (full residual) to 1 (shared decoder only)
$\Delta\alpha_g = \alpha_g - \pi_g$
protein-feature gate +0.30 +0.13 −0.16 −0.34 +0.13
per-gene readout +0.28 +0.09 −0.12 −0.31 0
gray: imputed from protein features
below the prior
pattern or level parts between species
genes
above the prior
pattern shared across species
−0.4
0
+0.4
$\Delta\alpha_g$ of one model: liver, human + monkey (8,430 genes)
(d)
Applications of the readouts
A separate axis
beside protein identity and dN/dS
Genes, programs, cell types
above or below the protein prior
Eight tissues, one reading
liver, testis, bone marrow, kidney, islet, eye, cortex, lung
Independent tests
held-out species, atlases, benchmark
Disease genes, drug targets
placed in healthy tissue (exploratory)
Protein-space atlas
deployment over ESM-2 neighborhoods; residue attribution (exploratory)
Ortholog audits
mapping and annotation gaps
(e)

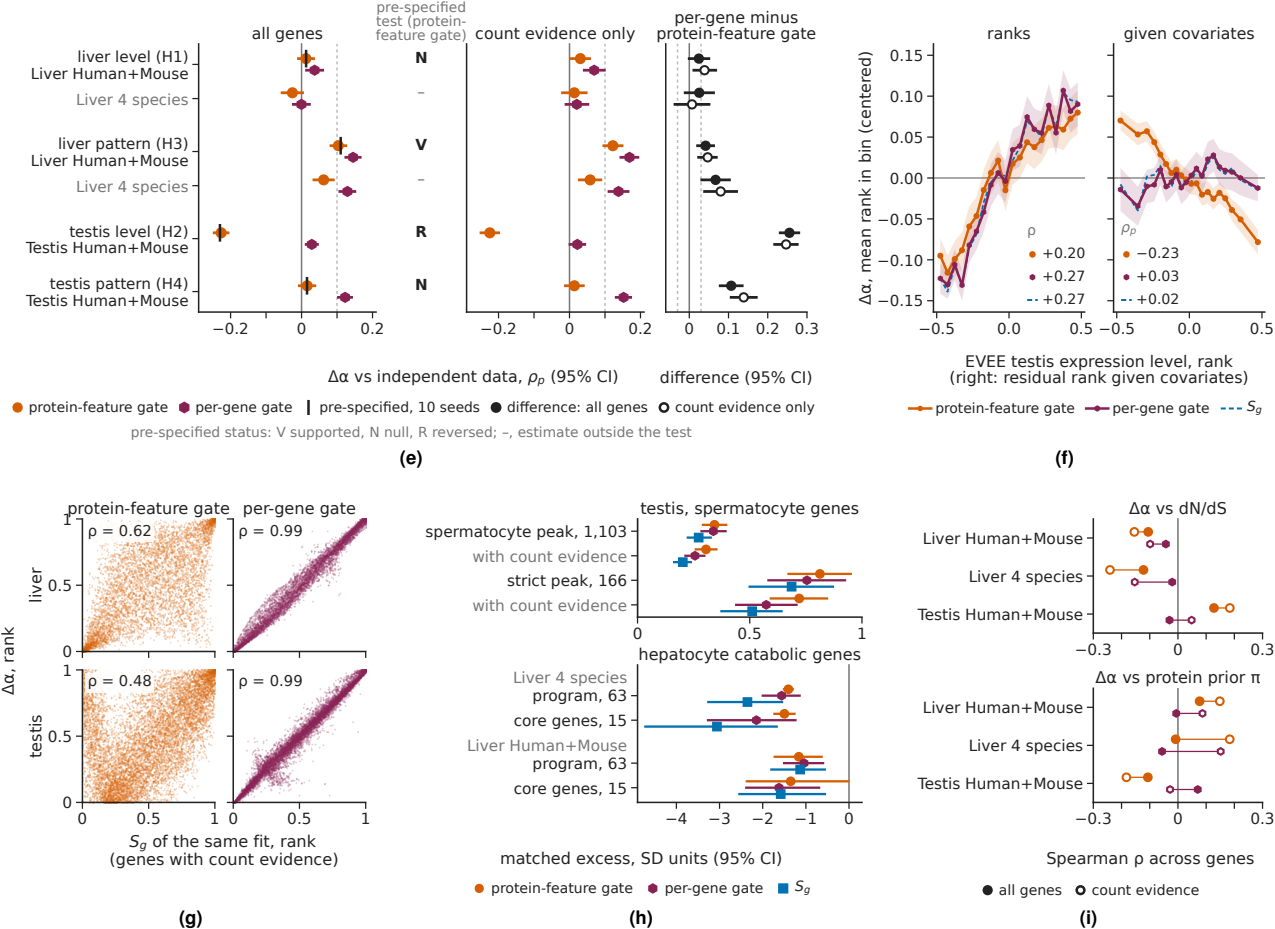

Spearman ρ = 0.770
α
protein prior π
(a)
Δα > 0: 48% of genes; < 0: 52%
number of genes
← 115 genes
Δα
(b)
protein prior π
α
n = 8,430 genes
mean α = 0.784
SD = 0.142
density
α, protein prior π
(c)

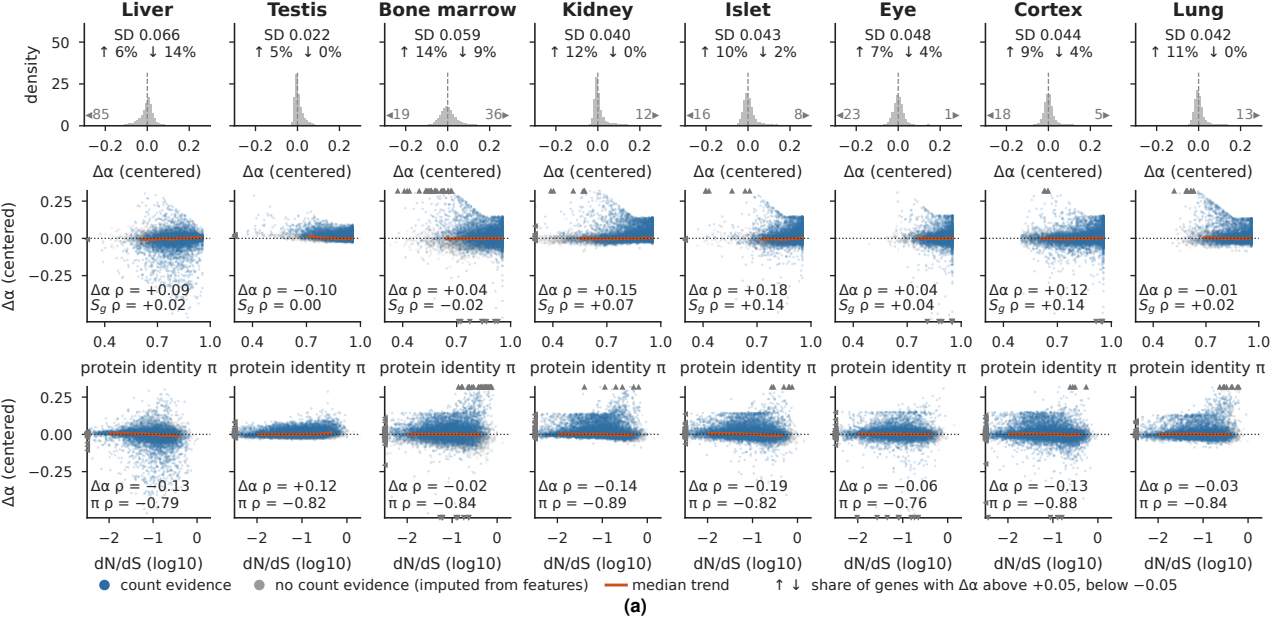

no cell-type labels
PCA 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.52 0.43 0.51 0.40 0.25 0.46 0.31 0.31 0.26 0.31 0.24 0.15 0.28 0.19
Harmony 0.13 0.01 0.00 0.13 0.20 0.17 0.46 0.23 0.23 0.51 0.29 0.10 0.31 0.15 0.19 0.14 0.31 0.23 0.14 0.26 0.27
scVI (species batch) 0.31 0.20 0.23 0.44 0.45 0.36 0.45 0.63 0.61 0.57 0.74 0.39 0.38 0.25 0.50 0.44 0.43 0.62 0.41 0.37 0.33
scVI (donor batch) 0.27 0.21 0.24 0.43 0.49 0.31 0.50 0.61 0.57 0.56 0.76 0.43 0.50 0.57 0.48 0.43 0.43 0.63 0.46 0.43 0.54
sysVI 0.24 0.21 0.12 0.31 0.58 0.37 0.61 0.20 0.43 0.19 0.26 0.26 0.26 0.43 0.21 0.34 0.17 0.28 0.39 0.31 0.50
CAMEX 1.00 1.00 1.00 1.00 0.99 1.00 1.00 0.33 0.28 0.33 0.58 0.17 0.31 0.29 0.60 0.57 0.60 0.75 0.49 0.59 0.57
within-species clusters
SATURN (Leiden clusters) 0.33 0.36 0.52 0.56 0.48 0.53 0.56 0.64 0.68 0.20 0.29 0.18 0.43 0.36 0.52 0.55 0.33 0.40 0.30 0.47 0.44
cell-type labels
scANVI 0.35 0.26 0.23 0.60 0.52 0.33 0.49 0.78 0.71 0.73 0.90 0.58 0.53 0.66 0.60 0.53 0.53 0.78 0.56 0.45 0.60
SATURN 0.31 0.30 0.25 0.33 0.58 0.35 0.59 0.98 1.00 1.00 0.74 1.00 1.00 1.00 0.71 0.72 0.70 0.58 0.83 0.74 0.84
no labels on liver; labels on testis (dashed)
Chimera 0.78 0.51 0.29 0.35 0.93 0.79 0.88 0.57 0.41 0.25 0.76 0.37 0.46 0.48 0.65 0.45 0.26 0.60 0.59 0.59 0.64
Liver Human+Monkey
Liver Human+Mouse
Liver Human+Zebrafish
Liver 4 species
Testis Human+Macaque
Testis Human+Mouse
Testis 3 species
0.0 0.2 0.4 0.6 0.8 1.0
species mixing score
(scaled within cohort, 1 = best)
(a)
cell-type conservation score
(scaled within cohort, 1 = best)
(b)
composite score
(scaled within cohort, 1 = best)
(c)


ECM: other (n = 55)
Homeobox TFs (n = 58)
Signaling kinases (n = 105)
Forkhead and other TFs (n = 112)
Microtubules (n = 56)
DNA replication and repair (n = 112)
Cell cycle (n = 100)
Zinc-finger TFs (n = 114)
Nuclear receptors (n = 121)
Splicing factors (n = 70)
OXPHOS, nuclear-encoded (n = 66)
Mitochondrial ribosome (n = 58)
2
1
2
1
3
8
1
1
−0.2
−0.1
0
+0.1
Δα (liver 4 species)
(a)
shift
amplitude
+0.51
−0.20
+0.19
−0.52
−0.23
+0.18
−0.15
+0.14
−0.49
+0.13
−0.53
+0.18
−0.40
+0.17
−0.53
+0.06
−0.49
−0.16
−0.60
+0.14
−0.78
−0.06
−1.38
−0.76
1
0.5
0
−0.5
−1
effect size
Cohen's d of Δα and of |Δα| (signed)
(b)
F3†
VCAM1†
PDGFRB†
CR2†
KIT†
C1QC†
DDX4
NDUFA1
EEF2
COX4I1
HNF4A†
RPL3
MT-CYB
ASL
MT-ND2
MT-CO3
MT-CO1
MT-ND4
ARG1†
PCK1
CPS1†
MT-ATP6
MT-CO2
PAH†
0.2
0.1
0.0
−0.1
−0.2
−0.3
−0.4
−0.5
Δα (liver 4 species)
0.2
0.4
0.6
0.8
1.0
protein prior π
urea cycle / gluconeogenesis
mitochondrially encoded OXPHOS
supporting tissue
housekeeping
† no margin support: Δα not backed by counts
(c)

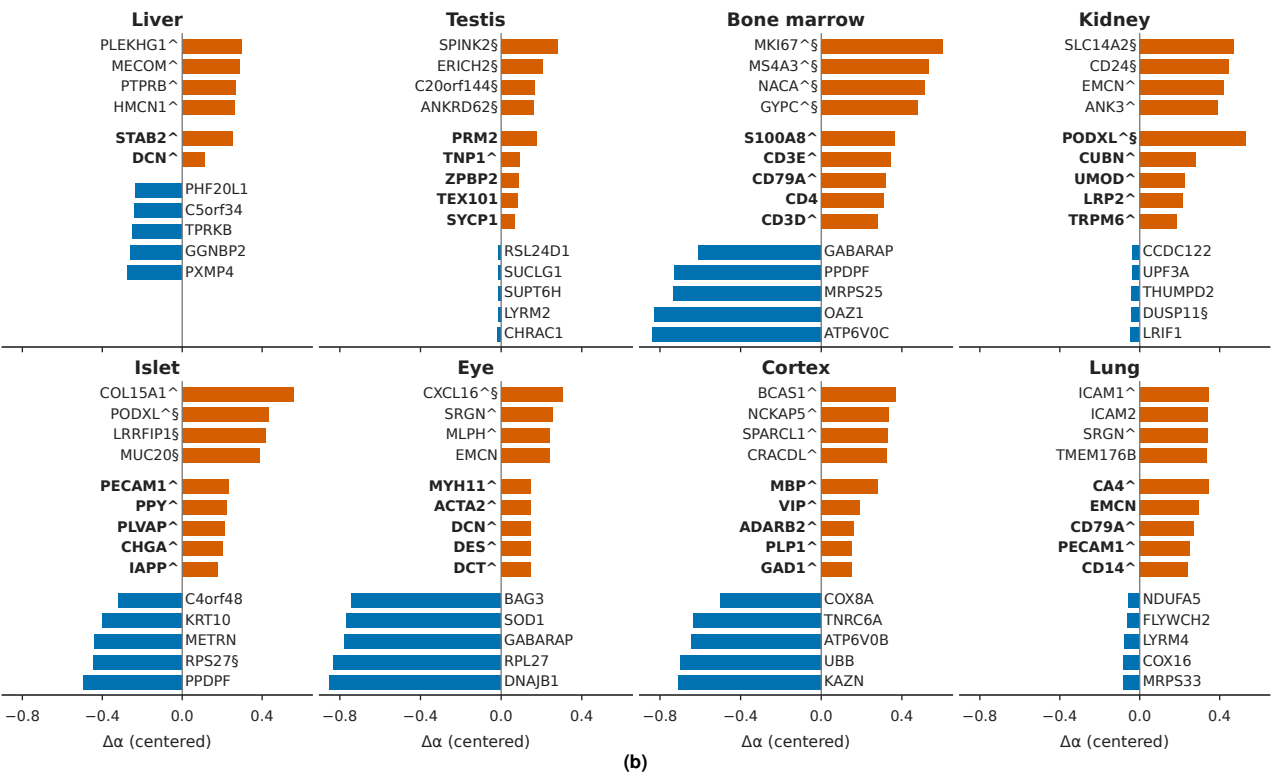

MT-CO2 (-0.51)
MT-ATP6 (-0.48)
PAH† (-0.46)
MT-CO3 (-0.45)
CPS1† (-0.44)
MT-CYB (-0.44)
ARG1† (-0.44)
ASL (-0.43)
MT-ND4 (-0.43)
PCK1 (-0.43)
MT-CO1 (-0.40)
MT-ND2 (-0.37)
RPL3 (-0.18)
HNF4A† (-0.10)
COX4I1 (-0.07)
EEF2 (-0.05)
NDUFA1 (-0.02)
DDX4 (+0.01)
C1QC† (+0.13)
KIT† (+0.13)
PDGFRB† (+0.14)
VCAM1† (+0.15)
CR2† (+0.15)
F3† (+0.17)
(Δα in parentheses) † no margin support
−0.4 −0.3 −0.2 −0.1 0.0 0.1
contribution to count evidence S_g, by cell type
(liver 4 species; tick: total S_g)
hepatocytes (weight 0.74)
endothelial (weight 0.12)
stellate / fibroblasts (weight 0.07)
Kupffer cells and macrophages (weight 0.03)
other cell types
(d)
negative tail (Δα < −0.05)
positive tail (Δα > +0.05)
Liver Human+Monkey
0.41 (n = 1,892)
0.41 (n = 1,327)
Liver Human+Mouse
0.45 (n = 2,102)
0.23 (n = 315)
Liver Human+Zebrafish
0.00 (n = 1)
0.79 (n = 522)
Liver 4 species
0.47 (n = 3,177)
0.03 (n = 683)
Testis Human+Macaque
0.00 (n = 1)
0.80 (n = 1,200)
Testis Human+Mouse
no genes
0.79 (n = 1,067)
Testis 3 species
0.03 (n = 247)
0.61 (n = 2,939)
Bone marrow 3 species
0.09 (n = 1,129)
0.85 (n = 2,359)
Kidney Human+Mouse
0.07 (n = 412)
0.78 (n = 2,486)
Islet Human+Mouse
0.31 (n = 609)
0.81 (n = 1,937)
Eye Human+Mouse
0.53 (n = 716)
0.82 (n = 792)
Eye Human+M. mulatta
0.39 (n = 1,299)
0.63 (n = 1,209)
Eye primates
0.12 (n = 964)
0.45 (n = 2,048)
Eye 5 species
0.21 (n = 1,430)
0.47 (n = 2,406)
Eye Human+M. fascicularis
0.13 (n = 131)
0.91 (n = 1,151)
Eye Human+Pig
0.17 (n = 597)
0.74 (n = 2,283)
0.00 0.25 0.50 0.75 1.00
fraction of tail genes whose Δα is backed by
count evidence of the same sign (margin support)
(e)
protein identity vs dN/dS
protein identity vs α
protein identity vs Δα
dN/dS vs Δα
−1.00 −0.75 −0.50 −0.25 0.00 0.25 0.50 0.75 1.00
Spearman ρ across genes, one point per cohort
Liver
Bone marrow
Islet
open: count evidence only
Testis
Kidney
Eye
(f)
liver Human+Monkey vs testis Human+Macaque
mitochondrially encoded genes
Δα in testis (raw prior, centered)
Spearman ρ = -0.002, n = 8,366
sign differs in 51% of genes
(expected if unrelated: 50%)
Δα in liver (raw prior, centered)
liver Human+Mouse vs testis Human+Mouse
Spearman ρ = -0.260, n = 8,233
sign differs in 60% of genes
(expected if unrelated: 50%)
Δα in liver (raw prior, centered)
(g)


status: V supported, N null, R reversed; –, outside the test
0.10
status
liver level (H1)
Liver Human+Mouse
Liver 4 species
liver pattern (H3)
Liver Human+Mouse
Liver 4 species
testis level (H2)
Testis Human+Mouse
Testis 3 species
testis pattern (H4)
Testis Human+Mouse
Testis 3 species
N
–
V
–
R
–
N
–
−0.3 −0.2 −0.1 0.0 0.1 0.2
Δα vs independent data, ρ_p (95% CI)
primary cohort
multi-species cohort
(a)
q
liver: housekeeping genes ↑
liver: ribosome, proteasome, spliceosome ↑
liver: xenobiotic and bile-acid metabolism ↓
liver: interferon and inflammatory response ↓
liver: secreted (extracellular region) ↓
testis: housekeeping genes ↑
testis: ribosome, proteasome, spliceosome ↑
testis: interferon and inflammatory response ↓
testis: secreted (extracellular region) ↓
1.00
1.00
0.56
1.00
1.00
1.00
1.00
0.85
1.00
0.2 0.4 0.6
set vs matched background, AUROC of Δα (95% CI)
arrow: predicted direction (↑ above 0.5, ↓ below)
(b)
liver level (H1)
liver pattern (H3)
testis level (H2)
testis pattern (H4)
profile correlation
rank-matched level per cell type
specificity similarity
gene features alone
profile correlation, clusters of Chimera's latent
gene-level tools (vs Δα mean of 3 seeds, on the genes each tool scores)
SATURN gene score, native genes
SATURN gene score, 1:1 orthologs
CAMEX gene-embedding cosine
−0.4 −0.2 0.0 0.2
difference in ρ_p, Δα minus comparator (95% CI; thin: simultaneous)
← comparator closer to independent data
Δα closer →
(c)
liver level (H1)
n = 4,168
liver pattern (H3)
n = 5,402
testis level (H2)
n = 8,097
testis pattern (H4)
n = 6,925
Δα (Chimera)
count-evidence margin S_g (Chimera)
direct margin, cell-type labels, no model
profile correlation
rank-matched level per cell type
specificity similarity
profile correlation, clusters of Chimera's latent
gene features alone
−0.2 0.0 0.2
ρ_p with independent data, given covariates and all other simple baselines (95% CI)
(d)


all genes
pre-specified test (protein-feature gate)
count evidence only
per-gene minus protein-feature gate
liver level (H1)
Liver Human+Mouse
Liver 4 species
liver pattern (H3)
Liver Human+Mouse
Liver 4 species
testis level (H2)
Testis Human+Mouse
testis pattern (H4)
Testis Human+Mouse
N
–
V
–
R
N
−0.2
0
0.2
0
0.1
0.2
0.3
Δα vs independent data, ρ_p (95% CI)
difference (95% CI)
protein-feature gate
per-gene gate
pre-specified, 10 seeds
difference: all genes
count evidence only
pre-specified status: V supported, N null, R reversed; –, estimate outside the test
(e)
ranks
given covariates
Δα, mean rank in bin (centered)
0.15
0.10
0.05
0.00
−0.05
−0.10
−0.15
ρ
+0.20
+0.27
+0.27
ρ_p
−0.23
+0.03
+0.02
−0.5
0.0
0.5
EVEE testis expression level, rank
(right: residual rank given covariates)
protein-feature gate
per-gene gate
S_g
(f)
protein-feature gate
per-gene gate
liver
testis
ρ = 0.62
ρ = 0.99
ρ = 0.48
ρ = 0.99
Δα, rank
S_g of the same fit, rank
(genes with count evidence)
(g)
testis, spermatocyte genes
spermatocyte peak, 1,103
with count evidence
strict peak, 166
with count evidence
hepatocyte catabolic genes
Liver 4 species
program, 63
core genes, 15
Liver Human+Mouse
program, 63
core genes, 15
−4
−3
−2
−1
0
matched excess, SD units (95% CI)
protein-feature gate
per-gene gate
S_g
(h)
Δα vs dN/dS
Liver Human+Mouse
Liver 4 species
Testis Human+Mouse
−0.3
0
0.3
Δα vs protein prior π
Liver Human+Mouse
Liver 4 species
Testis Human+Mouse
Spearman ρ across genes
all genes
count evidence
(i)


Liver
SD 0.066
↑ 6% ↓ 14%
Testis
SD 0.022
↑ 5% ↓ 0%
Bone marrow
SD 0.059
↑ 14% ↓ 9%
Kidney
SD 0.040
↑ 12% ↓ 0%
Islet
SD 0.043
↑ 10% ↓ 2%
Eye
SD 0.048
↑ 7% ↓ 4%
Cortex
SD 0.044
↑ 9% ↓ 4%
Lung
SD 0.042
↑ 11% ↓ 0%
density
Δα (centered)
protein identity π
dN/dS (log10)
Δα ρ = +0.09
S_g ρ = +0.02
Δα ρ = −0.10
S_g ρ = 0.00
Δα ρ = +0.04
S_g ρ = −0.02
Δα ρ = +0.15
S_g ρ = +0.07
Δα ρ = +0.18
S_g ρ = +0.14
Δα ρ = +0.04
S_g ρ = +0.04
Δα ρ = +0.12
S_g ρ = +0.14
Δα ρ = −0.01
S_g ρ = +0.02
Δα ρ = −0.13
π ρ = −0.79
Δα ρ = +0.12
π ρ = −0.82
Δα ρ = −0.02
π ρ = −0.84
Δα ρ = −0.14
π ρ = −0.89
Δα ρ = −0.19
π ρ = −0.82
Δα ρ = −0.06
π ρ = −0.76
Δα ρ = −0.13
π ρ = −0.88
Δα ρ = −0.03
π ρ = −0.84
count evidence
no count evidence (imputed from features)
median trend
↑ ↓ share of genes with Δα above +0.05, below −0.05

(a)


Liver
PLEKHG1^
MECOM^
PTPRB^
HMCN1^
STAB2^
DCN^
PHF20L1
C5orf34
TPRKB
GGNBP2
PXMP4
Testis
SPINK2§
ERICH2§
C20orf144§
ANKRD62§
PRM2
TNP1^
ZPBP2
TEX101
SYCP1
RSL24D1
SUCLG1
SUPT6H
LYRM2
CHRAC1
Bone marrow
MKI67^§
MS4A3^§
NACA^§
GYPC^§
S100A8^
CD3E^
CD79A^
CD4
CD3D^
GABARAP
PPDPF
MRPS25
OAZ1
ATP6V0C
Kidney
SLC14A2§
CD24§
EMCN^
ANK3^
PODXL^§
CUBN^
UMOD^
LRP2^
TRPM6^
CCDC122
UPF3A
THUMPD2
DUSP11§
LRIF1
Islet
COL15A1^
PODXL^§
LRRFIP1§
MUC20§
PECAM1^
PPY^
PLVAP^
CHGA^
IAPP^
C4orf48
KRT10
METRN
RPS27§
PPDPF
Eye
CXCL16^§
SRGN^
MLPH^
EMCN
MYH11^
ACTA2^
DCN^
DES^
DCT^
BAG3
SOD1
GABARAP
RPL27
DNAJB1
Cortex
BCAS1^
NCKAP5^
SPARCL1^
CRACDL^
MBP^
VIP^
ADARB2^
PLP1^
GAD1^
COX8A
TNRC6A
ATP6V0B
UBB
KAZN
Lung
ICAM1^
ICAM2
SRGN^
TMEM176B
CA4^
EMCN
CD79A^
PECAM1^
CD14^
NDUFA5
FLYWCH2
LYRM4
COX16
MRPS33
Δα (centered)

(b)

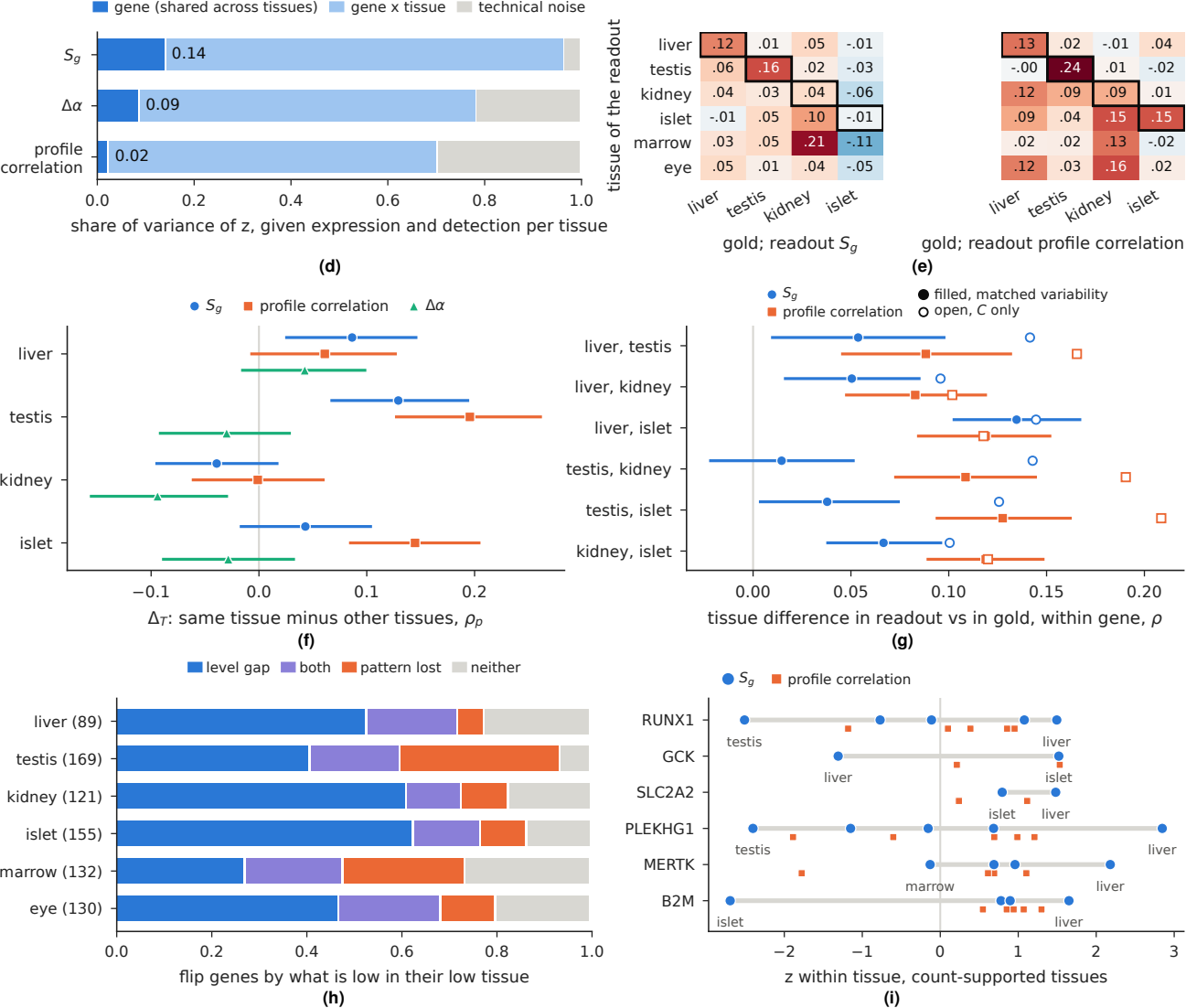
	Liver	Testis	Bone marrow	Kidney	Islet	Eye	Cortex	Lung	q	S_g	profile correlation	π	dN/dS
Recurrent, above the prior													
Receptors (class)	+7.1*	+7.3*	+9.5*	+15.0*	+13.6*	+7.2*	+13.1*	+14.1*	<.001	8↑0↓	7↑0↓	−2.2	+0.9
Epithelial Mesenchymal Transition	+7.8*	+7.6*	+4.0*	+12.1*	+16.7*	+10.0*	+8.1*	+11.6*	<.001	8↑0↓	6↑0↓	+0.4	0.0
Cytoskeleton & ECM (category)	+5.4*	+10.1*	+5.5*	+8.5*	+10.1*	+7.8*	+3.3*	+9.5*	<.001	8↑0↓	6↑1↓	+2.8	−2.9
KRAS Signaling Up	+2.9*	+5.1*	+5.2*	+6.5*	+9.8*	+6.1*	+4.5*	+8.9*	<.001	6↑0↓	5↑0↓	−2.7	+1.5
Inflammatory Response	+3.7*	+3.4*	+6.4*	+6.3*	+9.9*	+4.0*	+6.1*	+9.1*	<.001	7↑0↓	6↑0↓	−4.2	+3.4
Apical Junction	+4.0*	+2.6*	+3.6*	+7.1*	+9.2*	+6.4*	+5.0*	+7.6*	<.001	8↑0↓	6↑0↓	+2.0	−3.2
TNF-alpha Signaling via NF-kB	+1.3	+3.7*	+5.0*	+6.7*	+11.1*	+2.8*	+5.0*	+9.6*	<.001	6↑0↓	6↑0↓	−1.4	+1.5
GPCRs (class)	+3.7*	+4.5*	+1.9	+7.6*	+8.0*	+3.9*	+7.5*	+5.9*	<.001	7↑0↓	6↑0↓	−0.1	−0.5
Coagulation	+4.7*	+4.5*	+2.5*	+6.7*	+8.2*	+4.2*	+4.7*	+6.5*	<.001	7↑0↓	6↑0↓	−1.8	+1.4
IL-2/STAT5 Signaling	+2.7*	+3.2*	+5.1*	+5.5*	+7.4*	+5.6*	+4.1*	+7.3*	<.001	7↑0↓	5↑0↓	−1.5	+1.0
Myogenesis	+4.1*	+4.3*	+2.3*	+6.2*	+7.4*	+5.5*	+4.8*	+6.3*	<.001	8↑0↓	6↑0↓	+0.5	−1.2
UV Response Dn	+3.0*	+4.1*	+1.3	+5.2*	+8.5*	+5.3*	+5.9*	+5.3*	<.001	6↑0↓	7↑0↓	+2.4	−1.9
Complement	+4.5*	+1.4	+4.8*	+5.5*	+7.5*	+3.8*	+4.5*	+6.3*	<.001	7↑0↓	3↑0↓	−0.9	+0.6
Hypoxia	+3.0*	+2.1	+3.8*	+6.1*	+8.0*	+3.9*	+3.5*	+6.3*	<.001	6↑0↓	4↑0↓	+0.7	−1.3
Estrogen Response Early	−0.5	+1.8	+2.3*	+7.7*	+7.5*	+5.1*	+4.2*	+7.1*	<.001	5↑0↓	6↑0↓	0.0	−0.3
Allograft Rejection	+2.6*	+1.3	+6.7*	+4.5*	+6.3*	+2.9*	+2.2*	+7.0*	<.001	5↑0↓	7↑0↓	−3.0	+2.6
Angiogenesis		+3.1*	+3.5*	+5.1*	+6.1*	+2.6*	+4.4*	+6.3*	<.001	7↑0↓	4↑0↓	−0.1	+0.1
Signaling pathways (category)	+4.8*	+4.4*	+1.1	+2.7*	+5.9*	+4.4*	+2.4*	+7.4*	<.001	5↑0↓	6↑0↓	+6.4	−7.7
Estrogen Response Late	+0.2	+0.8	+3.4*	+7.0*	+6.3*	+4.2*	+3.4*	+6.9*	<.001	6↑0↓	3↑0↓	−0.2	−0.6
Immunity & inflammation (category)	+2.3*	+3.0*	+3.9*	+4.3*	+3.6*	+3.6*	+2.8*	+7.2*	<.001	4↑0↓	6↑0↓	−8.3	+7.0
Apoptosis	+1.5	+2.7*	+4.5*	+4.0*	+6.3*	+2.1	+2.8*	+4.9*	<.001	6↑0↓	4↑0↓	−0.9	+1.2
IL-6/JAK/STAT3 Signaling	+2.6*	+1.2	+3.9*	+4.0*	+4.4*	+2.7*	+1.5	+5.0*	<.001	6↑0↓	6↑0↓	−4.8	+3.7
Homeodomain TFs (class)	+1.8	+2.5*	−0.3	+4.5*	+5.4*	+2.0	+5.8*	+3.0*	<.001	5↑0↓	6↑0↓	+3.1	−2.7
Interferon Gamma Response	+1.7	+2.1	+4.3*	+3.4*	+5.3*	+1.8	+0.4	+5.6*	<.001	4↑0↓	4↑0↓	−6.6	+5.3
Androgen Response	+2.2	+1.8	+2.2*	+4.6*	+3.2*	+1.1	+4.6*	+4.0*	<.001	4↑0↓	0↑0↓	+0.4	−0.6
TGF-beta Signaling	+2.4*	+1.4	+0.7	+1.2	+5.7*	+1.5	+2.3*	+4.3*	<.001	3↑0↓	5↑0↓	+2.0	−3.3
Recurrent, below the prior													
E2F Targets	+0.1	−3.5*	+9.0*	−6.4*	−2.7*	−4.0*	−3.4*	−4.1*	<.001	1↑4↓	1↑3↓	+1.9	−1.7
Mitochondria & OXPHOS (category)	−3.2*	−3.1*	−4.7*	+1.7	−3.1*	+1.2	−5.0*	−7.6*	<.001	1↑6↓	0↑3↓	−5.0	+3.1
Protein homeostasis (category)	−1.9	−4.7*	−1.8	−3.4*	−1.6	−5.5*	−5.8*	−2.7*	<.001	0↑4↓	0↑4↓	+4.1	−3.8
Oxidative Phosphorylation	−2.6*	−9.3*	−3.1*	+1.6	−3.0*	+0.7	−5.3*	−9.4*	<.001	1↑5↓	0↑4↓	−1.7	+2.7
Enzymes (class)	−6.2*	−2.0	+0.2	−4.7*	−7.3*	−1.4	−5.3*	−4.8*	<.001	0↑5↓	0↑6↓	+5.0	−2.8
Translation & ribosome (category)	+0.4	−15.8*	0.0	+2.0	−3.9*	−5.6*	−4.2*	−11.2*	<.001	2↑4↓	1↑3↓	−0.5	+1.8
Chromatin regulators (class)	−3.7*	−1.1	−3.4*	−12.0*	−5.5*	−4.6*	−6.5*	−3.4*	<.001	0↑7↓	1↑6↓	+5.3	−4.9
Chromatin & DNA processing (category)	−1.7	−7.2*	−0.8	−9.0*	−6.5*	−5.7*	−8.7*	−8.8*	<.001	0↑5↓	0↑5↓	+4.3	−4.1
Mitochondrial proteins (class)	−5.0*	−14.2*	−6.0*	−0.4	−10.6*	−2.9*	−8.8*	−21.0*	<.001	0↑7↓	0↑6↓	−8.3	+9.0
Called in one tissue only													
Heme Metabolism	−1.5	−1.4	+8.2*	+0.3	+0.5	+1.1	+0.4	+0.1	0.351	1↑2↓	0↑1↓	+0.7	+0.4
Bile Acid Metabolism	+0.1	+1.0	+1.4	+2.6*	−0.9	+0.1	−1.5	−0.5	0.351	1↑1↓	1↑0↓	−2.5	+2.2
Protein Secretion	+0.4	−2.5*	+0.8	−0.6	+0.7	−0.5	−0.8	−1.2	0.351	0↑0↓	0↑0↓	+2.7	−2.8
Opposite signs between tissues													
G2-M Checkpoint	+1.0	+0.1	+7.2*	−6.1*	+0.5	−1.2	−4.1*	−3.2*	0.002	1↑1↓	1↑1↓	+3.0	−2.4
Unfolded Protein Response	−0.7	−5.3*	+2.7*	−3.4*	+1.9	−2.2	−3.3*	−3.0*	<.001	2↑2↓	2↑3↓	+3.6	−2.5
Myc Targets V1	−0.6	−7.2*	+4.3*	−4.4*	−2.3*	−4.5*	−5.0*	−5.0*	<.001	1↑3↓	2↑4↓	+7.0	−6.6

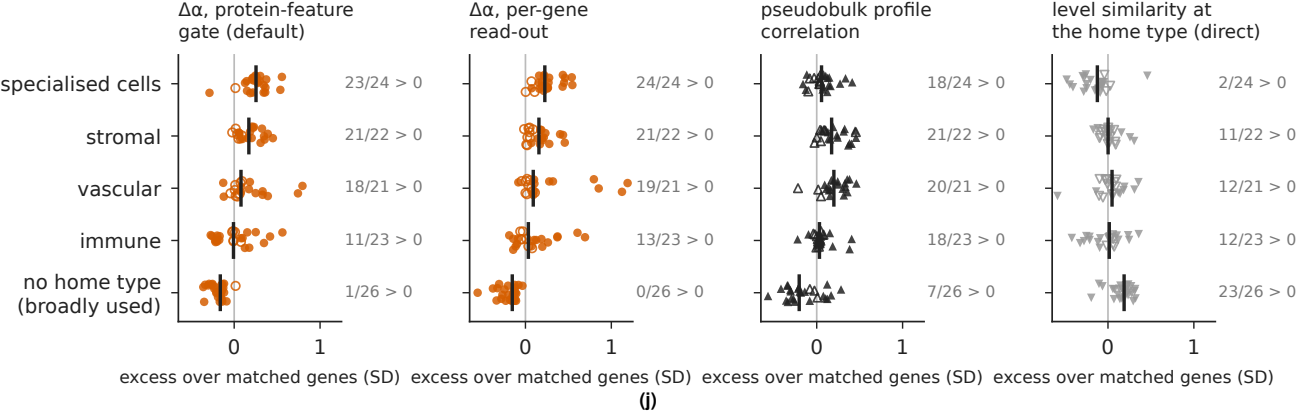
≤ −8 −4 0 +4 ≥ +8

set z (color clipped at ±8)

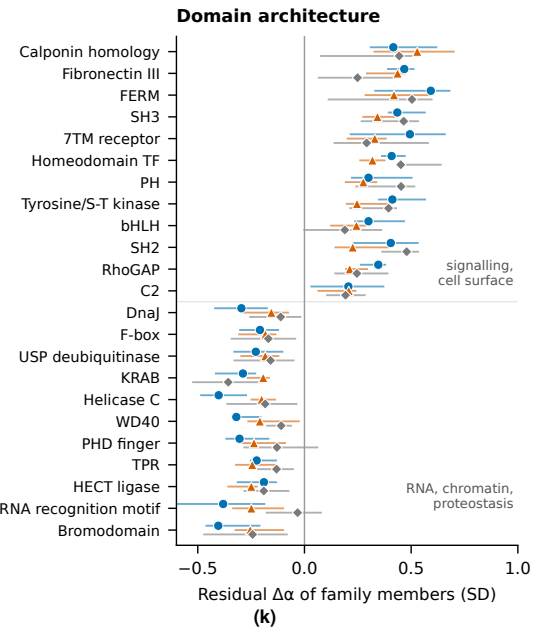
Cells: z of the set against matched genes of the tissue; * $q < 0.05$; gray cell, fewer than 15 genes.
Right: q, recurrence across tissues; S_g and profile correlation, tissues with the set above (↑) and below (↓);
π and dN/dS, mean z of the set over tissues. Gray type: not recurrent, or |mean z| below 1.96.

(c)

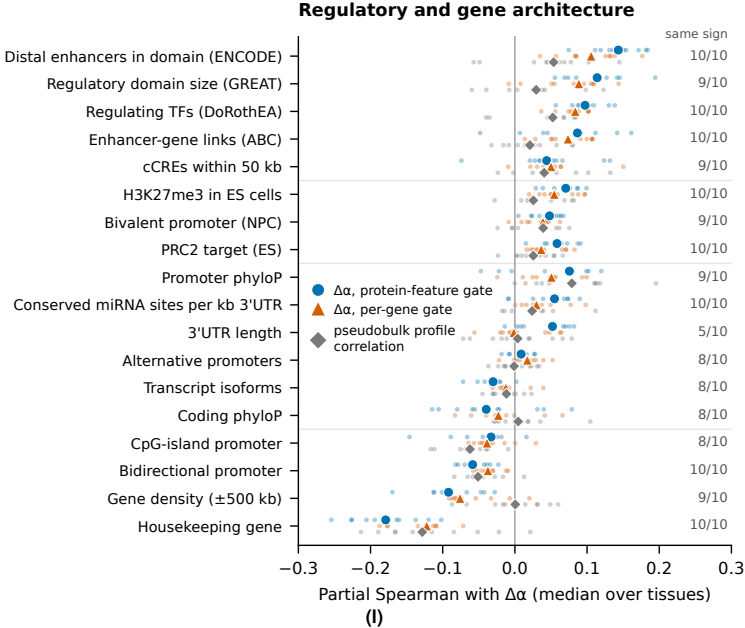

gene (shared across tissues)
gene x tissue
technical noise
S_g
$\Delta\alpha$
profile correlation
0.14
0.09
0.02
0.0
0.2
0.4
0.6
0.8
1.0
share of variance of z, given expression and detection per tissue
(d)
tissue of the readout
liver .12 .01 .05 -.01
testis .06 .16 .02 -.03
kidney .04 .03 .04 -.06
islet -.01 .05 .10 -.01
marrow .03 .05 .21 -.11
eye .05 .01 .04 -.05
liver testis kidney islet
gold; readout S_g
.13 .02 -.01 .04
-.00 .24 .01 -.02
.12 .09 .09 .01
.09 .04 .15 .15
.02 .02 .13 -.02
.12 .03 .16 .02
liver testis kidney islet
gold; readout profile correlation
(e)
S_g
profile correlation
$\Delta\alpha$
liver
testis
kidney
islet
−0.1
0.0
0.1
0.2
Δ_T: same tissue minus other tissues, ρ_p
(f)
S_g
profile correlation
filled, matched variability
open, C only
liver, testis
liver, kidney
liver, islet
testis, kidney
testis, islet
kidney, islet
0.00
0.05
0.10
0.15
0.20
tissue difference in readout vs in gold, within gene, ρ
(g)
level gap
both
pattern lost
neither
liver (89)
testis (169)
kidney (121)
islet (155)
marrow (132)
eye (130)
0.0
0.2
0.4
0.6
0.8
1.0
flip genes by what is low in their low tissue
(h)
S_g
profile correlation
RUNX1
GCK
SLC2A2
PLEKHG1
MERTK
B2M
testis
liver
liver
islet
islet
liver
testis
liver
marrow
liver
islet
liver
−2
−1
0
1
2
3
z within tissue, count-supported tissues
(i)


Δα, protein-feature gate (default)
Δα, per-gene read-out
pseudobulk profile correlation
level similarity at the home type (direct)
specialised cells
stromal
vascular
immune
no home type (broadly used)
23/24 > 0
21/22 > 0
18/21 > 0
11/23 > 0
1/26 > 0
24/24 > 0
21/22 > 0
19/21 > 0
13/23 > 0
0/26 > 0
18/24 > 0
21/22 > 0
20/21 > 0
18/23 > 0
7/26 > 0
2/24 > 0
11/22 > 0
12/21 > 0
12/23 > 0
23/26 > 0
0
1
excess over matched genes (SD)

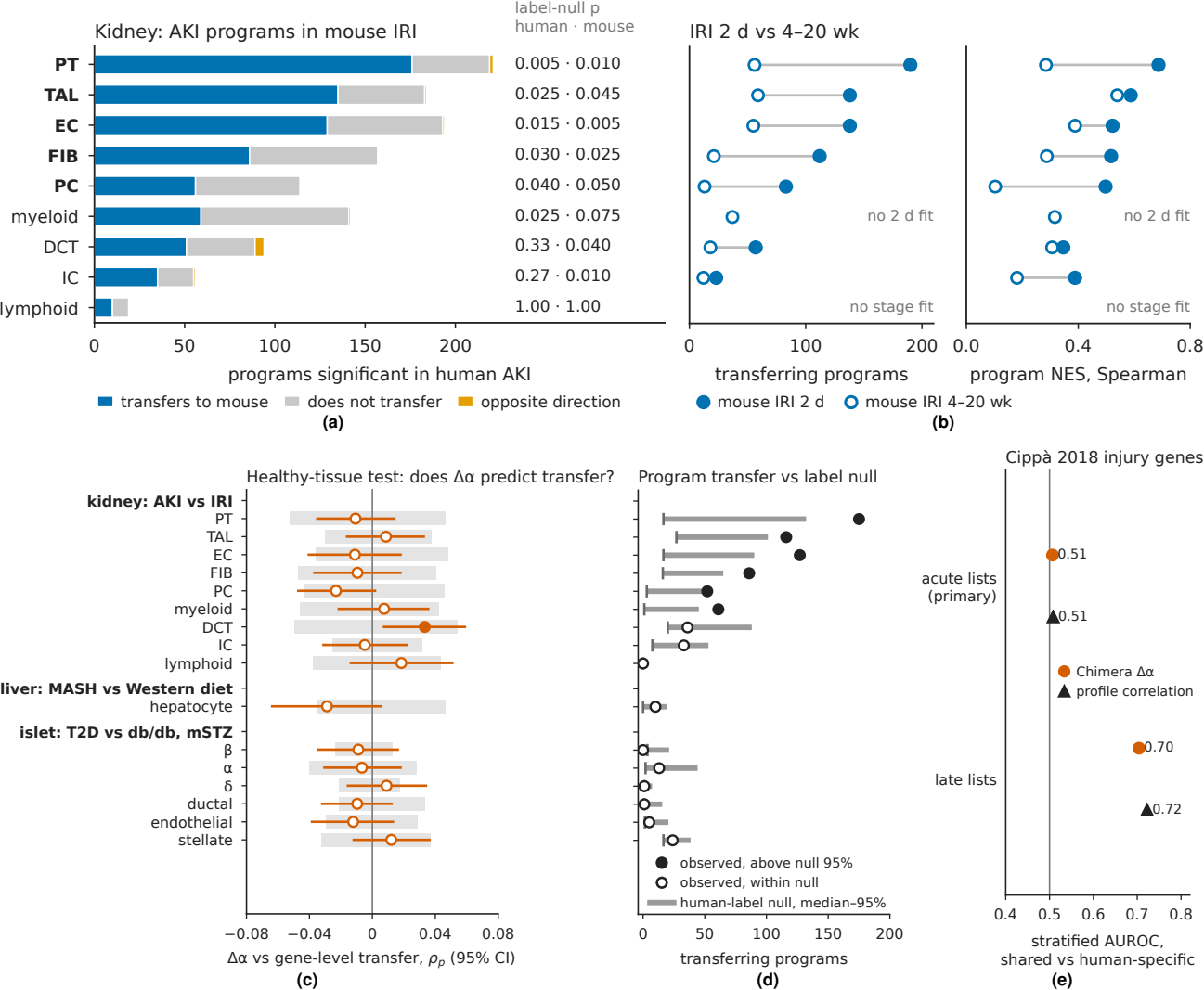
(j)

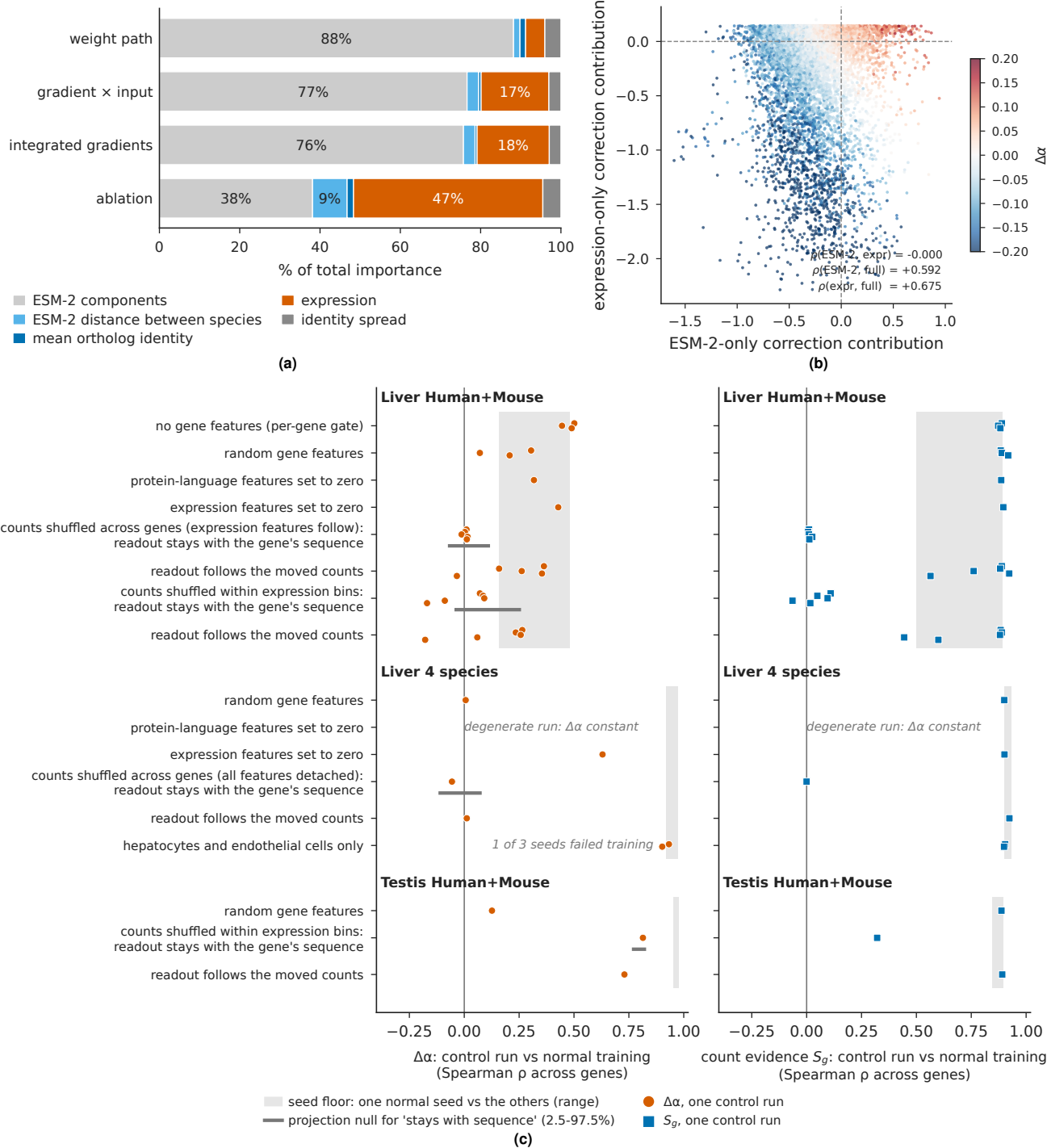

Domain architecture
Calponin homology
Fibronectin III
FERM
SH3
7TM receptor
Homeodomain TF
PH
Tyrosine/S-T kinase
bHLH
SH2
RhoGAP
C2
DnaJ
F-box
USP deubiquitinase
KRAB
Helicase C
WD40
PHD finger
TPR
HECT ligase
RNA recognition motif
Bromodomain
signalling, cell surface
RNA, chromatin, proteostasis
−0.5
0.0
0.5
1.0
Residual Δα of family members (SD)

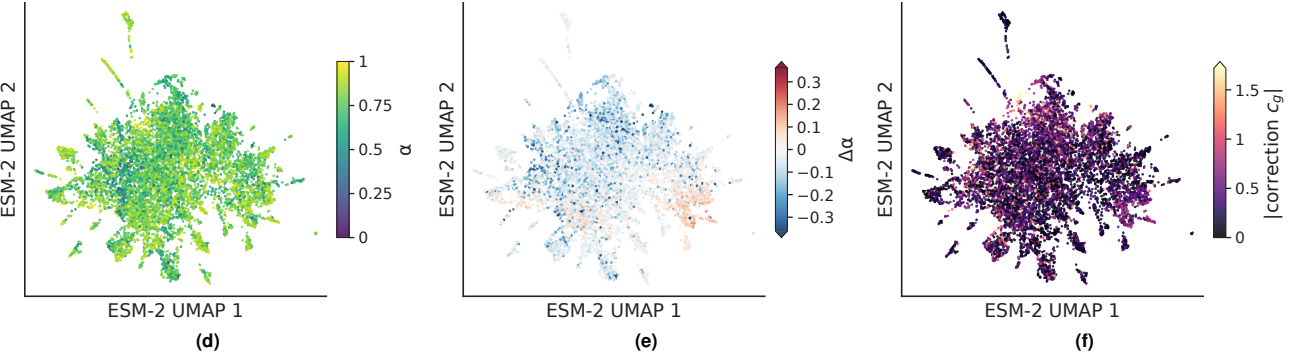
(k)

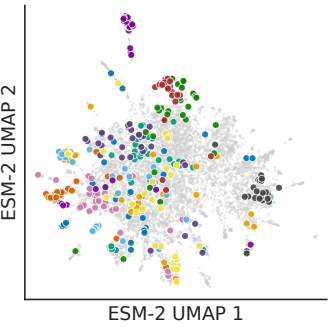

Regulatory and gene architecture
same sign
Distal enhancers in domain (ENCODE)
Regulatory domain size (GREAT)
Regulating TFs (DoRothEA)
Enhancer-gene links (ABC)
cCREs within 50 kb
H3K27me3 in ES cells
Bivalent promoter (NPC)
PRC2 target (ES)
Promoter phyloP
Conserved miRNA sites per kb 3'UTR
3'UTR length
Alternative promoters
Transcript isoforms
Coding phyloP
CpG-island promoter
Bidirectional promoter
Gene density (±500 kb)
Housekeeping gene
10/10
9/10
10/10
10/10
9/10
10/10
9/10
10/10
9/10
10/10
5/10
8/10
8/10
8/10
8/10
10/10
9/10
10/10
Δα, protein-feature gate
Δα, per-gene gate
pseudobulk profile correlation
−0.3
−0.2
−0.1
0.0
0.1
0.2
0.3
Partial Spearman with Δα (median over tissues)

(l)

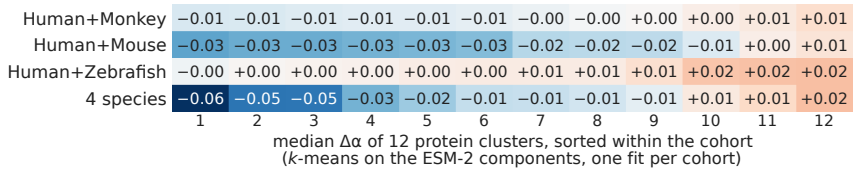

Kidney: AKI programs in mouse IRI
label-null p
human · mouse
PT
TAL
EC
FIB
PC
myeloid
DCT
IC
lymphoid
0.005 · 0.010
0.025 · 0.045
0.015 · 0.005
0.030 · 0.025
0.040 · 0.050
0.025 · 0.075
0.33 · 0.040
0.27 · 0.010
1.00 · 1.00
0
50
100
150
200
programs significant in human AKI
transfers to mouse
does not transfer
opposite direction
(a)
IRI 2 d vs 4–20 wk
no 2 d fit
no stage fit
no 2 d fit
no stage fit
0
100
200
transferring programs
0.0
0.4
0.8
program NES, Spearman
mouse IRI 2 d
mouse IRI 4–20 wk
(b)
Healthy-tissue test: does Δα predict transfer?
kidney: AKI vs IRI
PT
TAL
EC
FIB
PC
myeloid
DCT
IC
lymphoid
liver: MASH vs Western diet
hepatocyte
islet: T2D vs db/db, mSTZ
β
α
δ
ductal
endothelial
stellate
−0.08
−0.04
0
0.04
0.08
Δα vs gene-level transfer, ρ_p (95% CI)
(c)
Program transfer vs label null
observed, above null 95%
observed, within null
human-label null, median–95%
0
50
100
150
200
transferring programs
(d)
Cippà 2018 injury genes
acute lists
(primary)
late lists
0.51
0.51
Chimera Δα
profile correlation
0.70
0.72
0.4
0.5
0.6
0.7
0.8
stratified AUROC,
shared vs human-specific
(e)


weight path
gradient × input
integrated gradients
ablation
88%
77%
17%
76%
18%
38%
9%
47%
0
20
40
60
80
100
% of total importance
ESM-2 components
ESM-2 distance between species
mean ortholog identity
expression
identity spread
(a)
expression-only correction contribution
ESM-2-only correction contribution
ρ(ESM-2, expr) = -0.000
ρ(ESM-2, full) = +0.592
ρ(expr, full) = +0.675
Δα
(b)
Liver Human+Mouse
no gene features (per-gene gate)
random gene features
protein-language features set to zero
expression features set to zero
counts shuffled across genes (expression features follow): readout stays with the gene's sequence
readout follows the moved counts
counts shuffled within expression bins: readout stays with the gene's sequence
readout follows the moved counts
Liver 4 species
random gene features
protein-language features set to zero
degenerate run: Δα constant
expression features set to zero
counts shuffled across genes (all features detached): readout stays with the gene's sequence
readout follows the moved counts
hepatocytes and endothelial cells only
1 of 3 seeds failed training
Testis Human+Mouse
random gene features
counts shuffled within expression bins: readout stays with the gene's sequence
readout follows the moved counts
−0.25
0.00
0.25
0.50
0.75
1.00
Δα: control run vs normal training (Spearman ρ across genes)
count evidence S_g: control run vs normal training (Spearman ρ across genes)
seed floor: one normal seed vs the others (range)
projection null for 'stays with sequence' (2.5-97.5%)
Δα, one control run
S_g, one control run
(c)

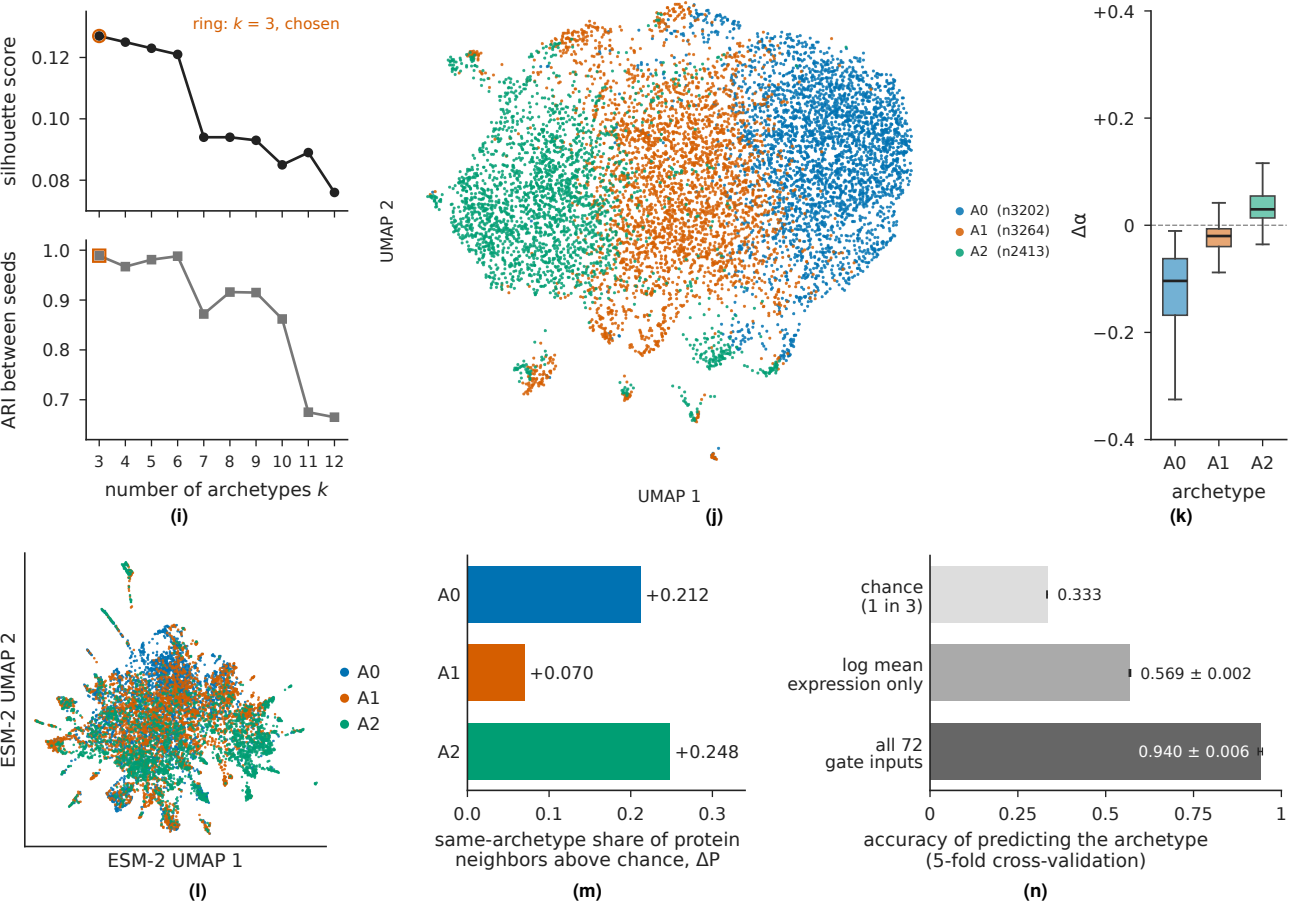


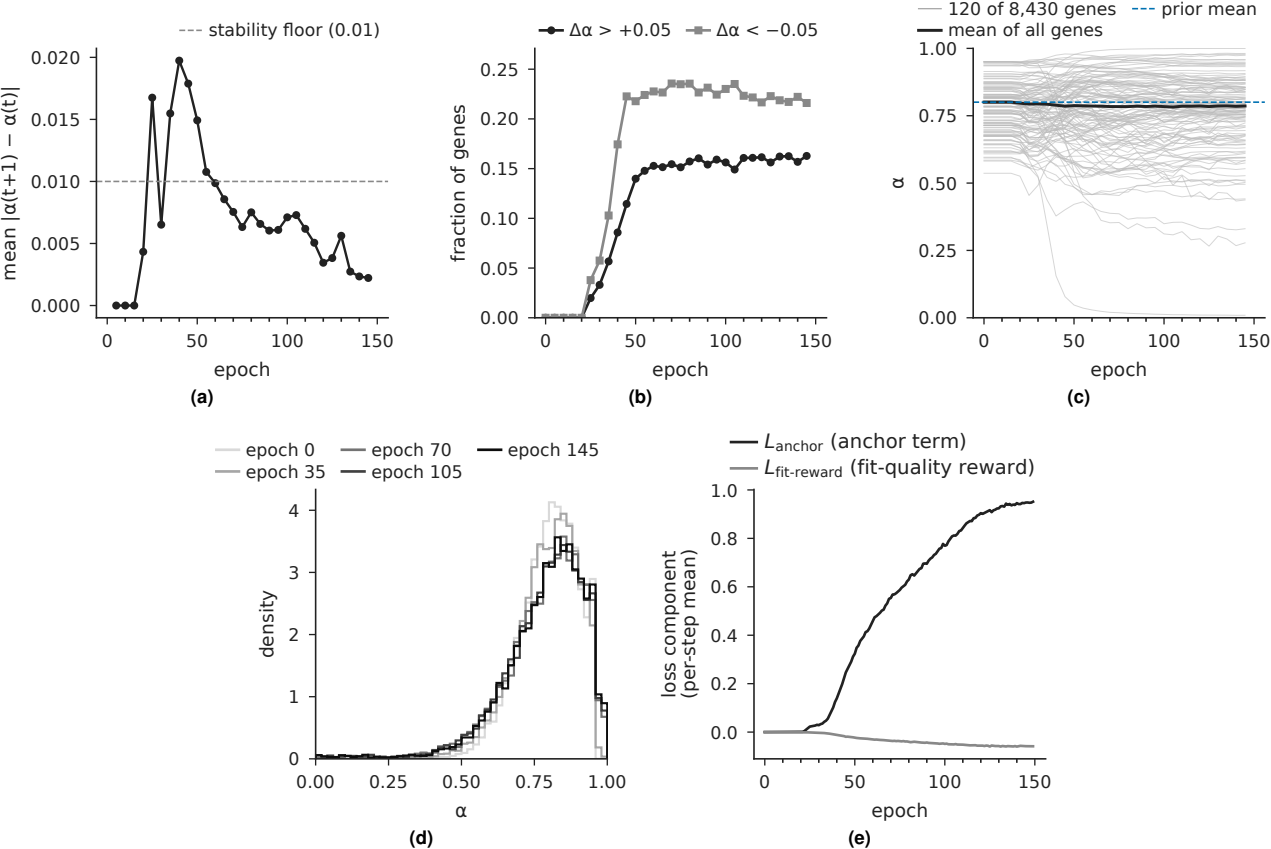


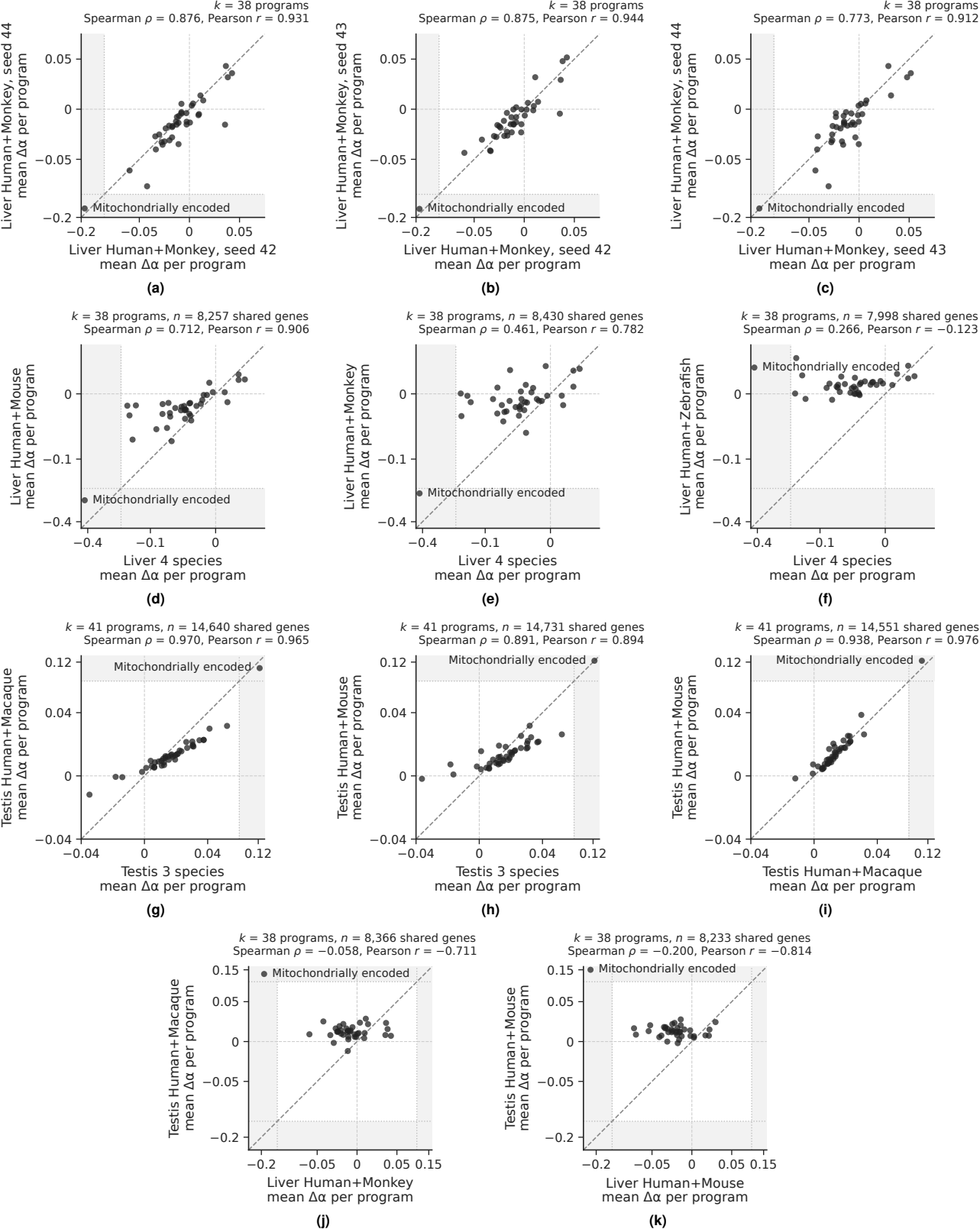
- Nuclear receptors (n = 121)
- Zinc-finger TFs (n = 114)
- DNA replication and repair (n = 112)
- Forkhead and other TFs (n = 112)
- Signaling kinases (n = 105)
- Cell cycle (n = 100)
- Splicing factors (n = 70)
- OXPHOS, nuclear-encoded (n = 66)
- Homeobox TFs (n = 58)
- Mitochondrial ribosome (n = 58)
- Microtubules (n = 56)
- ECM: other (n = 55)

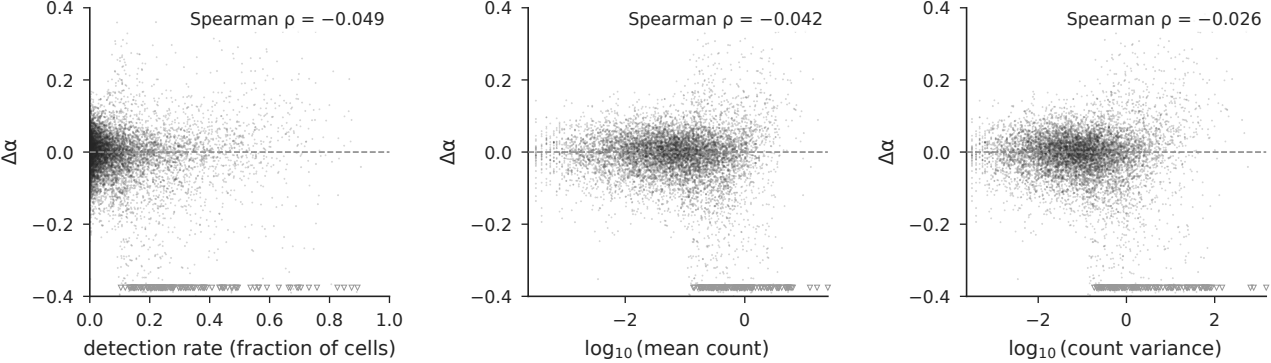


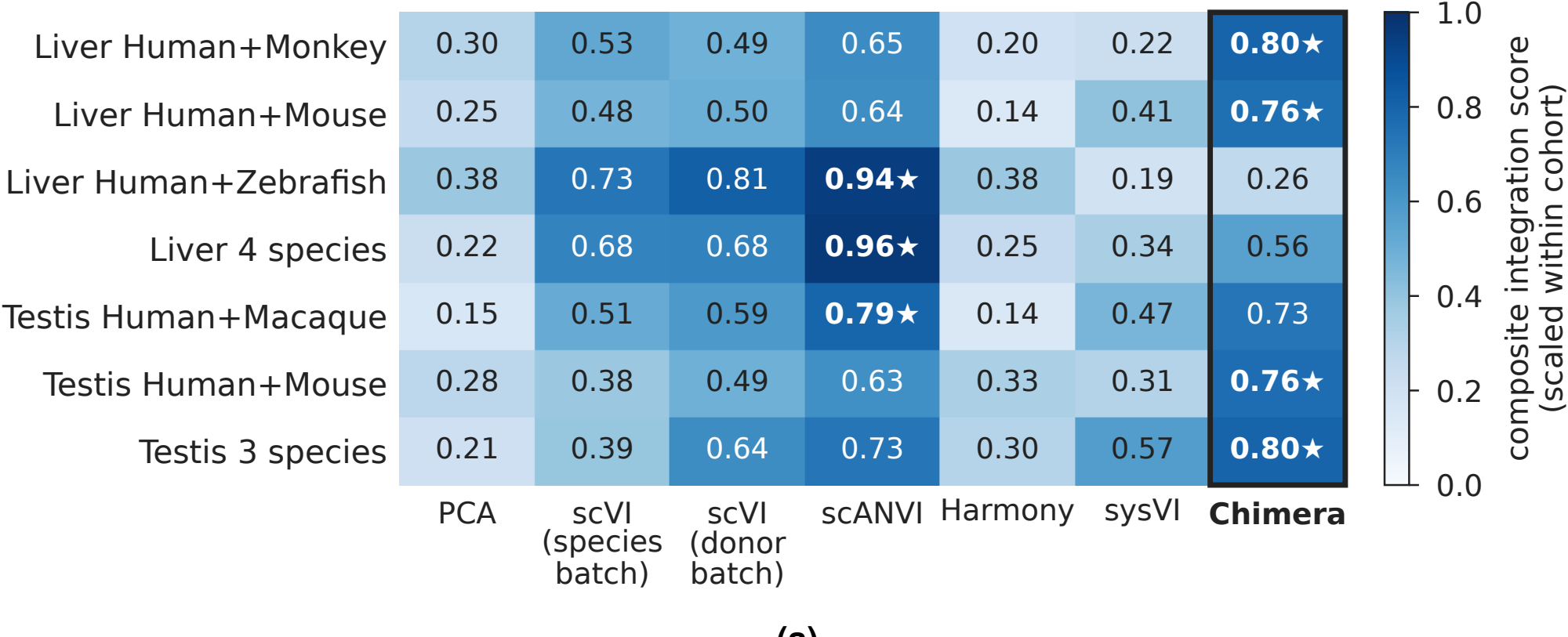
(h)


ring: k = 3, chosen
silhouette score
0.12
0.10
0.08
ARI between seeds
1.0
0.9
0.8
0.7
3 4 5 6 7 8 9 10 11 12
number of archetypes k
(i)
UMAP 2
UMAP 1
A0 (n3202)
A1 (n3264)
A2 (n2413)
(j)
+0.4
+0.2
0
−0.2
−0.4
Δα
A0 A1 A2
archetype
(k)
ESM-2 UMAP 2
ESM-2 UMAP 1
A0
A1
A2
(l)
A0
A1
A2
+0.212
+0.070
+0.248
0.0 0.1 0.2 0.3
same-archetype share of protein neighbors above chance, ΔP
(m)
chance (1 in 3)
log mean expression only
all 72 gate inputs
0.333
0.569 ± 0.002
0.940 ± 0.006
0 0.25 0.5 0.75 1
accuracy of predicting the archetype (5-fold cross-validation)
(n)

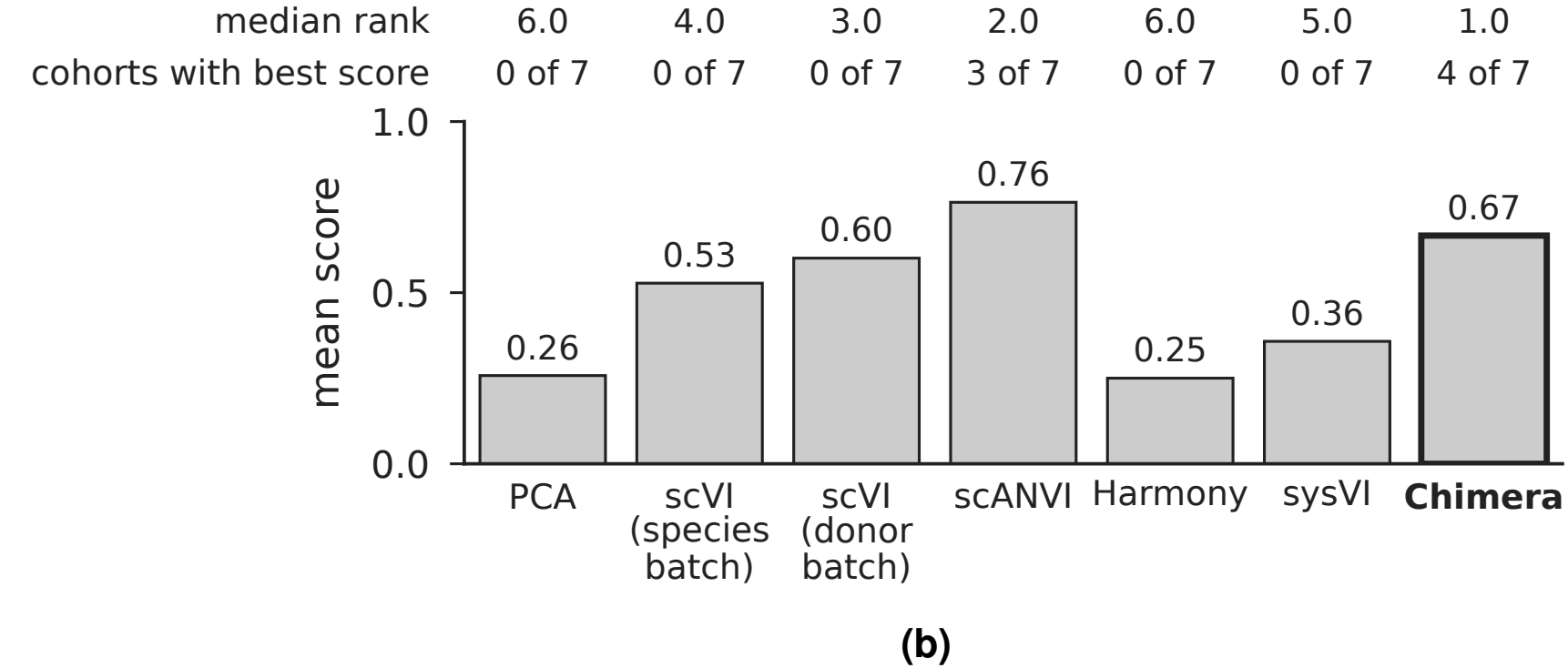

stability floor (0.01)
mean |α(t+1) − α(t)|
epoch
(a)
Δα > +0.05
Δα < −0.05
fraction of genes
epoch
(b)
120 of 8,430 genes
prior mean
mean of all genes
α
epoch
(c)
epoch 0
epoch 35
epoch 70
epoch 105
epoch 145
density
α
(d)
L_{anchor} (anchor term)
$L_{\text{fit-reward}}$ (fit-quality reward)
loss component (per-step mean)
epoch
(e)


k = 38 programs
Spearman ρ = 0.876, Pearson r = 0.931
Liver Human+Monkey, seed 44 mean Δα per program
Liver Human+Monkey, seed 42 mean Δα per program
Mitochondrially encoded
(a)
k = 38 programs
Spearman ρ = 0.875, Pearson r = 0.944
Liver Human+Monkey, seed 43 mean Δα per program
Liver Human+Monkey, seed 42 mean Δα per program
Mitochondrially encoded
(b)
k = 38 programs
Spearman ρ = 0.773, Pearson r = 0.912
Liver Human+Monkey, seed 44 mean Δα per program
Liver Human+Monkey, seed 43 mean Δα per program
Mitochondrially encoded
(c)
k = 38 programs, n = 8,257 shared genes
Spearman ρ = 0.712, Pearson r = 0.906
Liver Human+Mouse mean Δα per program
Liver 4 species mean Δα per program
Mitochondrially encoded
(d)
k = 38 programs, n = 8,430 shared genes
Spearman ρ = 0.461, Pearson r = 0.782
Liver Human+Monkey mean Δα per program
Liver 4 species mean Δα per program
Mitochondrially encoded
(e)
k = 38 programs, n = 7,998 shared genes
Spearman ρ = 0.266, Pearson r = −0.123
Liver Human+Zebrafish mean Δα per program
Liver 4 species mean Δα per program
Mitochondrially encoded
(f)
k = 41 programs, n = 14,640 shared genes
Spearman ρ = 0.970, Pearson r = 0.965
Testis Human+Macaque mean Δα per program
Testis 3 species mean Δα per program
Mitochondrially encoded
(g)
k = 41 programs, n = 14,731 shared genes
Spearman ρ = 0.891, Pearson r = 0.894
Testis Human+Mouse mean Δα per program
Testis 3 species mean Δα per program
Mitochondrially encoded
(h)
k = 41 programs, n = 14,551 shared genes
Spearman ρ = 0.938, Pearson r = 0.976
Testis Human+Mouse mean Δα per program
Testis Human+Macaque mean Δα per program
Mitochondrially encoded
(i)
k = 38 programs, n = 8,366 shared genes
Spearman ρ = −0.058, Pearson r = −0.711
Testis Human+Macaque mean Δα per program
Liver Human+Monkey mean Δα per program
Mitochondrially encoded
(j)
k = 38 programs, n = 8,233 shared genes
Spearman ρ = −0.200, Pearson r = −0.814
Testis Human+Mouse mean Δα per program
Liver Human+Mouse mean Δα per program
Mitochondrially encoded
(k)

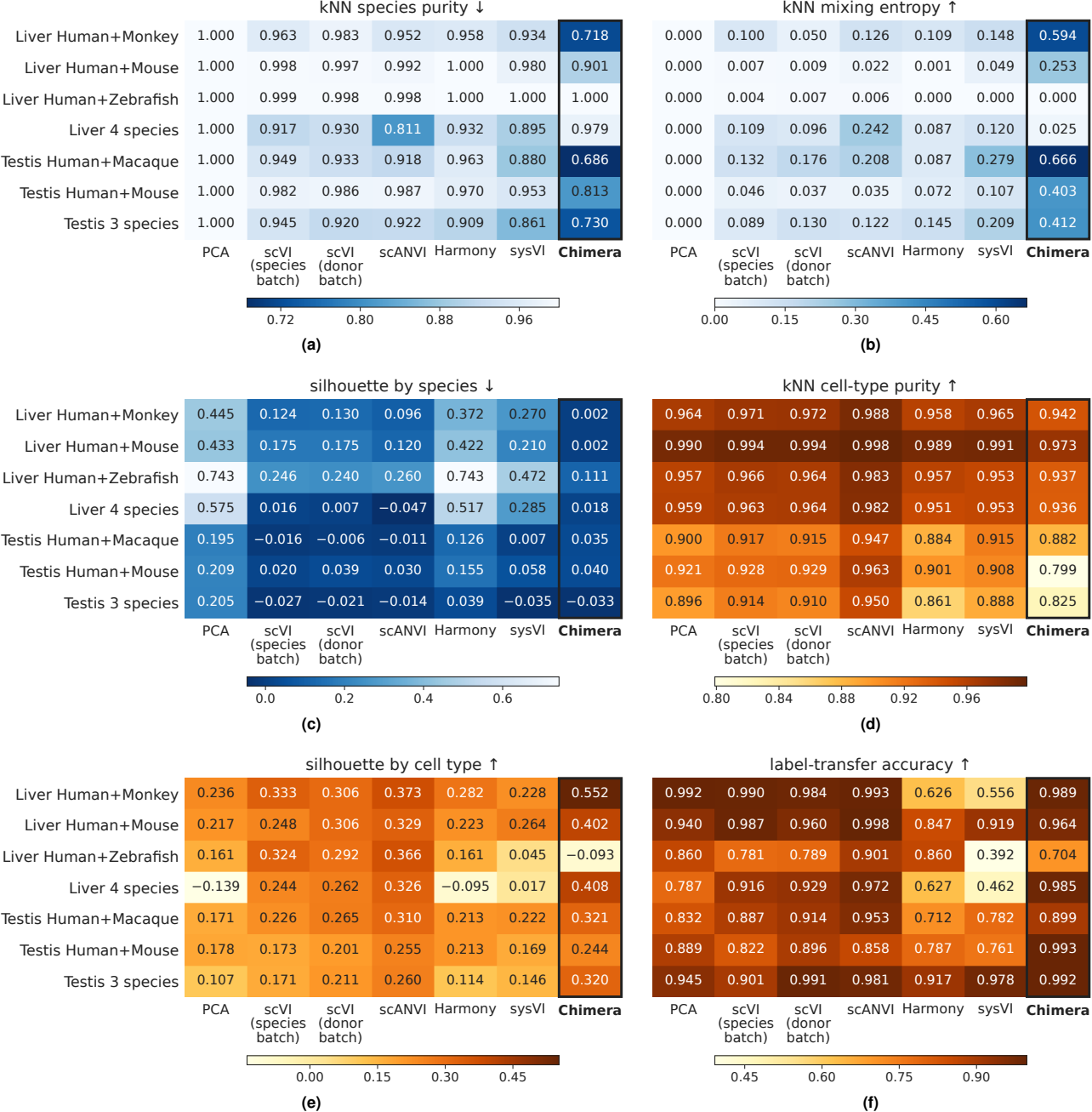

Spearman ρ = −0.049
Δα
0.4
0.2
0.0
−0.2
−0.4
0.0
0.2
0.4
0.6
0.8
1.0
detection rate (fraction of cells)
(a)
Spearman ρ = −0.042
Δα
−2
0
log10 (mean count)
(b)
Spearman ρ = −0.026
Δα
−2
0
2
log10 (count variance)
(c)

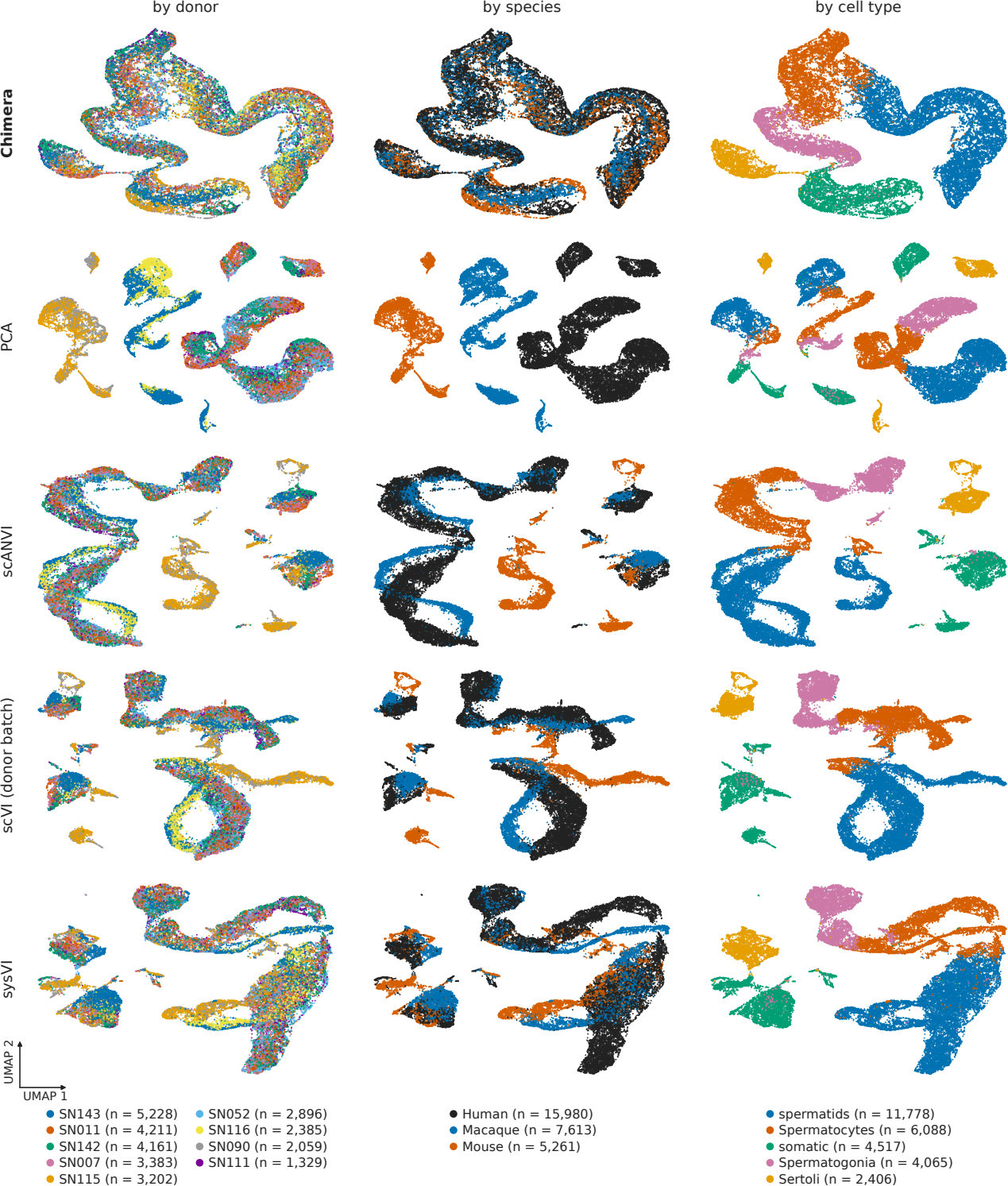

Liver Human+Monkey
Liver Human+Mouse
Liver Human+Zebrafish
Liver 4 species
Testis Human+Macaque
Testis Human+Mouse
Testis 3 species
0.30 0.53 0.49 0.65 0.20 0.22 0.80★
0.25 0.48 0.50 0.64 0.14 0.41 0.76★
0.38 0.73 0.81 0.94★ 0.38 0.19 0.26
0.22 0.68 0.68 0.96★ 0.25 0.34 0.56
0.15 0.51 0.59 0.79★ 0.14 0.47 0.73
0.28 0.38 0.49 0.63 0.33 0.31 0.76★
0.21 0.39 0.64 0.73 0.30 0.57 0.80★
PCA
scVI (species batch)
scVI (donor batch)
scANVI
Harmony
sysVI
Chimera
composite integration score (scaled within cohort)
1.0
0.8
0.6
0.4
0.2
0.0

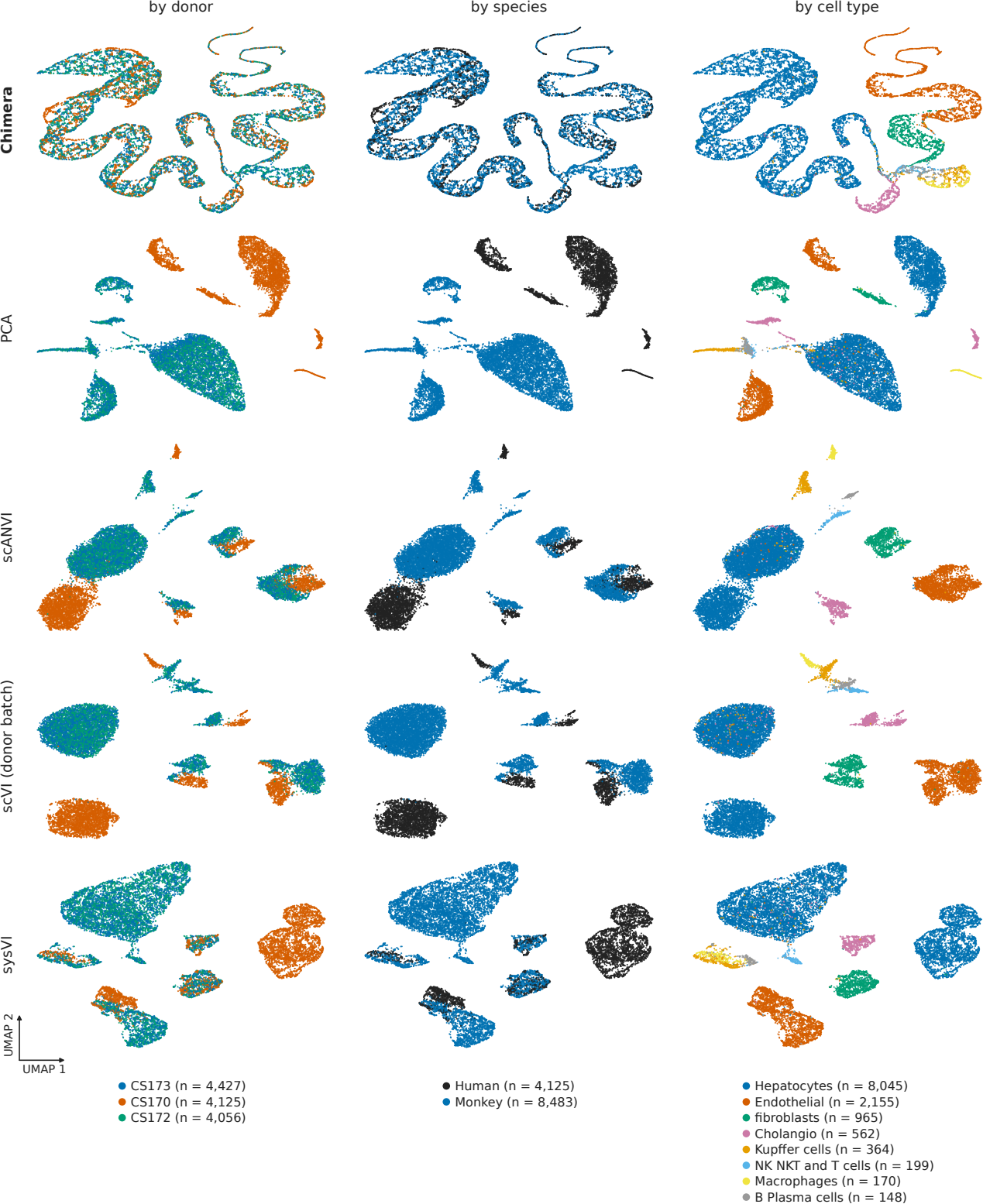
(a)

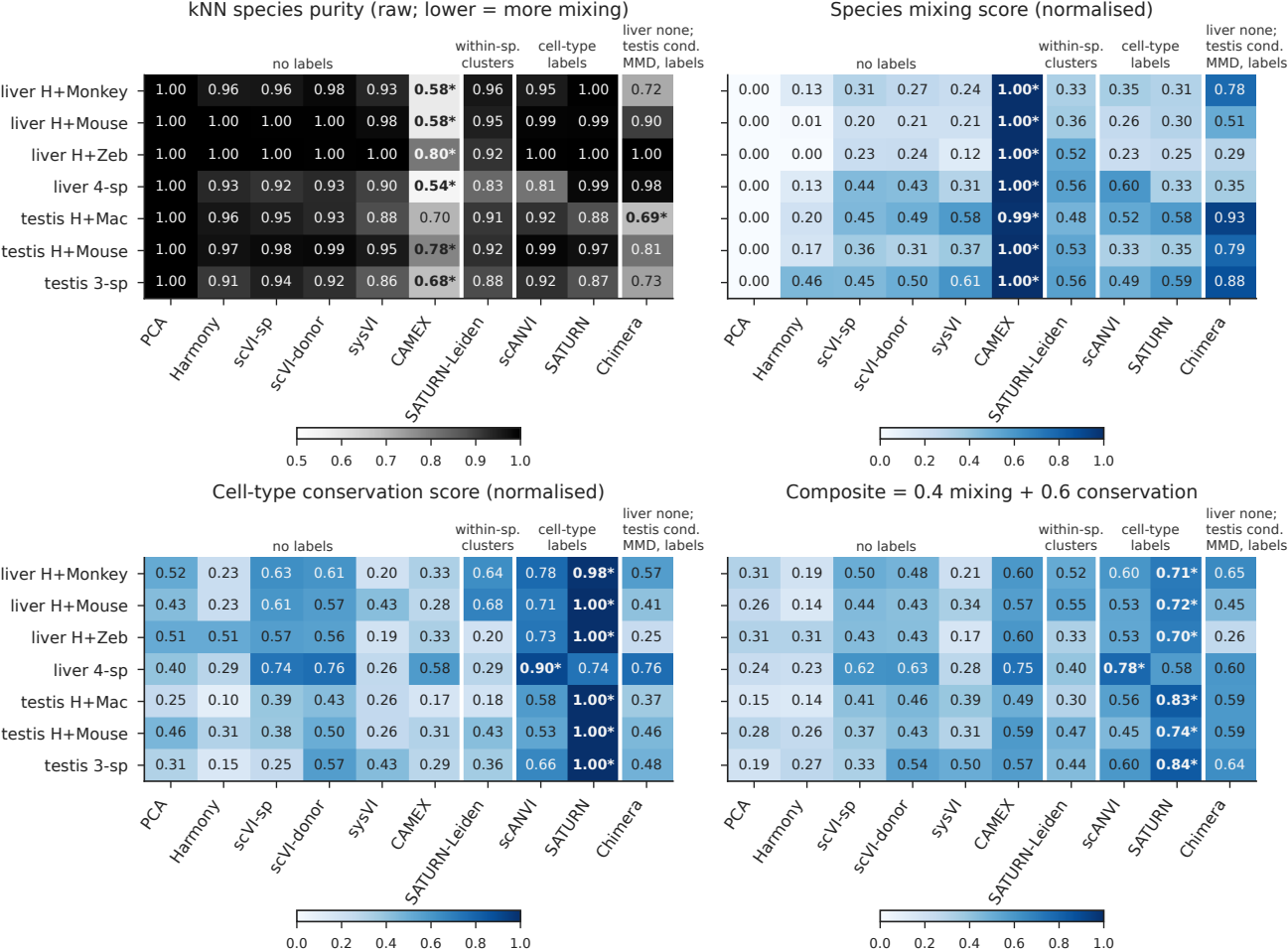

median rank 6.0 4.0 3.0 2.0 6.0 5.0 1.0
cohorts with best score 0 of 7 0 of 7 0 of 7 3 of 7 0 of 7 0 of 7 4 of 7
mean score
1.0
0.5
0.0
0.26
0.53
0.60
0.76
0.25
0.36
0.67
PCA
scVI (species batch)
scVI (donor batch)
scANVI
Harmony
sysVI
Chimera

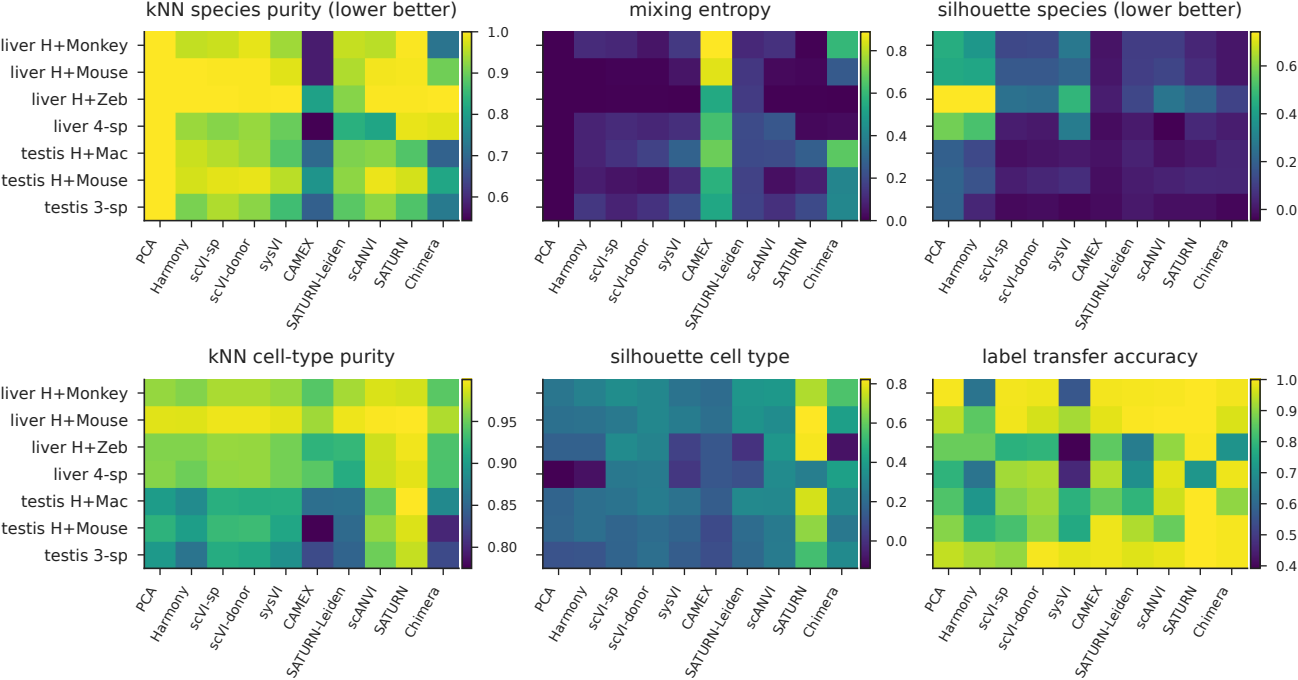
(b)

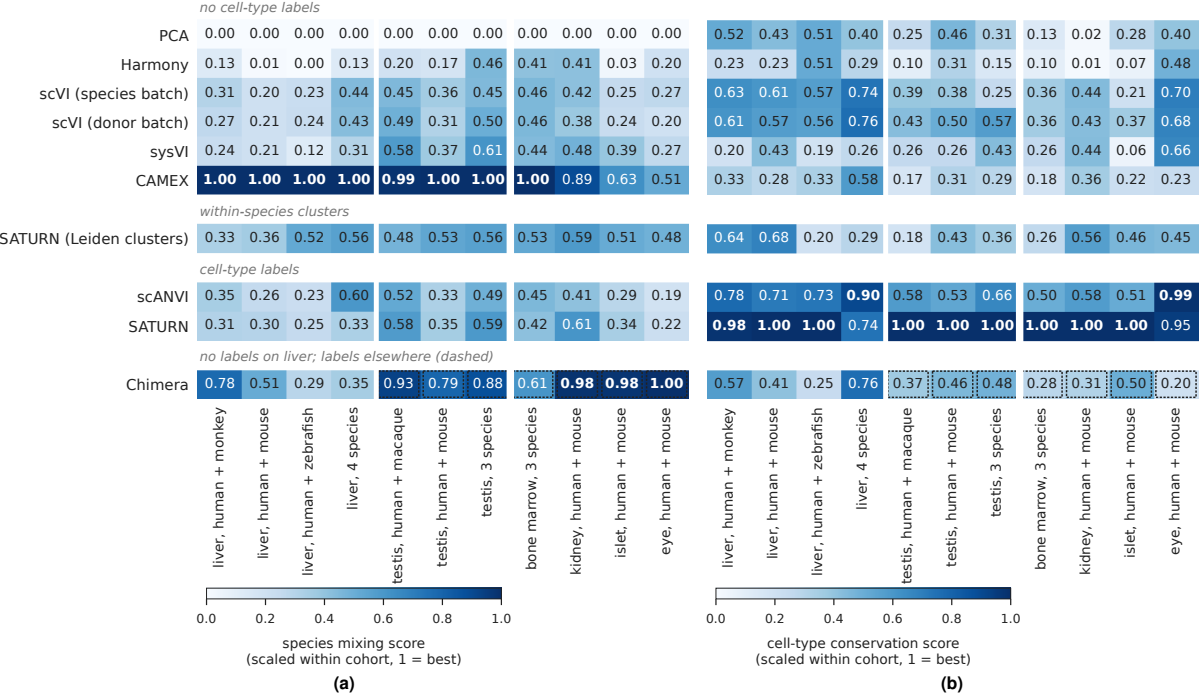

kNN species purity ↓
Liver Human+Monkey 1.000 0.963 0.983 0.952 0.958 0.934 0.718
Liver Human+Mouse 1.000 0.998 0.997 0.992 1.000 0.980 0.901
Liver Human+Zebrafish 1.000 0.999 0.998 0.998 1.000 1.000 1.000
Liver 4 species 1.000 0.917 0.930 0.811 0.932 0.895 0.979
Testis Human+Macaque 1.000 0.949 0.933 0.918 0.963 0.880 0.686
Testis Human+Mouse 1.000 0.982 0.986 0.987 0.970 0.953 0.813
Testis 3 species 1.000 0.945 0.920 0.922 0.909 0.861 0.730
PCA
scVI (species batch)
scVI (donor batch)
scANVI
Harmony
sysVI
Chimera
0.72 0.80 0.88 0.96
(a)
kNN mixing entropy ↑
0.000 0.100 0.050 0.126 0.109 0.148 0.594
0.000 0.007 0.009 0.022 0.001 0.049 0.253
0.000 0.004 0.007 0.006 0.000 0.000 0.000
0.000 0.109 0.096 0.242 0.087 0.120 0.025
0.000 0.132 0.176 0.208 0.087 0.279 0.666
0.000 0.046 0.037 0.035 0.072 0.107 0.403
0.000 0.089 0.130 0.122 0.145 0.209 0.412
PCA
scVI (species batch)
scVI (donor batch)
scANVI
Harmony
sysVI
Chimera
0.00 0.15 0.30 0.45 0.60
(b)
silhouette by species ↓
Liver Human+Monkey 0.445 0.124 0.130 0.096 0.372 0.270 0.002
Liver Human+Mouse 0.433 0.175 0.175 0.120 0.422 0.210 0.002
Liver Human+Zebrafish 0.743 0.246 0.240 0.260 0.743 0.472 0.111
Liver 4 species 0.575 0.016 0.007 −0.047 0.517 0.285 0.018
Testis Human+Macaque 0.195 −0.016 −0.006 −0.011 0.126 0.007 0.035
Testis Human+Mouse 0.209 0.020 0.039 0.030 0.155 0.058 0.040
Testis 3 species 0.205 −0.027 −0.021 −0.014 0.039 −0.035 −0.033
PCA
scVI (species batch)
scVI (donor batch)
scANVI
Harmony
sysVI
Chimera
0.0 0.2 0.4 0.6
(c)
kNN cell-type purity ↑
0.964 0.971 0.972 0.988 0.958 0.965 0.942
0.990 0.994 0.994 0.998 0.989 0.991 0.973
0.957 0.966 0.964 0.983 0.957 0.953 0.937
0.959 0.963 0.964 0.982 0.951 0.953 0.936
0.900 0.917 0.915 0.947 0.884 0.915 0.882
0.921 0.928 0.929 0.963 0.901 0.908 0.799
0.896 0.914 0.910 0.950 0.861 0.888 0.825
PCA
scVI (species batch)
scVI (donor batch)
scANVI
Harmony
sysVI
Chimera
0.80 0.84 0.88 0.92 0.96
(d)
silhouette by cell type ↑
Liver Human+Monkey 0.236 0.333 0.306 0.373 0.282 0.228 0.552
Liver Human+Mouse 0.217 0.248 0.306 0.329 0.223 0.264 0.402
Liver Human+Zebrafish 0.161 0.324 0.292 0.366 0.161 0.045 −0.093
Liver 4 species −0.139 0.244 0.262 0.326 −0.095 0.017 0.408
Testis Human+Macaque 0.171 0.226 0.265 0.310 0.213 0.222 0.321
Testis Human+Mouse 0.178 0.173 0.201 0.255 0.213 0.169 0.244
Testis 3 species 0.107 0.171 0.211 0.260 0.114 0.146 0.320
PCA
scVI (species batch)
scVI (donor batch)
scANVI
Harmony
sysVI
Chimera
0.00 0.15 0.30 0.45
(e)
label-transfer accuracy ↑
0.992 0.990 0.984 0.993 0.626 0.556 0.989
0.940 0.987 0.960 0.998 0.847 0.919 0.964
0.860 0.781 0.789 0.901 0.860 0.392 0.704
0.787 0.916 0.929 0.972 0.627 0.462 0.985
0.832 0.887 0.914 0.953 0.712 0.782 0.899
0.889 0.822 0.896 0.858 0.787 0.761 0.993
0.945 0.901 0.991 0.981 0.917 0.978 0.992
PCA
scVI (species batch)
scVI (donor batch)
scANVI
Harmony
sysVI
Chimera
0.45 0.60 0.75 0.90
(f)

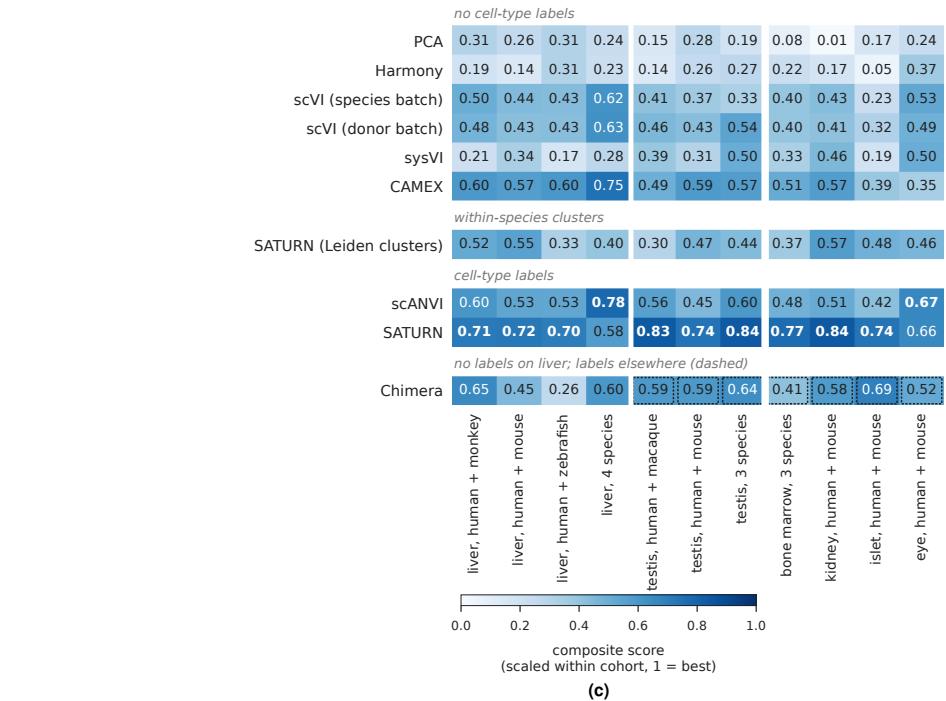

by donor
by species
by cell type
Chimera
PCA
scANVI
scVI (donor batch)
sysVI
UMAP 2
UMAP 1
SN143 (n = 5,228)
SN011 (n = 4,211)
SN142 (n = 4,161)
SN007 (n = 3,383)
SN115 (n = 3,202)
SN052 (n = 2,896)
SN116 (n = 2,385)
SN090 (n = 2,059)
SN111 (n = 1,329)
Human (n = 15,980)
Macaque (n = 7,613)
Mouse (n = 5,261)
spermatids (n = 11,778)
Spermatocytes (n = 6,088)
somatic (n = 4,517)
Spermatogonia (n = 4,065)
Sertoli (n = 2,406)

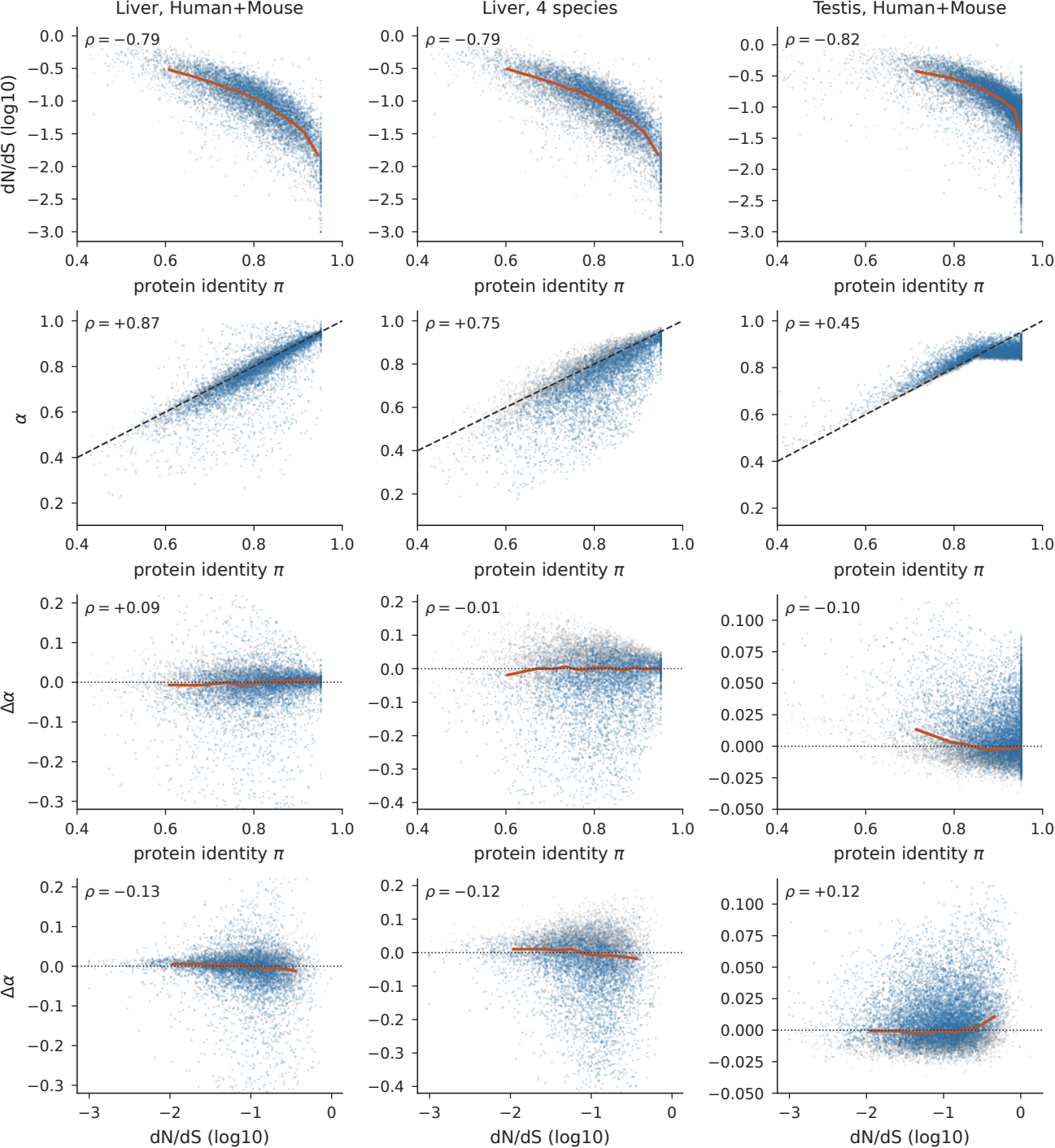

by donor
by species
by cell type
Chimera
PCA
scANVI
scVI (donor batch)
sysVI
UMAP 2
UMAP 1
CS173 (n = 4,427)
CS170 (n = 4,125)
CS172 (n = 4,056)
Human (n = 4,125)
Monkey (n = 8,483)
Hepatocytes (n = 8,045)
Endothelial (n = 2,155)
fibroblasts (n = 965)
Cholangio (n = 562)
Kupffer cells (n = 364)
NK NKT and T cells (n = 199)
Macrophages (n = 170)
B Plasma cells (n = 148)

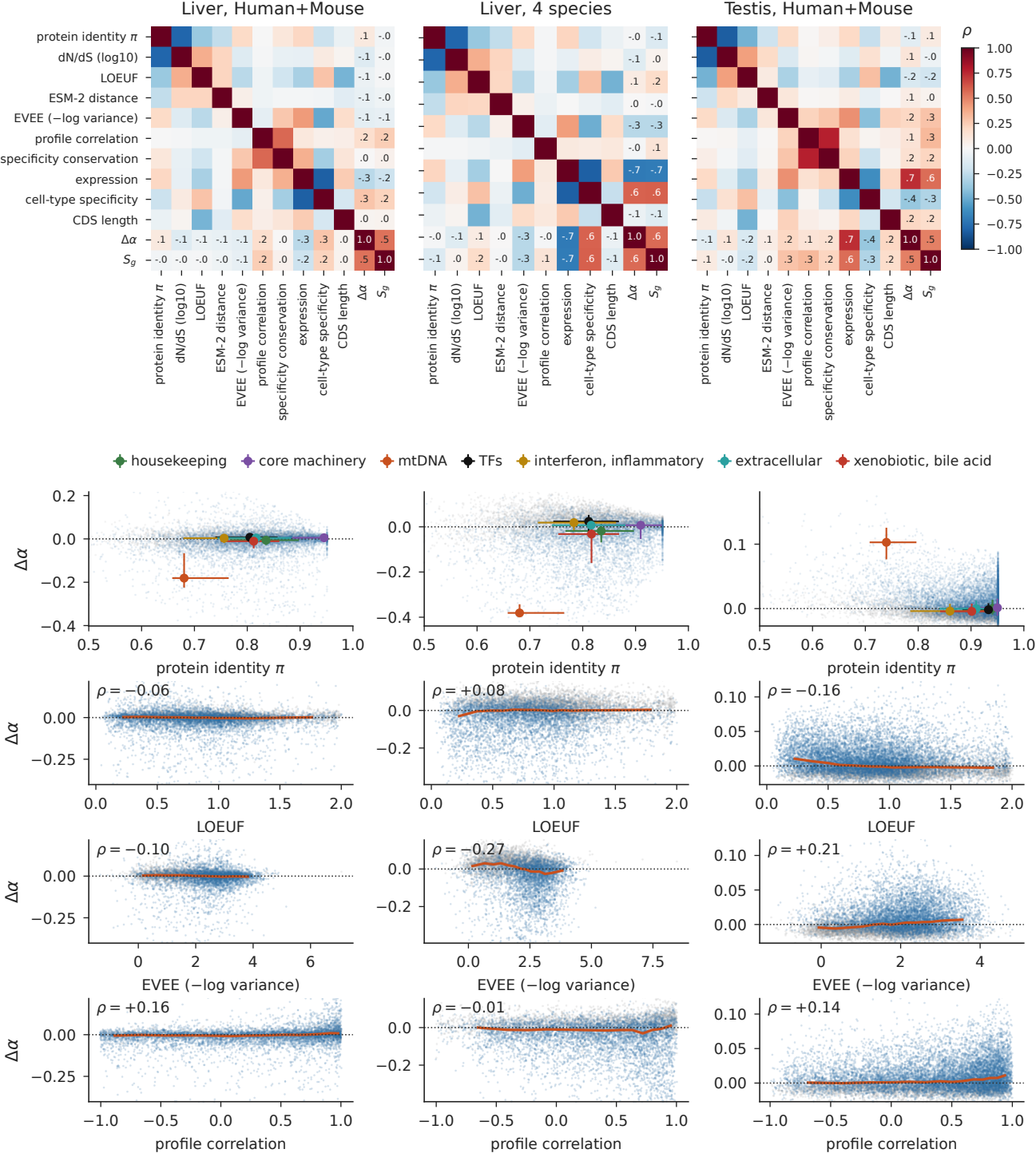

kNN species purity (raw; lower = more mixing)
Species mixing score (normalised)
Cell-type conservation score (normalised)
Composite = 0.4 mixing + 0.6 conservation
no labels
within-sp. clusters
cell-type labels
liver none; testis cond. MMD, labels
liver H+Monkey
liver H+Mouse
liver H+Zeb
liver 4-sp
testis H+Mac
testis H+Mouse
testis 3-sp
PCA
Harmony
scVI-sp
scVI-donor
sysVI
CAMEX
SATURN-Leiden
scANVI
SATURN
Chimera
0.5 0.6 0.7 0.8 0.9 1.0
0.0 0.2 0.4 0.6 0.8 1.0

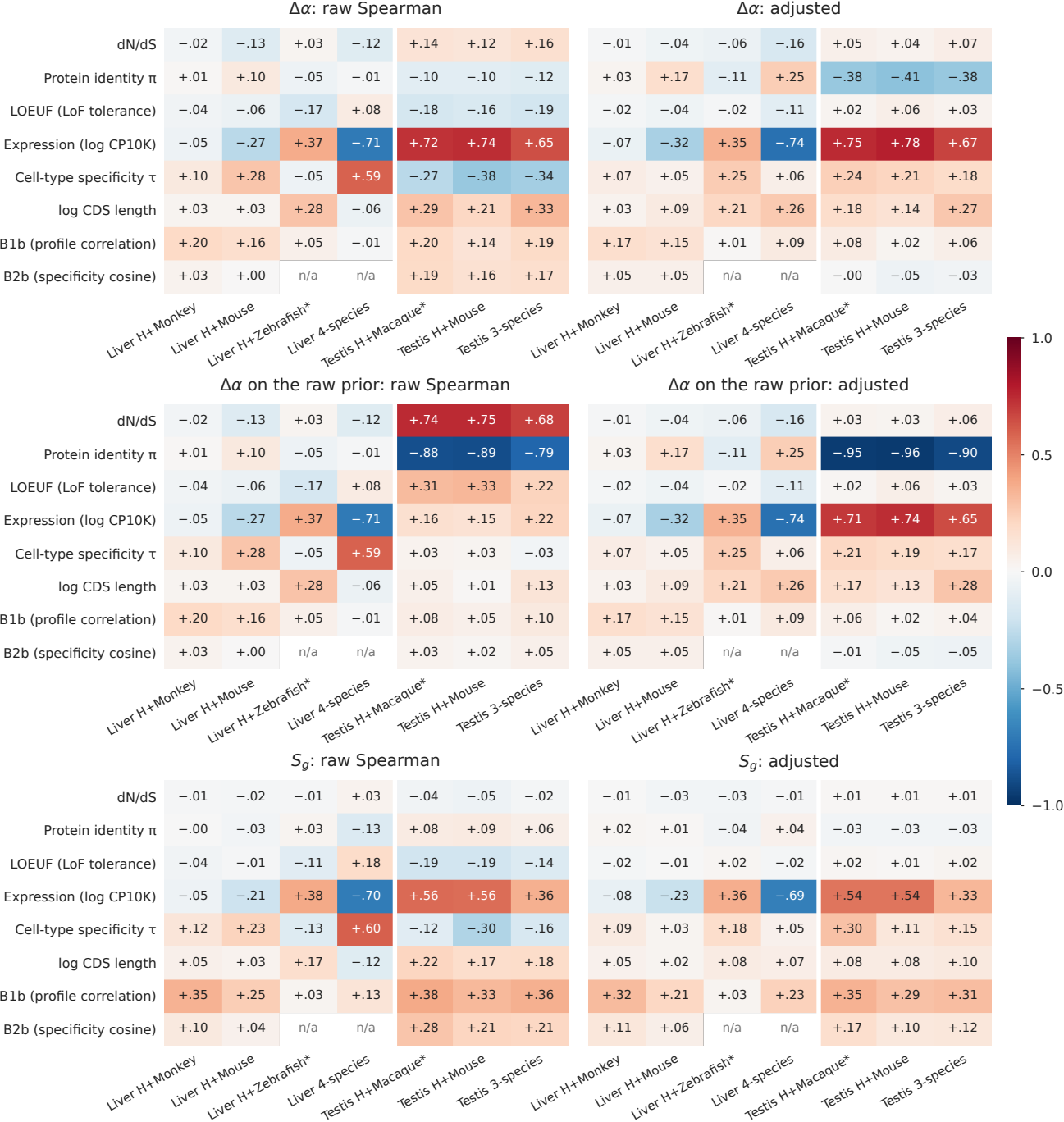

kNN species purity (lower better)
mixing entropy
silhouette species (lower better)
kNN cell-type purity
silhouette cell type
label transfer accuracy
liver H+Monkey
liver H+Mouse
liver H+Zeb
liver 4-sp
testis H+Mac
testis H+Mouse
testis 3-sp
PCA
Harmony
scVI-sp
scVI-donor
sysVI
CAMEX
SATURN-Leiden
scANVI
SATURN
Chimera


no cell-type labels
PCA 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.52 0.43 0.51 0.40 0.25 0.46 0.31 0.13 0.02 0.28 0.40
Harmony 0.13 0.01 0.00 0.13 0.20 0.17 0.46 0.41 0.41 0.03 0.20 0.23 0.23 0.51 0.29 0.10 0.31 0.15 0.10 0.01 0.07 0.48
scVI (species batch) 0.31 0.20 0.23 0.44 0.45 0.36 0.45 0.46 0.42 0.25 0.27 0.63 0.61 0.57 0.74 0.39 0.38 0.25 0.36 0.44 0.21 0.70
scVI (donor batch) 0.27 0.21 0.24 0.43 0.49 0.31 0.50 0.46 0.38 0.24 0.20 0.61 0.57 0.56 0.76 0.43 0.50 0.57 0.36 0.43 0.37 0.68
sysVI 0.24 0.21 0.12 0.31 0.58 0.37 0.61 0.44 0.48 0.39 0.27 0.20 0.43 0.19 0.26 0.26 0.26 0.43 0.26 0.44 0.06 0.66
CAMEX 1.00 1.00 1.00 1.00 0.99 1.00 1.00 1.00 0.89 0.63 0.51 0.33 0.28 0.33 0.58 0.17 0.31 0.29 0.18 0.36 0.22 0.23
within-species clusters
SATURN (Leiden clusters) 0.33 0.36 0.52 0.56 0.48 0.53 0.56 0.53 0.59 0.51 0.48 0.64 0.68 0.20 0.29 0.18 0.43 0.36 0.26 0.56 0.46 0.45
cell-type labels
scANVI 0.35 0.26 0.23 0.60 0.52 0.33 0.49 0.45 0.41 0.29 0.19 0.78 0.71 0.73 0.90 0.58 0.53 0.66 0.50 0.58 0.51 0.99
SATURN 0.31 0.30 0.25 0.33 0.58 0.35 0.59 0.42 0.61 0.34 0.22 0.98 1.00 1.00 0.74 1.00 1.00 1.00 1.00 1.00 1.00 0.95
no labels on liver; labels elsewhere (dashed)
Chimera 0.78 0.51 0.29 0.35 0.93 0.79 0.88 0.61 0.98 0.98 1.00 0.57 0.41 0.25 0.76 0.37 0.46 0.48 0.28 0.31 0.50 0.20
liver, human + monkey
liver, human + mouse
liver, human + zebrafish
liver, 4 species
testis, human + macaque
testis, human + mouse
testis, 3 species
bone marrow, 3 species
kidney, human + mouse
islet, human + mouse
eye, human + mouse
0.0 0.2 0.4 0.6 0.8 1.0
species mixing score
(scaled within cohort, 1 = best)
(a)
0.0 0.2 0.4 0.6 0.8 1.0
cell-type conservation score
(scaled within cohort, 1 = best)
(b)

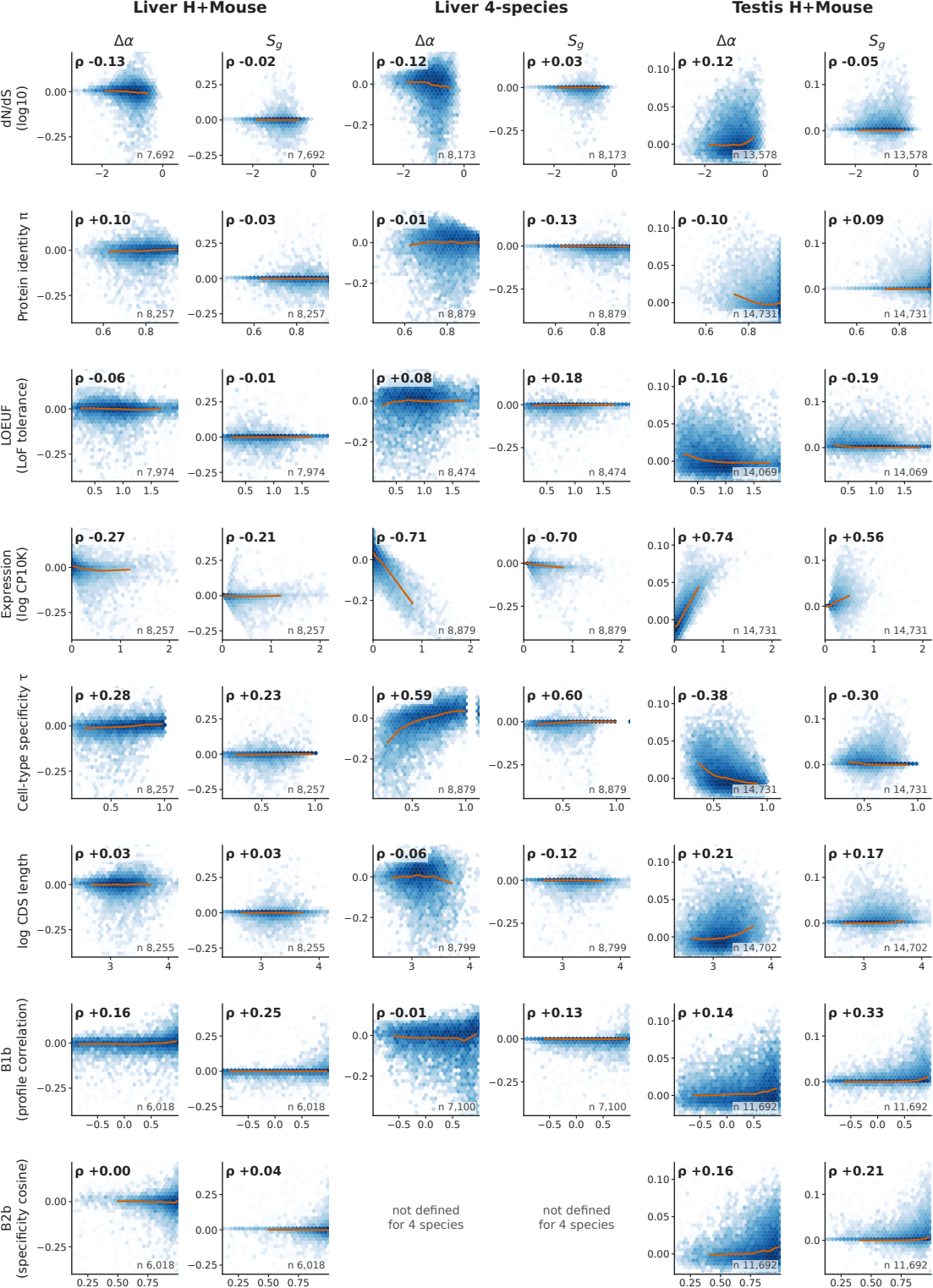

no cell-type labels
PCA 0.31 0.26 0.31 0.24 0.15 0.28 0.19 0.08 0.01 0.17 0.24
Harmony 0.19 0.14 0.31 0.23 0.14 0.26 0.27 0.22 0.17 0.05 0.37
scVI (species batch) 0.50 0.44 0.43 0.62 0.41 0.37 0.33 0.40 0.43 0.23 0.53
scVI (donor batch) 0.48 0.43 0.43 0.63 0.46 0.43 0.54 0.40 0.41 0.32 0.49
sysVI 0.21 0.34 0.17 0.28 0.39 0.31 0.50 0.33 0.46 0.19 0.50
CAMEX 0.60 0.57 0.60 0.75 0.49 0.59 0.57 0.51 0.57 0.39 0.35
within-species clusters
SATURN (Leiden clusters) 0.52 0.55 0.33 0.40 0.30 0.47 0.44 0.37 0.57 0.48 0.46
cell-type labels
scANVI 0.60 0.53 0.53 0.78 0.56 0.45 0.60 0.48 0.51 0.42 0.67
SATURN 0.71 0.72 0.70 0.58 0.83 0.74 0.84 0.77 0.84 0.74 0.66
no labels on liver; labels elsewhere (dashed)
Chimera 0.65 0.45 0.26 0.60 0.59 0.59 0.64 0.41 0.58 0.69 0.52
liver, human + monkey
liver, human + mouse
liver, human + zebrafish
liver, 4 species
testis, human + macaque
testis, human + mouse
testis, 3 species
bone marrow, 3 species
kidney, human + mouse
islet, human + mouse
eye, human + mouse
0.0 0.2 0.4 0.6 0.8 1.0
composite score
(scaled within cohort, 1 = best)
(c)

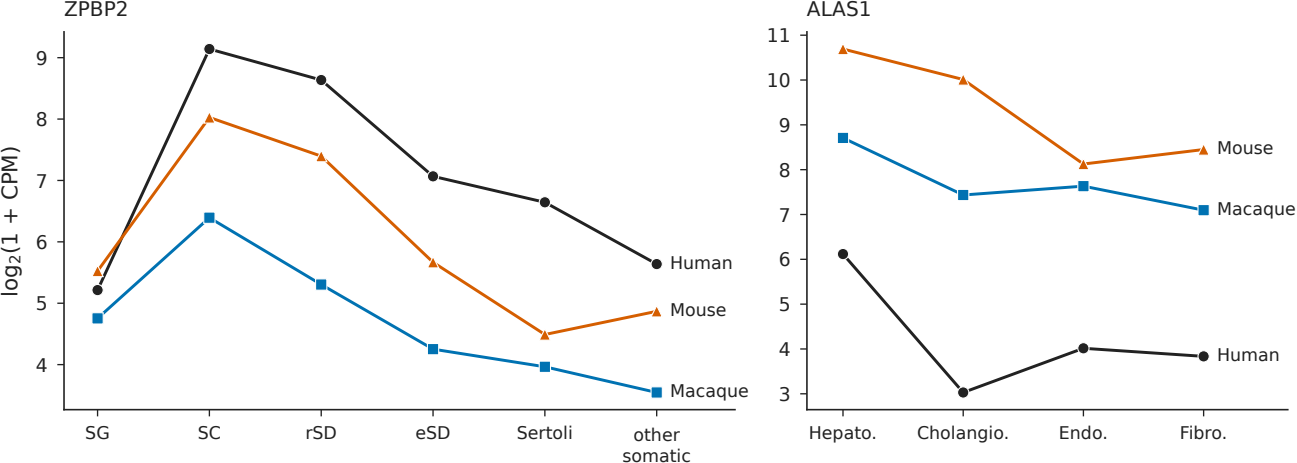

Liver, Human+Mouse
Liver, 4 species
Testis, Human+Mouse
dN/dS (log10)
α
Δα
ρ = −0.79
ρ = −0.79
ρ = −0.82
ρ = +0.87
ρ = +0.75
ρ = +0.45
ρ = +0.09
ρ = −0.01
ρ = −0.10
ρ = −0.13
ρ = −0.12
ρ = +0.12
protein identity π
dN/dS (log10)

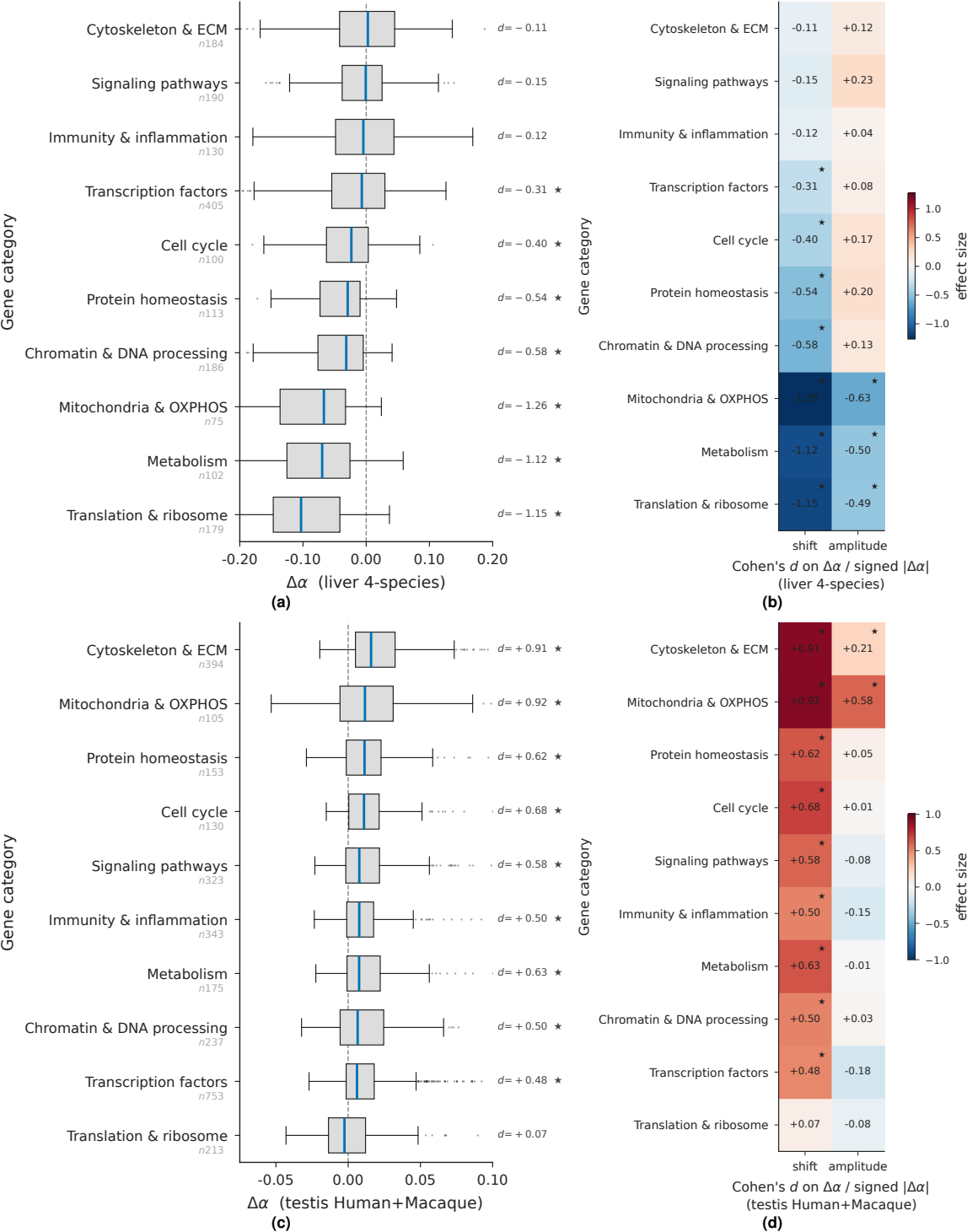

Liver, Human+Mouse
Liver, 4 species
Testis, Human+Mouse
protein identity π
dN/dS (log10)
LOEUF
ESM-2 distance
EVEE (−log variance)
profile correlation
specificity conservation
expression
cell-type specificity
CDS length
Δα
S_g
ρ
housekeeping
core machinery
mtDNA
TFs
interferon, inflammatory
extracellular
xenobiotic, bile acid
ρ = −0.06
ρ = +0.08
ρ = −0.16
ρ = −0.10
ρ = −0.27
ρ = +0.21
ρ = +0.16
ρ = −0.01
ρ = +0.14

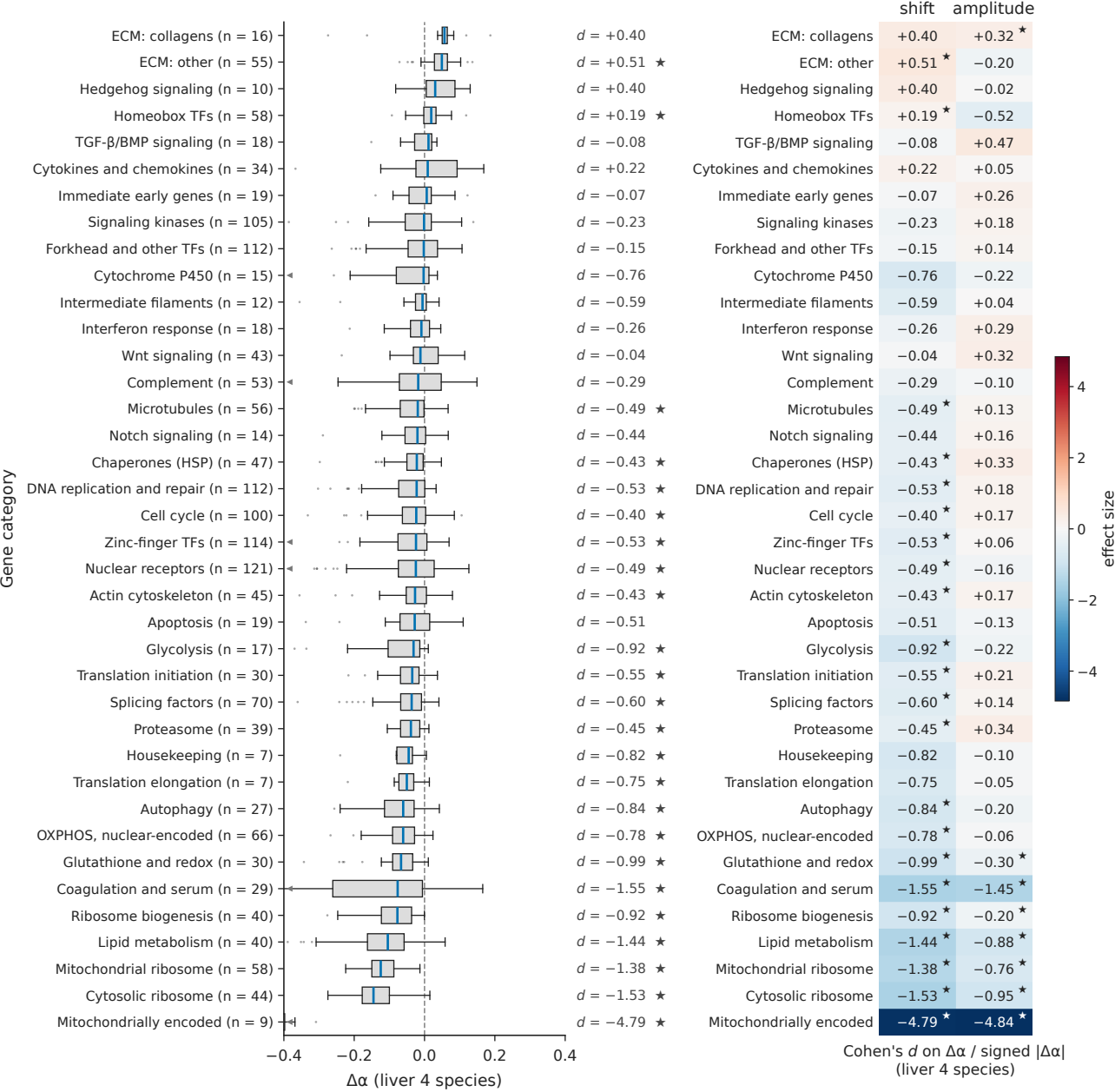


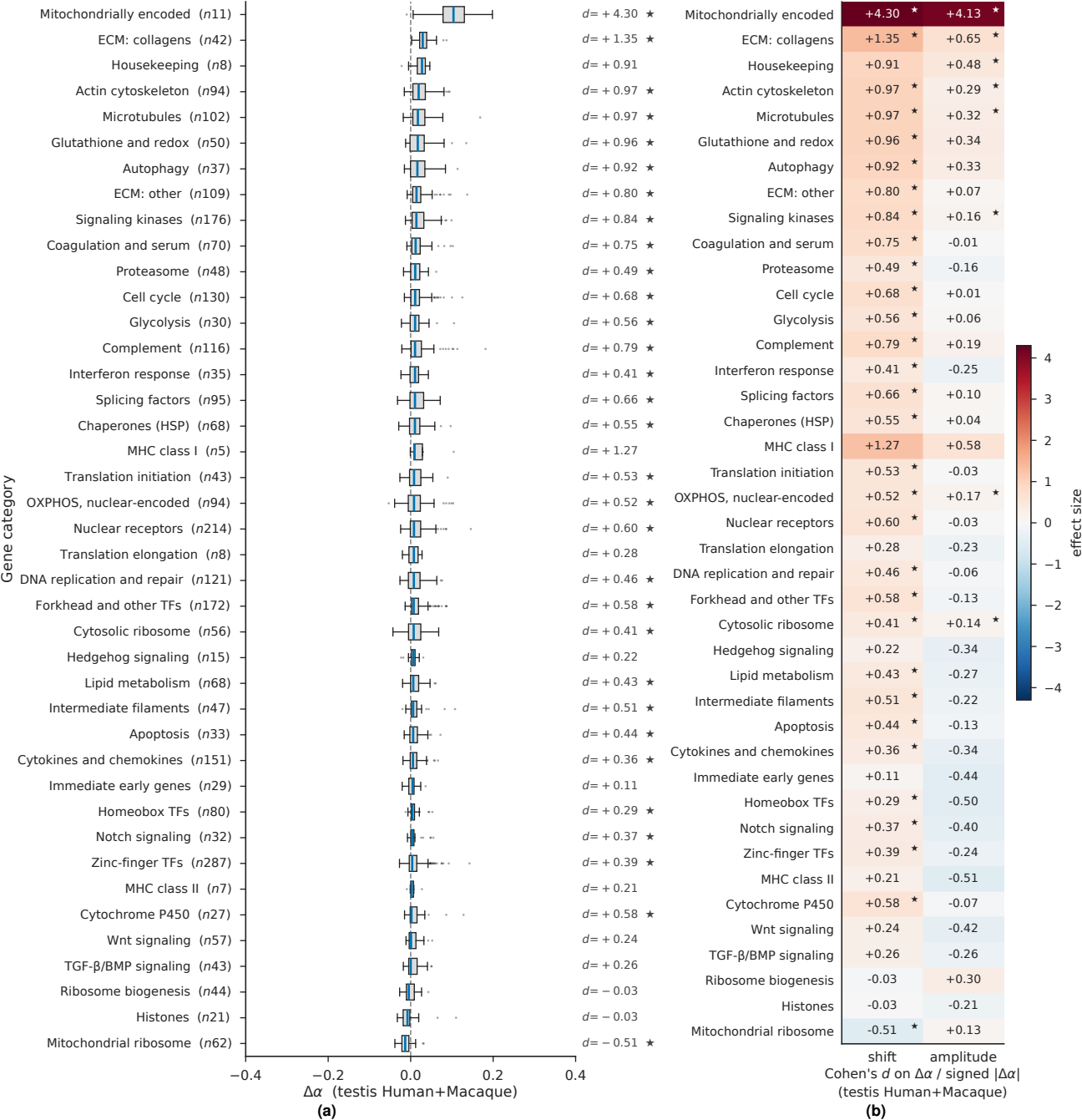
Adjusted: rank partial correlation given the other covariates (identity, expression, detection, CDS length). * one seed. n/a: B2b is not defined for these cohorts.

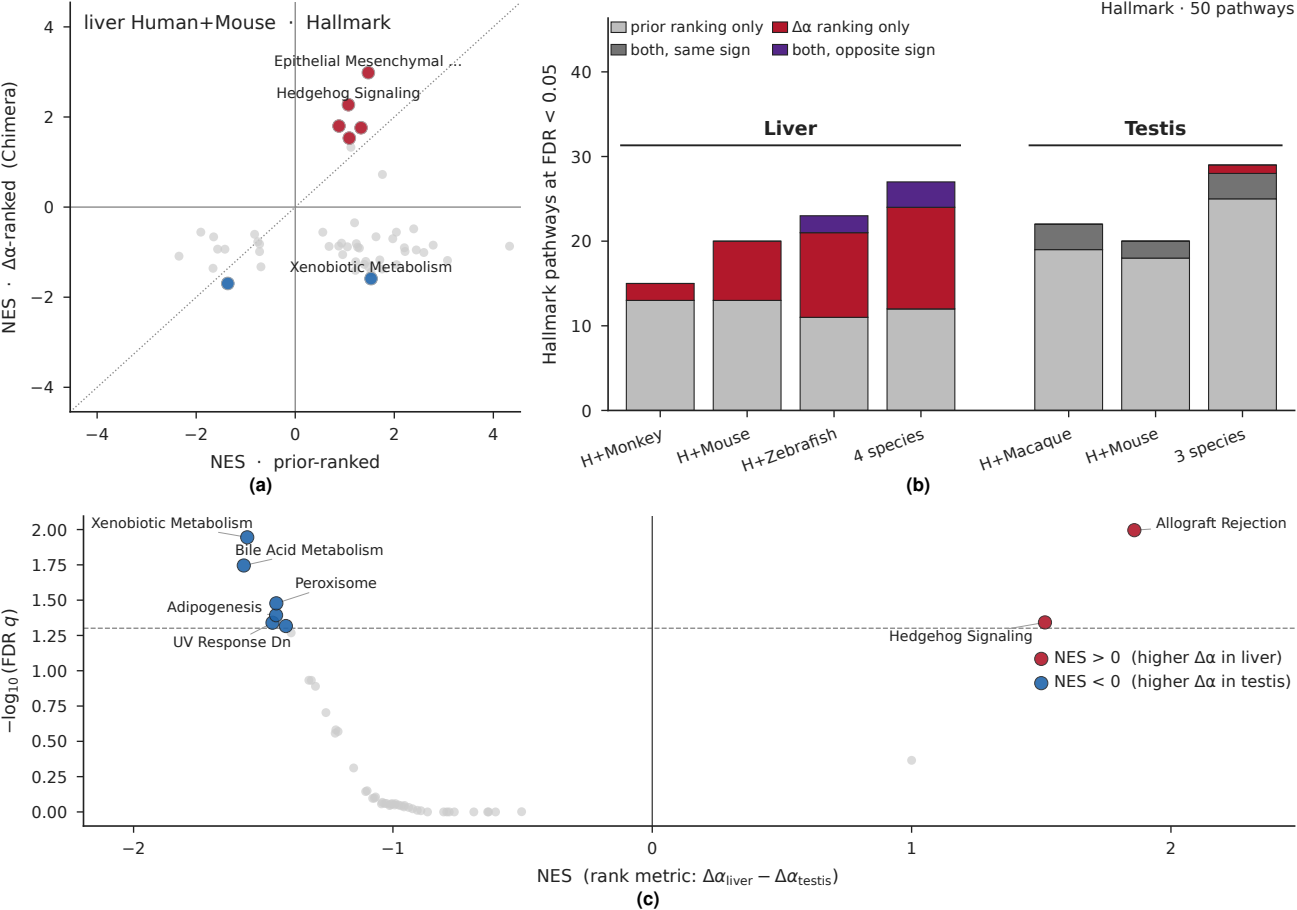

Liver H+Mouse
Liver 4-species
Testis H+Mouse
Δα
S_g
dN/dS (log10)
Protein identity π
LOEUF (LoF tolerance)
Expression (log CP10K)
Cell-type specificity τ
log CDS length
B1b (profile correlation)
B2b (specificity cosine)
ρ -0.13
n 7,692
ρ -0.02
n 7,692
ρ -0.12
n 8,173
ρ +0.03
n 8,173
ρ +0.12
n 13,578
ρ -0.05
n 13,578
ρ +0.10
n 8,257
ρ -0.03
n 8,257
ρ -0.01
n 8,879
ρ -0.13
n 8,879
ρ -0.10
n 14,731
ρ +0.09
n 14,731
ρ -0.06
n 7,974
ρ -0.01
n 7,974
ρ +0.08
n 8,474
ρ +0.18
n 8,474
ρ -0.16
n 14,069
ρ -0.19
n 14,069
ρ -0.27
n 8,257
ρ -0.21
n 8,257
ρ -0.71
n 8,879
ρ -0.70
n 8,879
ρ +0.74
n 14,731
ρ +0.56
n 14,731
ρ +0.28
n 8,257
ρ +0.23
n 8,257
ρ +0.59
n 8,879
ρ +0.60
n 8,879
ρ -0.38
n 14,731
ρ -0.30
n 14,731
ρ +0.03
n 8,255
ρ +0.03
n 8,255
ρ -0.06
n 8,799
ρ -0.12
n 8,799
ρ +0.21
n 14,702
ρ +0.17
n 14,702
ρ +0.16
n 6,018
ρ +0.25
n 6,018
ρ -0.01
n 7,100
ρ +0.13
n 7,100
ρ +0.14
n 11,692
ρ +0.33
n 11,692
ρ +0.00
n 6,018
ρ +0.04
n 6,018
not defined for 4 species
not defined for 4 species
ρ +0.16
n 11,692
ρ +0.21
n 11,692

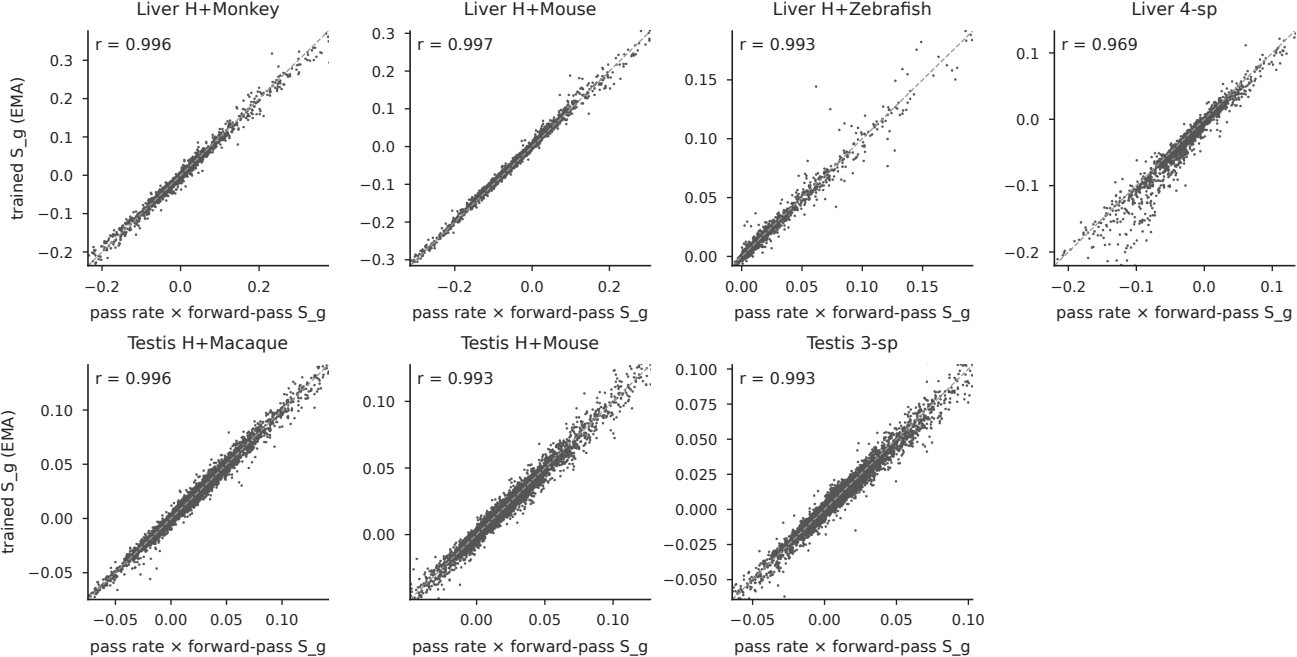

ZPBP2
log2(1 + CPM)
9
8
7
6
5
4
Human
Mouse
Macaque
SG
SC
rSD
eSD
Sertoli
other somatic
ALAS1
11
10
9
8
7
6
5
4
3
Mouse
Macaque
Human
Hepato.
Cholangio.
Endo.
Fibro.

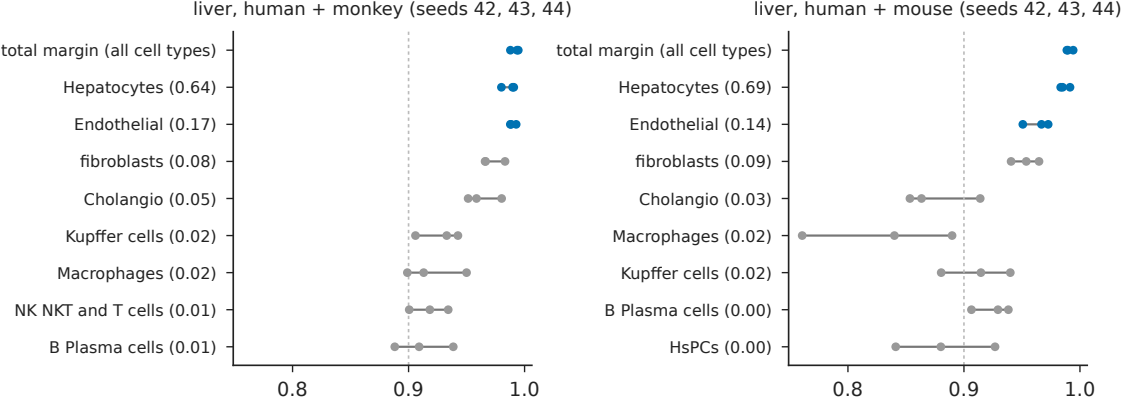

Gene category
Cytoskeleton & ECM
n184
d = −0.11
Signaling pathways
n190
d = −0.15
Immunity & inflammation
n130
d = −0.12
Transcription factors
n405
d = −0.31 ★
Cell cycle
n100
d = −0.40 ★
Protein homeostasis
n113
d = −0.54 ★
Chromatin & DNA processing
n186
d = −0.58 ★
Mitochondria & OXPHOS
n75
d = −1.26 ★
Metabolism
n102
d = −1.12 ★
Translation & ribosome
n179
d = −1.15 ★
-0.20
-0.10
0.00
0.10
0.20
Δα (liver 4-species)
(a)
Gene category
Cytoskeleton & ECM
-0.11
+0.12
Signaling pathways
-0.15
+0.23
Immunity & inflammation
-0.12
+0.04
Transcription factors
★
-0.31
+0.08
Cell cycle
★
-0.40
+0.17
Protein homeostasis
★
-0.54
+0.20
Chromatin & DNA processing
★
-0.58
+0.13
Mitochondria & OXPHOS
★
-1.26
★
-0.63
Metabolism
★
-1.12
★
-0.50
Translation & ribosome
★
-1.15
★
-0.49
shift
amplitude
1.0
0.5
0.0
−0.5
−1.0
effect size
Cohen's d on Δα / signed |Δα|
(liver 4-species)
(b)
Gene category
Cytoskeleton & ECM
n394
d = +0.91 ★
Mitochondria & OXPHOS
n105
d = +0.92 ★
Protein homeostasis
n153
d = +0.62 ★
Cell cycle
n130
d = +0.68 ★
Signaling pathways
n323
d = +0.58 ★
Immunity & inflammation
n343
d = +0.50 ★
Metabolism
n175
d = +0.63 ★
Chromatin & DNA processing
n237
d = +0.50 ★
Transcription factors
n753
d = +0.48 ★
Translation & ribosome
n213
d = +0.07
-0.05
0.00
0.05
0.10
Δα (testis Human+Macaque)
(c)
Gene category
Cytoskeleton & ECM
★
+0.91
★
+0.21
Mitochondria & OXPHOS
★
+0.92
★
+0.58
Protein homeostasis
★
+0.62
+0.05
Cell cycle
★
+0.68
+0.01
Signaling pathways
★
+0.58
-0.08
Immunity & inflammation
★
+0.50
-0.15
Metabolism
★
+0.63
-0.01
Chromatin & DNA processing
★
+0.50
+0.03
Transcription factors
★
+0.48
-0.18
Translation & ribosome
+0.07
-0.08
shift
amplitude
1.0
0.5
0.0
−0.5
−1.0
effect size
Cohen's d on Δα / signed |Δα|
(testis Human+Macaque)
(d)


Gene category
ECM: collagens (n = 16)
ECM: other (n = 55)
Hedgehog signaling (n = 10)
Homeobox TFs (n = 58)
TGF-β/BMP signaling (n = 18)
Cytokines and chemokines (n = 34)
Immediate early genes (n = 19)
Signaling kinases (n = 105)
Forkhead and other TFs (n = 112)
Cytochrome P450 (n = 15)
Intermediate filaments (n = 12)
Interferon response (n = 18)
Wnt signaling (n = 43)
Complement (n = 53)
Microtubules (n = 56)
Notch signaling (n = 14)
Chaperones (HSP) (n = 47)
DNA replication and repair (n = 112)
Cell cycle (n = 100)
Zinc-finger TFs (n = 114)
Nuclear receptors (n = 121)
Actin cytoskeleton (n = 45)
Apoptosis (n = 19)
Glycolysis (n = 17)
Translation initiation (n = 30)
Splicing factors (n = 70)
Proteasome (n = 39)
Housekeeping (n = 7)
Translation elongation (n = 7)
Autophagy (n = 27)
OXPHOS, nuclear-encoded (n = 66)
Glutathione and redox (n = 30)
Coagulation and serum (n = 29)
Ribosome biogenesis (n = 40)
Lipid metabolism (n = 40)
Mitochondrial ribosome (n = 58)
Cytosolic ribosome (n = 44)
Mitochondrially encoded (n = 9)
d = +0.40
d = +0.51 ★
d = +0.40
d = +0.19 ★
d = −0.08
d = +0.22
d = −0.07
d = −0.23
d = −0.15
d = −0.76
d = −0.59
d = −0.26
d = −0.04
d = −0.29
d = −0.49 ★
d = −0.44
d = −0.43 ★
d = −0.53 ★
d = −0.40 ★
d = −0.53 ★
d = −0.49 ★
d = −0.43 ★
d = −0.51
d = −0.92 ★
d = −0.55 ★
d = −0.60 ★
d = −0.45 ★
d = −0.82 ★
d = −0.75 ★
d = −0.84 ★
d = −0.78 ★
d = −0.99 ★
d = −1.55 ★
d = −0.92 ★
d = −1.44 ★
d = −1.38 ★
d = −1.53 ★
d = −4.79 ★
−0.4
−0.2
0.0
0.2
0.4
Δα (liver 4 species)
(a)
shift
amplitude
+0.40
+0.32 ★
+0.51 ★
−0.20
+0.40
−0.02
+0.19 ★
−0.52
−0.08
+0.47
+0.22
+0.05
−0.07
+0.26
−0.23
+0.18
−0.15
+0.14
−0.76
−0.22
−0.59
+0.04
−0.26
+0.29
−0.04
+0.32
−0.29
−0.10
−0.49 ★
+0.13
−0.44
+0.16
−0.43 ★
+0.33
−0.53 ★
+0.18
−0.40 ★
+0.17
−0.53 ★
+0.06
−0.49 ★
−0.16
−0.43 ★
+0.17
−0.51
−0.13
−0.92 ★
−0.22
−0.55 ★
+0.21
−0.60 ★
+0.14
−0.45 ★
+0.34
−0.82
−0.10
−0.75
−0.05
−0.84 ★
−0.20
−0.78 ★
−0.06
−0.99 ★
−0.30 ★
−1.55 ★
−1.45 ★
−0.92 ★
−0.20 ★
−1.44 ★
−0.88 ★
−1.38 ★
−0.76 ★
−1.53 ★
−0.95 ★
−4.79 ★
−4.84 ★
4
2
0
−2
−4
effect size
Cohen's d on Δα / signed |Δα|
(liver 4 species)
(b)

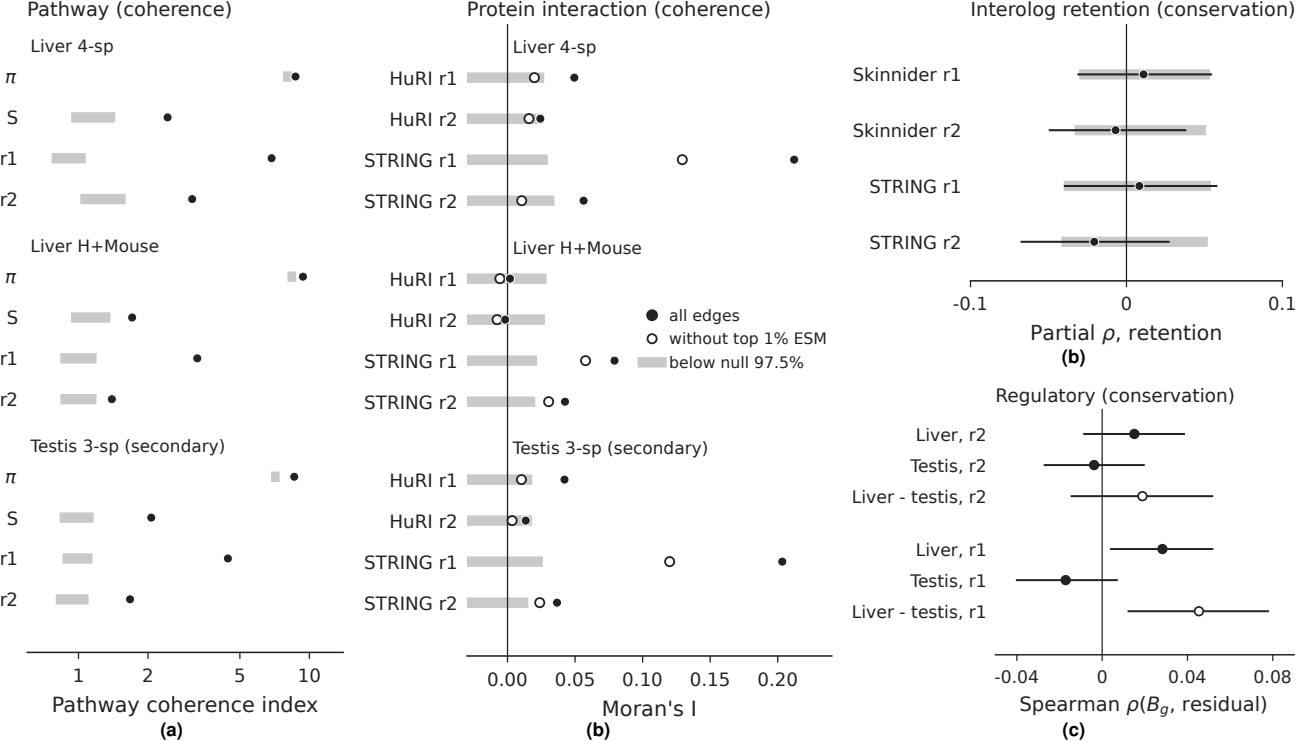

Gene category
Mitochondrially encoded (n11) d = +4.30 ★
ECM: collagens (n42) d = +1.35 ★
Housekeeping (n8) d = +0.91
Actin cytoskeleton (n94) d = +0.97 ★
Microtubules (n102) d = +0.97 ★
Glutathione and redox (n50) d = +0.96 ★
Autophagy (n37) d = +0.92 ★
ECM: other (n109) d = +0.80 ★
Signaling kinases (n176) d = +0.84 ★
Coagulation and serum (n70) d = +0.75 ★
Proteasome (n48) d = +0.49 ★
Cell cycle (n130) d = +0.68 ★
Glycolysis (n30) d = +0.56 ★
Complement (n116) d = +0.79 ★
Interferon response (n35) d = +0.41 ★
Splicing factors (n95) d = +0.66 ★
Chaperones (HSP) (n68) d = +0.55 ★
MHC class I (n5) d = +1.27
Translation initiation (n43) d = +0.53 ★
OXPHOS, nuclear-encoded (n94) d = +0.52 ★
Nuclear receptors (n214) d = +0.60 ★
Translation elongation (n8) d = +0.28
DNA replication and repair (n121) d = +0.46 ★
Forkhead and other TFs (n172) d = +0.58 ★
Cytosolic ribosome (n56) d = +0.41 ★
Hedgehog signaling (n15) d = +0.22
Lipid metabolism (n68) d = +0.43 ★
Intermediate filaments (n47) d = +0.51 ★
Apoptosis (n33) d = +0.44 ★
Cytokines and chemokines (n151) d = +0.36 ★
Immediate early genes (n29) d = +0.11
Homeobox TFs (n80) d = +0.29 ★
Notch signaling (n32) d = +0.37 ★
Zinc-finger TFs (n287) d = +0.39 ★
MHC class II (n7) d = +0.21
Cytochrome P450 (n27) d = +0.58 ★
Wnt signaling (n57) d = +0.24
TGF-β/BMP signaling (n43) d = +0.26
Ribosome biogenesis (n44) d = −0.03
Histones (n21) d = −0.03
Mitochondrial ribosome (n62) d = −0.51 ★
−0.4
−0.2
0.0
0.2
0.4
Δα (testis Human+Macaque)
(a)
Mitochondrially encoded +4.30 ★ +4.13 ★
ECM: collagens +1.35 ★ +0.65 ★
Housekeeping +0.91 +0.48 ★
Actin cytoskeleton +0.97 ★ +0.29 ★
Microtubules +0.97 ★ +0.32 ★
Glutathione and redox +0.96 ★ +0.34
Autophagy +0.92 ★ +0.33
ECM: other +0.80 ★ +0.07
Signaling kinases +0.84 ★ +0.16 ★
Coagulation and serum +0.75 ★ -0.01
Proteasome +0.49 ★ -0.16
Cell cycle +0.68 ★ +0.01
Glycolysis +0.56 ★ +0.06
Complement +0.79 ★ +0.19
Interferon response +0.41 ★ -0.25
Splicing factors +0.66 ★ +0.10
Chaperones (HSP) +0.55 ★ +0.04
MHC class I +1.27 +0.58
Translation initiation +0.53 ★ -0.03
OXPHOS, nuclear-encoded +0.52 ★ +0.17 ★
Nuclear receptors +0.60 ★ -0.03
Translation elongation +0.28 -0.23
DNA replication and repair +0.46 ★ -0.06
Forkhead and other TFs +0.58 ★ -0.13
Cytosolic ribosome +0.41 ★ +0.14 ★
Hedgehog signaling +0.22 -0.34
Lipid metabolism +0.43 ★ -0.27
Intermediate filaments +0.51 ★ -0.22
Apoptosis +0.44 ★ -0.13
Cytokines and chemokines +0.36 ★ -0.34
Immediate early genes +0.11 -0.44
Homeobox TFs +0.29 ★ -0.50
Notch signaling +0.37 ★ -0.40
Zinc-finger TFs +0.39 ★ -0.24
MHC class II +0.21 -0.51
Cytochrome P450 +0.58 ★ -0.07
Wnt signaling +0.24 -0.42
TGF-β/BMP signaling +0.26 -0.26
Ribosome biogenesis -0.03 +0.30
Histones -0.03 -0.21
Mitochondrial ribosome -0.51 ★ +0.13
shift
amplitude
Cohen's d on Δα / signed |Δα|
(testis Human+Macaque)
(b)
effect size
4
3
2
1
0
−1
−2
−3
−4

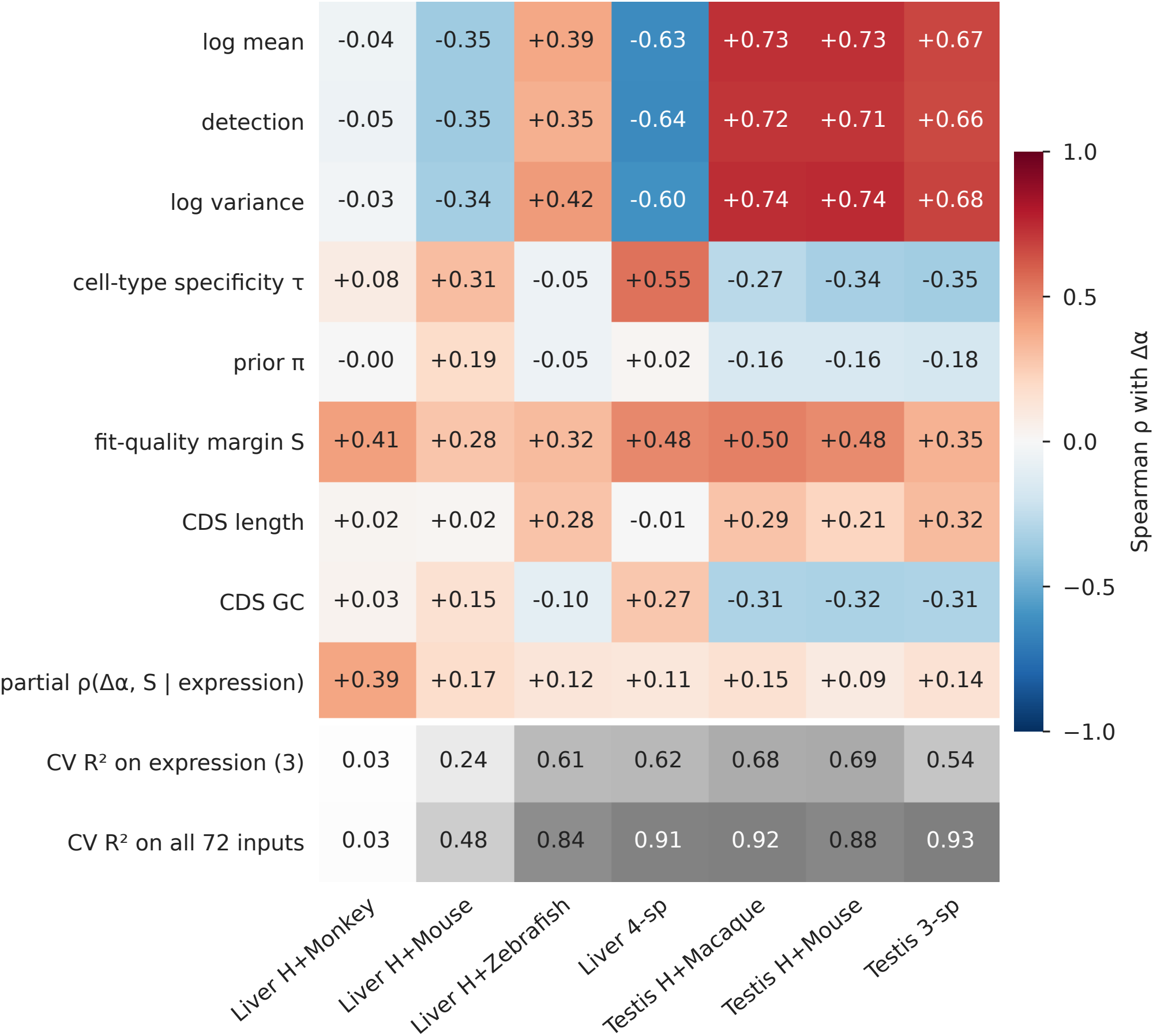

liver Human+Mouse · Hallmark
Epithelial Mesenchymal ...
Hedgehog Signaling
Xenobiotic Metabolism
NES · Δα-ranked (Chimera)
NES · prior-ranked
(a)
Hallmark · 50 pathways
prior ranking only
Δα ranking only
both, same sign
both, opposite sign
Liver
Testis
Hallmark pathways at FDR < 0.05
H+Monkey
H+Mouse
H+Zebrafish
4 species
H+Macaque
H+Mouse
3 species
(b)
Xenobiotic Metabolism
Bile Acid Metabolism
Peroxisome
Adipogenesis
UV Response Dn
Allograft Rejection
Hedgehog Signaling
NES > 0 (higher Δα in liver)
NES < 0 (higher Δα in testis)
−log10 (FDR q)
NES (rank metric: Δα_liver − Δα_testis)
(c)

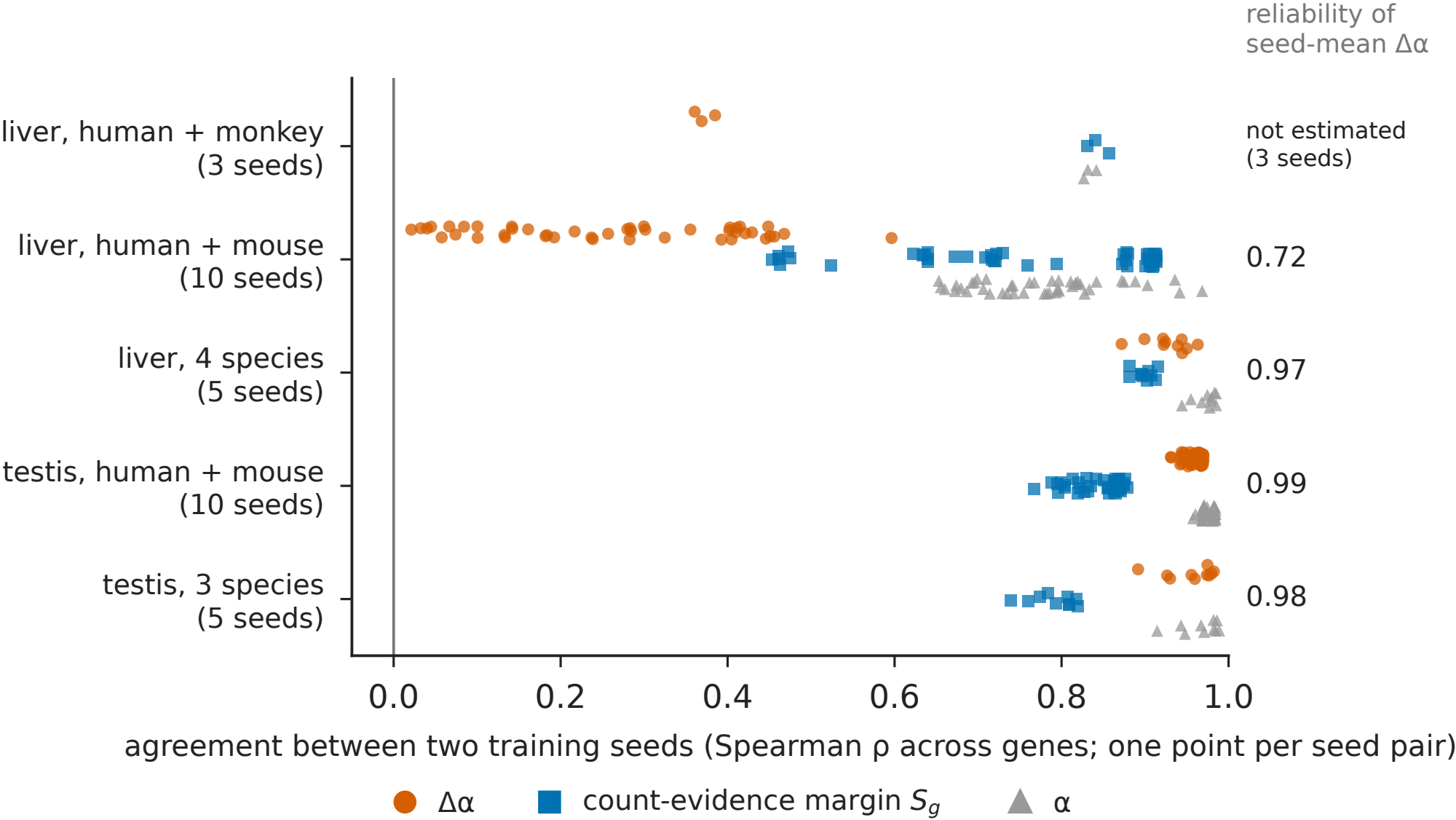

Liver H+Monkey
r = 0.996
Liver H+Mouse
r = 0.997
Liver H+Zebrafish
r = 0.993
Liver 4-sp
r = 0.969
Testis H+Macaque
r = 0.996
Testis H+Mouse
r = 0.993
Testis 3-sp
r = 0.993
trained S_g (EMA)
pass rate × forward-pass S_g

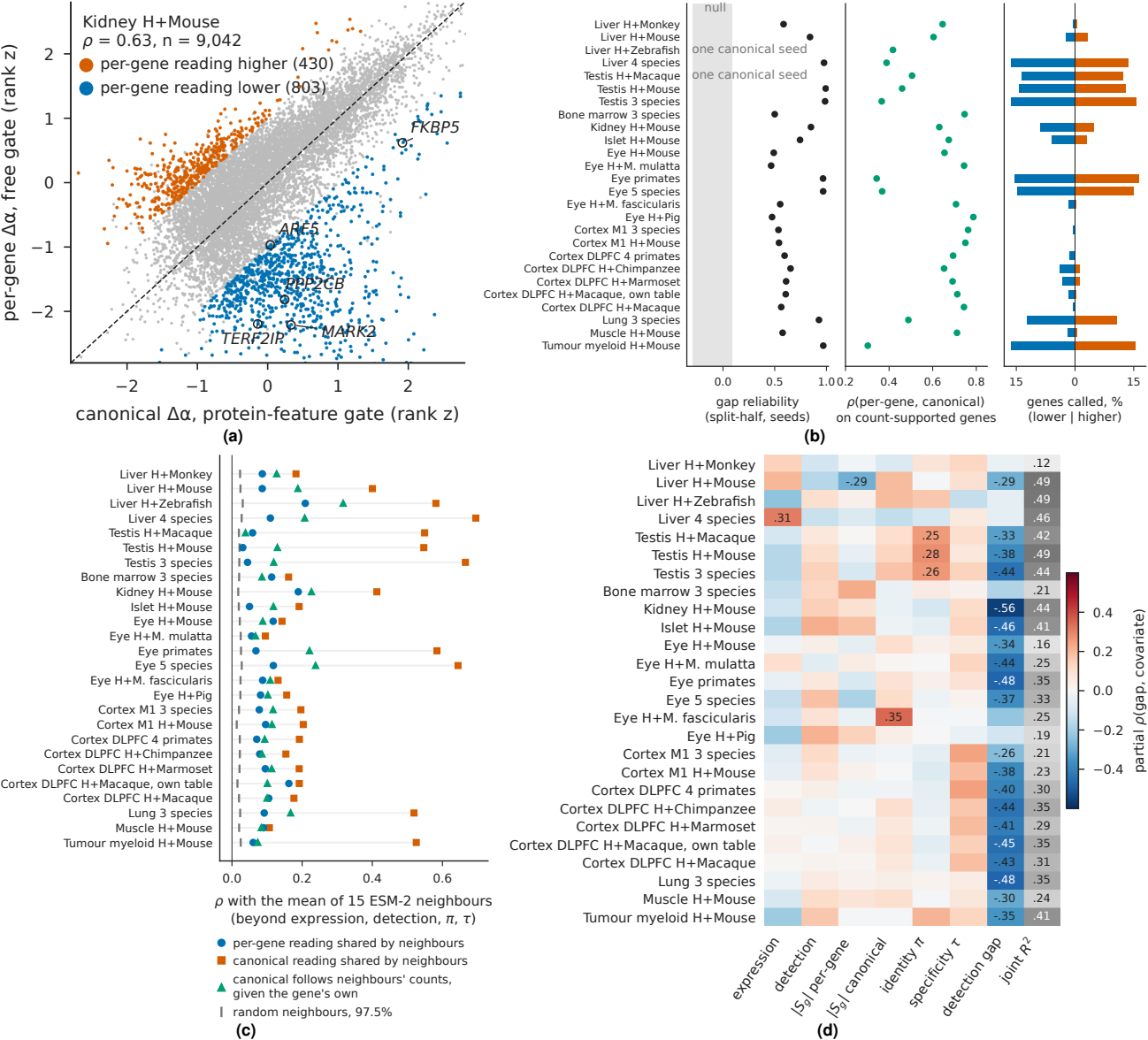


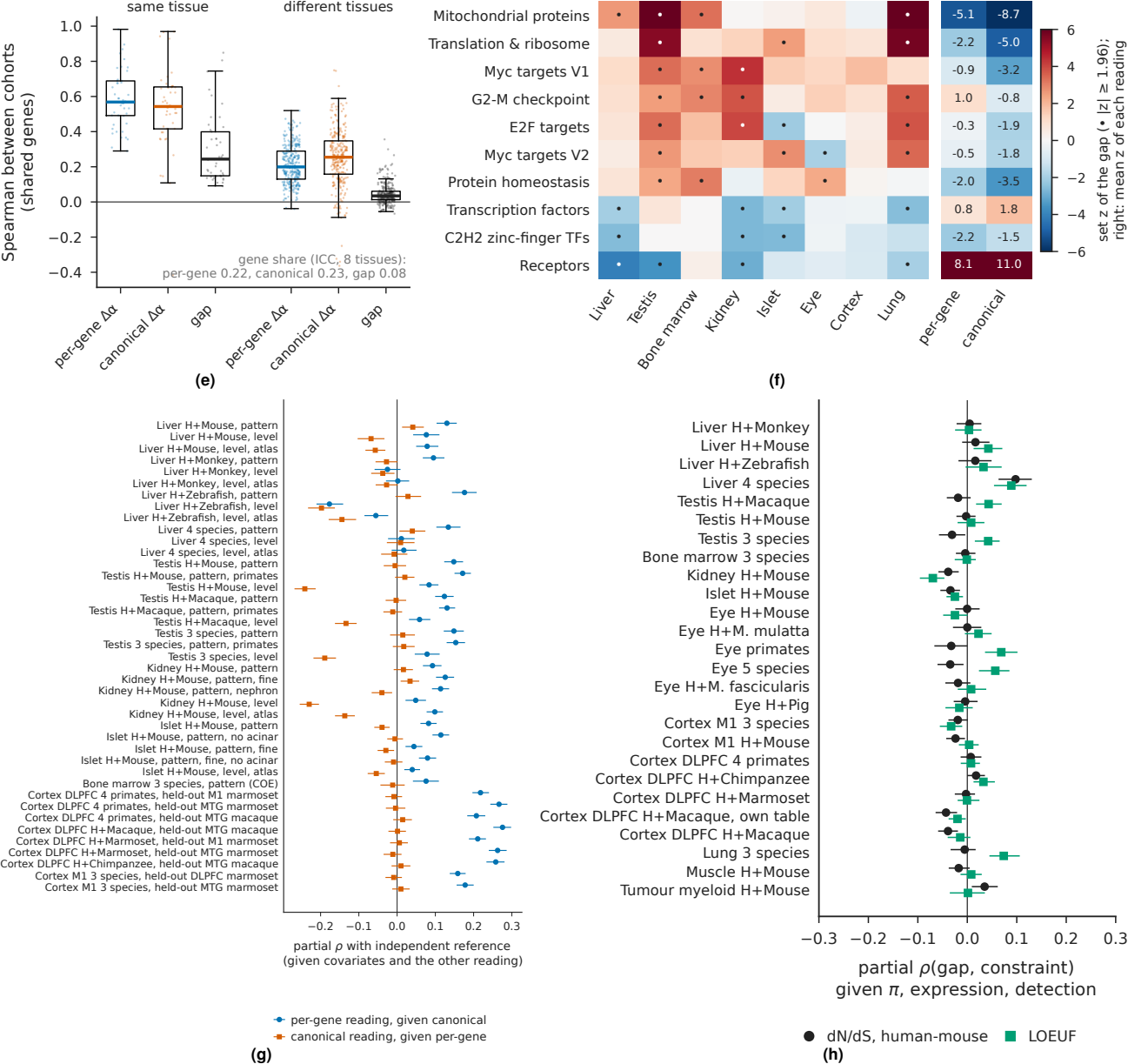
agreement between seeds of a cell type's contribution to S_g (Spearman ρ across genes;
points: seed pairs; in brackets: cell-type weight; blue: weight at least 0.10)

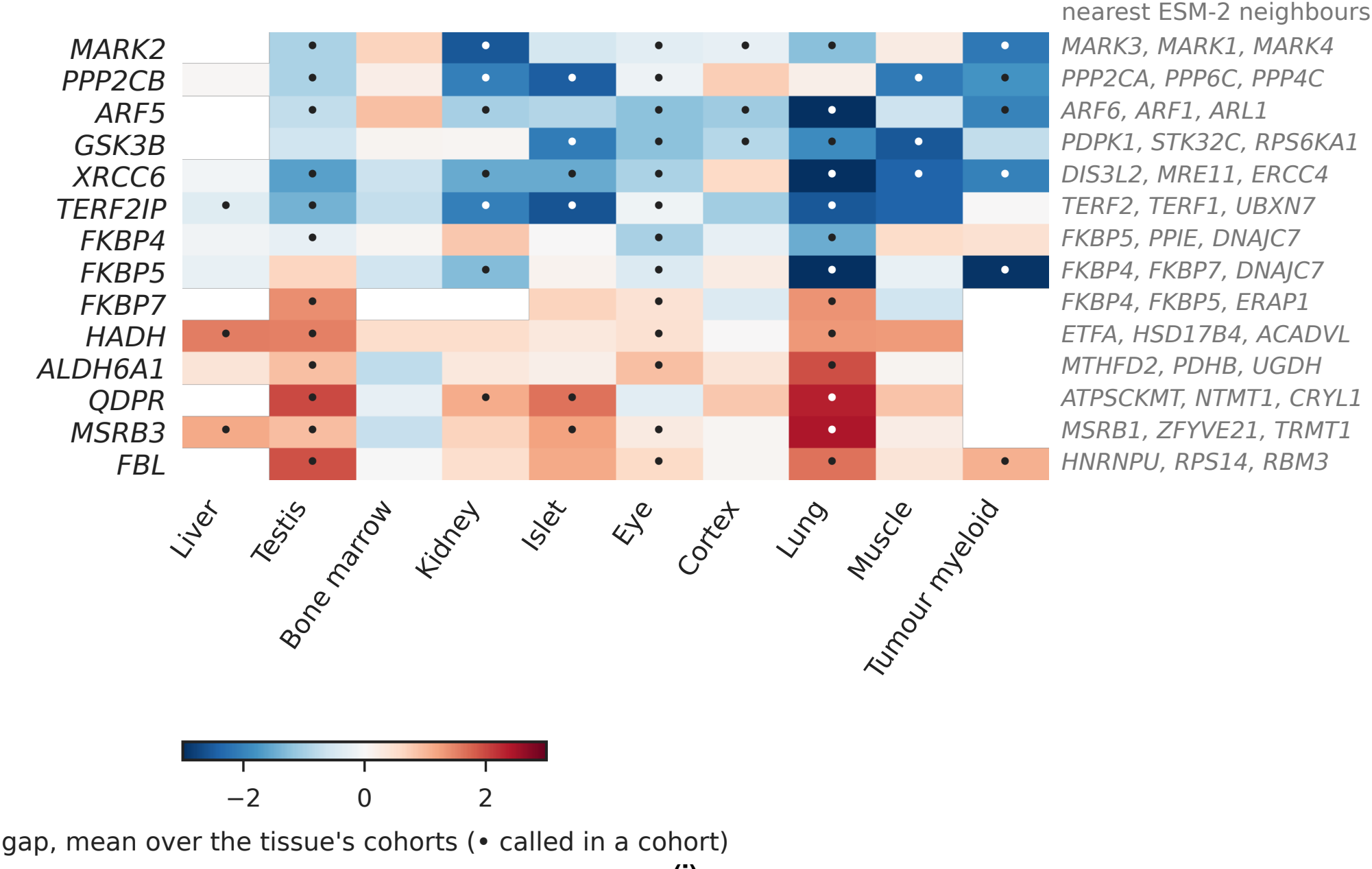

Pathway (coherence)
Liver 4-sp
π
S
r1
r2
Liver H+Mouse
π
S
r1
r2
Testis 3-sp (secondary)
π
S
r1
r2
1
2
5
10
Pathway coherence index
(a)
Protein interaction (coherence)
Liver 4-sp
HuRI r1
HuRI r2
STRING r1
STRING r2
Liver H+Mouse
HuRI r1
HuRI r2
STRING r1
STRING r2
Testis 3-sp (secondary)
HuRI r1
HuRI r2
STRING r1
STRING r2
all edges
without top 1% ESM
below null 97.5%
0.00
0.05
0.10
0.15
0.20
Moran's I
(b)
Interolog retention (conservation)
Skinnider r1
Skinnider r2
STRING r1
STRING r2
-0.1
0
0.1
Partial ρ, retention
(b)
Regulatory (conservation)
Liver, r2
Testis, r2
Liver - testis, r2
Liver, r1
Testis, r1
Liver - testis, r1
-0.04
0
0.04
0.08
Spearman ρ(B_g, residual)
(c)


log mean
detection
log variance
cell-type specificity τ
prior π
fit-quality margin S
CDS length
CDS GC
partial ρ(Δα, S | expression)
CV R² on expression (3)
CV R² on all 72 inputs
-0.04 -0.35 +0.39 -0.63 +0.73 +0.73 +0.67
-0.05 -0.35 +0.35 -0.64 +0.72 +0.71 +0.66
-0.03 -0.34 +0.42 -0.60 +0.74 +0.74 +0.68
+0.08 +0.31 -0.05 +0.55 -0.27 -0.34 -0.35
-0.00 +0.19 -0.05 +0.02 -0.16 -0.16 -0.18
+0.41 +0.28 +0.32 +0.48 +0.50 +0.48 +0.35
+0.02 +0.02 +0.28 -0.01 +0.29 +0.21 +0.32
+0.03 +0.15 -0.10 +0.27 -0.31 -0.32 -0.31
+0.39 +0.17 +0.12 +0.11 +0.15 +0.09 +0.14
0.03 0.24 0.61 0.62 0.68 0.69 0.54
0.03 0.48 0.84 0.91 0.92 0.88 0.93
Liver H+Monkey
Liver H+Mouse
Liver H+Zebrafish
Liver 4-sp
Testis H+Macaque
Testis H+Mouse
Testis 3-sp
1.0
0.5
0.0
−0.5
−1.0
Spearman ρ with Δα


reliability of
seed-mean Δα
liver, human + monkey
(3 seeds)
not estimated
(3 seeds)
liver, human + mouse
(10 seeds)
0.72
liver, 4 species
(5 seeds)
0.97
testis, human + mouse
(10 seeds)
0.99
testis, 3 species
(5 seeds)
0.98
0.0
0.2
0.4
0.6
0.8
1.0
agreement between two training seeds (Spearman ρ across genes; one point per seed pair)
Δα
count-evidence margin S_g
α

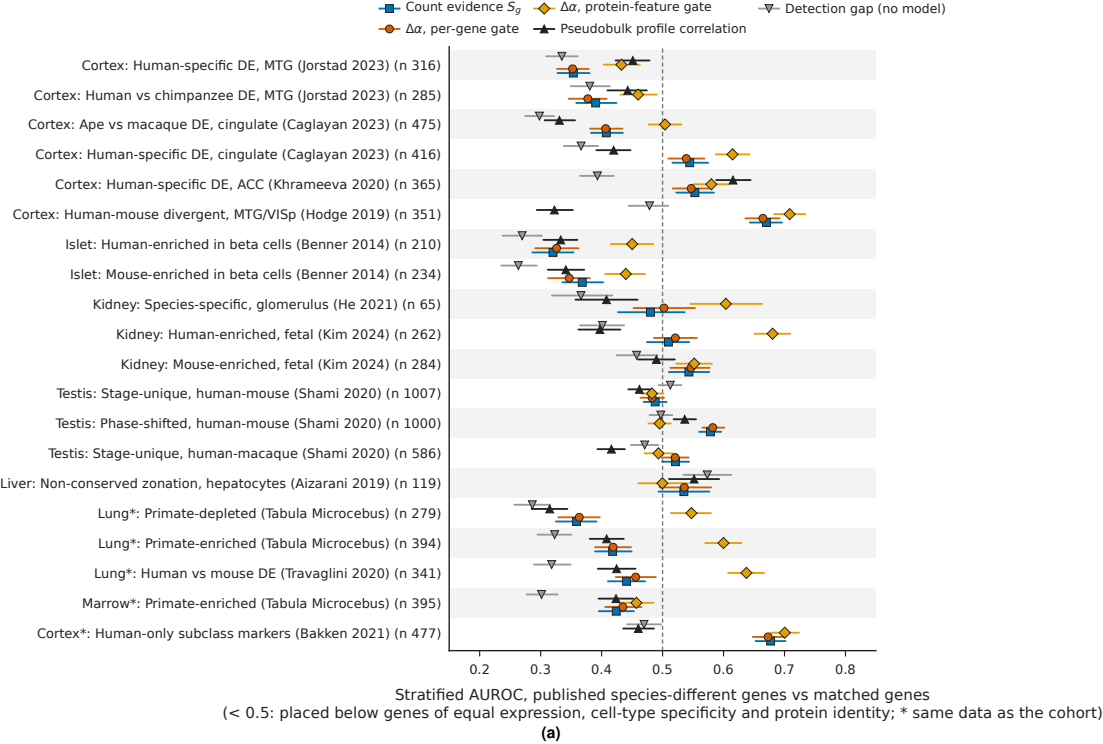

Kidney H+Mouse
ρ = 0.63, n = 9,042
per-gene reading higher (430)
per-gene reading lower (803)
FKBP5
ARL5
PPP2CB
TERF2IP
MARK2
per-gene Δα, free gate (rank z)
canonical Δα, protein-feature gate (rank z)
(a)
null
one canonical seed
one canonical seed
Liver H+Monkey
Liver H+Mouse
Liver H+Zebrafish
Liver 4 species
Testis H+Macaque
Testis H+Mouse
Testis 3 species
Bone marrow 3 species
Kidney H+Mouse
Islet H+Mouse
Eye H+Mouse
Eye H+M. mulatta
Eye primates
Eye 5 species
Eye H+M. fascicularis
Eye H+Pig
Cortex M1 3 species
Cortex M1 H+Mouse
Cortex DLPFC 4 primates
Cortex DLPFC H+Chimpanzee
Cortex DLPFC H+Marmoset
Cortex DLPFC H+Macaque, own table
Cortex DLPFC H+Macaque
Lung 3 species
Muscle H+Mouse
Tumour myeloid H+Mouse
gap reliability (split-half, seeds)
ρ(per-gene, canonical) on count-supported genes
genes called, % (lower | higher)
(b)
ρ with the mean of 15 ESM-2 neighbours (beyond expression, detection, π, τ)
per-gene reading shared by neighbours
canonical reading shared by neighbours
canonical follows neighbours' counts, given the gene's own
random neighbours, 97.5%
(c)
expression
detection
|S_g| per-gene
|S_g| canonical
identity π
specificity τ
detection gap
joint R²
partial ρ(gap, covariate)
(d)


same tissue
different tissues
Spearman between cohorts (shared genes)
gene share (ICC, 8 tissues):
per-gene 0.22, canonical 0.23, gap 0.08
per-gene Δα
canonical Δα
gap
(e)
Mitochondrial proteins
Translation & ribosome
Myc targets V1
G2-M checkpoint
E2F targets
Myc targets V2
Protein homeostasis
Transcription factors
C2H2 zinc-finger TFs
Receptors
Liver
Testis
Bone marrow
Kidney
Islet
Eye
Cortex
Lung
per-gene
canonical
set z of the gap (• |z| ≥ 1.96);
right: mean z of each reading
(f)
partial ρ with independent reference
(given covariates and the other reading)
per-gene reading, given canonical
canonical reading, given per-gene
(g)
partial ρ(gap, constraint)
given π, expression, detection
dN/dS, human-mouse
LOEUF
(h)

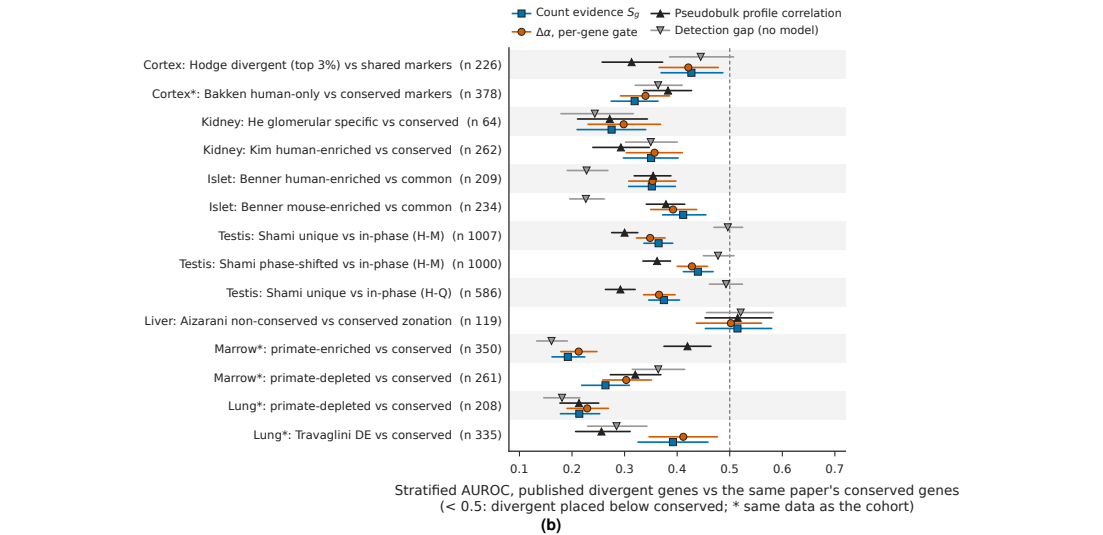


gap, mean over the tissue's cohorts (• called in a cohort)

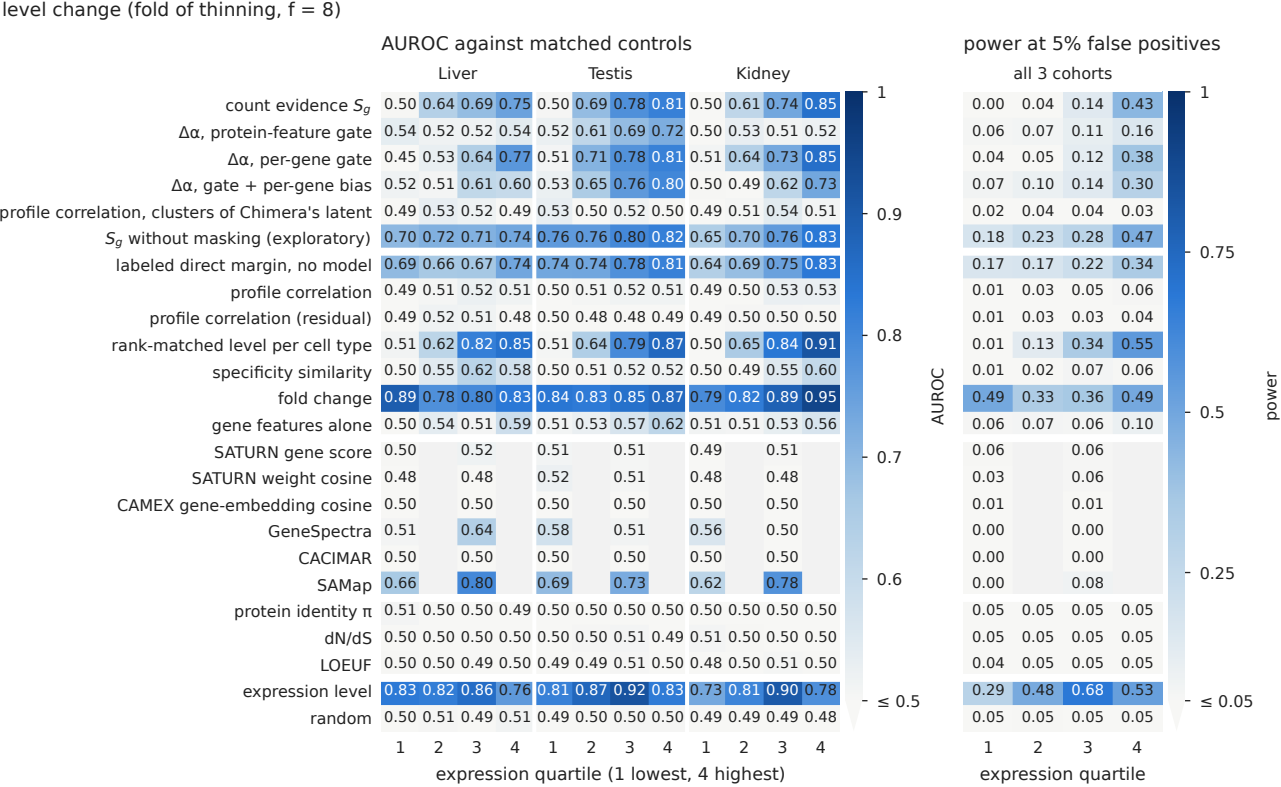
(i)

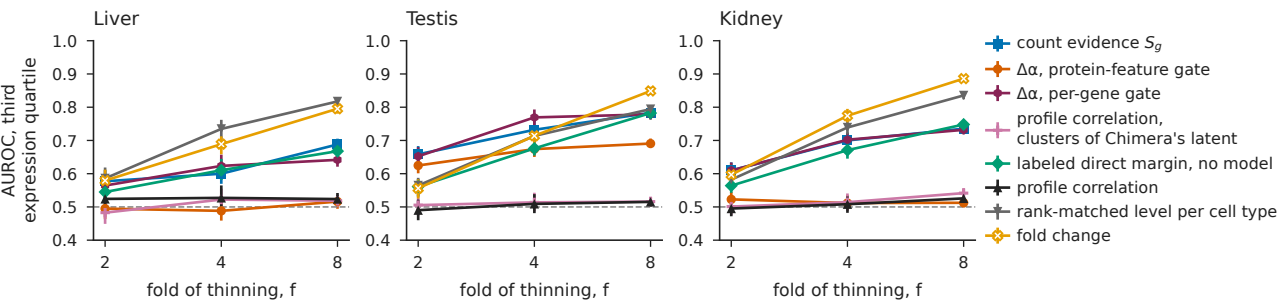

Count evidence S_g
$\Delta\alpha$, protein-feature gate
Detection gap (no model)
$\Delta\alpha$, per-gene gate
Pseudobulk profile correlation
Cortex: Human-specific DE, MTG (Jorstad 2023) (n 316)
Cortex: Human vs chimpanzee DE, MTG (Jorstad 2023) (n 285)
Cortex: Ape vs macaque DE, cingulate (Caglayan 2023) (n 475)
Cortex: Human-specific DE, cingulate (Caglayan 2023) (n 416)
Cortex: Human-specific DE, ACC (Khrameeva 2020) (n 365)
Cortex: Human-mouse divergent, MTG/VISp (Hodge 2019) (n 351)
Islet: Human-enriched in beta cells (Benner 2014) (n 210)
Islet: Mouse-enriched in beta cells (Benner 2014) (n 234)
Kidney: Species-specific, glomerulus (He 2021) (n 65)
Kidney: Human-enriched, fetal (Kim 2024) (n 262)
Kidney: Mouse-enriched, fetal (Kim 2024) (n 284)
Testis: Stage-unique, human-mouse (Shami 2020) (n 1007)
Testis: Phase-shifted, human-mouse (Shami 2020) (n 1000)
Testis: Stage-unique, human-macaque (Shami 2020) (n 586)
Liver: Non-conserved zonation, hepatocytes (Aizarani 2019) (n 119)
Lung*: Primate-depleted (Tabula Microcebus) (n 279)
Lung*: Primate-enriched (Tabula Microcebus) (n 394)
Lung*: Human vs mouse DE (Travaglini 2020) (n 341)
Marrow*: Primate-enriched (Tabula Microcebus) (n 395)
Cortex*: Human-only subclass markers (Bakken 2021) (n 477)
0.2
0.3
0.4
0.5
0.6
0.7
0.8
Stratified AUROC, published species-different genes vs matched genes
(< 0.5: placed below genes of equal expression, cell-type specificity and protein identity; * same data as the cohort)

(a)

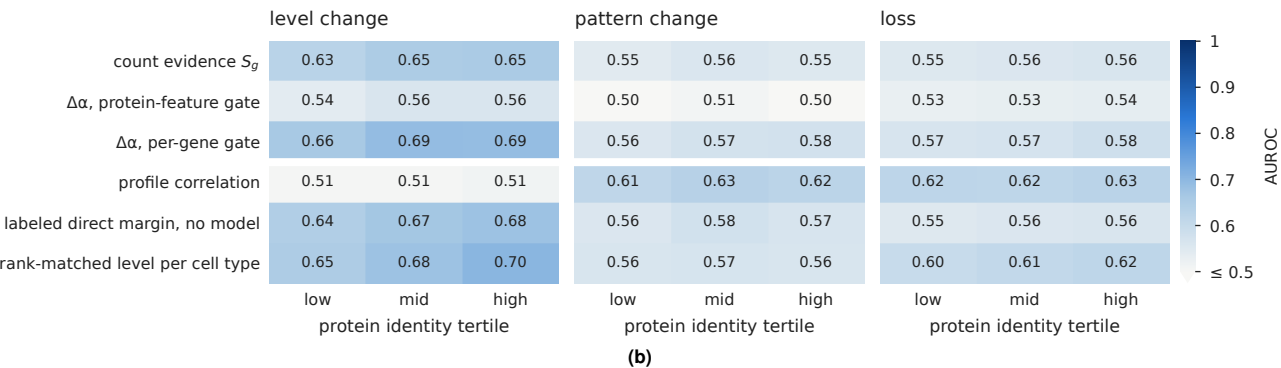

Count evidence S_g
Pseudobulk profile correlation
$\Delta\alpha$, per-gene gate
Detection gap (no model)
Cortex: Hodge divergent (top 3%) vs shared markers (n 226)
Cortex*: Bakken human-only vs conserved markers (n 378)
Kidney: He glomerular specific vs conserved (n 64)
Kidney: Kim human-enriched vs conserved (n 262)
Islet: Benner human-enriched vs common (n 209)
Islet: Benner mouse-enriched vs common (n 234)
Testis: Shami unique vs in-phase (H-M) (n 1007)
Testis: Shami phase-shifted vs in-phase (H-M) (n 1000)
Testis: Shami unique vs in-phase (H-Q) (n 586)
Liver: Aizarani non-conserved vs conserved zonation (n 119)
Marrow*: primate-enriched vs conserved (n 350)
Marrow*: primate-depleted vs conserved (n 261)
Lung*: primate-depleted vs conserved (n 208)
Lung*: Travaglini DE vs conserved (n 335)
0.1
0.2
0.3
0.4
0.5
0.6
0.7
Stratified AUROC, published divergent genes vs the same paper's conserved genes
(< 0.5: divergent placed below conserved; * same data as the cohort)

(b)

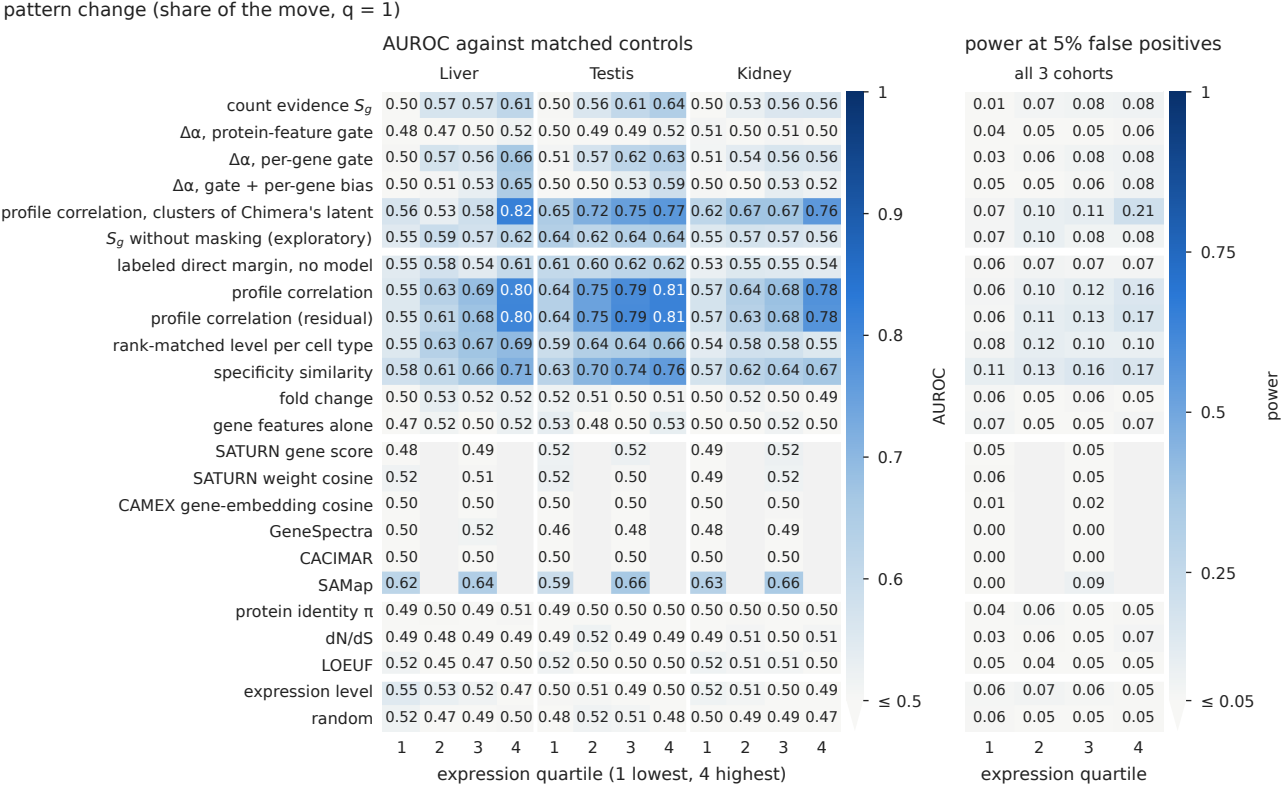

level change (fold of thinning, f = 8)
AUROC against matched controls
power at 5% false positives
Liver
Testis
Kidney
all 3 cohorts
count evidence S_g
Δα, protein-feature gate
Δα, per-gene gate
Δα, gate + per-gene bias
profile correlation, clusters of Chimera's latent
S_g without masking (exploratory)
labeled direct margin, no model
profile correlation
profile correlation (residual)
rank-matched level per cell type
specificity similarity
fold change
gene features alone
SATURN gene score
SATURN weight cosine
CAMEX gene-embedding cosine
GeneSpectra
CACIMAR
SAMap
protein identity π
dN/dS
LOEUF
expression level
random
AUROC
power
expression quartile (1 lowest, 4 highest)
expression quartile

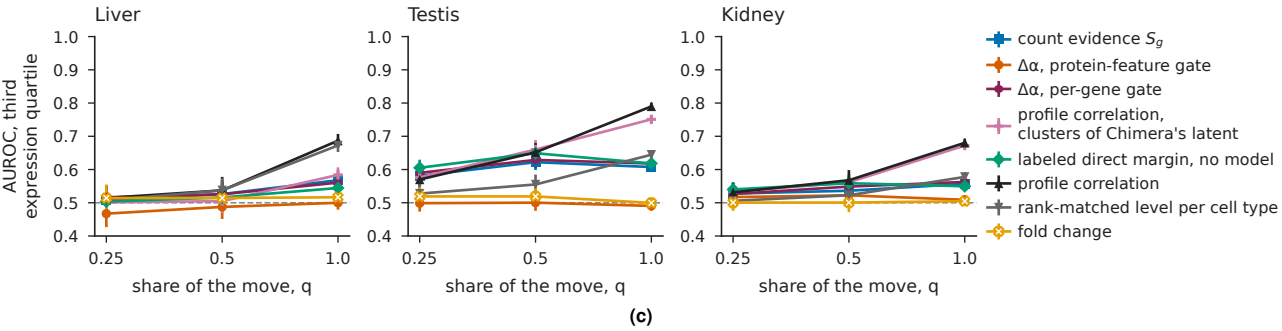

Liver
Testis
Kidney
AUROC, third expression quartile
fold of thinning, f
count evidence S_g
Δα, protein-feature gate
Δα, per-gene gate
profile correlation, clusters of Chimera's latent
labeled direct margin, no model
profile correlation
rank-matched level per cell type
fold change

(a)

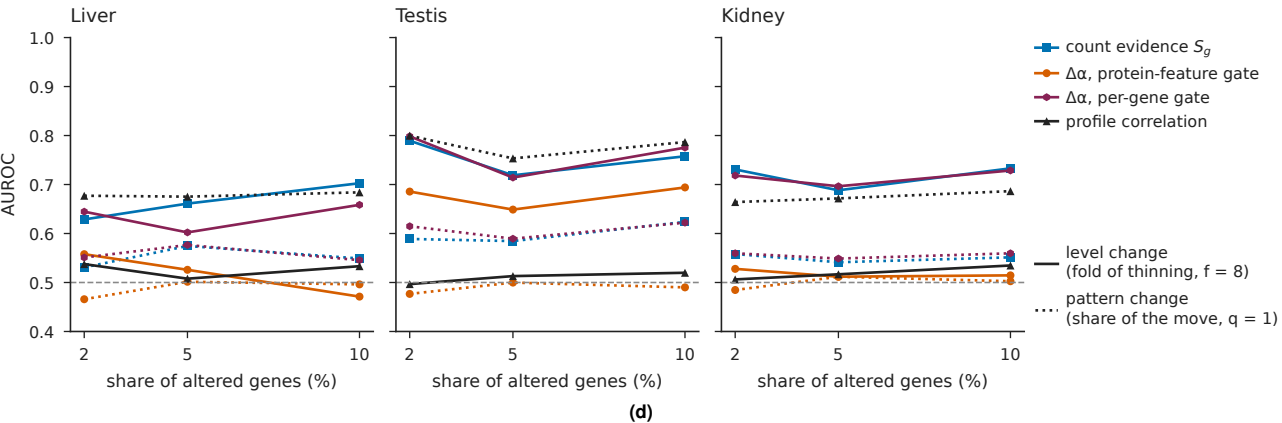

level change
pattern change
loss
count evidence S_g
Δα, protein-feature gate
Δα, per-gene gate
profile correlation
labeled direct margin, no model
rank-matched level per cell type
low
mid
high
protein identity tertile
AUROC

(b)

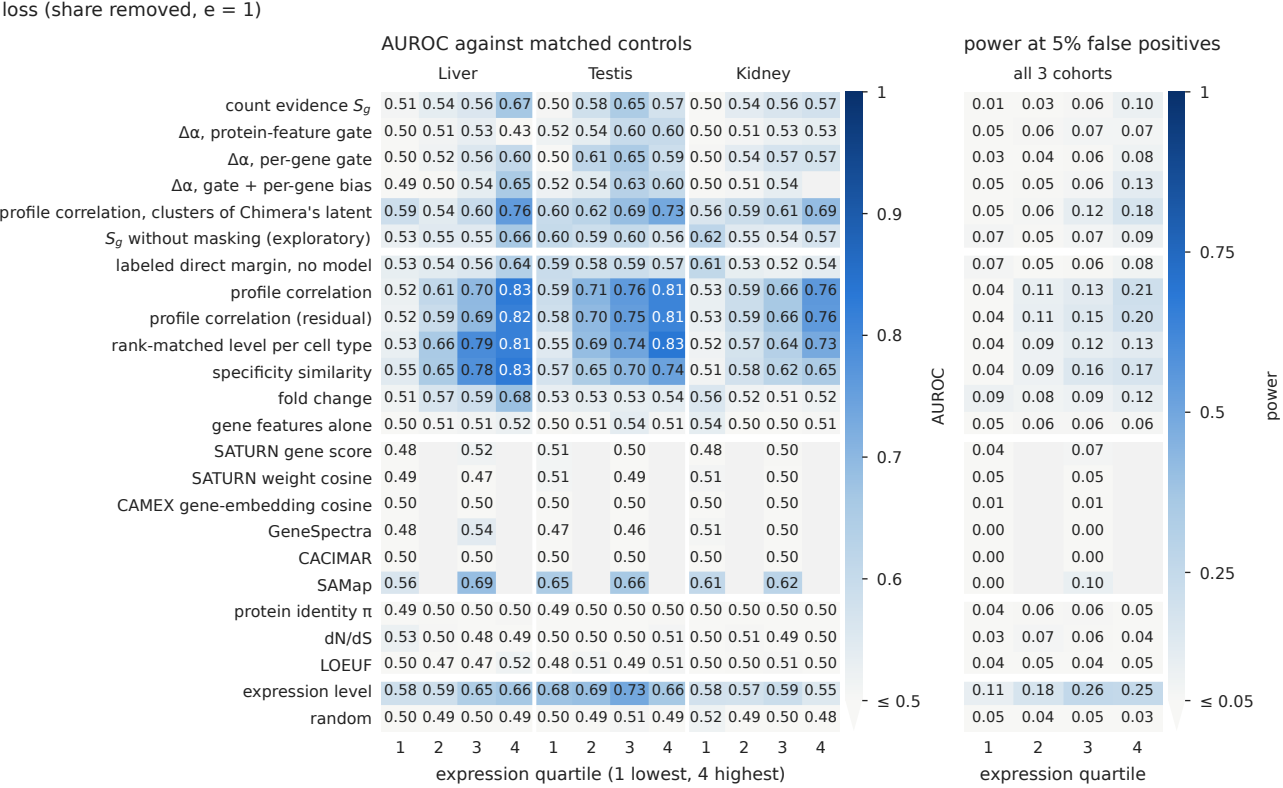

pattern change (share of the move, q = 1)
AUROC against matched controls
power at 5% false positives
Liver
Testis
Kidney
all 3 cohorts
count evidence S_g
Δα, protein-feature gate
Δα, per-gene gate
Δα, gate + per-gene bias
profile correlation, clusters of Chimera's latent
S_g without masking (exploratory)
labeled direct margin, no model
profile correlation
profile correlation (residual)
rank-matched level per cell type
specificity similarity
fold change
gene features alone
SATURN gene score
SATURN weight cosine
CAMEX gene-embedding cosine
GeneSpectra
CACIMAR
SAMap
protein identity π
dN/dS
LOEUF
expression level
random
AUROC
power
expression quartile (1 lowest, 4 highest)
expression quartile

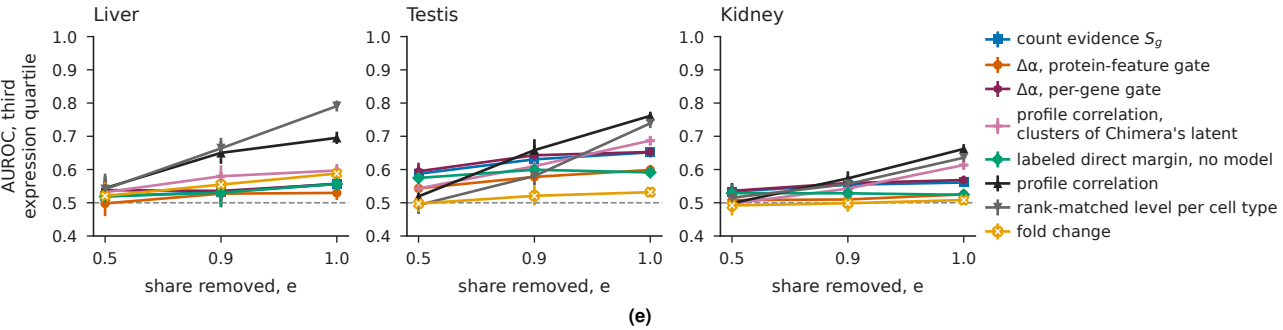

Liver
Testis
Kidney
AUROC, third expression quartile
share of the move, q
count evidence S_g
Δα, protein-feature gate
Δα, per-gene gate
profile correlation, clusters of Chimera's latent
labeled direct margin, no model
profile correlation
rank-matched level per cell type
fold change

(c)

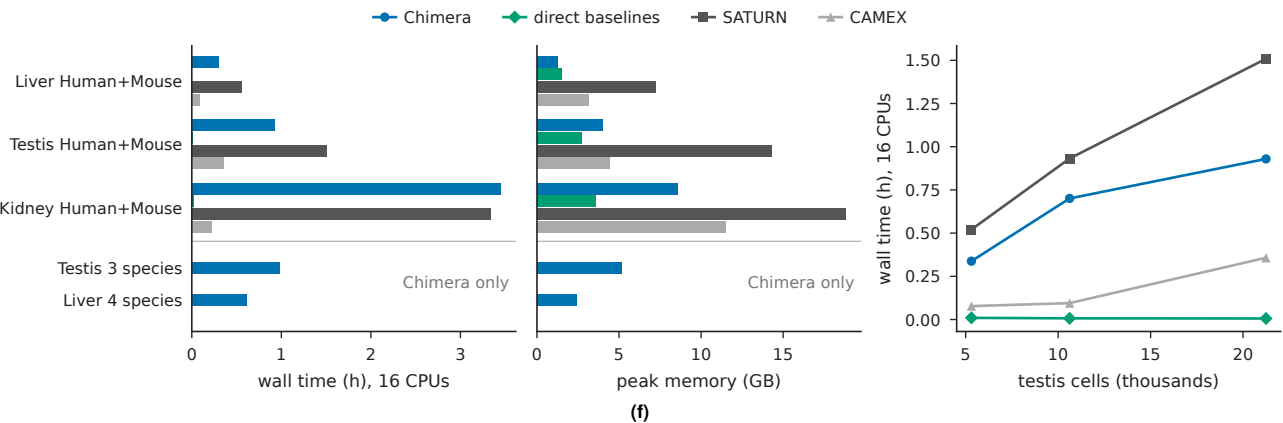

Liver
Testis
Kidney
AUROC
share of altered genes (%)
count evidence S_g
Δα, protein-feature gate
Δα, per-gene gate
profile correlation
level change (fold of thinning, f = 8)
pattern change (share of the move, q = 1)

(d)

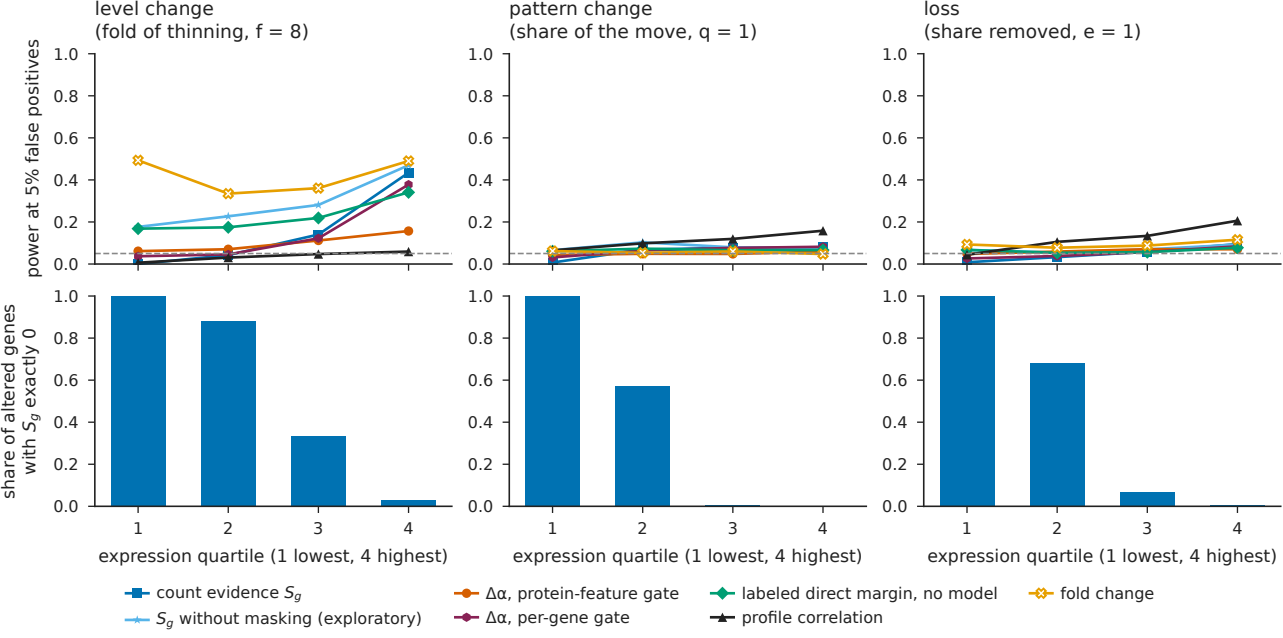

loss (share removed, e = 1)
AUROC against matched controls
Liver
Testis
Kidney
power at 5% false positives
all 3 cohorts
count evidence S_g 0.51 0.54 0.56 0.67 0.50 0.58 0.65 0.57 0.50 0.54 0.56 0.57 0.01 0.03 0.06 0.10
Δα, protein-feature gate 0.50 0.51 0.53 0.43 0.52 0.54 0.60 0.60 0.50 0.51 0.53 0.53 0.05 0.06 0.07 0.07
Δα, per-gene gate 0.50 0.52 0.56 0.60 0.50 0.61 0.65 0.59 0.50 0.54 0.57 0.57 0.03 0.04 0.06 0.08
Δα, gate + per-gene bias 0.49 0.50 0.54 0.65 0.52 0.54 0.63 0.60 0.50 0.51 0.54 0.05 0.05 0.06 0.13
profile correlation, clusters of Chimera's latent 0.59 0.54 0.60 0.76 0.60 0.62 0.69 0.73 0.56 0.59 0.61 0.69 0.05 0.06 0.12 0.18
S_g without masking (exploratory) 0.53 0.55 0.55 0.66 0.60 0.59 0.60 0.56 0.62 0.55 0.54 0.57 0.07 0.05 0.07 0.09
labeled direct margin, no model 0.53 0.54 0.56 0.64 0.59 0.58 0.59 0.57 0.61 0.53 0.52 0.54 0.07 0.05 0.06 0.08
profile correlation 0.52 0.61 0.70 0.83 0.59 0.71 0.76 0.81 0.53 0.59 0.66 0.76 0.04 0.11 0.13 0.21
profile correlation (residual) 0.52 0.59 0.69 0.82 0.58 0.70 0.75 0.81 0.53 0.59 0.66 0.76 0.04 0.11 0.15 0.20
rank-matched level per cell type 0.53 0.66 0.79 0.81 0.55 0.69 0.74 0.83 0.52 0.57 0.64 0.73 0.04 0.09 0.12 0.13
specificity similarity 0.55 0.65 0.78 0.83 0.57 0.65 0.70 0.74 0.51 0.58 0.62 0.65 0.04 0.09 0.16 0.17
fold change 0.51 0.57 0.59 0.68 0.53 0.53 0.53 0.54 0.56 0.52 0.51 0.52 0.09 0.08 0.09 0.12
gene features alone 0.50 0.51 0.51 0.52 0.50 0.51 0.54 0.51 0.54 0.50 0.50 0.51 0.05 0.06 0.06 0.06
SATURN gene score 0.48 0.52 0.51 0.50 0.48 0.50 0.04 0.07
SATURN weight cosine 0.49 0.47 0.51 0.49 0.51 0.50 0.05 0.05
CAMEX gene-embedding cosine 0.50 0.50 0.50 0.50 0.50 0.50 0.01 0.01
GeneSpectra 0.48 0.54 0.47 0.46 0.51 0.50 0.00 0.00
CACIMAR 0.50 0.50 0.50 0.50 0.50 0.50 0.00 0.00
SAMap 0.56 0.69 0.65 0.66 0.61 0.62 0.00 0.10
protein identity π 0.49 0.50 0.50 0.50 0.49 0.50 0.50 0.50 0.50 0.50 0.50 0.50 0.04 0.06 0.06 0.05
dN/dS 0.53 0.50 0.48 0.49 0.50 0.50 0.50 0.51 0.50 0.51 0.49 0.50 0.03 0.07 0.06 0.04
LOEUF 0.50 0.47 0.47 0.52 0.48 0.51 0.49 0.51 0.50 0.50 0.51 0.50 0.04 0.05 0.04 0.05
expression level 0.58 0.59 0.65 0.66 0.68 0.69 0.73 0.66 0.58 0.57 0.59 0.55 0.11 0.18 0.26 0.25
random 0.50 0.49 0.50 0.49 0.50 0.49 0.51 0.49 0.52 0.49 0.50 0.48 0.05 0.04 0.05 0.03
1 2 3 4 1 2 3 4 1 2 3 4
expression quartile (1 lowest, 4 highest)
1 2 3 4
expression quartile
AUROC
1
0.9
0.8
0.7
0.6
≤ 0.5
power
1
0.75
0.5
0.25
≤ 0.05


Liver
Testis
Kidney
AUROC, third expression quartile
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.5
0.9
1.0
share removed, e
count evidence S_g
Δα, protein-feature gate
Δα, per-gene gate
profile correlation, clusters of Chimera's latent
labeled direct margin, no model
profile correlation
rank-matched level per cell type
fold change

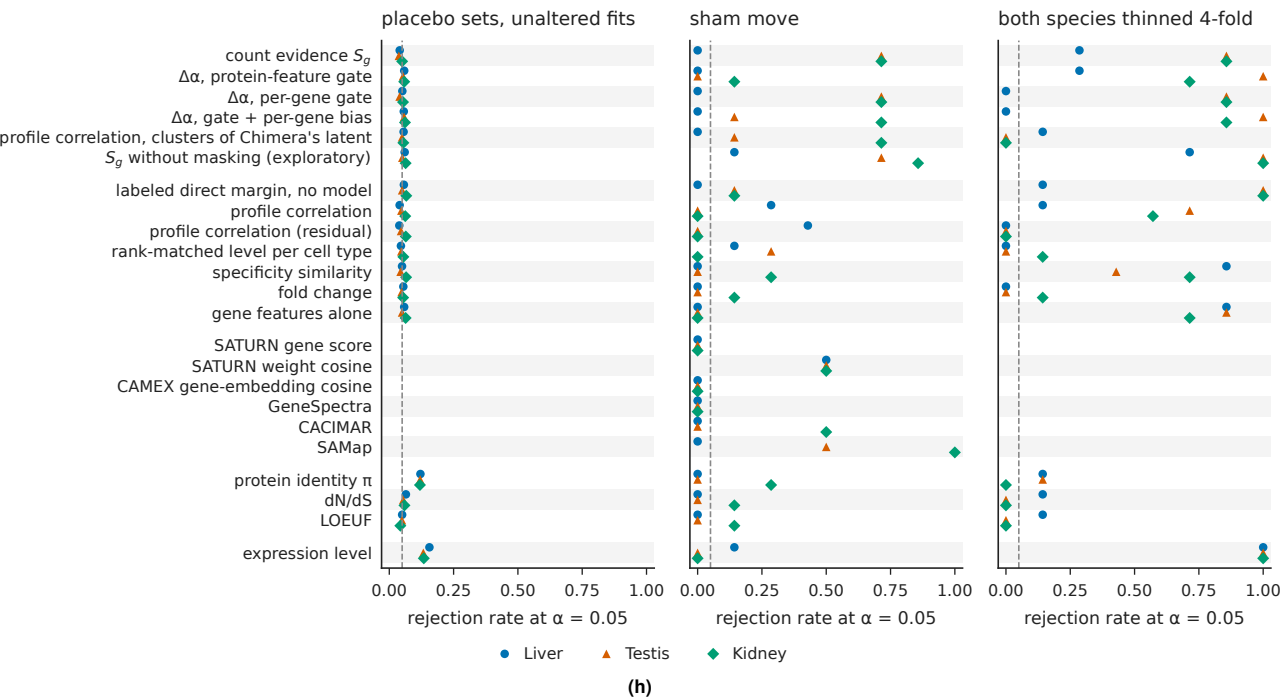
(e)


Chimera
direct baselines
SATURN
CAMEX
Liver Human+Mouse
Testis Human+Mouse
Kidney Human+Mouse
Testis 3 species
Liver 4 species
Chimera only
Chimera only
0
1
2
3
wall time (h), 16 CPUs
0
5
10
15
peak memory (GB)
wall time (h), 16 CPUs
1.50
1.25
1.00
0.75
0.50
0.25
0.00
5
10
15
20
testis cells (thousands)

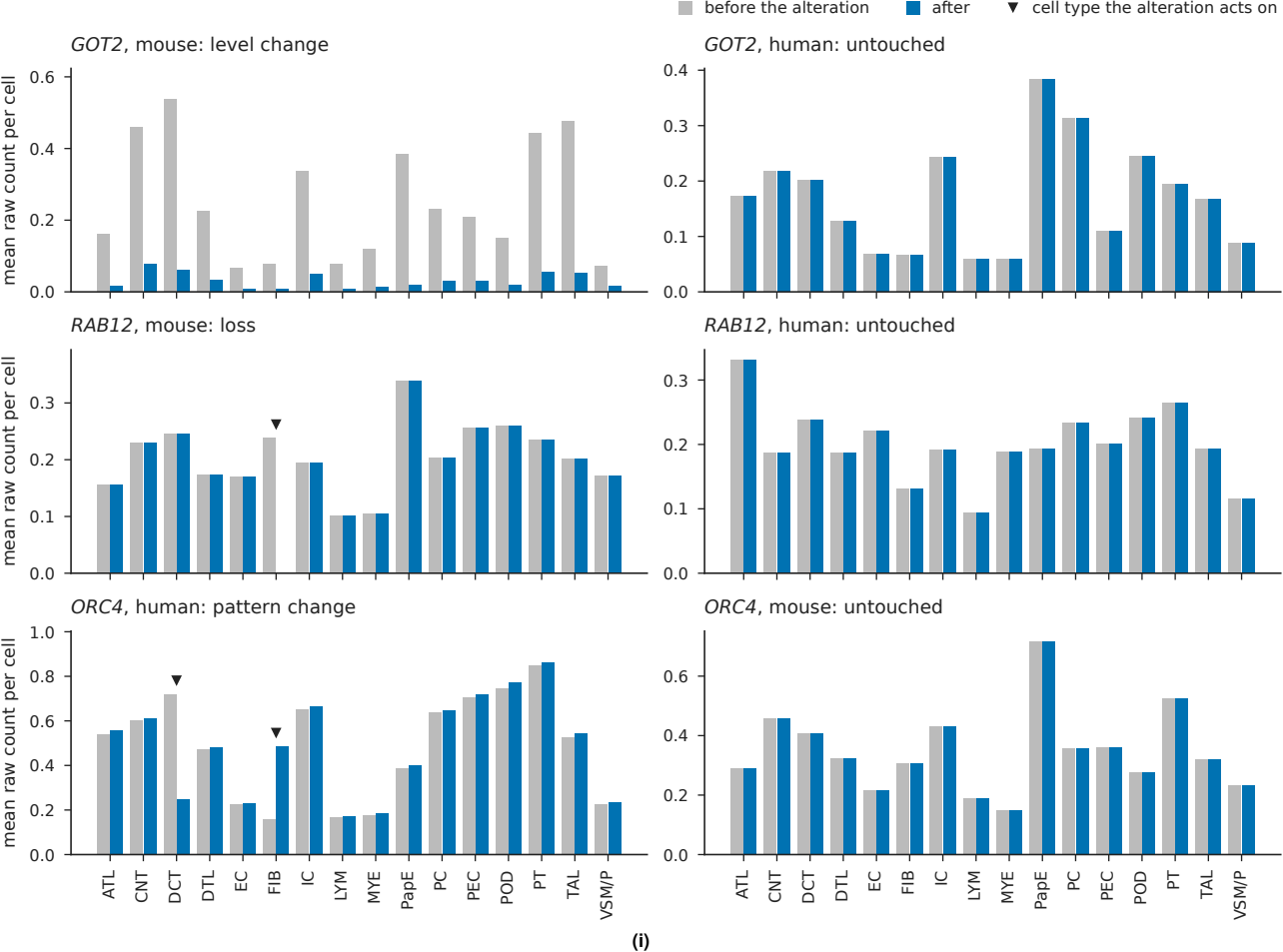
(f)


level change
(fold of thinning, f = 8)
pattern change
(share of the move, q = 1)
loss
(share removed, e = 1)
power at 5% false positives
share of altered genes with S_g exactly 0
expression quartile (1 lowest, 4 highest)
count evidence S_g
S_g without masking (exploratory)
Δα, protein-feature gate
Δα, per-gene gate
labeled direct margin, no model
profile correlation
fold change

(g)

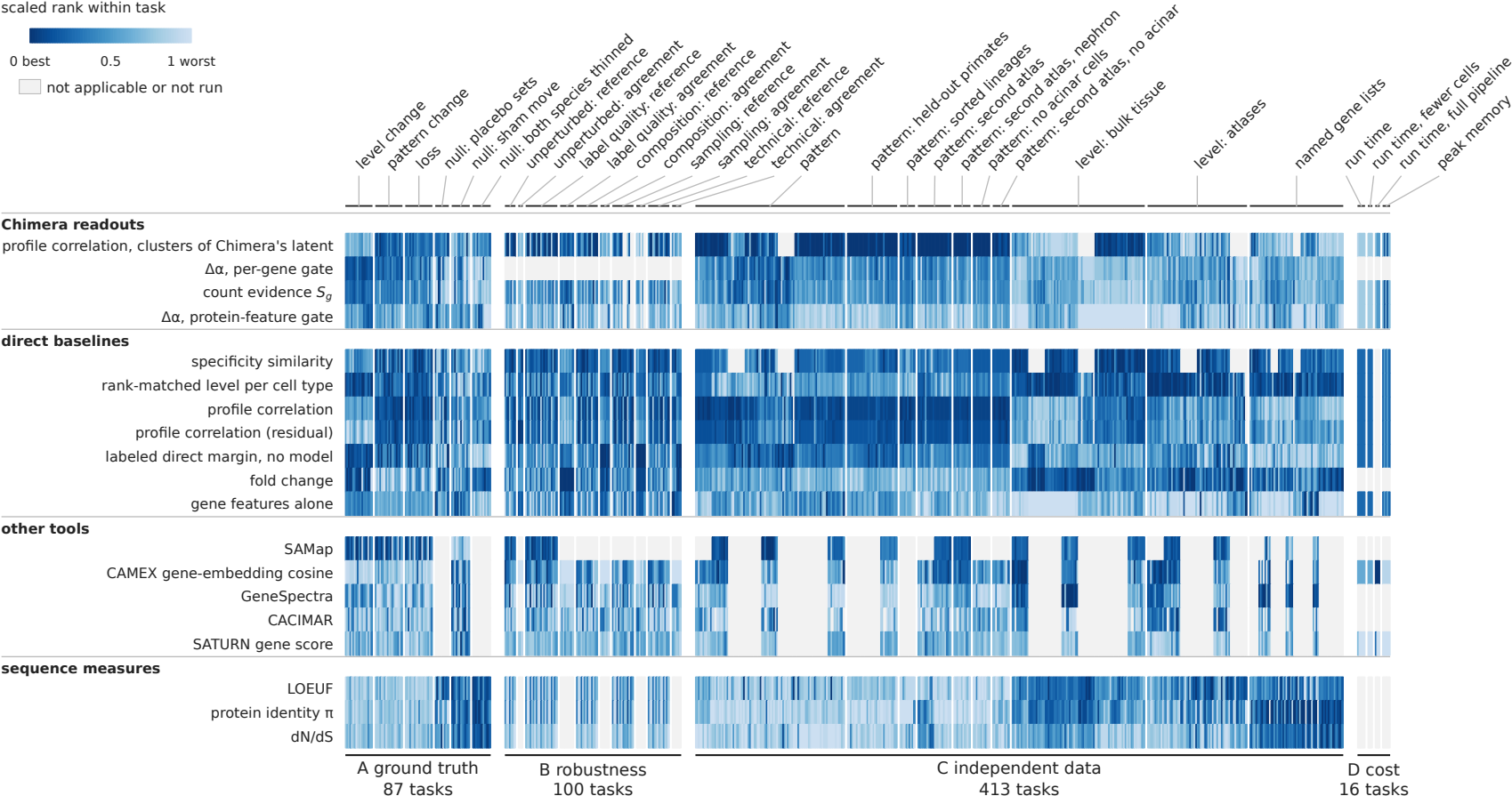

placebo sets, unaltered fits
sham move
both species thinned 4-fold
count evidence S_g
Δα, protein-feature gate
Δα, per-gene gate
Δα, gate + per-gene bias
profile correlation, clusters of Chimera's latent
S_g without masking (exploratory)
labeled direct margin, no model
profile correlation
profile correlation (residual)
rank-matched level per cell type
specificity similarity
fold change
gene features alone
SATURN gene score
SATURN weight cosine
CAMEX gene-embedding cosine
GeneSpectra
CACIMAR
SAMap
protein identity π
dN/dS
LOEUF
expression level
rejection rate at α = 0.05
Liver
Testis
Kidney

(h)

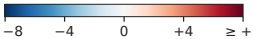

before the alteration
after
cell type the alteration acts on
GOT2, mouse: level change
GOT2, human: untouched
RAB12, mouse: loss
RAB12, human: untouched
ORC4, human: pattern change
ORC4, mouse: untouched
mean raw count per cell
ATL
CNT
DCT
DTL
EC
FIB
IC
LYM
MYE
PapE
PC
PEC
POD
PT
TAL
VSM/P

(i)

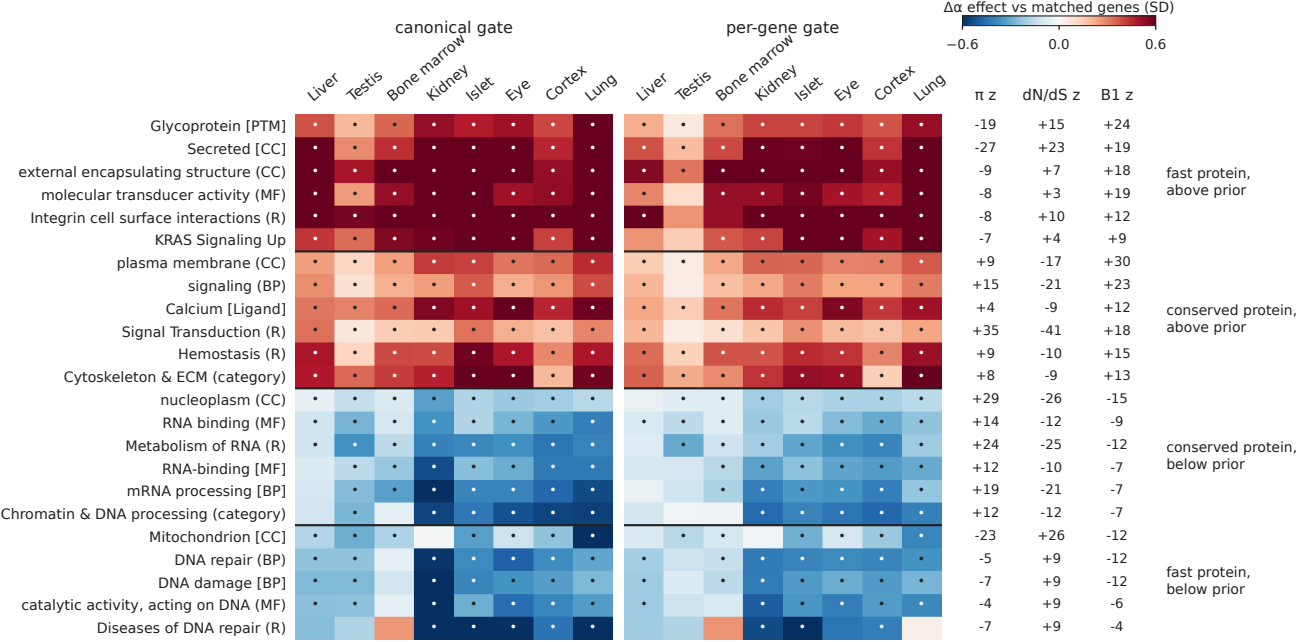
mean scaled rank within task
0 best
0.5
1 worst
not applicable or not run
5/15 runs finished / runs tried
level change
pattern change
loss
null: placebo sets
null: sham move
null: both species thinned
unperturbed: reference
unperturbed: agreement
label quality: reference
label quality: agreement
composition: reference
composition: agreement
sampling: reference
sampling: agreement
technical: reference
technical: agreement
pattern
pattern: held-out primates
pattern: sorted lineages
pattern: second atlas
pattern: second atlas, nephron
pattern: no acinar cells
pattern: second atlas, no acinar
level: bulk tissue
level: atlases
named gene lists
run time
run time, fewer cells
run time, full pipeline
peak memory
number of tasks
18 18 18 9 12 12
A ground truth
7 3 21 9 14 6 14 6 14 6
B robustness
99 33 10 22 11 11 11 88 66 62
C independent data
5 3 3 5
D cost
overall scaled rank
0 best; bar = 1 − rank
Chimera readouts
profile correlation, clusters of Chimera's latent
Δα, per-gene gate
count evidence S_g
Δα, protein-feature gate
5/15 2/6
direct baselines
specificity similarity
rank-matched level per cell type
profile correlation
profile correlation (residual)
labeled direct margin, no model
fold change
gene features alone
other tools
SAMap
CAMEX gene-embedding cosine
GeneSpectra
CACIMAR
SATURN gene score
sequence measures
LOEUF
protein identity π
dN/dS
0.45
0.47
0.56
0.68
0.29
0.32
0.32
0.36
0.39
0.43
0.50
0.27
0.57
0.61
0.65
0.74
0.60
0.62
0.68


scaled rank within task
0 best
0.5
1 worst
not applicable or not run
level change
pattern change
loss
null: placebo sets
null: sham move
null: both species thinned
unperturbed: reference
unperturbed: agreement
label quality: reference
label quality: agreement
composition: reference
composition: agreement
sampling: reference
sampling: agreement
technical: reference
technical: agreement
pattern
pattern: held-out primates
pattern: sorted lineages
pattern: second atlas
pattern: second atlas, nephron
pattern: no acinar cells
pattern: second atlas, no acinar
level: bulk tissue
level: atlases
named gene lists
run time
run time, fewer cells
run time, full pipeline
peak memory
Chimera readouts
profile correlation, clusters of Chimera's latent
Δα, per-gene gate
count evidence S_g
Δα, protein-feature gate
direct baselines
specificity similarity
rank-matched level per cell type
profile correlation
profile correlation (residual)
labeled direct margin, no model
fold change
gene features alone
other tools
SAMap
CAMEX gene-embedding cosine
GeneSpectra
CACIMAR
SATURN gene score
sequence measures
LOEUF
protein identity π
dN/dS
A ground truth
87 tasks
B robustness
100 tasks
C independent data
413 tasks
D cost
16 tasks

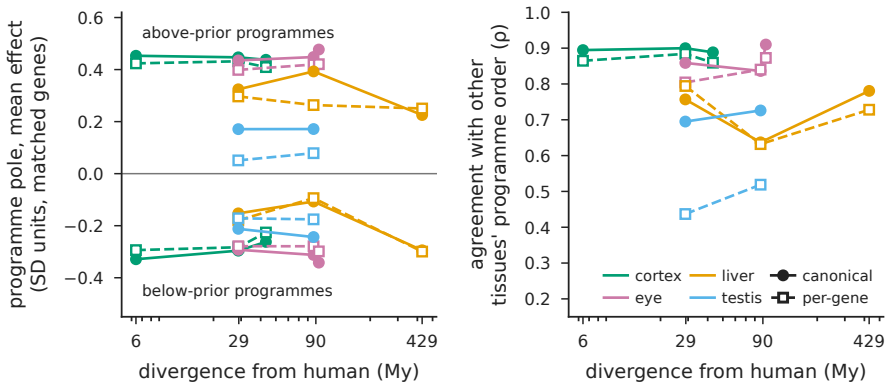
	Liver	Testis	Bone marrow	Kidney	Islet	Eye	Cortex	Lung	q	S_g	profile correlation	π	dN/dS
Recurrent, above the prior													
Receptors (class)	+7.1*	+7.3*	+9.5*	+15.0*	+13.6*	+7.2*	+13.1*	+14.1*	<.001	8↑0↓	7↑0↓	−2.2	+0.9
Epithelial Mesenchymal Transition	+7.8*	+7.6*	+4.0*	+12.1*	+16.7*	+10.0*	+8.1*	+11.6*	<.001	8↑0↓	6↑0↓	+0.4	0.0
Cytoskeleton & ECM (category)	+5.4*	+10.1*	+5.5*	+8.5*	+10.1*	+7.8*	+3.3*	+9.5*	<.001	8↑0↓	6↑1↓	+2.8	−2.9
KRAS Signaling Up	+2.9*	+5.1*	+5.2*	+6.5*	+9.8*	+6.1*	+4.5*	+8.9*	<.001	6↑0↓	5↑0↓	−2.7	+1.5
Inflammatory Response	+3.7*	+3.4*	+6.4*	+6.3*	+9.9*	+4.0*	+6.1*	+9.1*	<.001	7↑0↓	6↑0↓	−4.2	+3.4
Apical Junction	+4.0*	+2.6*	+3.6*	+7.1*	+9.2*	+6.4*	+5.0*	+7.6*	<.001	8↑0↓	6↑0↓	+2.0	−3.2
TNF-alpha Signaling via NF-kB	+1.3	+3.7*	+5.0*	+6.7*	+11.1*	+2.8*	+5.0*	+9.6*	<.001	6↑0↓	6↑0↓	−1.4	+1.5
GPCRs (class)	+3.7*	+4.5*	+1.9	+7.6*	+8.0*	+3.9*	+7.5*	+5.9*	<.001	7↑0↓	6↑0↓	−0.1	−0.5
Coagulation	+4.7*	+4.5*	+2.5*	+6.7*	+8.2*	+4.2*	+4.7*	+6.5*	<.001	7↑0↓	6↑0↓	−1.8	+1.4
IL-2/STAT5 Signaling	+2.7*	+3.2*	+5.1*	+5.5*	+7.4*	+5.6*	+4.1*	+7.3*	<.001	7↑0↓	5↑0↓	−1.5	+1.0
Myogenesis	+4.1*	+4.3*	+2.3*	+6.2*	+7.4*	+5.5*	+4.8*	+6.3*	<.001	8↑0↓	6↑0↓	+0.5	−1.2
UV Response Dn	+3.0*	+4.1*	+1.3	+5.2*	+8.5*	+5.3*	+5.9*	+5.3*	<.001	6↑0↓	7↑0↓	+2.4	−1.9
Complement	+4.5*	+1.4	+4.8*	+5.5*	+7.5*	+3.8*	+4.5*	+6.3*	<.001	7↑0↓	3↑0↓	−0.9	+0.6
Hypoxia	+3.0*	+2.1	+3.8*	+6.1*	+8.0*	+3.9*	+3.5*	+6.3*	<.001	6↑0↓	4↑0↓	+0.7	−1.3
Estrogen Response Early	−0.5	+1.8	+2.3*	+7.7*	+7.5*	+5.1*	+4.2*	+7.1*	<.001	5↑0↓	6↑0↓	0.0	−0.3
Allograft Rejection	+2.6*	+1.3	+6.7*	+4.5*	+6.3*	+2.9*	+2.2*	+7.0*	<.001	5↑0↓	7↑0↓	−3.0	+2.6
Angiogenesis		+3.1*	+3.5*	+5.1*	+6.1*	+2.6*	+4.4*	+6.3*	<.001	7↑0↓	4↑0↓	−0.1	+0.1
Signaling pathways (category)	+4.8*	+4.4*	+1.1	+2.7*	+5.9*	+4.4*	+2.4*	+7.4*	<.001	5↑0↓	6↑0↓	+6.4	−7.7
Ion channels (class)	−0.7	+7.0*	0.0	+5.9*	+4.9*	+2.6*	+9.4*	+3.2*	<.001	4↑0↓	3↑1↓	+2.9	−5.0
Estrogen Response Late	+0.2	+0.8	+3.4*	+7.0*	+6.3*	+4.2*	+3.4*	+6.9*	<.001	6↑0↓	3↑0↓	−0.2	−0.6
Immunity & inflammation (category)	+2.3*	+3.0*	+3.9*	+4.3*	+3.6*	+3.6*	+2.8*	+7.2*	<.001	4↑0↓	6↑0↓	−8.3	+7.0
Apoptosis	+1.5	+2.7*	+4.5*	+4.0*	+6.3*	+2.1	+2.8*	+4.9*	<.001	6↑0↓	4↑0↓	−0.9	+1.2
Protein kinases (class)	+2.9*	+5.0*	+3.2*	+2.1	+5.2*	+0.3	+1.2	+8.3*	<.001	3↑0↓	6↑0↓	+7.0	−7.4
IL-6/JAK/STAT3 Signaling	+2.6*	+1.2	+3.9*	+4.0*	+4.4*	+2.7*	+1.5	+5.0*	<.001	6↑0↓	6↑0↓	−4.8	+3.7
Homeodomain TFs (class)	+1.8	+2.5*	−0.3	+4.5*	+5.4*	+2.0	+5.8*	+3.0*	<.001	5↑0↓	6↑0↓	+3.1	−2.7
Interferon Gamma Response	+1.7	+2.1	+4.3*	+3.4*	+5.3*	+1.8	+0.4	+5.6*	<.001	4↑0↓	4↑0↓	−6.6	+5.3
Androgen Response	+2.2	+1.8	+2.2*	+4.6*	+3.2*	+1.1	+4.6*	+4.0*	<.001	4↑0↓	0↑0↓	+0.4	−0.6
Glycolysis	+2.6*	+1.3	+2.1	+4.3*	+3.3*	+3.4*	+2.1	+2.3*	<.001	5↑0↓	1↑0↓	+1.6	−1.2
Xenobiotic Metabolism	−1.1	−0.2	+3.2*	+6.9*	+4.2*	+2.5*	+2.2*	+2.8*	<.001	5↑0↓	1↑0↓	−1.1	+0.2
Hedgehog Signaling		+2.5*	+1.5	+2.8*	+4.0*	+2.1	+3.0*	+3.1*	<.001	4↑0↓	1↑0↓	+1.7	−2.0
Apical Surface		+1.3	+2.7*	+3.6*	+3.3*		+3.6*	+3.0*	<.001	5↑0↓	2↑0↓	−1.2	+0.5
Mitotic Spindle	+2.9*	+2.5*	+4.4*	−0.7	+4.3*	+1.6	+1.1	+4.0*	<.001	2↑0↓	3↑0↓	+1.1	−2.0
TGF-beta Signaling	+2.4*	+1.4	+0.7	+1.2	+5.7*	+1.5	+2.3*	+4.3*	<.001	3↑0↓	5↑0↓	+2.0	−3.3
Nuclear receptors (class)	+1.3	+4.7*	+0.6	+5.9*	+0.8	+1.9	+0.6	+3.6*	0.002	2↑0↓	1↑0↓	+3.0	−2.8
SLC transporters (class)	−0.3	+5.3*	+2.0	+8.8*	+1.0	+1.3	+1.1	−0.2	0.002	2↑0↓	2↑0↓	+0.5	+0.5
KRAS Signaling Dn	+1.3	+2.2	+1.6	+5.6*	+0.1	+1.4	+3.1*	+2.4*	<.001	4↑0↓	3↑0↓	+0.2	−1.2
p53 Pathway	+1.9	−0.6	+2.9*	+1.9	+4.2*	+0.3	+2.4*	+4.4*	<.001	5↑0↓	1↑0↓	+0.2	+0.1
Cholesterol Homeostasis	−0.5	+0.9	+1.1	+2.5*	+2.7*	+3.2*	+2.7*	+1.9	<.001	4↑0↓	0↑2↓	−1.8	+1.3
Transcription factors (class)	+5.9*	+3.9*	−2.2*	−0.6	+1.0	+0.4	+1.1	+4.4*	0.002	3↑0↓	5↑0↓	+1.4	−1.4
Notch Signaling		−0.1	+1.1	+1.5	+3.0*	+2.7*	+0.9	+2.4*	0.002	2↑0↓	0↑0↓	+2.7	−4.5
mTORC1 Signaling	−1.7	−1.4	+5.6*	+2.3*	+3.8*	−0.1	+0.6	0.0	0.002	2↑0↓	1↑1↓	+3.4	−2.7
UV Response Up	−0.3	−1.4	+2.7*	+2.6*	+1.6	+0.5	+0.3	+2.1*	0.002	2↑0↓	0↑0↓	+1.5	−1.8
Recurrent, below the prior													
G2-M Checkpoint	+1.0	+0.1	+7.2*	−6.1*	+0.5	−1.2	−4.1*	−3.2*	0.002	1↑1↓	1↑1↓	+3.0	−2.4
C2H2 zinc-finger TFs (class)	+3.2*	+0.1	−4.2*	−3.2*	−4.4*	−1.3	−2.9*	+0.6	<.001	0↑4↓	0↑1↓	−1.9	+1.0
Unfolded Protein Response	−0.7	−5.3*	+2.7*	−3.4*	+1.9	−2.2	−3.3*	−3.0*	<.001	2↑2↓	2↑3↓	+3.6	−2.5
Myc Targets V2	−0.9	−4.6*	+2.5*	−2.9*	−0.5	−1.7	−1.8	−3.8*	0.002	1↑1↓	1↑2↓	−1.7	+1.7
E2F Targets	+0.1	−3.5*	+9.0*	−6.4*	−2.7*	−4.0*	−3.4*	−4.1*	<.001	1↑4↓	1↑3↓	+1.9	−1.7
Mitochondria & OXPHOS (category)	−3.2*	−3.1*	−4.7*	+1.7	−3.1*	+1.2	−5.0*	−7.6*	<.001	1↑6↓	0↑3↓	−5.0	+3.1
DNA Repair	0.0	−6.7*	−0.1	−2.8*	−3.6*	−2.2	−4.5*	−4.8*	<.001	0↑5↓	0↑4↓	+2.2	−3.2
Myc Targets V1	−0.6	−7.2*	+4.3*	−4.4*	−2.3*	−4.5*	−5.0*	−5.0*	<.001	1↑3↓	2↑4↓	+7.0	−6.6
Protein homeostasis (category)	−1.9	−4.7*	−1.8	−3.4*	−1.6	−5.5*	−5.8*	−2.7*	<.001	0↑4↓	0↑4↓	+4.1	−3.8
Oxidative Phosphorylation	−2.6*	−9.3*	−3.1*	+1.6	−3.0*	+0.7	−5.3*	−9.4*	<.001	1↑5↓	0↑4↓	−1.7	+2.7
Enzymes (class)	−6.2*	−2.0	+0.2	−4.7*	−7.3*	−1.4	−5.3*	−4.8*	<.001	0↑5↓	0↑6↓	+5.0	−2.8
Translation & ribosome (category)	+0.4	−15.8*	0.0	+2.0	−3.9*	−5.6*	−4.2*	−11.2*	<.001	2↑4↓	1↑3↓	−0.5	+1.8
Chromatin regulators (class)	−3.7*	−1.1	−3.4*	−12.0*	−5.5*	−4.6*	−6.5*	−3.4*	<.001	0↑7↓	1↑6↓	+5.3	−4.9
Chromatin & DNA processing (category)	−1.7	−7.2*	−0.8	−9.0*	−6.5*	−5.7*	−8.7*	−8.8*	<.001	0↑5↓	0↑5↓	+4.3	−4.1
Mitochondrial proteins (class)	−5.0*	−14.2*	−6.0*	−0.4	−10.6*	−2.9*	−8.8*	−21.0*	<.001	0↑7↓	0↑6↓	−8.3	+9.0
Called in one tissue only													
Heme Metabolism	−1.5	−1.4	+8.2*	+0.3	+0.5	+1.1	+0.4	+0.1	0.351	1↑2↓	0↑1↓	+0.7	+0.4
Bile Acid Metabolism	+0.1	+1.0	+1.4	+2.6*	−0.9	+0.1	−1.5	−0.5	0.351	1↑1↓	1↑0↓	−2.5	+2.2
Protein Secretion	+0.4	−2.5*	+0.8	−0.6	+0.7	−0.5	−0.8	−1.2	0.351	0↑0↓	0↑0↓	+2.7	−2.8



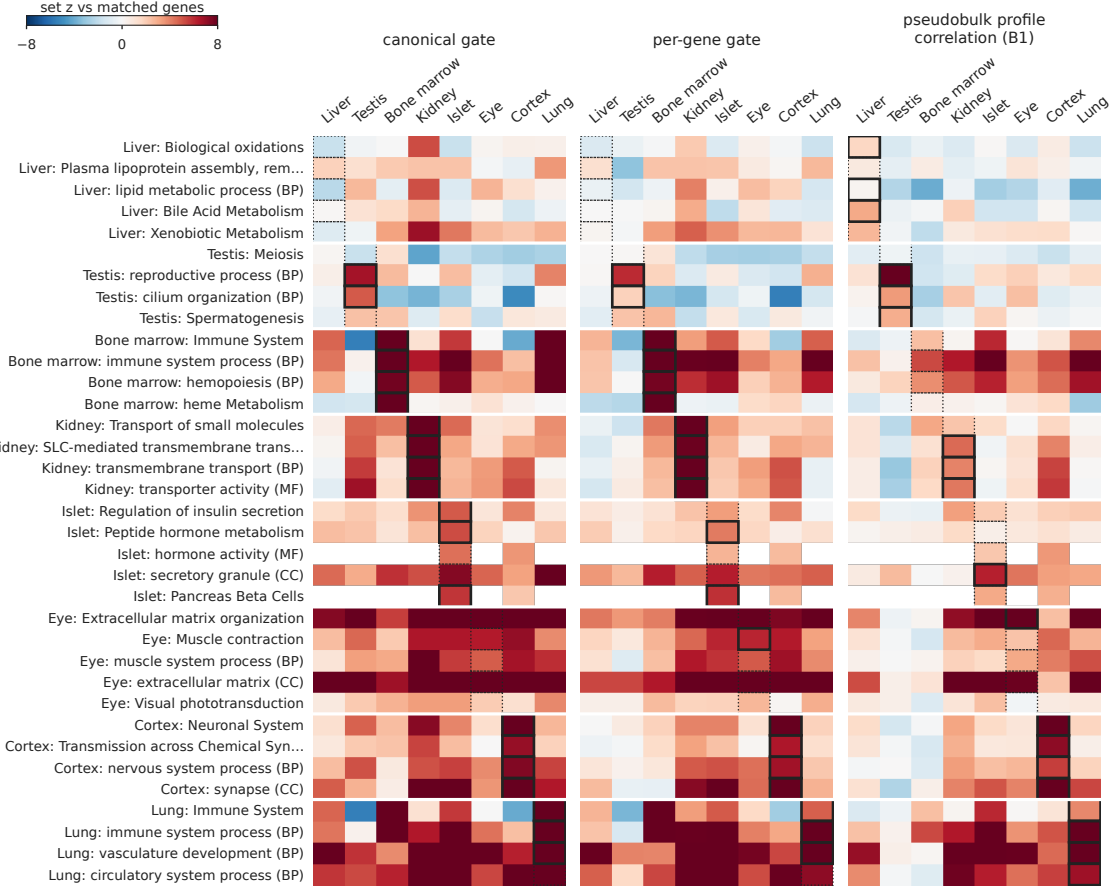
Cells: z of the set against matched genes of the tissue; * $q < 0.05$; gray cell, fewer than 15 genes.
Right: q, recurrence across tissues; S_g and profile correlation, tissues with the set above (↑) and below (↓);
π and dN/dS, mean z of the set over tissues. Gray type: not recurrent, or |mean z| below 1.96.


canonical gate
per-gene gate
Δα effect vs matched genes (SD)
−0.6
0.0
0.6
Liver
Testis
Bone marrow
Kidney
Islet
Eye
Cortex
Lung
π z
dN/dS z
B1 z
Glycoprotein [PTM] -19 +15 +24
Secreted [CC] -27 +23 +19
external encapsulating structure (CC) -9 +7 +18
molecular transducer activity (MF) -8 +3 +19
Integrin cell surface interactions (R) -8 +10 +12
KRAS Signaling Up -7 +4 +9
plasma membrane (CC) +9 -17 +30
signaling (BP) +15 -21 +23
Calcium [Ligand] +4 -9 +12
Signal Transduction (R) +35 -41 +18
Hemostasis (R) +9 -10 +15
Cytoskeleton & ECM (category) +8 -9 +13
nucleoplasm (CC) +29 -26 -15
RNA binding (MF) +14 -12 -9
Metabolism of RNA (R) +24 -25 -12
RNA-binding [MF] +12 -10 -7
mRNA processing [BP] +19 -21 -7
Chromatin & DNA processing (category) +12 -12 -7
Mitochondrion [CC] -23 +26 -12
DNA repair (BP) -5 +9 -12
DNA damage [BP] -7 +9 -12
catalytic activity, acting on DNA (MF) -4 +9 -6
Diseases of DNA repair (R) -7 +9 -4
fast protein, above prior
conserved protein, above prior
conserved protein, below prior
fast protein, below prior

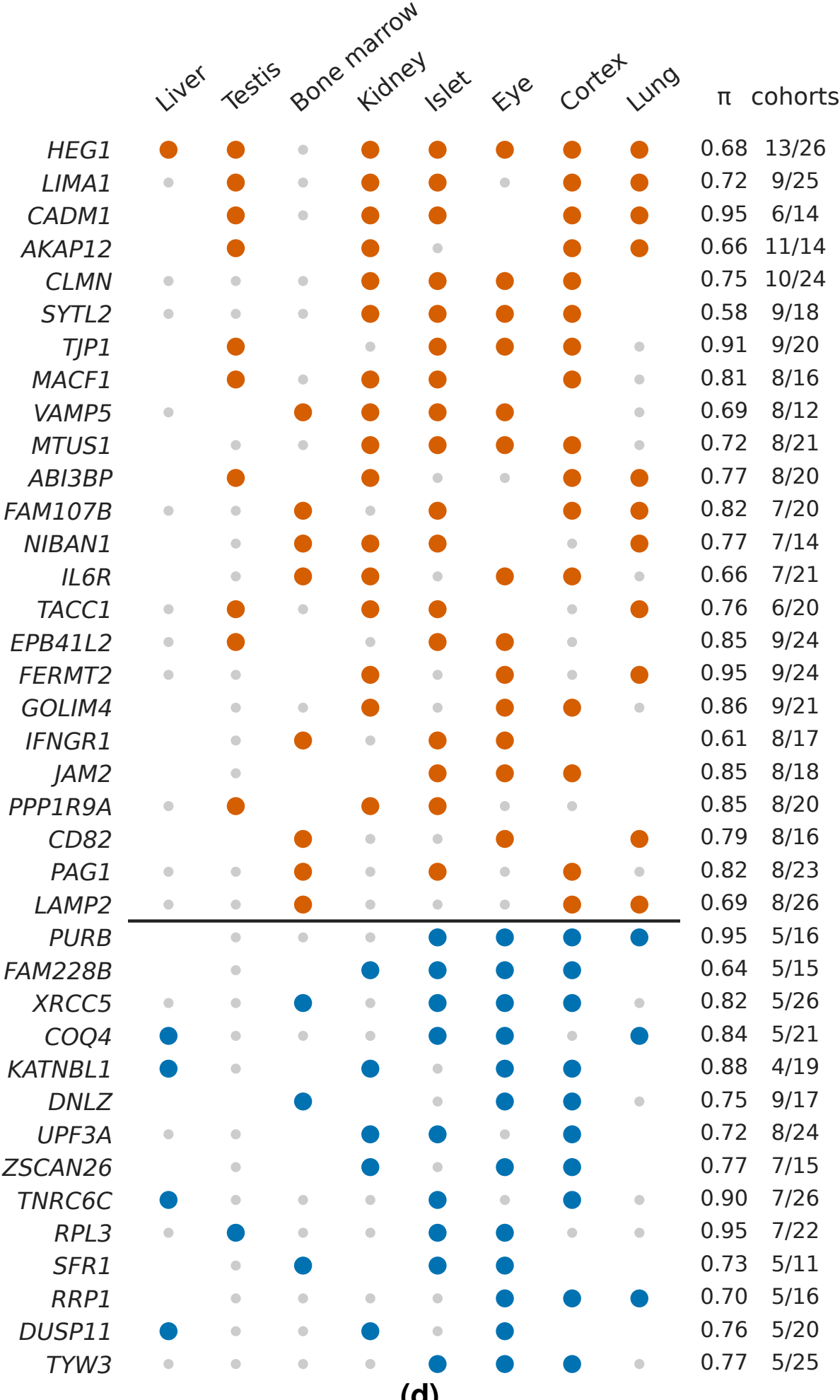
(a)


programme pole, mean effect (SD units, matched genes)
above-prior programmes
below-prior programmes
0.6
0.4
0.2
0.0
−0.2
−0.4
6
29
90
429
divergence from human (My)
agreement with other tissues' programme order (ρ)
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
cortex
liver
eye
testis
canonical
per-gene

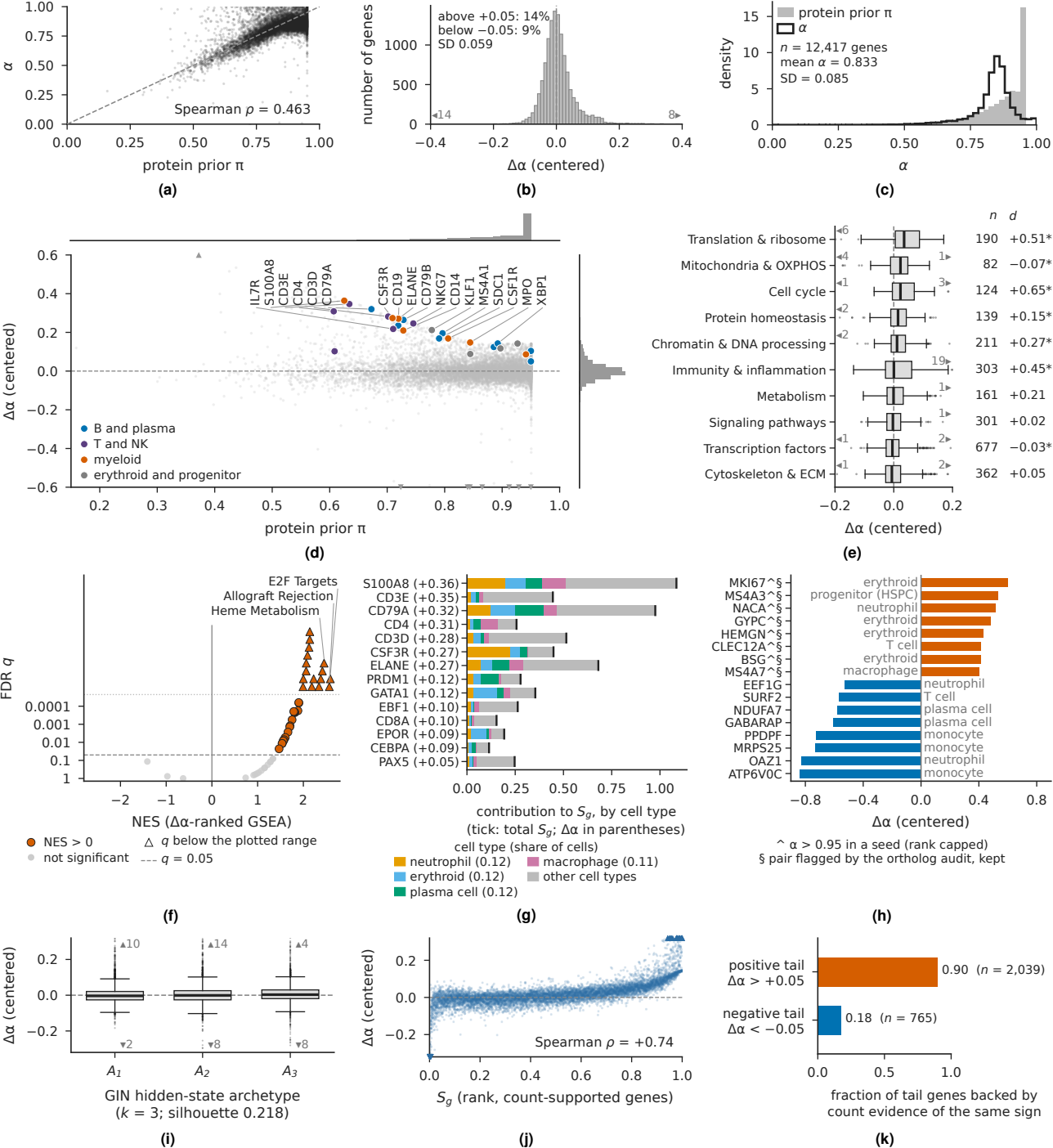
(b)

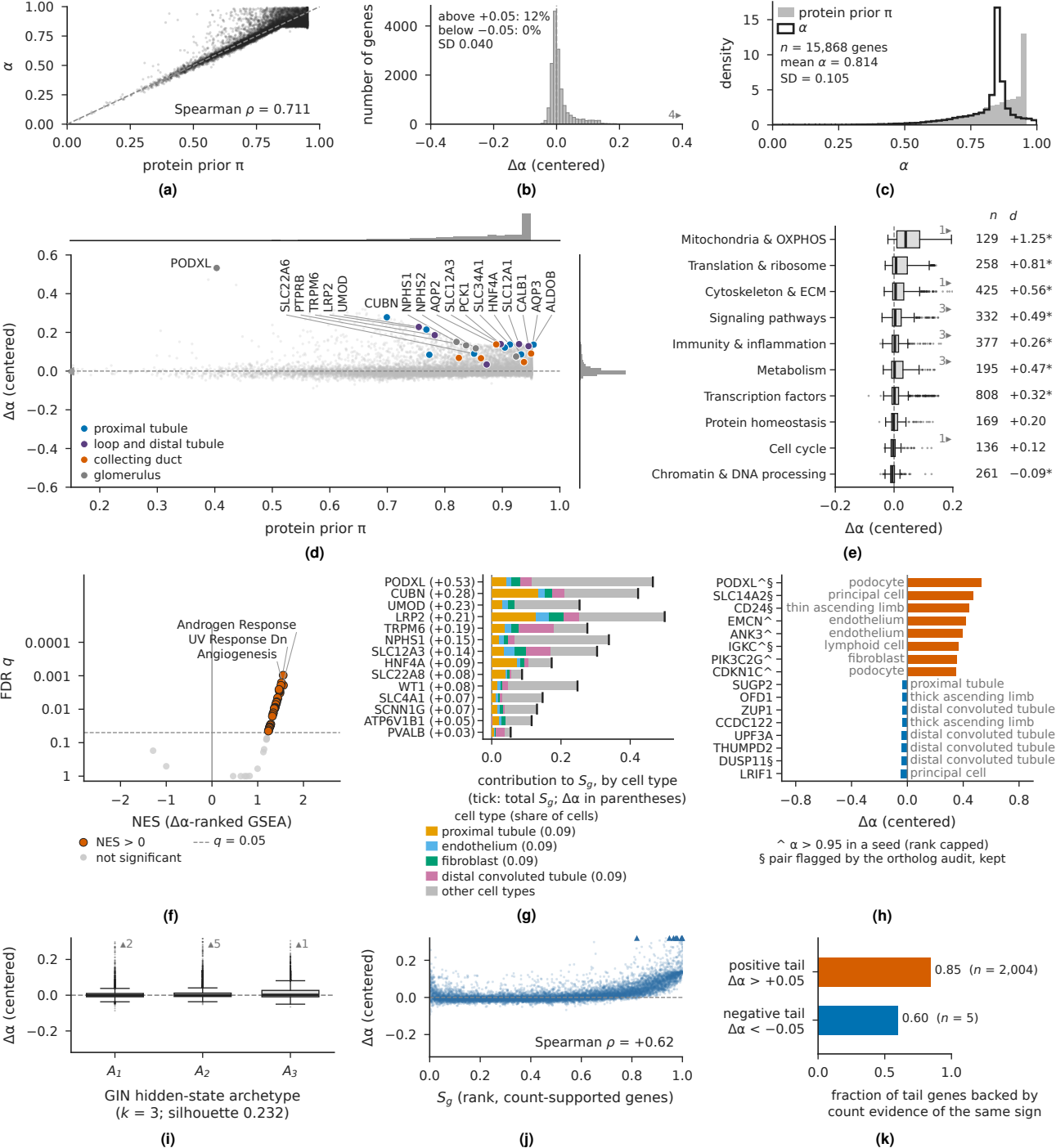

set z vs matched genes
−8
0
8
canonical gate
per-gene gate
pseudobulk profile correlation (B1)
Liver
Testis
Bone marrow
Kidney
Islet
Eye
Cortex
Lung
Liver: Biological oxidations
Liver: Plasma lipoprotein assembly, rem...
Liver: lipid metabolic process (BP)
Liver: Bile Acid Metabolism
Liver: Xenobiotic Metabolism
Testis: Meiosis
Testis: reproductive process (BP)
Testis: cilium organization (BP)
Testis: Spermatogenesis
Bone marrow: Immune System
Bone marrow: immune system process (BP)
Bone marrow: hemopoiesis (BP)
Bone marrow: heme Metabolism
Kidney: Transport of small molecules
Kidney: SLC-mediated transmembrane trans...
Kidney: transmembrane transport (BP)
Kidney: transporter activity (MF)
Islet: Regulation of insulin secretion
Islet: Peptide hormone metabolism
Islet: hormone activity (MF)
Islet: secretory granule (CC)
Islet: Pancreas Beta Cells
Eye: Extracellular matrix organization
Eye: Muscle contraction
Eye: muscle system process (BP)
Eye: extracellular matrix (CC)
Eye: Visual phototransduction
Cortex: Neuronal System
Cortex: Transmission across Chemical Syn...
Cortex: nervous system process (BP)
Cortex: synapse (CC)
Lung: Immune System
Lung: immune system process (BP)
Lung: vasculature development (BP)
Lung: circulatory system process (BP)

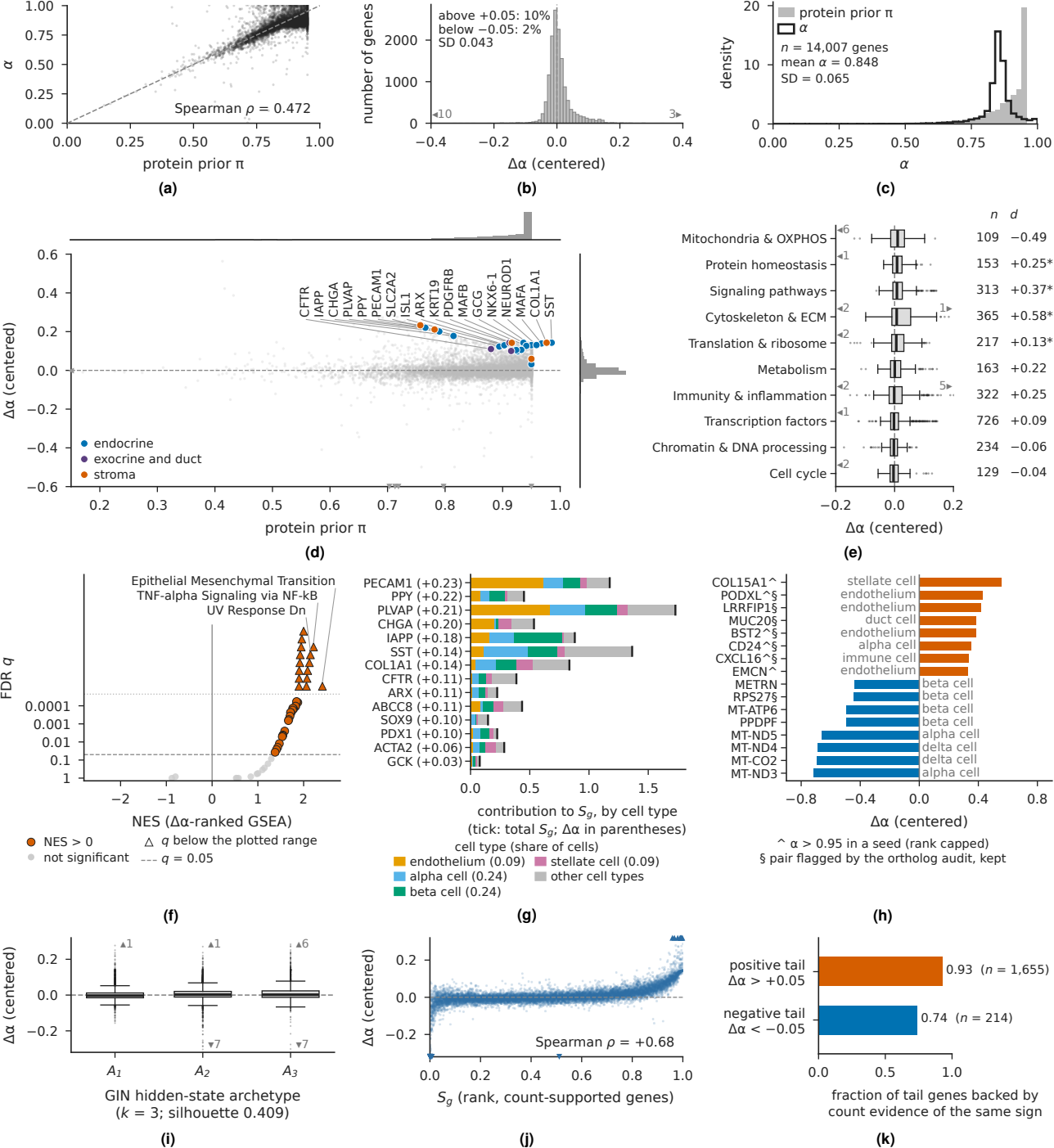
(c)

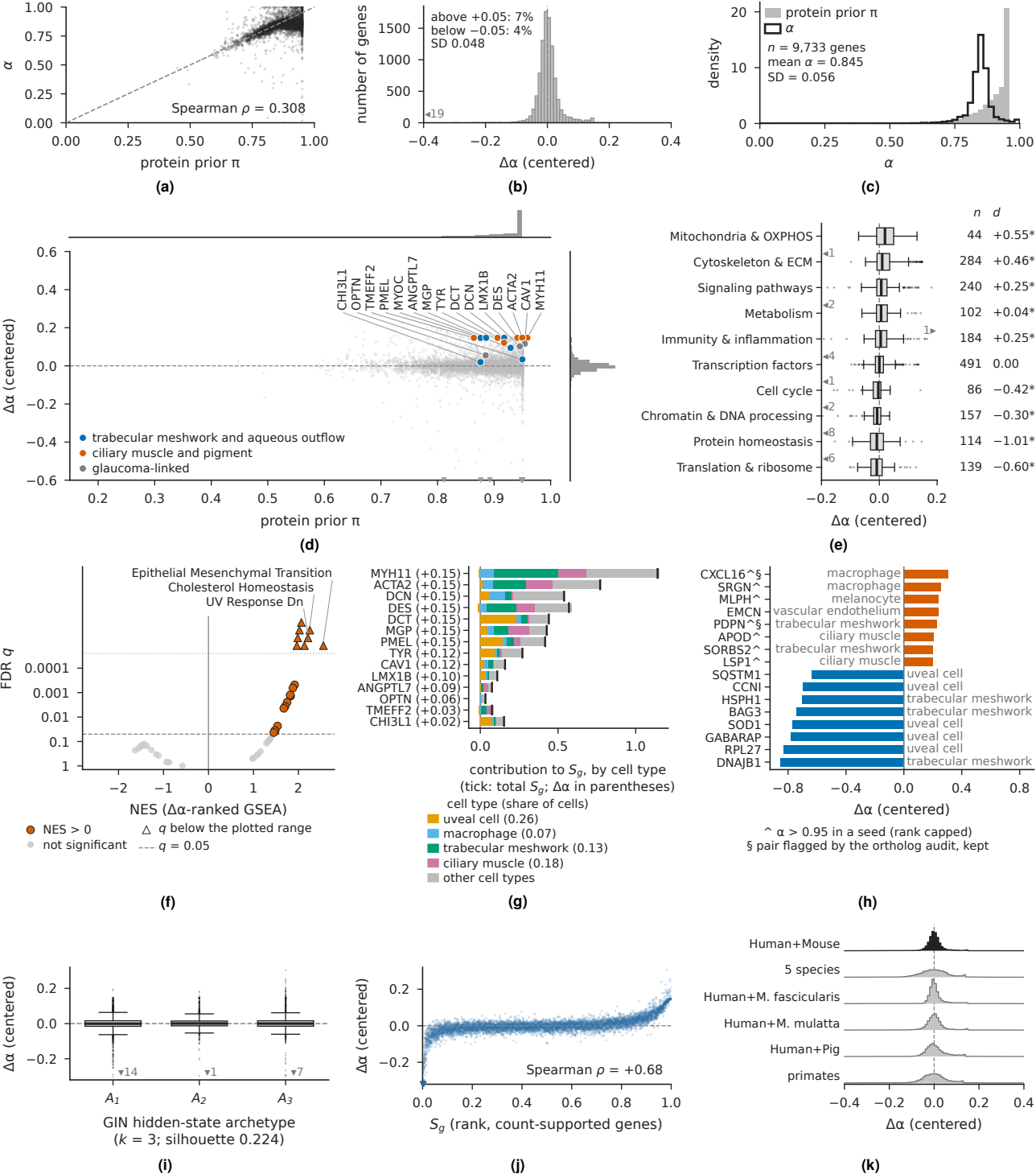

Liver
Testis
Bone marrow
Kidney
Islet
Eye
Cortex
Lung
π
cohorts
HEG1 0.68 13/26
LIMA1 0.72 9/25
CADM1 0.95 6/14
AKAP12 0.66 11/14
CLMN 0.75 10/24
SYTL2 0.58 9/18
TJP1 0.91 9/20
MACF1 0.81 8/16
VAMP5 0.69 8/12
MTUS1 0.72 8/21
ABI3BP 0.77 8/20
FAM107B 0.82 7/20
NIBAN1 0.77 7/14
IL6R 0.66 7/21
TACC1 0.76 6/20
EPB41L2 0.85 9/24
FERMT2 0.95 9/24
GOLIM4 0.86 9/21
IFNGR1 0.61 8/17
JAM2 0.85 8/18
PPP1R9A 0.85 8/20
CD82 0.79 8/16
PAG1 0.82 8/23
LAMP2 0.69 8/26
PURB 0.95 5/16
FAM228B 0.64 5/15
XRCC5 0.82 5/26
COQ4 0.84 5/21
KATNBL1 0.88 4/19
DNLZ 0.75 9/17
UPF3A 0.72 8/24
ZSCAN26 0.77 7/15
TNRC6C 0.90 7/26
RPL3 0.95 7/22
SFR1 0.73 5/11
RRP1 0.70 5/16
DUSP11 0.76 5/20
TYW3 0.77 5/25

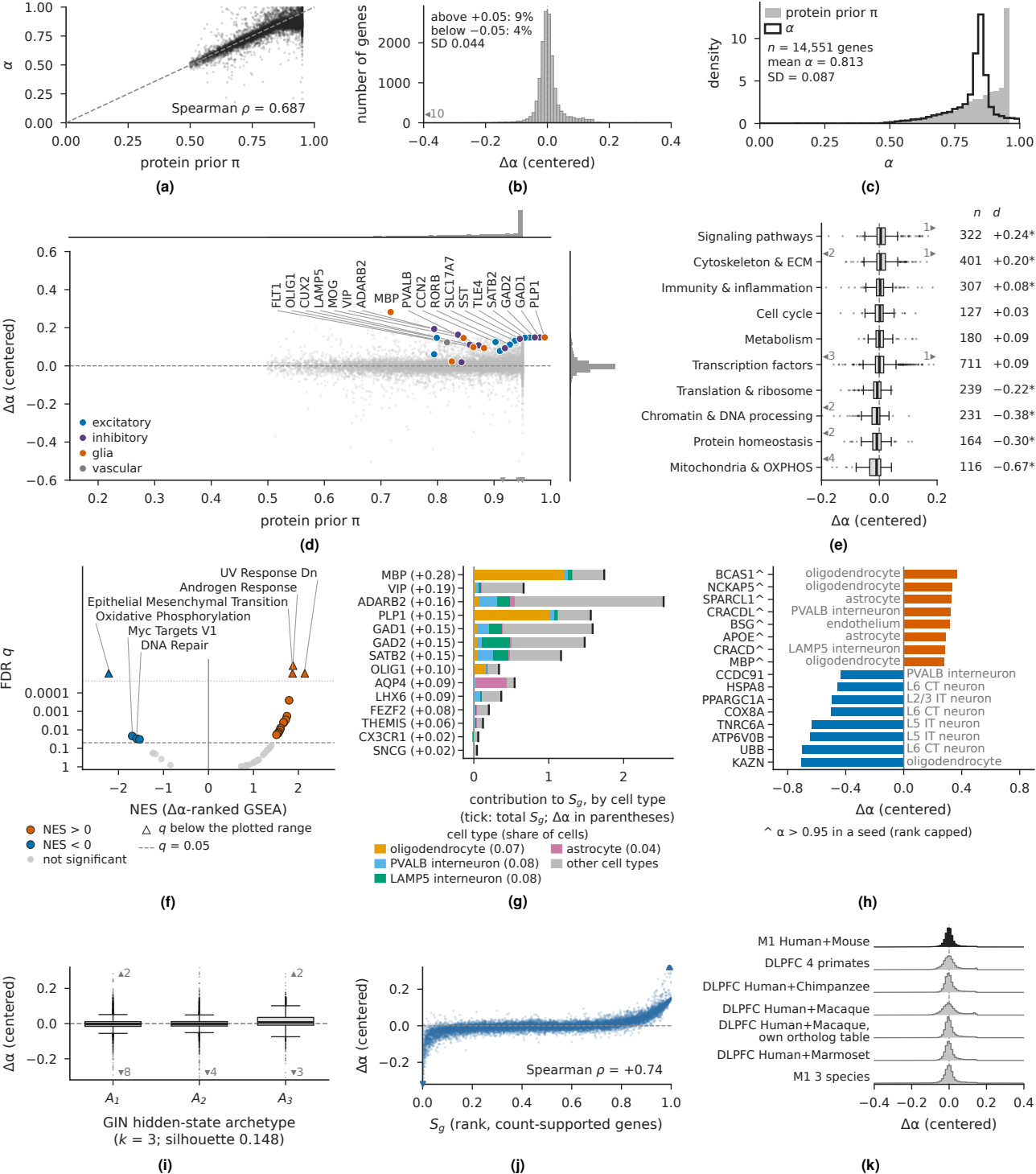
(d)

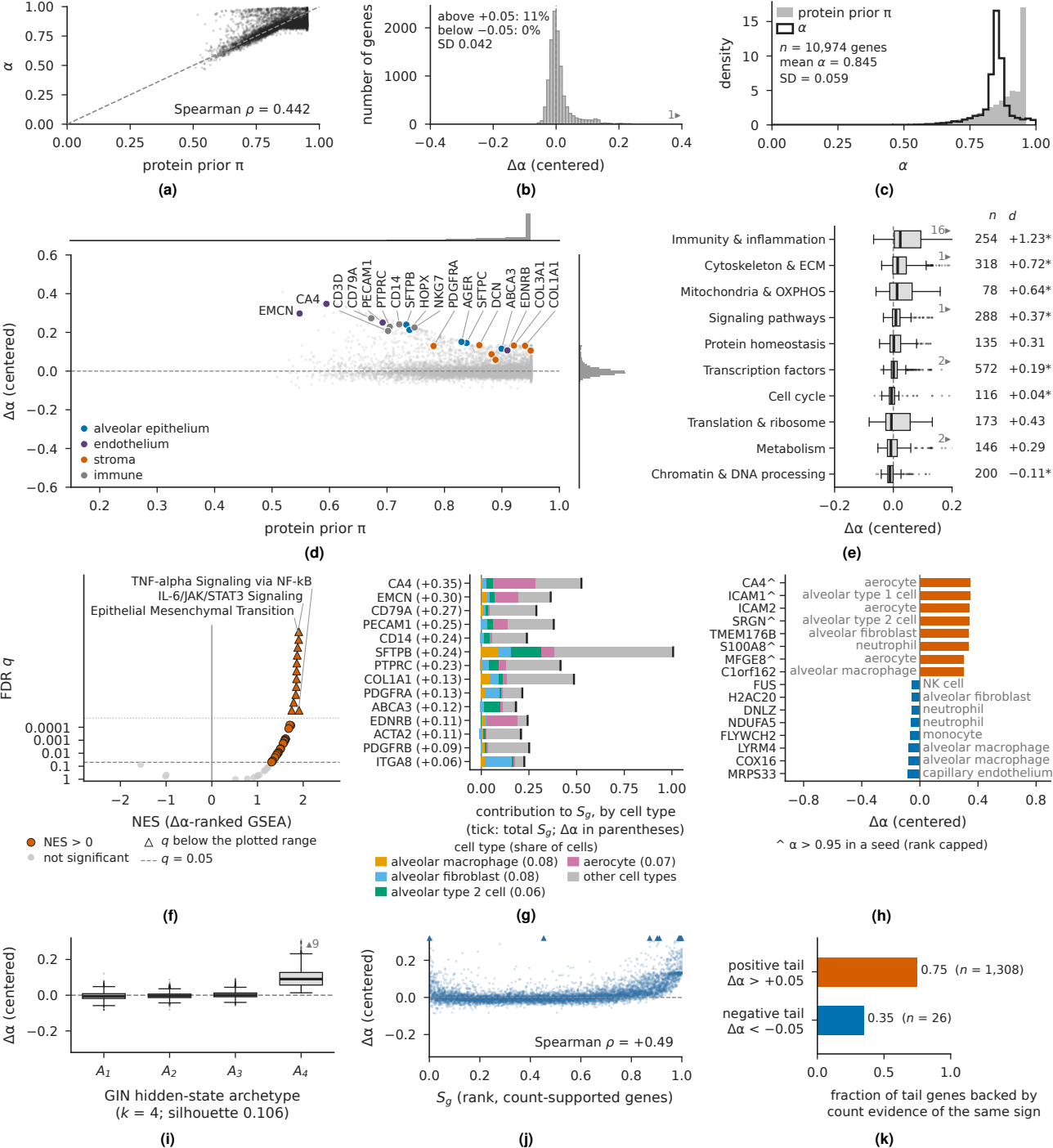

Spearman ρ = 0.463
protein prior π
α
(a)
above +0.05: 14%
below −0.05: 9%
SD 0.059
number of genes
Δα (centered)
(b)
protein prior π
α
n = 12,417 genes
mean α = 0.833
SD = 0.085
density
(c)
IL7R
S100A8
CD3E
CD4
CD3D
CD79A
CSF3R
CD19
ELANE
CD79B
NKG7
CD14
KLF1
MS4A1
SDC1
CSF1R
MPO
XBP1
B and plasma
T and NK
myeloid
erythroid and progenitor
Δα (centered)
protein prior π
(d)
n
d
Translation & ribosome 190 +0.51*
Mitochondria & OXPHOS 82 −0.07*
Cell cycle 124 +0.65*
Protein homeostasis 139 +0.15*
Chromatin & DNA processing 211 +0.27*
Immunity & inflammation 303 +0.45*
Metabolism 161 +0.21
Signaling pathways 301 +0.02
Transcription factors 677 −0.03*
Cytoskeleton & ECM 362 +0.05
Δα (centered)
(e)
E2F Targets
Allograft Rejection
Heme Metabolism
FDR q
NES (Δα-ranked GSEA)
NES > 0
not significant
△ q below the plotted range
q = 0.05
(f)
S100A8 (+0.36)
CD3E (+0.35)
CD79A (+0.32)
CD4 (+0.31)
CD3D (+0.28)
CSF3R (+0.27)
ELANE (+0.27)
PRDM1 (+0.12)
GATA1 (+0.12)
EBF1 (+0.10)
CD8A (+0.10)
EPOR (+0.09)
CEBPA (+0.09)
PAX5 (+0.05)
contribution to S_g, by cell type
(tick: total S_g; Δα in parentheses)
cell type (share of cells)
neutrophil (0.12)
macrophage (0.11)
erythroid (0.12)
other cell types
plasma cell (0.12)
(g)
MKI67^§ erythroid
MS4A3^§ progenitor (HSPC)
NACA^§ neutrophil
GYPC^§ erythroid
HEMGN^§ erythroid
CLEC12A^§ T cell
BSG^§ erythroid
MS4A7^§ macrophage
EEF1G neutrophil
SURF2 T cell
NDUFA7 plasma cell
GABARAP plasma cell
PPDPF monocyte
MRPS25 monocyte
OAZ1 neutrophil
ATP6V0C monocyte
Δα (centered)
^ α > 0.95 in a seed (rank capped)
§ pair flagged by the ortholog audit, kept
(h)
Δα (centered)
A_1
A_2
A_3
GIN hidden-state archetype
(k = 3; silhouette 0.218)
(i)
Spearman ρ = +0.74
S_g (rank, count-supported genes)
(j)
positive tail
Δα > +0.05
0.90 (n = 2,039)
negative tail
Δα < −0.05
0.18 (n = 765)
fraction of tail genes backed by
count evidence of the same sign
(k)

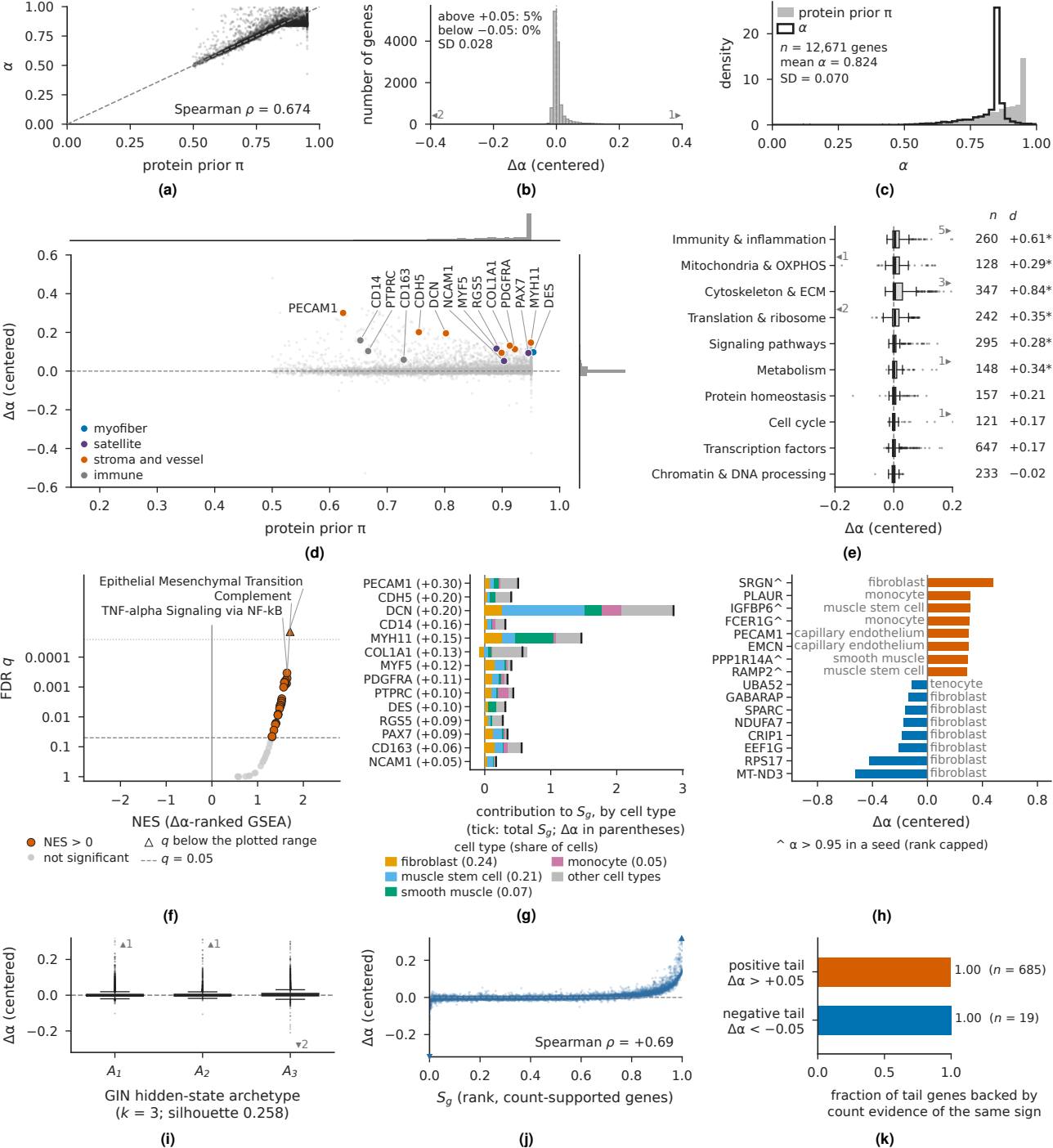

Spearman ρ = 0.711
protein prior π
α
(a)
above +0.05: 12%
below −0.05: 0%
SD 0.040
number of genes
Δα (centered)
(b)
protein prior π
α
n = 15,868 genes
mean α = 0.814
SD = 0.105
density
(c)
PODXL
CUBN
SLC22A6
PTPRB
TRPM6
LRP2
UMOD
NPHS1
NPHS2
AQP2
SLC12A3
PCK1
SLC34A1
HNF4A
SLC12A1
CALB1
AQP3
ALDOB
proximal tubule
loop and distal tubule
collecting duct
glomerulus
Δα (centered)
protein prior π
(d)
n d
Mitochondria & OXPHOS 129 +1.25*
Translation & ribosome 258 +0.81*
Cytoskeleton & ECM 425 +0.56*
Signaling pathways 332 +0.49*
Immunity & inflammation 377 +0.26*
Metabolism 195 +0.47*
Transcription factors 808 +0.32*
Protein homeostasis 169 +0.20
Cell cycle 136 +0.12
Chromatin & DNA processing 261 −0.09*
Δα (centered)
(e)
Androgen Response
UV Response Dn
Angiogenesis
FDR q
NES (Δα-ranked GSEA)
NES > 0
q = 0.05
not significant
(f)
PODXL (+0.53)
CUBN (+0.28)
UMOD (+0.23)
LRP2 (+0.21)
TRPM6 (+0.19)
NPHS1 (+0.15)
SLC12A3 (+0.14)
HNF4A (+0.09)
SLC22A8 (+0.08)
WT1 (+0.08)
SLC4A1 (+0.07)
SCNN1G (+0.07)
ATP6V1B1 (+0.05)
PVALB (+0.03)
contribution to S_g, by cell type
(tick: total S_g; Δα in parentheses)
cell type (share of cells)
proximal tubule (0.09)
endothelium (0.09)
fibroblast (0.09)
distal convoluted tubule (0.09)
other cell types
(g)
PODXL^§ podocyte
SLC14A2§ principal cell
CD24§ thin ascending limb
EMCN^ endothelium
ANK3^ endothelium
IGKC^§ lymphoid cell
PIK3C2G^ fibroblast
CDKN1C^ podocyte
SUGP2 proximal tubule
OFD1 thick ascending limb
ZUP1 distal convoluted tubule
CCDC122 thick ascending limb
UPF3A distal convoluted tubule
THUMPD2 distal convoluted tubule
DUSP11§ distal convoluted tubule
LRIF1 principal cell
Δα (centered)
^ α > 0.95 in a seed (rank capped)
§ pair flagged by the ortholog audit, kept
(h)
Δα (centered)
GIN hidden-state archetype
(k = 3; silhouette 0.232)
(i)
Δα (centered)
Spearman ρ = +0.62
S_g (rank, count-supported genes)
(j)
positive tail
Δα > +0.05
0.85 (n = 2,004)
negative tail
Δα < −0.05
0.60 (n = 5)
fraction of tail genes backed by
count evidence of the same sign
(k)

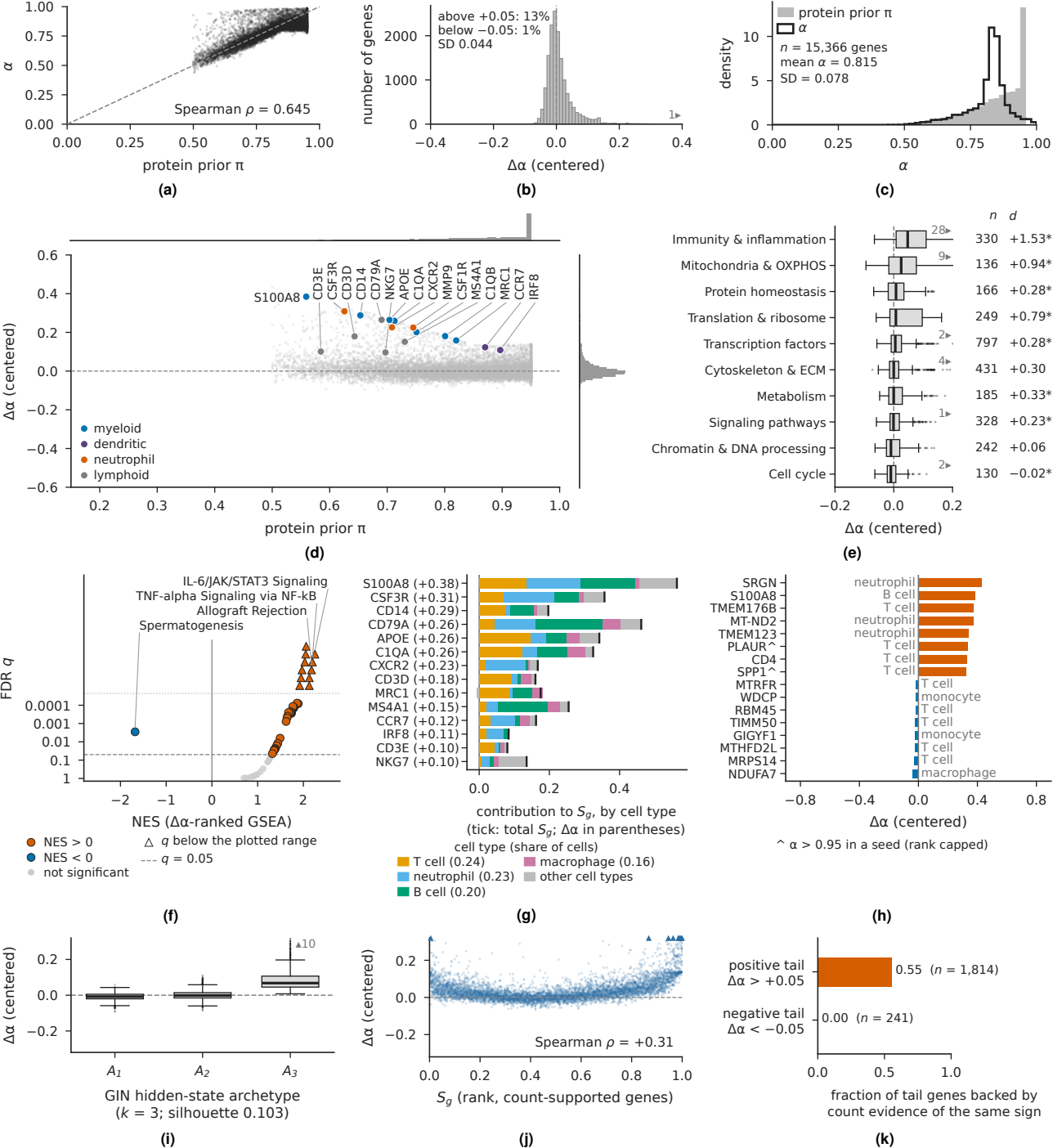

Spearman ρ = 0.472
protein prior π
α
(a)
above +0.05: 10%
below −0.05: 2%
SD 0.043
number of genes
Δα (centered)
(b)
protein prior π
α
n = 14,007 genes
mean α = 0.848
SD = 0.065
density
(c)
CFTR
IAPP
CHGA
PLVAP
PPY
PECAM1
SLC2A2
ISL1
ARX
KRT19
PDGFRB
MAFB
GCG
NKX6-1
NEUROD1
MAFA
COL1A1
SST
endocrine
exocrine and duct
stroma
Δα (centered)
protein prior π
(d)
n d
Mitochondria & OXPHOS 109 −0.49
Protein homeostasis 153 +0.25*
Signaling pathways 313 +0.37*
Cytoskeleton & ECM 365 +0.58*
Translation & ribosome 217 +0.13*
Metabolism 163 +0.22
Immunity & inflammation 322 +0.25
Transcription factors 726 +0.09
Chromatin & DNA processing 234 −0.06
Cell cycle 129 −0.04
Δα (centered)
(e)
Epithelial Mesenchymal Transition
TNF-alpha Signaling via NF-kB
UV Response Dn
FDR q
NES (Δα-ranked GSEA)
NES > 0
not significant
q below the plotted range
q = 0.05
(f)
PECAM1 (+0.23)
PPY (+0.22)
PLVAP (+0.21)
CHGA (+0.20)
IAPP (+0.18)
SST (+0.14)
COL1A1 (+0.14)
CFTR (+0.11)
ARX (+0.11)
ABCC8 (+0.11)
SOX9 (+0.10)
PDX1 (+0.10)
ACTA2 (+0.06)
GCK (+0.03)
contribution to S_g, by cell type
(tick: total S_g; Δα in parentheses)
cell type (share of cells)
endothelium (0.09)
stellate cell (0.09)
alpha cell (0.24)
other cell types
beta cell (0.24)
(g)
COL15A1^
PODXL^§
LRRFIP1§
MUC20§
BST2^§
CD24^§
CXCL16^§
EMCN^
METRN
RPS27§
MT-ATP6
PPDPF
MT-ND5
MT-ND4
MT-CO2
MT-ND3
stellate cell
endothelium
endothelium
duct cell
endothelium
alpha cell
immune cell
endothelium
beta cell
beta cell
beta cell
beta cell
alpha cell
delta cell
delta cell
alpha cell
Δα (centered)
^ α > 0.95 in a seed (rank capped)
§ pair flagged by the ortholog audit, kept
(h)
Δα (centered)
A_1
A_2
A_3
GIN hidden-state archetype
(k = 3; silhouette 0.409)
(i)
Spearman ρ = +0.68
S_g (rank, count-supported genes)
(j)
positive tail
Δα > +0.05
0.93 (n = 1,655)
negative tail
Δα < −0.05
0.74 (n = 214)
fraction of tail genes backed by
count evidence of the same sign
(k)

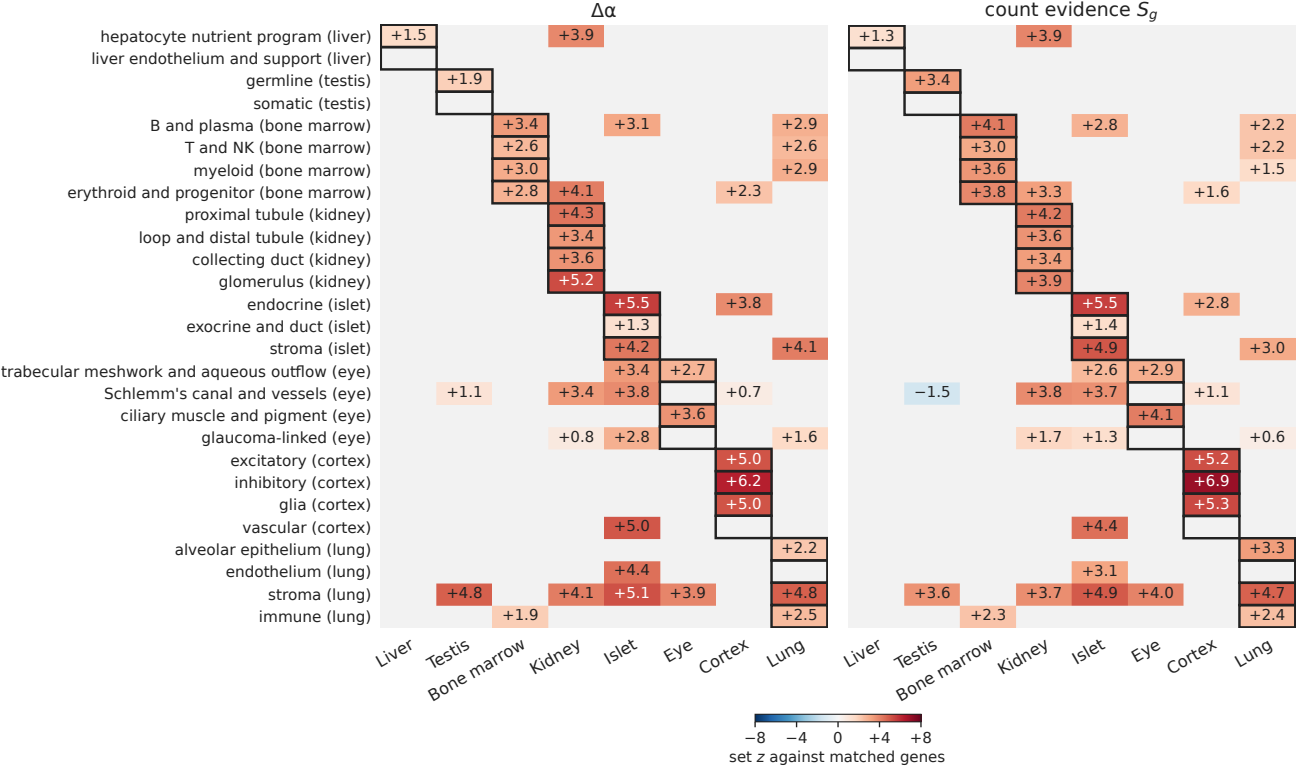

Spearman ρ = 0.308
protein prior π
α
(a)
above +0.05: 7%
below −0.05: 4%
SD 0.048
number of genes
Δα (centered)
(b)
protein prior π
α
n = 9,733 genes
mean α = 0.845
SD = 0.056
density
(c)
CHI3L1
OPTN
TMEFF2
PMEL
MYOC
ANGPTL7
MGP
TYR
DCT
DCN
LMX1B
DES
ACTA2
CAV1
MYH11
trabecular meshwork and aqueous outflow
ciliary muscle and pigment
glaucoma-linked
Δα (centered)
protein prior π
(d)
n
d
Mitochondria & OXPHOS 44 +0.55*
Cytoskeleton & ECM 284 +0.46*
Signaling pathways 240 +0.25*
Metabolism 102 +0.04*
Immunity & inflammation 184 +0.25*
Transcription factors 491 0.00
Cell cycle 86 −0.42*
Chromatin & DNA processing 157 −0.30*
Protein homeostasis 114 −1.01*
Translation & ribosome 139 −0.60*
Δα (centered)
(e)
Epithelial Mesenchymal Transition
Cholesterol Homeostasis
UV Response Dn
FDR q
NES (Δα-ranked GSEA)
NES > 0
not significant
q below the plotted range
q = 0.05
(f)
MYH11 (+0.15)
ACTA2 (+0.15)
DCN (+0.15)
DES (+0.15)
DCT (+0.15)
MGP (+0.15)
PMEL (+0.15)
TYR (+0.12)
CAV1 (+0.12)
LMX1B (+0.10)
ANGPTL7 (+0.09)
OPTN (+0.06)
TMEFF2 (+0.03)
CHI3L1 (+0.02)
contribution to S_g, by cell type
(tick: total S_g; Δα in parentheses)
cell type (share of cells)
uveal cell (0.26)
macrophage (0.07)
trabecular meshwork (0.13)
ciliary muscle (0.18)
other cell types
(g)
CXCL16^§ macrophage
SRGN^ macrophage
MLPH^ melanocyte
EMCN vascular endothelium
PDPN^§ trabecular meshwork
APOD^ ciliary muscle
SORBS2^ trabecular meshwork
LSP1^ ciliary muscle
SQSTM1 uveal cell
CCNI uveal cell
HSPH1 trabecular meshwork
BAG3 trabecular meshwork
SOD1 uveal cell
GABARAP uveal cell
RPL27 uveal cell
DNAJB1 trabecular meshwork
Δα (centered)
^ α > 0.95 in a seed (rank capped)
§ pair flagged by the ortholog audit, kept
(h)
Δα (centered)
GIN hidden-state archetype
(k = 3; silhouette 0.224)
(i)
Spearman ρ = +0.68
S_g (rank, count-supported genes)
(j)
Human+Mouse
5 species
Human+M. fascicularis
Human+M. mulatta
Human+Pig
primates
Δα (centered)
(k)

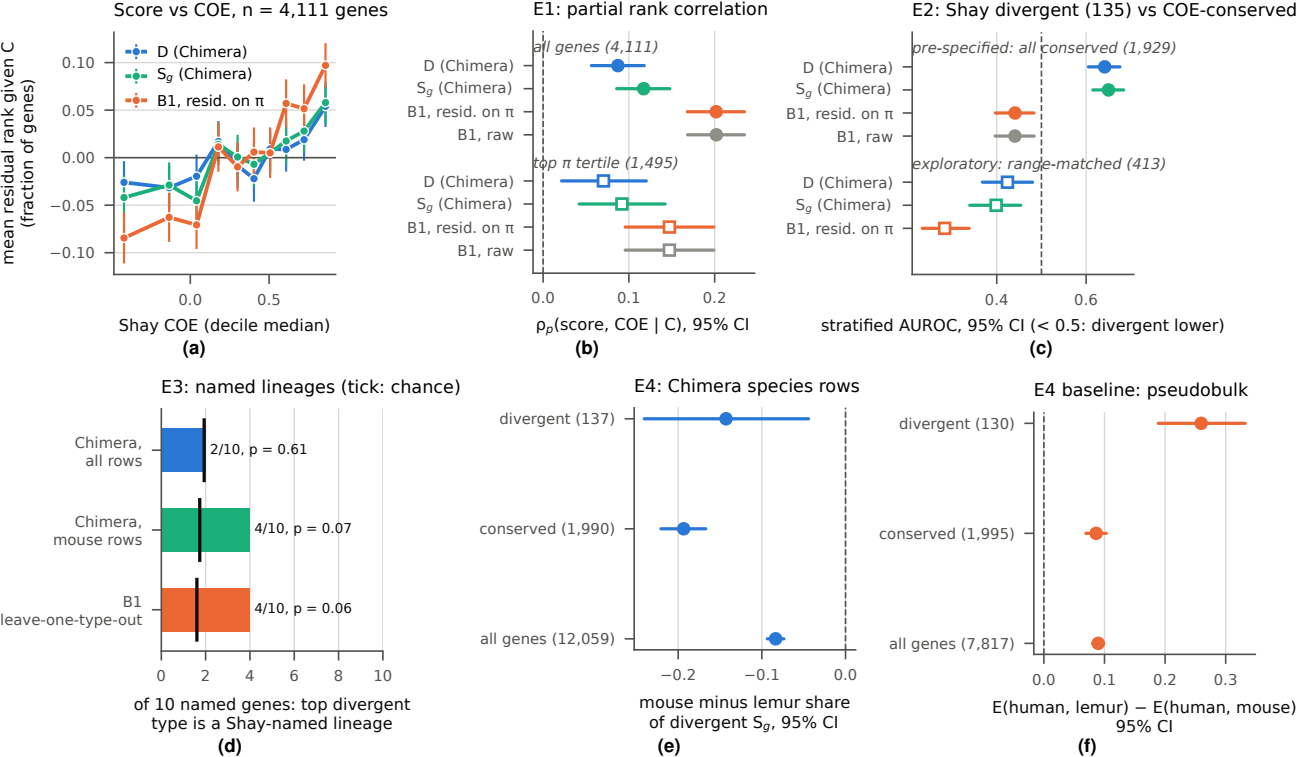

Spearman ρ = 0.687
protein prior π
α
(a)
above +0.05: 9%
below −0.05: 4%
SD 0.044
number of genes
Δα (centered)
(b)
protein prior π
α
n = 14,551 genes
mean α = 0.813
SD = 0.087
density
(c)
FLT1
OLIG1
CUX2
LAMP5
MOG
VIP
ADARB2
MBP
PVALB
CCN2
RORB
SLC17A7
SST
TLE4
SATB2
GAD2
GAD1
PLP1
excitatory
inhibitory
glia
vascular
Δα (centered)
protein prior π
(d)
n
d
Signaling pathways 322 +0.24*
Cytoskeleton & ECM 401 +0.20*
Immunity & inflammation 307 +0.08*
Cell cycle 127 +0.03
Metabolism 180 +0.09
Transcription factors 711 +0.09
Translation & ribosome 239 −0.22*
Chromatin & DNA processing 231 −0.38*
Protein homeostasis 164 −0.30*
Mitochondria & OXPHOS 116 −0.67*
Δα (centered)
(e)
UV Response Dn
Androgen Response
Epithelial Mesenchymal Transition
Oxidative Phosphorylation
Myc Targets V1
DNA Repair
FDR q
NES (Δα-ranked GSEA)
NES > 0
NES < 0
not significant
△ q below the plotted range
--- q = 0.05
(f)
MBP (+0.28)
VIP (+0.19)
ADARB2 (+0.16)
PLP1 (+0.15)
GAD1 (+0.15)
GAD2 (+0.15)
SATB2 (+0.15)
OLIG1 (+0.10)
AQP4 (+0.09)
LHX6 (+0.09)
FEZF2 (+0.08)
THEMIS (+0.06)
CX3CR1 (+0.02)
SNCG (+0.02)
contribution to S_g, by cell type
(tick: total S_g; Δα in parentheses)
cell type (share of cells)
oligodendrocyte (0.07)
astrocyte (0.04)
PVALB interneuron (0.08)
other cell types
LAMP5 interneuron (0.08)
(g)
BCAS1^ oligodendrocyte
NCKAP5^ oligodendrocyte
SPARCL1^ astrocyte
CRACDL^ PVALB interneuron
BSG^ endothelium
APOE^ astrocyte
CRACD^ LAMP5 interneuron
MBP^ oligodendrocyte
CCDC91 PVALB interneuron
HSPA8 L6 CT neuron
PPARGC1A L2/3 IT neuron
COX8A L6 CT neuron
TNRC6A L5 IT neuron
ATP6V0B L5 IT neuron
UBB L6 CT neuron
KAZN oligodendrocyte
Δα (centered)
^ α > 0.95 in a seed (rank capped)
(h)
Δα (centered)
A1
A2
A3
GIN hidden-state archetype
(k = 3; silhouette 0.148)
(i)
Spearman ρ = +0.74
S_g (rank, count-supported genes)
(j)
M1 Human+Mouse
DLPFC 4 primates
DLPFC Human+Chimpanzee
DLPFC Human+Macaque
DLPFC Human+Macaque, own ortholog table
DLPFC Human+Marmoset
M1 3 species
Δα (centered)
(k)

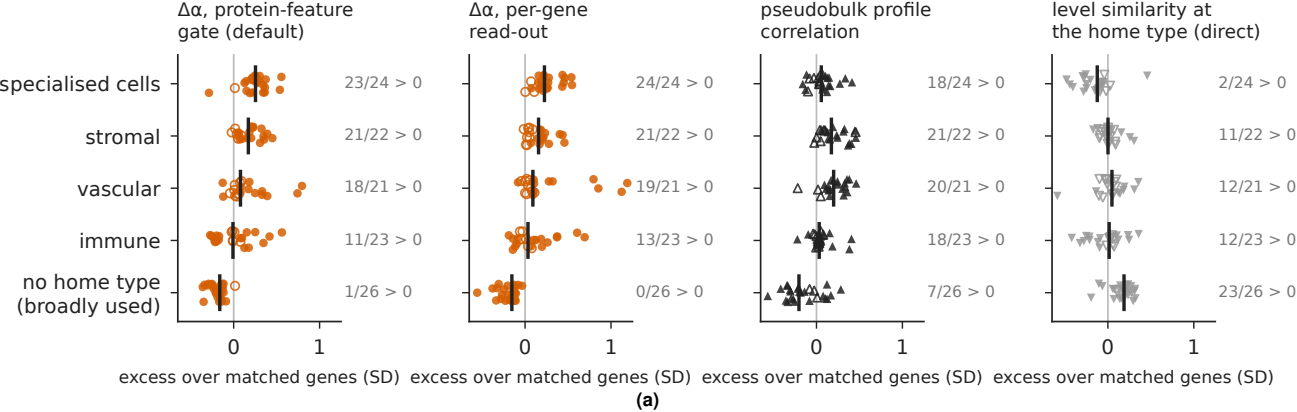

Spearman ρ = 0.442
protein prior π
α
(a)
above +0.05: 11%
below −0.05: 0%
SD 0.042
number of genes
Δα (centered)
(b)
protein prior π
α
n = 10,974 genes
mean α = 0.845
SD = 0.059
density
α
(c)
EMCN
CA4
CD3D
CD79A
PECAM1
PTPRC
CD14
SFTPB
HOPX
NKG7
PDGFRA
AGER
SFTPC
DCN
ABCA3
EDNRB
COL3A1
COL1A1
alveolar epithelium
endothelium
stroma
immune
Δα (centered)
protein prior π
(d)
n
d
Immunity & inflammation 254 +1.23*
Cytoskeleton & ECM 318 +0.72*
Mitochondria & OXPHOS 78 +0.64*
Signaling pathways 288 +0.37*
Protein homeostasis 135 +0.31
Transcription factors 572 +0.19*
Cell cycle 116 +0.04*
Translation & ribosome 173 +0.43
Metabolism 146 +0.29
Chromatin & DNA processing 200 −0.11*
Δα (centered)
(e)
TNF-alpha Signaling via NF-kB
IL-6/JAK/STAT3 Signaling
Epithelial Mesenchymal Transition
FDR q
NES (Δα-ranked GSEA)
NES > 0
not significant
△ q below the plotted range
q = 0.05
(f)
CA4 (+0.35)
EMCN (+0.30)
CD79A (+0.27)
PECAM1 (+0.25)
CD14 (+0.24)
SFTPB (+0.24)
PTPRC (+0.23)
COL1A1 (+0.13)
PDGFRA (+0.13)
ABCA3 (+0.12)
EDNRB (+0.11)
ACTA2 (+0.11)
PDGFRB (+0.09)
ITGA8 (+0.06)
contribution to S_g, by cell type
(tick: total S_g; Δα in parentheses)
cell type (share of cells)
alveolar macrophage (0.08)
aerocyte (0.07)
alveolar fibroblast (0.08)
other cell types
alveolar type 2 cell (0.06)
(g)
CA4^ aerocyte
ICAM1^ alveolar type 1 cell
ICAM2 aerocyte
SRGN^ alveolar type 2 cell
TMEM176B alveolar fibroblast
S100A8^ neutrophil
MFGE8^ aerocyte
C1orf162 alveolar macrophage
FUS NK cell
H2AC20 alveolar fibroblast
DNLZ neutrophil
NDUFA5 neutrophil
FLYWCH2 monocyte
LYRM4 alveolar macrophage
COX16 alveolar macrophage
MRPS33 capillary endothelium
Δα (centered)
^ α > 0.95 in a seed (rank capped)
(h)
Δα (centered)
A_1
A_2
A_3
A_4
GIN hidden-state archetype
(k = 4; silhouette 0.106)
(i)
Spearman ρ = +0.49
S_g (rank, count-supported genes)
(j)
positive tail
Δα > +0.05
0.75 (n = 1,308)
negative tail
Δα < −0.05
0.35 (n = 26)
fraction of tail genes backed by
count evidence of the same sign
(k)


Spearman ρ = 0.674
protein prior π
α
(a)
above +0.05: 5%
below −0.05: 0%
SD 0.028
number of genes
Δα (centered)
(b)
protein prior π
α
n = 12,671 genes
mean α = 0.824
SD = 0.070
density
(c)
PECAM1
CD14
PTPRC
CD163
CDH5
DCN
NCAM1
MYF5
RGS5
COL1A1
PDGFRA
PAX7
MYH11
DES
myofiber
satellite
stroma and vessel
immune
Δα (centered)
protein prior π
(d)
n
d
Immunity & inflammation 260 +0.61*
Mitochondria & OXPHOS 128 +0.29*
Cytoskeleton & ECM 347 +0.84*
Translation & ribosome 242 +0.35*
Signaling pathways 295 +0.28*
Metabolism 148 +0.34*
Protein homeostasis 157 +0.21
Cell cycle 121 +0.17
Transcription factors 647 +0.17
Chromatin & DNA processing 233 −0.02
Δα (centered)
(e)
Epithelial Mesenchymal Transition
Complement
TNF-alpha Signaling via NF-kB
FDR q
NES (Δα-ranked GSEA)
NES > 0
not significant
q below the plotted range
q = 0.05
(f)
PECAM1 (+0.30)
CDH5 (+0.20)
DCN (+0.20)
CD14 (+0.16)
MYH11 (+0.15)
COL1A1 (+0.13)
MYF5 (+0.12)
PDGFRA (+0.11)
PTPRC (+0.10)
DES (+0.10)
RGS5 (+0.09)
PAX7 (+0.09)
CD163 (+0.06)
NCAM1 (+0.05)
contribution to S_g, by cell type
(tick: total S_g; Δα in parentheses)
cell type (share of cells)
fibroblast (0.24)
monocyte (0.05)
muscle stem cell (0.21)
other cell types
smooth muscle (0.07)
(g)
SRGN^ fibroblast
PLAUR monocyte
IGFBP6^ muscle stem cell
FCER1G^ monocyte
PECAM1 capillary endothelium
EMCN capillary endothelium
PPP1R14A^ smooth muscle
RAMP2^ muscle stem cell
UBA52 tenocyte
GABARAP fibroblast
SPARC fibroblast
NDUFA7 fibroblast
CRIP1 fibroblast
EEF1G fibroblast
RPS17 fibroblast
MT-ND3 fibroblast
Δα (centered)
^ α > 0.95 in a seed (rank capped)
(h)
Δα (centered)
A_1
A_2
A_3
GIN hidden-state archetype
(k = 3; silhouette 0.258)
(i)
Spearman ρ = +0.69
S_g (rank, count-supported genes)
(j)
positive tail
Δα > +0.05
1.00 (n = 685)
negative tail
Δα < −0.05
1.00 (n = 19)
fraction of tail genes backed by
count evidence of the same sign
(k)

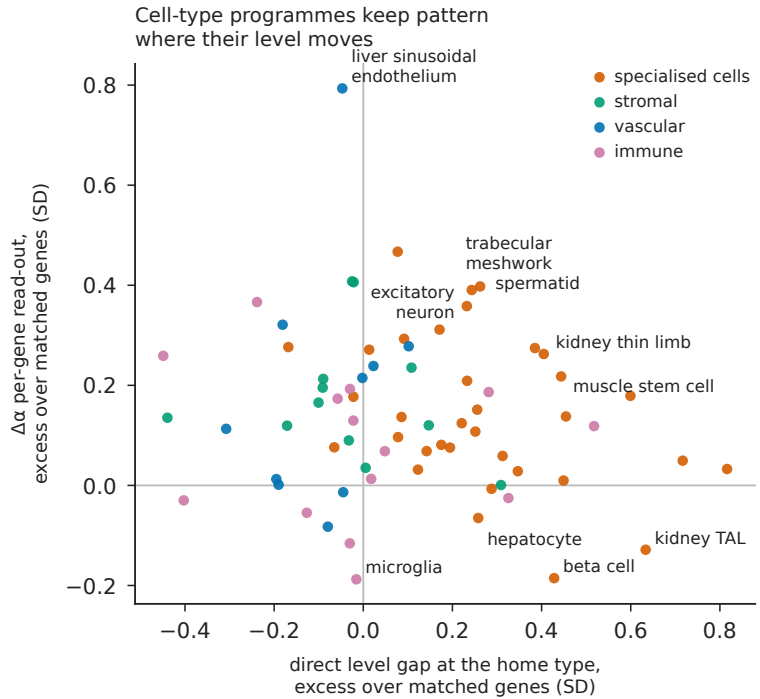

Spearman ρ = 0.645
protein prior π
α
(a)
above +0.05: 13%
below −0.05: 1%
SD 0.044
number of genes
Δα (centered)
(b)
protein prior π
α
n = 15,366 genes
mean α = 0.815
SD = 0.078
density
(c)
S100A8
CD3E
CSF3R
CD3D
CD14
CD79A
NKG7
APOE
C1QA
CXCR2
MMP9
CSF1R
MS4A1
C1QB
MRC1
CCR7
IRF8
myeloid
dendritic
neutrophil
lymphoid
Δα (centered)
protein prior π
(d)
n
d
Immunity & inflammation 330 +1.53*
Mitochondria & OXPHOS 136 +0.94*
Protein homeostasis 166 +0.28*
Translation & ribosome 249 +0.79*
Transcription factors 797 +0.28*
Cytoskeleton & ECM 431 +0.30
Metabolism 185 +0.33*
Signaling pathways 328 +0.23*
Chromatin & DNA processing 242 +0.06
Cell cycle 130 −0.02*
Δα (centered)
(e)
IL-6/JAK/STAT3 Signaling
TNF-alpha Signaling via NF-kB
Allograft Rejection
Spermatogenesis
FDR q
NES (Δα-ranked GSEA)
NES > 0
NES < 0
not significant
△ q below the plotted range
q = 0.05
(f)
S100A8 (+0.38)
CSF3R (+0.31)
CD14 (+0.29)
CD79A (+0.26)
APOE (+0.26)
C1QA (+0.26)
CXCR2 (+0.23)
CD3D (+0.18)
MRC1 (+0.16)
MS4A1 (+0.15)
CCR7 (+0.12)
IRF8 (+0.11)
CD3E (+0.10)
NKG7 (+0.10)
contribution to S_g, by cell type
(tick: total S_g; Δα in parentheses)
cell type (share of cells)
T cell (0.24)
macrophage (0.16)
neutrophil (0.23)
other cell types
B cell (0.20)
(g)
SRGN neutrophil
S100A8 B cell
TMEM176B T cell
MT-ND2 neutrophil
TMEM123 neutrophil
PLAUR^ T cell
CD4 T cell
SPP1^ T cell
MTRFR T cell
WDCP monocyte
RBM45 T cell
TIMM50 T cell
GIGYF1 monocyte
MTHFD2L T cell
MRPS14 T cell
NDUFA7 macrophage
Δα (centered)
^ α > 0.95 in a seed (rank capped)
(h)
Δα (centered)
GIN hidden-state archetype
(k = 3; silhouette 0.103)
(i)
Spearman ρ = +0.31
S_g (rank, count-supported genes)
(j)
positive tail
Δα > +0.05
0.55 (n = 1,814)
negative tail
Δα < −0.05
0.00 (n = 241)
fraction of tail genes backed by
count evidence of the same sign
(k)


Δα
count evidence S_g
hepatocyte nutrient program (liver)
liver endothelium and support (liver)
germline (testis)
somatic (testis)
B and plasma (bone marrow)
T and NK (bone marrow)
myeloid (bone marrow)
erythroid and progenitor (bone marrow)
proximal tubule (kidney)
loop and distal tubule (kidney)
collecting duct (kidney)
glomerulus (kidney)
endocrine (islet)
exocrine and duct (islet)
stroma (islet)
trabecular meshwork and aqueous outflow (eye)
Schlemm's canal and vessels (eye)
ciliary muscle and pigment (eye)
glaucoma-linked (eye)
excitatory (cortex)
inhibitory (cortex)
glia (cortex)
vascular (cortex)
alveolar epithelium (lung)
endothelium (lung)
stroma (lung)
immune (lung)
Liver
Testis
Bone marrow
Kidney
Islet
Eye
Cortex
Lung
−8 −4 0 +4 +8
set z against matched genes

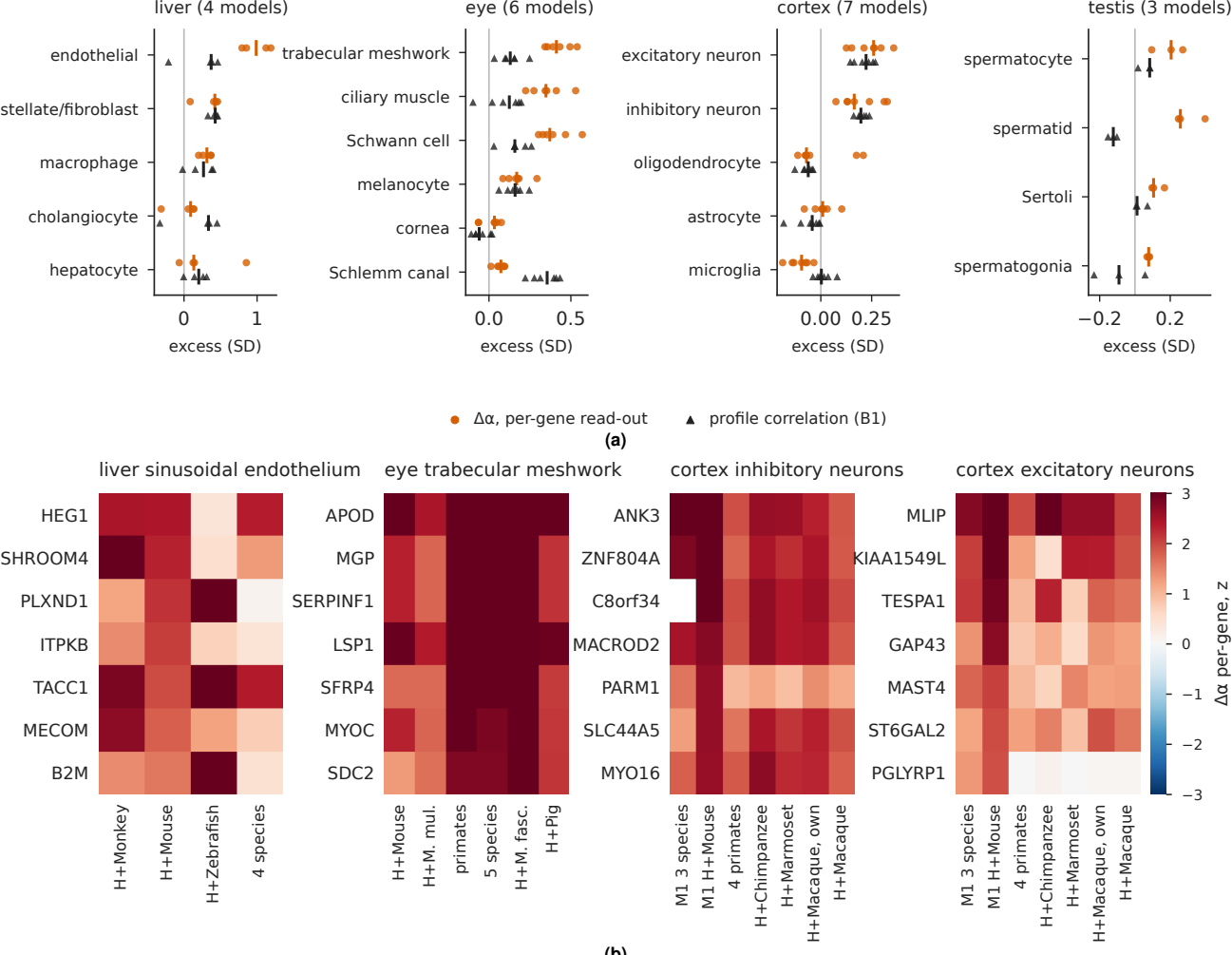

Score vs COE, n = 4,111 genes
D (Chimera)
S_g (Chimera)
B1, resid. on π
mean residual rank given C (fraction of genes)
Shay COE (decile median)
(a)
E1: partial rank correlation
all genes (4,111)
top π tertile (1,495)
D (Chimera)
S_g (Chimera)
B1, resid. on π
B1, raw
ρ_p(score, COE | C), 95% CI
(b)
E2: Shay divergent (135) vs COE-conserved
pre-specified: all conserved (1,929)
exploratory: range-matched (413)
stratified AUROC, 95% CI (< 0.5: divergent lower)
(c)
E3: named lineages (tick: chance)
Chimera, all rows
Chimera, mouse rows
B1 leave-one-type-out
2/10, p = 0.61
4/10, p = 0.07
4/10, p = 0.06
of 10 named genes: top divergent type is a Shay-named lineage
(d)
E4: Chimera species rows
divergent (137)
conserved (1,990)
all genes (12,059)
mouse minus lemur share of divergent S_g, 95% CI
(e)
E4 baseline: pseudobulk
divergent (130)
conserved (1,995)
all genes (7,817)
E(human, lemur) − E(human, mouse) 95% CI
(f)

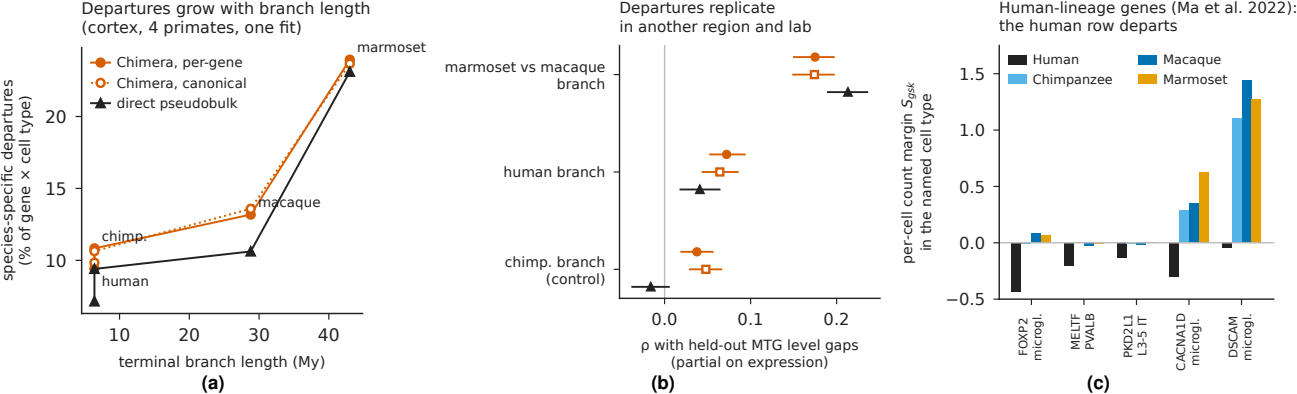

Δα, protein-feature gate (default)
Δα, per-gene read-out
pseudobulk profile correlation
level similarity at the home type (direct)
specialised cells
stromal
vascular
immune
no home type (broadly used)
23/24 > 0
21/22 > 0
18/21 > 0
11/23 > 0
1/26 > 0
24/24 > 0
21/22 > 0
19/21 > 0
13/23 > 0
0/26 > 0
18/24 > 0
21/22 > 0
20/21 > 0
18/23 > 0
7/26 > 0
2/24 > 0
11/22 > 0
12/21 > 0
12/23 > 0
23/26 > 0
0
1
excess over matched genes (SD)

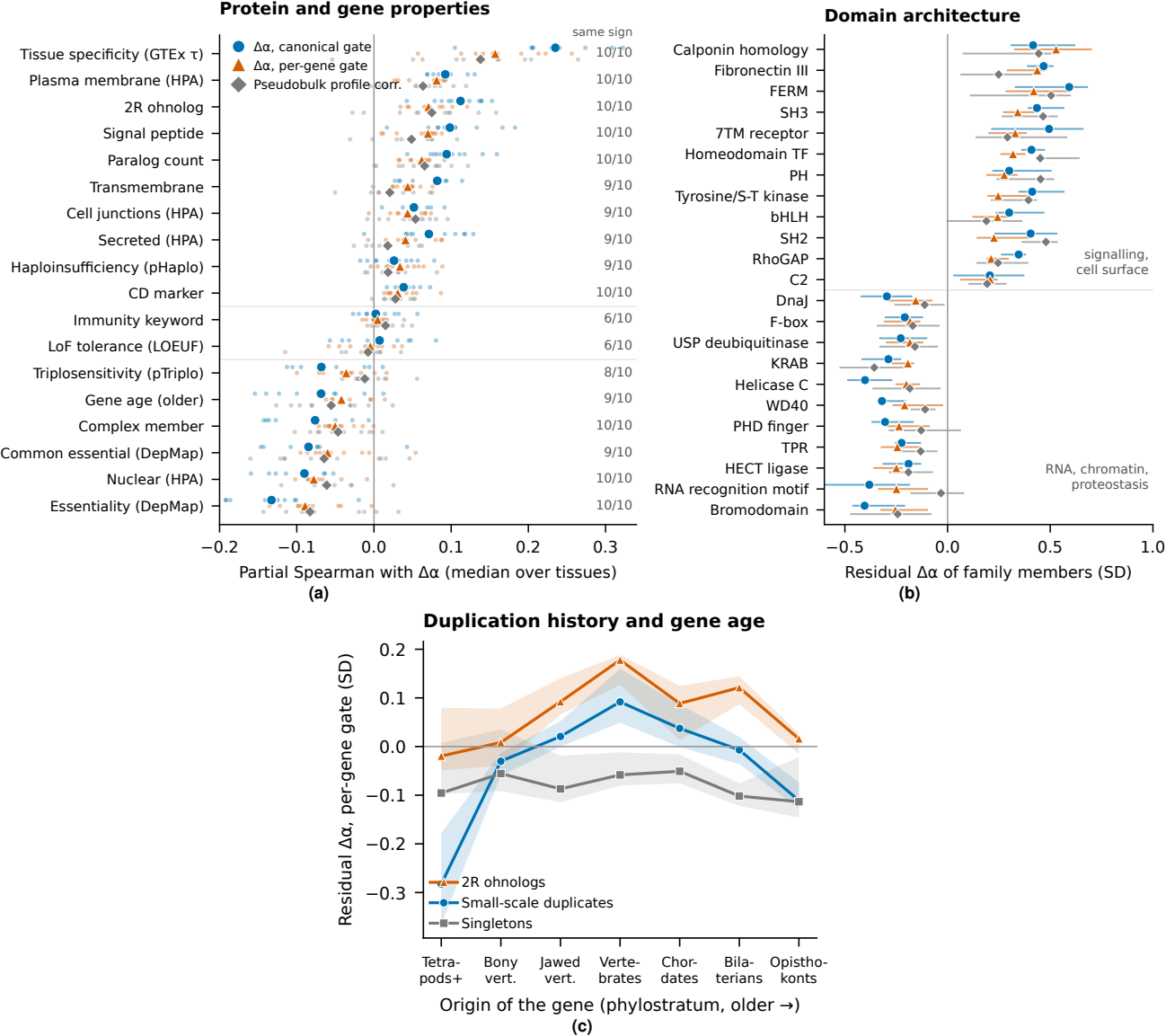
(a)

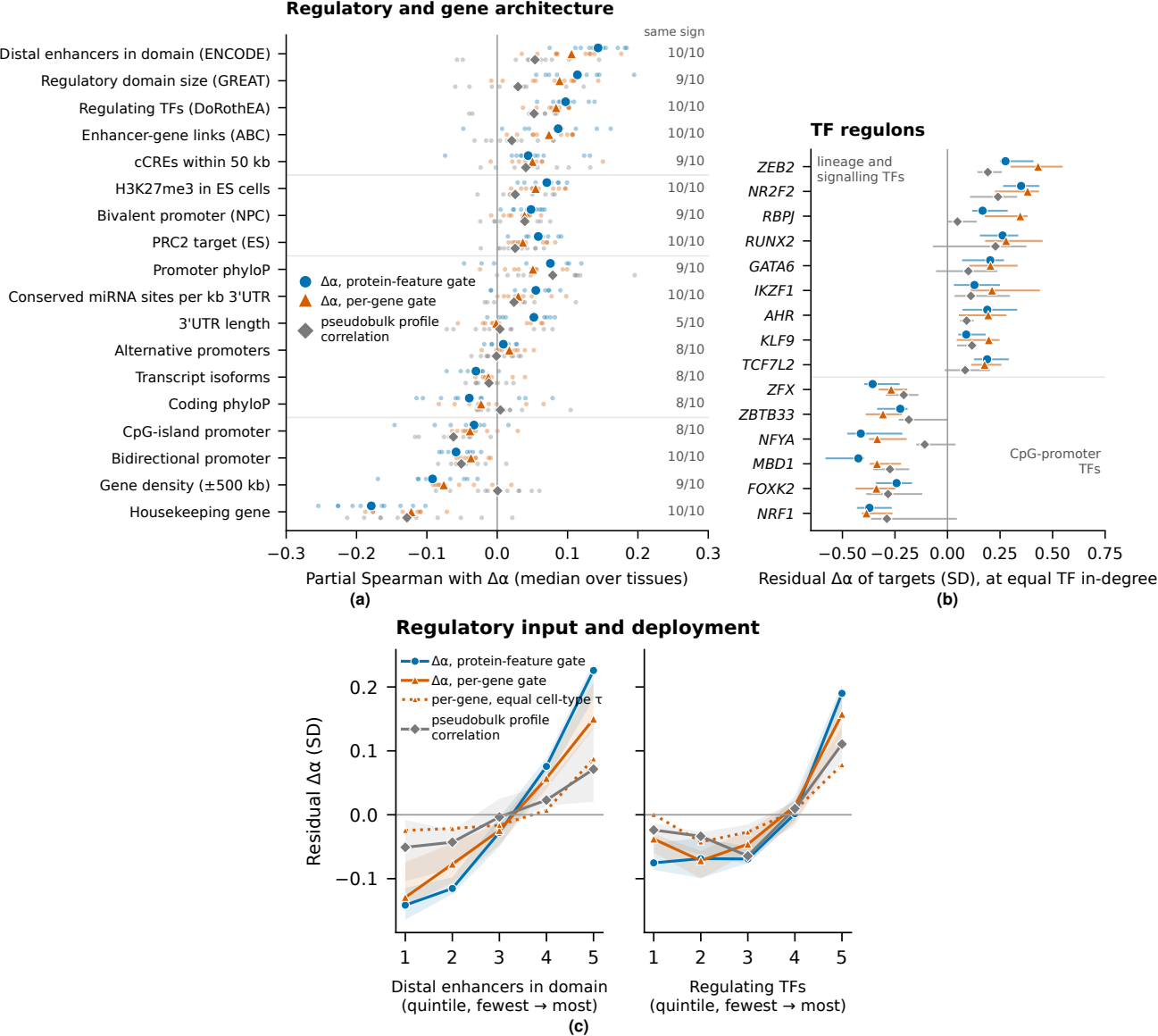

Cell-type programmes keep pattern where their level moves
liver sinusoidal endothelium
specialised cells
stromal
vascular
immune
trabecular meshwork
spermatid
excitatory neuron
kidney thin limb
muscle stem cell
hepatocyte
kidney TAL
beta cell
microglia
Δα per-gene read-out, excess over matched genes (SD)
direct level gap at the home type, excess over matched genes (SD)
0.8
0.6
0.4
0.2
0.0
−0.2
−0.4
−0.2
0.0
0.2
0.4
0.6
0.8

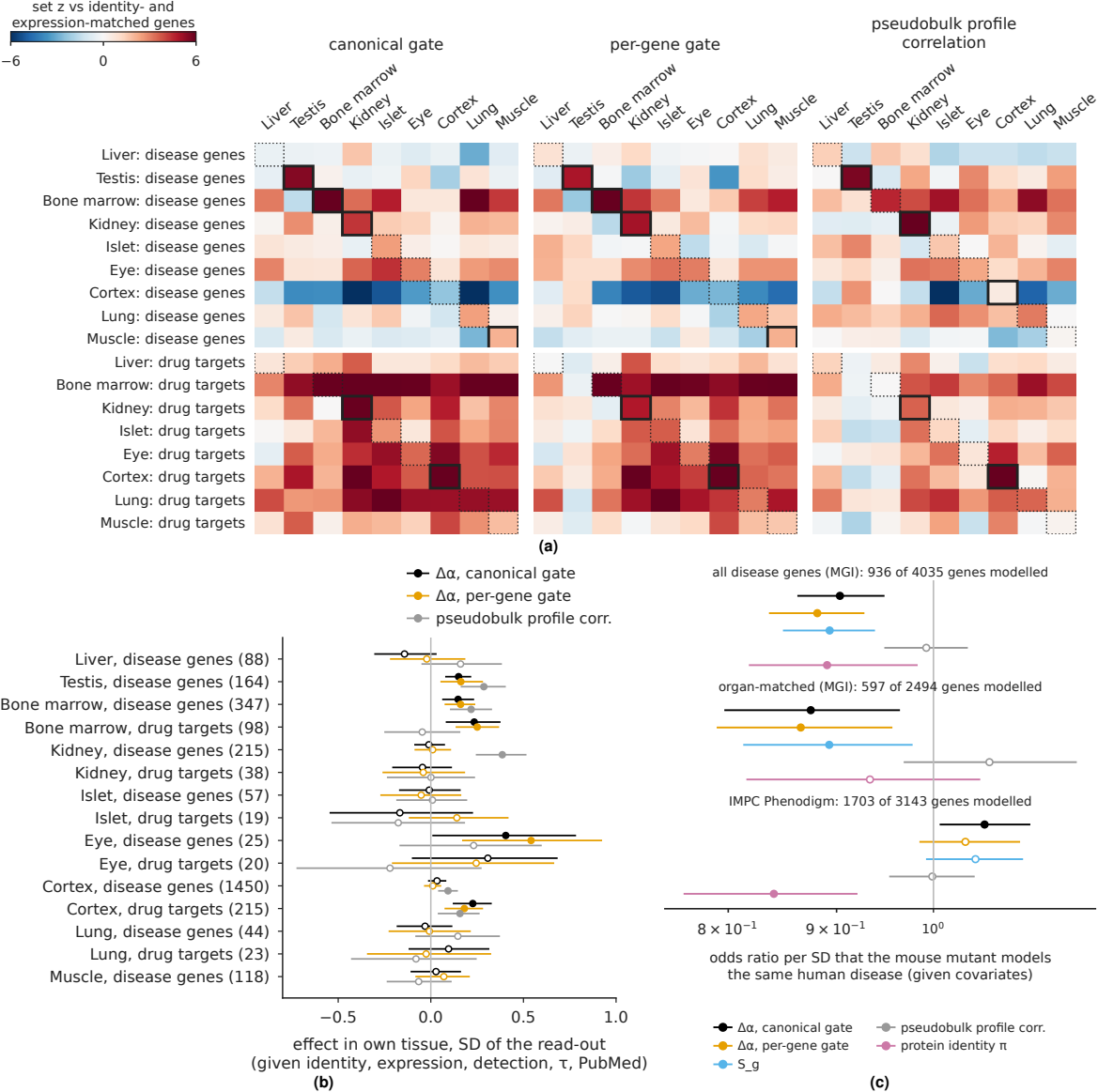
(b)

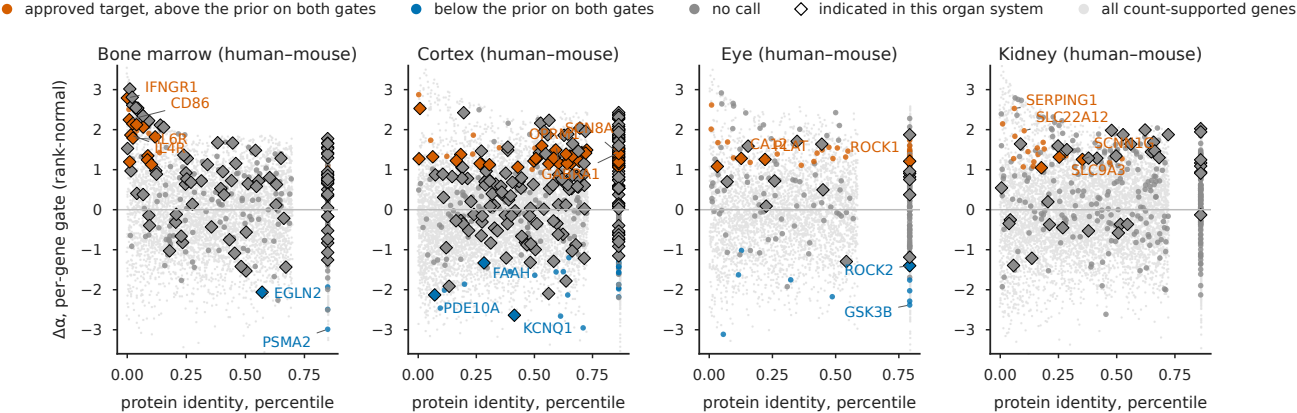

liver (4 models)
endothelial
stellate/fibroblast
macrophage
cholangiocyte
hepatocyte
0
1
excess (SD)
eye (6 models)
trabecular meshwork
ciliary muscle
Schwann cell
melanocyte
cornea
Schlemm canal
0.0
0.5
excess (SD)
cortex (7 models)
excitatory neuron
inhibitory neuron
oligodendrocyte
astrocyte
microglia
0.00
0.25
excess (SD)
testis (3 models)
spermatocyte
spermatid
Sertoli
spermatogonia
−0.2
0.2
excess (SD)
Δα, per-gene read-out
profile correlation (B1)
(a)
liver sinusoidal endothelium
HEG1
SHROOM4
PLXND1
ITPKB
TACC1
MECOM
B2M
H+Monkey
H+Mouse
H+Zebrafish
4 species
eye trabecular meshwork
APOD
MGP
SERPINF1
LSP1
SFRP4
MYOC
SDC2
H+Mouse
H+M. mul.
primates
5 species
H+M. fasc.
H+Pig
cortex inhibitory neurons
ANK3
ZNF804A
C8orf34
MACROD2
PARM1
SLC44A5
MYO16
M1 3 species
M1 H+Mouse
4 primates
H+Chimpanzee
H+Marmoset
H+Macaque, own
H+Macaque
cortex excitatory neurons
MLIP
KIAA1549L
TESPA1
GAP43
MAST4
ST6GAL2
PGLYRP1
M1 3 species
M1 H+Mouse
4 primates
H+Chimpanzee
H+Marmoset
H+Macaque, own
H+Macaque
3
2
1
0
−1
−2
−3
Δα per-gene, z
(b)


Departures grow with branch length
(cortex, 4 primates, one fit)
Chimera, per-gene
Chimera, canonical
direct pseudobulk
marmoset
macaque
chimp.
human
species-specific departures
(% of gene × cell type)
terminal branch length (My)
(a)
Departures replicate
in another region and lab
marmoset vs macaque
branch
human branch
chimp. branch
(control)
ρ with held-out MTG level gaps
(partial on expression)
(b)
Human-lineage genes (Ma et al. 2022):
the human row departs
Human
Chimpanzee
Macaque
Marmoset
per-cell count margin S_{gsk}
in the named cell type
FOXP2
microgl.
MELTF
PVALB
PKD2L1
L3-5 IT
CACNA1D
microgl.
DSCAM
microgl.
(c)

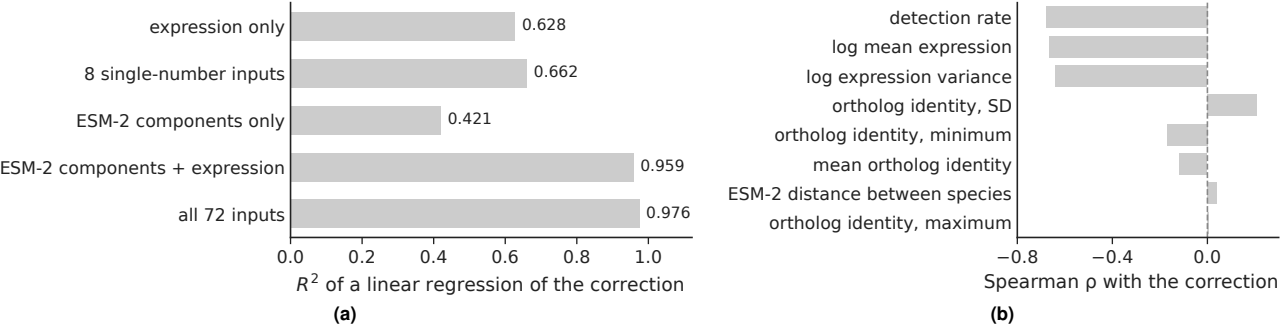

Protein and gene properties
same sign
Δα, canonical gate
Δα, per-gene gate
Pseudobulk profile corr.
Tissue specificity (GTEx τ)
Plasma membrane (HPA)
2R ohnolog
Signal peptide
Paralog count
Transmembrane
Cell junctions (HPA)
Secreted (HPA)
Haploinsufficiency (pHaplo)
CD marker
Immunity keyword
LoF tolerance (LOEUF)
Triplosensitivity (pTriplo)
Gene age (older)
Complex member
Common essential (DepMap)
Nuclear (HPA)
Essentiality (DepMap)
10/10
10/10
10/10
10/10
10/10
9/10
9/10
9/10
9/10
10/10
6/10
6/10
8/10
9/10
10/10
9/10
10/10
10/10
−0.2
−0.1
0.0
0.1
0.2
0.3
Partial Spearman with Δα (median over tissues)
(a)
Domain architecture
Calponin homology
Fibronectin III
FERM
SH3
7TM receptor
Homeodomain TF
PH
Tyrosine/S-T kinase
bHLH
SH2
RhoGAP
C2
DnaJ
F-box
USP deubiquitinase
KRAB
Helicase C
WD40
PHD finger
TPR
HECT ligase
RNA recognition motif
Bromodomain
signalling, cell surface
RNA, chromatin, proteostasis
−0.5
0.0
0.5
1.0
Residual Δα of family members (SD)
(b)
Duplication history and gene age
Residual Δα, per-gene gate (SD)
0.2
0.1
0.0
−0.1
−0.2
−0.3
2R ohnologs
Small-scale duplicates
Singletons
Tetra-pods+
Bony vert.
Jawed vert.
Verte-brates
Chor-dates
Bila-terians
Opistho-konts
Origin of the gene (phylostratum, older →)
(c)

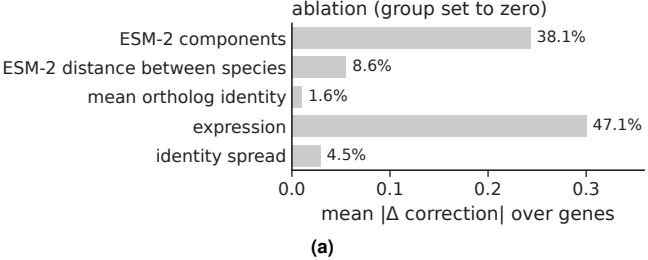

Regulatory and gene architecture
same sign
Distal enhancers in domain (ENCODE)
Regulatory domain size (GREAT)
Regulating TFs (DoRothEA)
Enhancer-gene links (ABC)
cCREs within 50 kb
H3K27me3 in ES cells
Bivalent promoter (NPC)
PRC2 target (ES)
Promoter phyloP
Conserved miRNA sites per kb 3'UTR
3'UTR length
Alternative promoters
Transcript isoforms
Coding phyloP
CpG-island promoter
Bidirectional promoter
Gene density (±500 kb)
Housekeeping gene
10/10
9/10
10/10
10/10
9/10
10/10
9/10
10/10
9/10
10/10
5/10
8/10
8/10
8/10
8/10
10/10
9/10
10/10
Δα, protein-feature gate
Δα, per-gene gate
pseudobulk profile correlation
−0.3
−0.2
−0.1
0.0
0.1
0.2
0.3
Partial Spearman with Δα (median over tissues)
(a)
TF regulons
lineage and signalling TFs
ZEB2
NR2F2
RBPJ
RUNX2
GATA6
IKZF1
AHR
KLF9
TCF7L2
ZFX
ZBTB33
NFYA
MBD1
FOXK2
NRF1
CpG-promoter TFs
−0.50
−0.25
0.00
0.25
0.50
0.75
Residual Δα of targets (SD), at equal TF in-degree
(b)
Regulatory input and deployment
Δα, protein-feature gate
Δα, per-gene gate
per-gene, equal cell-type τ
pseudobulk profile correlation
Residual Δα (SD)
0.2
0.1
0.0
−0.1
1
2
3
4
5
Distal enhancers in domain (quintile, fewest → most)
Regulating TFs (quintile, fewest → most)
(c)


set z vs identity- and expression-matched genes
−6
0
6
canonical gate
per-gene gate
pseudobulk profile correlation
Liver
Testis
Bone marrow
Kidney
Islet
Eye
Cortex
Lung
Muscle
Liver: disease genes
Testis: disease genes
Bone marrow: disease genes
Kidney: disease genes
Islet: disease genes
Eye: disease genes
Cortex: disease genes
Lung: disease genes
Muscle: disease genes
Liver: drug targets
Bone marrow: drug targets
Kidney: drug targets
Islet: drug targets
Eye: drug targets
Cortex: drug targets
Lung: drug targets
Muscle: drug targets
(a)
Δα, canonical gate
Δα, per-gene gate
pseudobulk profile corr.
Liver, disease genes (88)
Testis, disease genes (164)
Bone marrow, disease genes (347)
Bone marrow, drug targets (98)
Kidney, disease genes (215)
Kidney, drug targets (38)
Islet, disease genes (57)
Islet, drug targets (19)
Eye, disease genes (25)
Eye, drug targets (20)
Cortex, disease genes (1450)
Cortex, drug targets (215)
Lung, disease genes (44)
Lung, drug targets (23)
Muscle, disease genes (118)
−0.5
0.0
0.5
1.0
effect in own tissue, SD of the read-out
(given identity, expression, detection, τ, PubMed)
(b)
all disease genes (MGI): 936 of 4035 genes modelled
organ-matched (MGI): 597 of 2494 genes modelled
IMPC Phenodigm: 1703 of 3143 genes modelled
8 × 10−1
9 × 10−1
100
odds ratio per SD that the mouse mutant models
the same human disease (given covariates)
Δα, canonical gate
Δα, per-gene gate
S_g
pseudobulk profile corr.
protein identity π
(c)

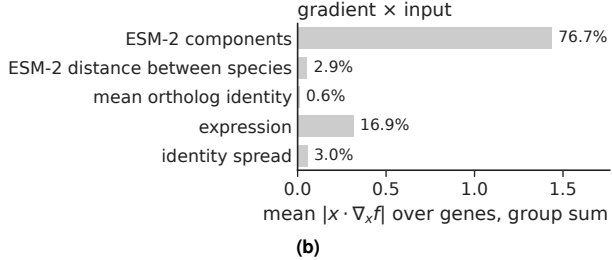

approved target, above the prior on both gates
below the prior on both gates
no call
indicated in this organ system
all count-supported genes
Bone marrow (human–mouse)
Cortex (human–mouse)
Eye (human–mouse)
Kidney (human–mouse)
Δα, per-gene gate (rank-normal)
protein identity, percentile
IFNGR1
CD86
IL6R
IL4R
EGLN2
PSMA2
CACNA1H
FAAH
PDE10A
KCNQ1
CA12
CLCN1
ROCK1
ROCK2
GSK3B
SERPING1
SLC22A12
SCNN1G
SLC9A3

(d)

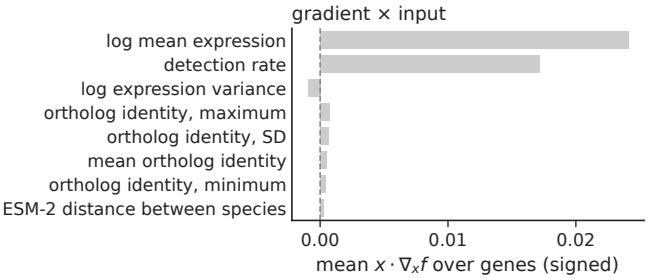

expression only
8 single-number inputs
ESM-2 components only
ESM-2 components + expression
all 72 inputs
0.628
0.662
0.421
0.959
0.976
0.0
0.2
0.4
0.6
0.8
1.0
R^2 of a linear regression of the correction
(a)
detection rate
log mean expression
log expression variance
ortholog identity, SD
ortholog identity, minimum
mean ortholog identity
ESM-2 distance between species
ortholog identity, maximum
−0.8
−0.4
0.0
Spearman ρ with the correction
(b)


ablation (group set to zero)
ESM-2 components
ESM-2 distance between species
mean ortholog identity
expression
identity spread
38.1%
8.6%
1.6%
47.1%
4.5%
0.0
0.1
0.2
0.3
mean |Δ correction| over genes

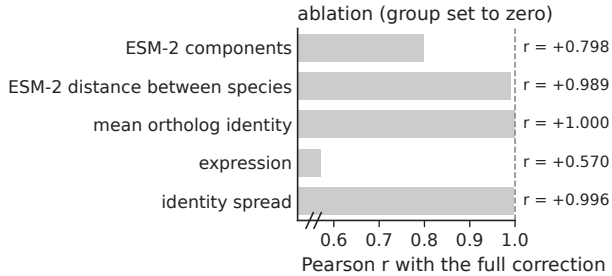
(a)


gradient × input
ESM-2 components
ESM-2 distance between species
mean ortholog identity
expression
identity spread
76.7%
2.9%
0.6%
16.9%
3.0%
0.0
0.5
1.0
1.5
mean |x · ∇ₓf| over genes, group sum

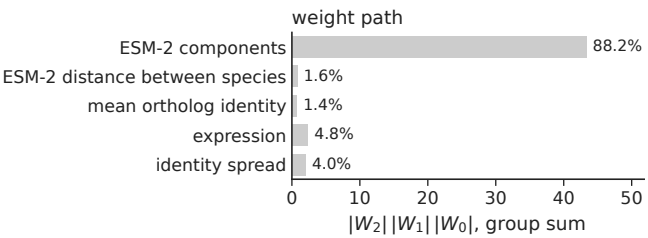
(b)


gradient × input
log mean expression
detection rate
log expression variance
ortholog identity, maximum
ortholog identity, SD
mean ortholog identity
ortholog identity, minimum
ESM-2 distance between species
0.00
0.01
0.02
mean x · ∇ₓf over genes (signed)

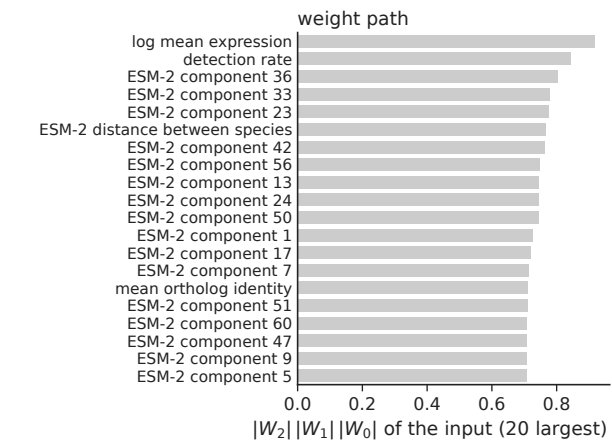
(c)


ablation (group set to zero)
ESM-2 components
ESM-2 distance between species
mean ortholog identity
expression
identity spread
r = +0.798
r = +0.989
r = +1.000
r = +0.570
r = +0.996
0.6
0.7
0.8
0.9
1.0
Pearson r with the full correction

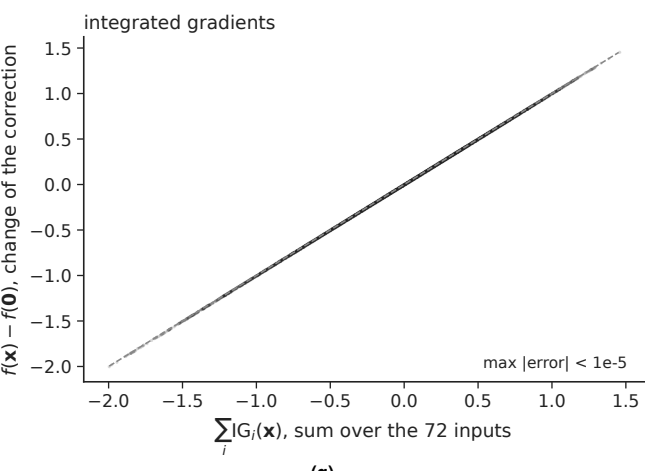
(d)


weight path
ESM-2 components
ESM-2 distance between species
mean ortholog identity
expression
identity spread
88.2%
1.6%
1.4%
4.8%
4.0%
0
10
20
30
40
50
|W₂||W₁||W₀|, group sum

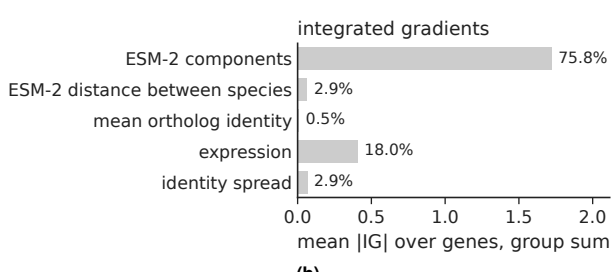
(e)


weight path
log mean expression
detection rate
ESM-2 component 36
ESM-2 component 33
ESM-2 component 23
ESM-2 distance between species
ESM-2 component 42
ESM-2 component 56
ESM-2 component 13
ESM-2 component 24
ESM-2 component 50
ESM-2 component 1
ESM-2 component 17
ESM-2 component 7
mean ortholog identity
ESM-2 component 51
ESM-2 component 60
ESM-2 component 47
ESM-2 component 9
ESM-2 component 5
0.0
0.2
0.4
0.6
0.8
|W₂||W₁||W₀| of the input (20 largest)

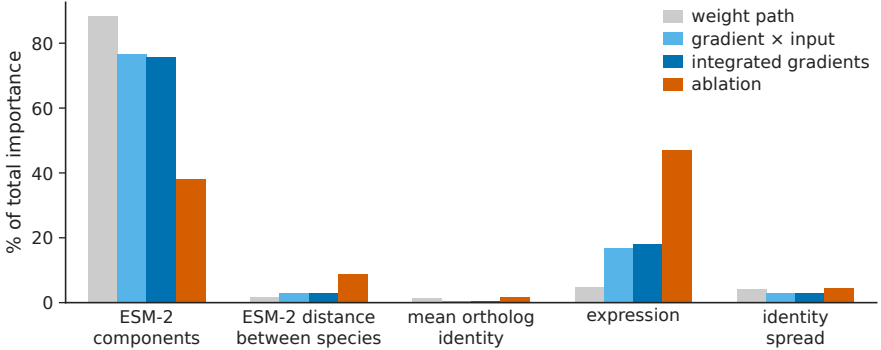
(f)


integrated gradients
1.5
1.0
0.5
0.0
−0.5
−1.0
−1.5
−2.0
f(x) − f(0), change of the correction
max |error| < 1e-5
−2.0
−1.5
−1.0
−0.5
0.0
0.5
1.0
1.5
∑ᵢ IGᵢ(x), sum over the 72 inputs

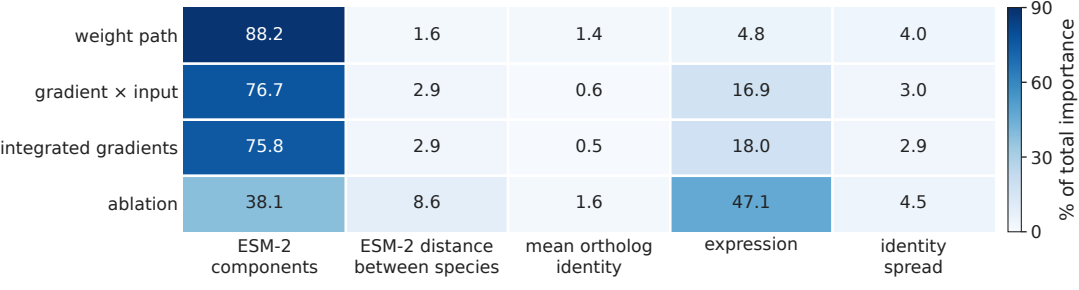
(g)


integrated gradients
ESM-2 components
ESM-2 distance between species
mean ortholog identity
expression
identity spread
75.8%
2.9%
0.5%
18.0%
2.9%
0.0
0.5
1.0
1.5
2.0
mean |IG| over genes, group sum

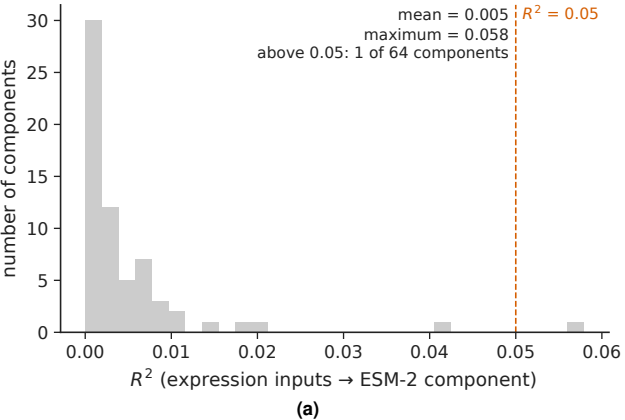
(h)


weight path
gradient × input
integrated gradients
ablation
% of total importance
0
20
40
60
80
ESM-2 components
ESM-2 distance between species
mean ortholog identity
expression
identity spread

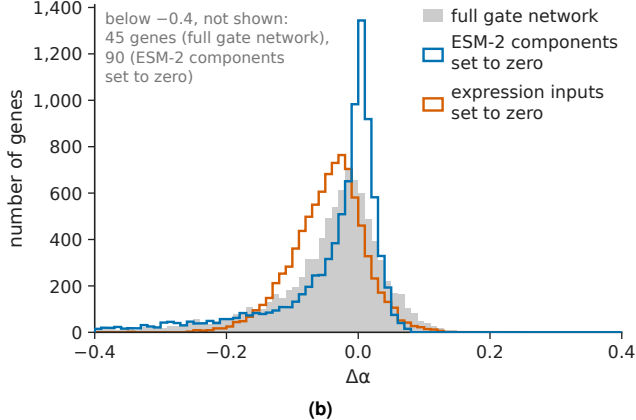
(a)


weight path
gradient × input
integrated gradients
ablation
88.2
1.6
1.4
4.8
4.0
76.7
2.9
0.6
16.9
3.0
75.8
2.9
0.5
18.0
2.9
38.1
8.6
1.6
47.1
4.5
ESM-2 components
ESM-2 distance between species
mean ortholog identity
expression
identity spread
90
60
30
0
% of total importance

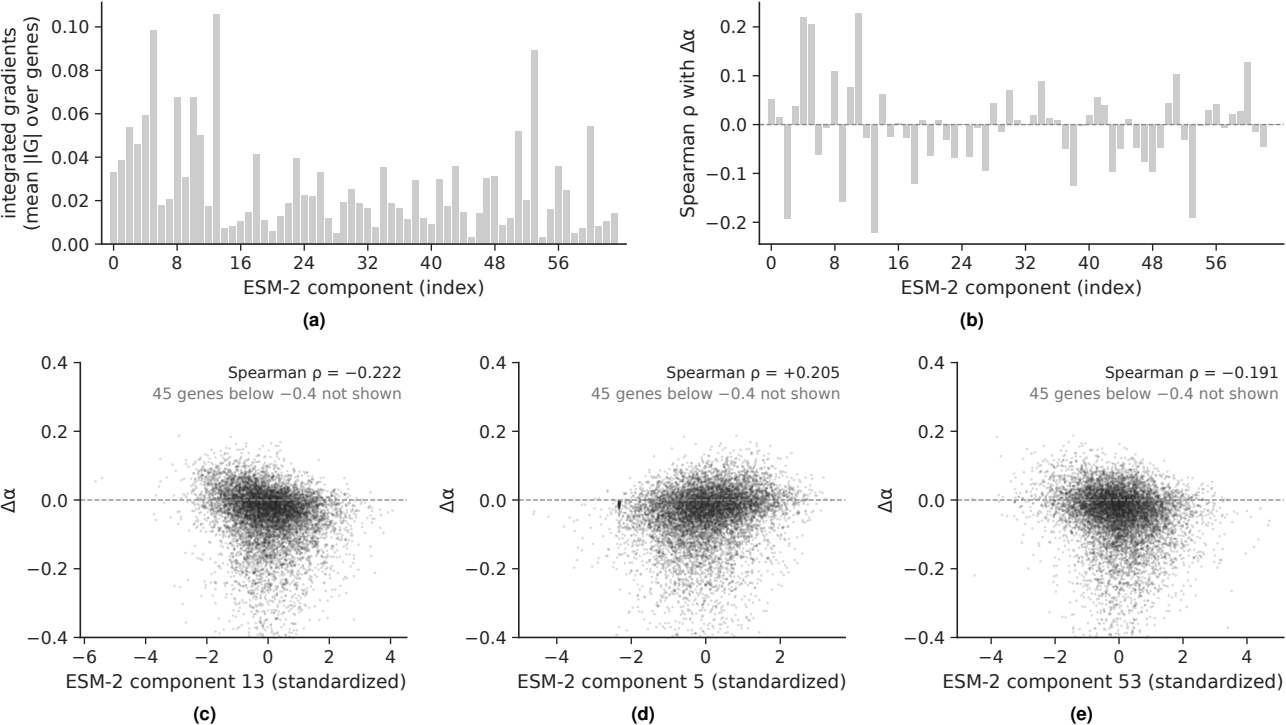
(b)

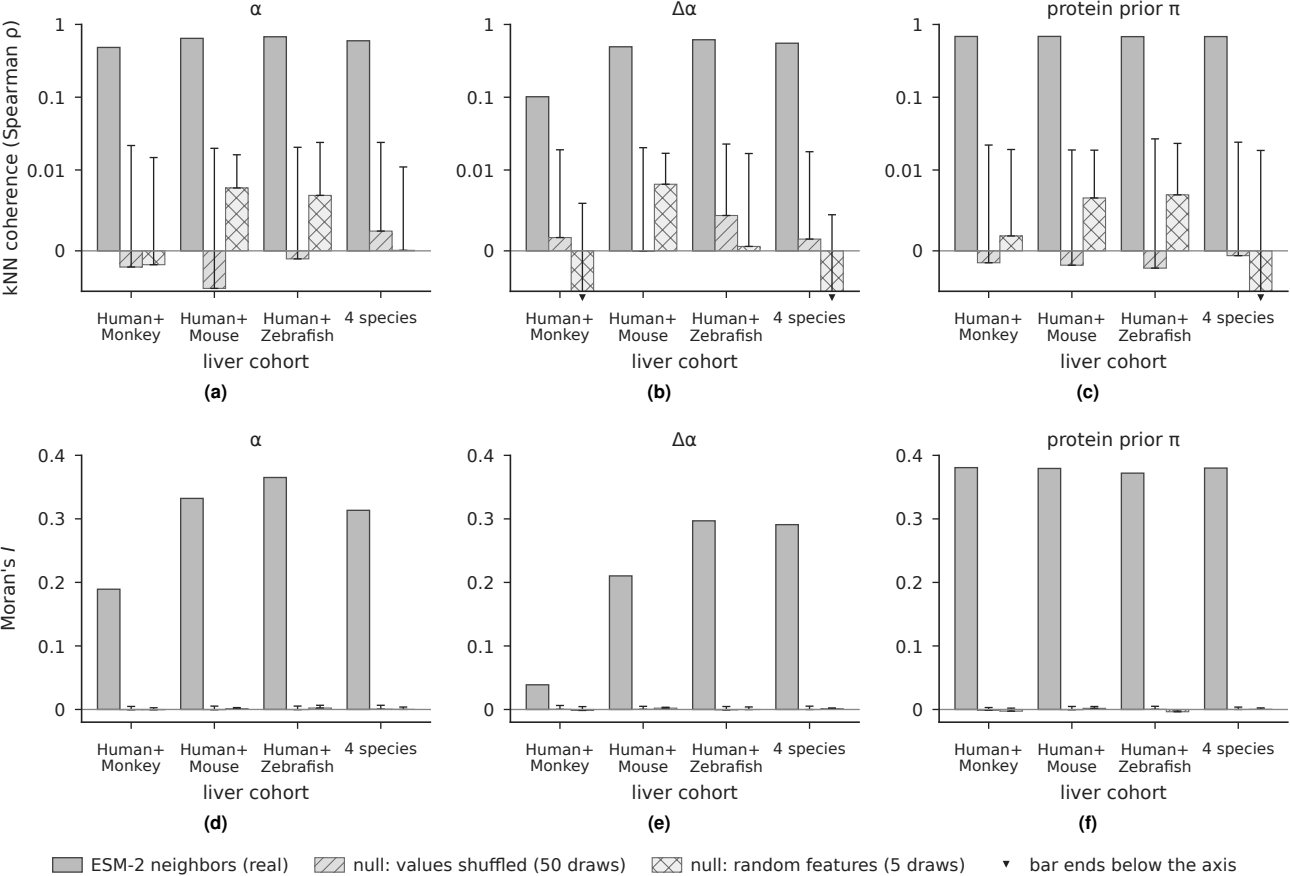

mean = 0.005
maximum = 0.058
above 0.05: 1 of 64 components
R^2 = 0.05
number of components
0
5
10
15
20
25
30
0.00
0.01
0.02
0.03
0.04
0.05
0.06
R^2 (expression inputs → ESM-2 component)

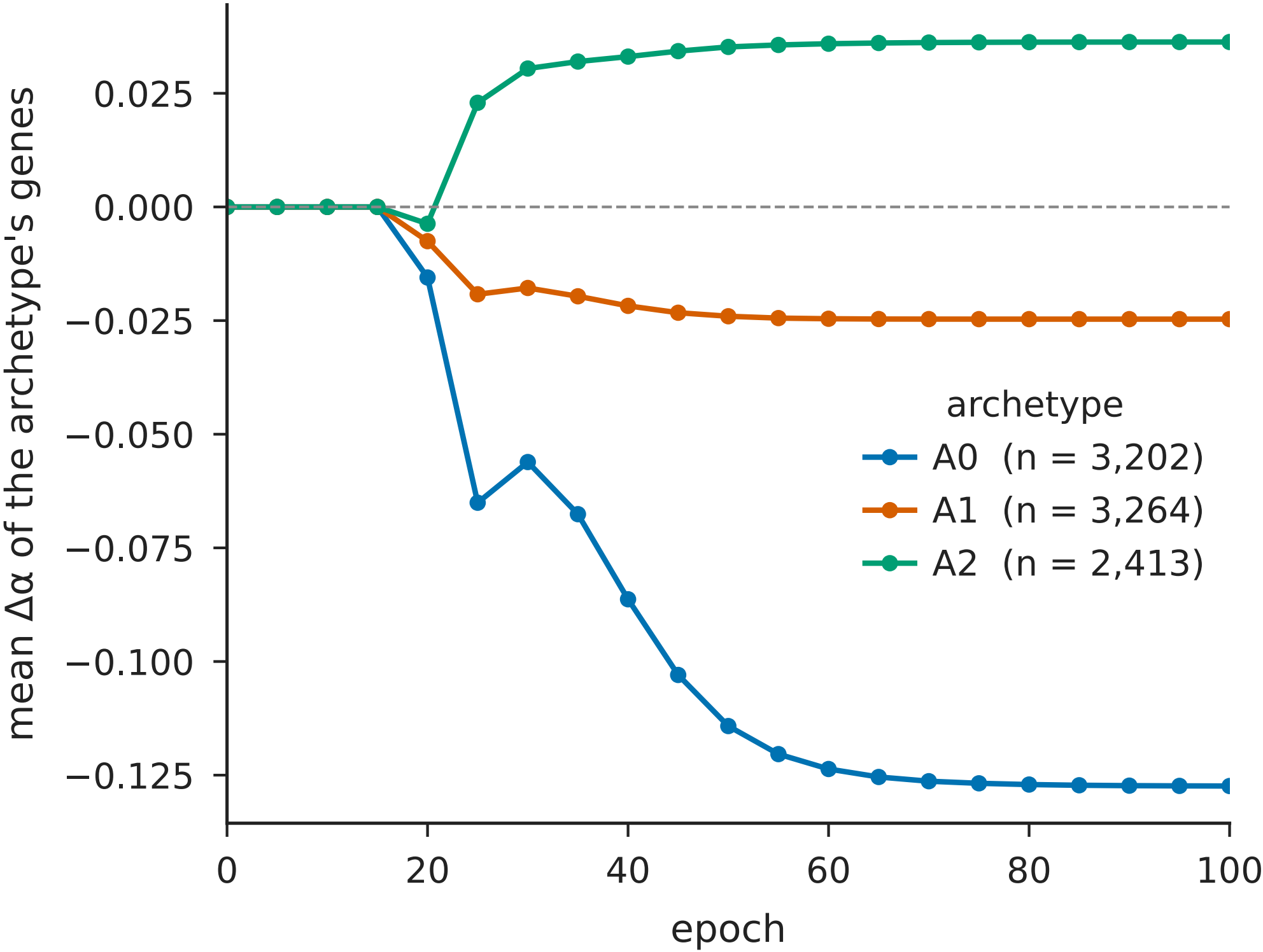
(a)

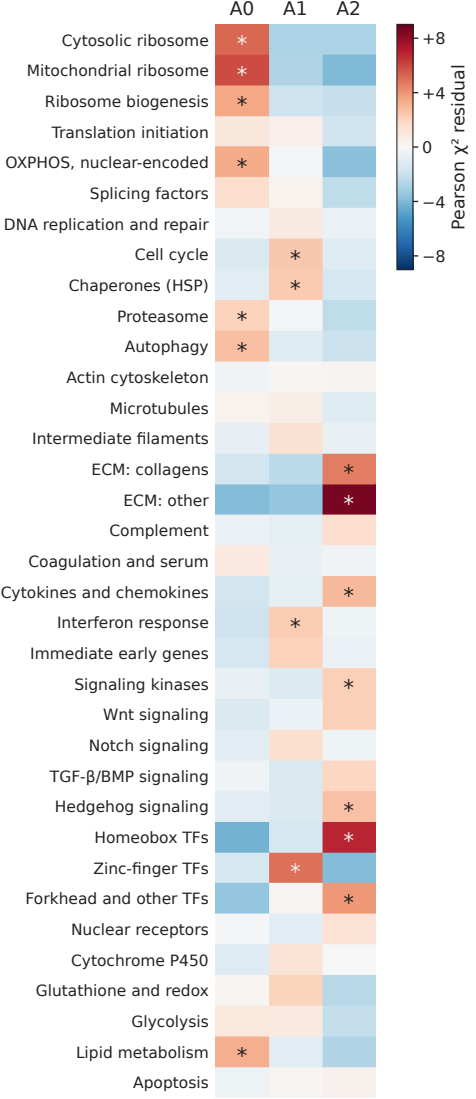

below −0.4, not shown:
45 genes (full gate network),
90 (ESM-2 components
set to zero)
full gate network
ESM-2 components
set to zero
expression inputs
set to zero
number of genes
0
200
400
600
800
1,000
1,200
1,400
−0.4
−0.2
0.0
0.2
0.4
Δα

(b)

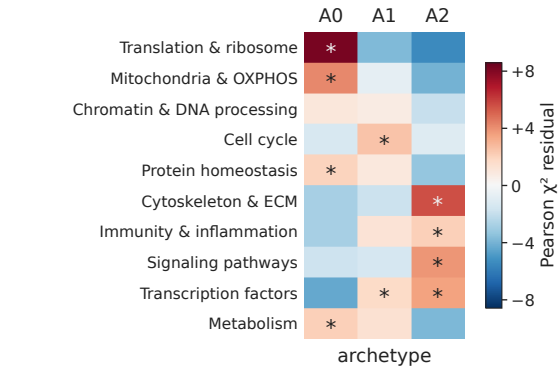

integrated gradients
(mean |IG| over genes)
0.10
0.08
0.06
0.04
0.02
0.00
0
8
16
24
32
40
48
56
ESM-2 component (index)
(a)
Spearman ρ with Δα
0.2
0.1
0.0
−0.1
−0.2
0
8
16
24
32
40
48
56
ESM-2 component (index)
(b)
Spearman ρ = −0.222
45 genes below −0.4 not shown
Δα
0.4
0.2
0.0
−0.2
−0.4
−6
−4
−2
0
2
4
ESM-2 component 13 (standardized)
(c)
Spearman ρ = +0.205
45 genes below −0.4 not shown
Δα
0.4
0.2
0.0
−0.2
−0.4
−4
−2
0
2
ESM-2 component 5 (standardized)
(d)
Spearman ρ = −0.191
45 genes below −0.4 not shown
Δα
0.4
0.2
0.0
−0.2
−0.4
−4
−2
0
2
4
ESM-2 component 53 (standardized)
(e)


α
Δα
protein prior π
kNN coherence (Spearman ρ)
1
0.1
0.01
0
Human+ Monkey
Human+ Mouse
Human+ Zebrafish
4 species
liver cohort
(a)
(b)
(c)
Moran's I
0.4
0.3
0.2
0.1
(d)
(e)
(f)
ESM-2 neighbors (real)
null: values shuffled (50 draws)
null: random features (5 draws)
bar ends below the axis

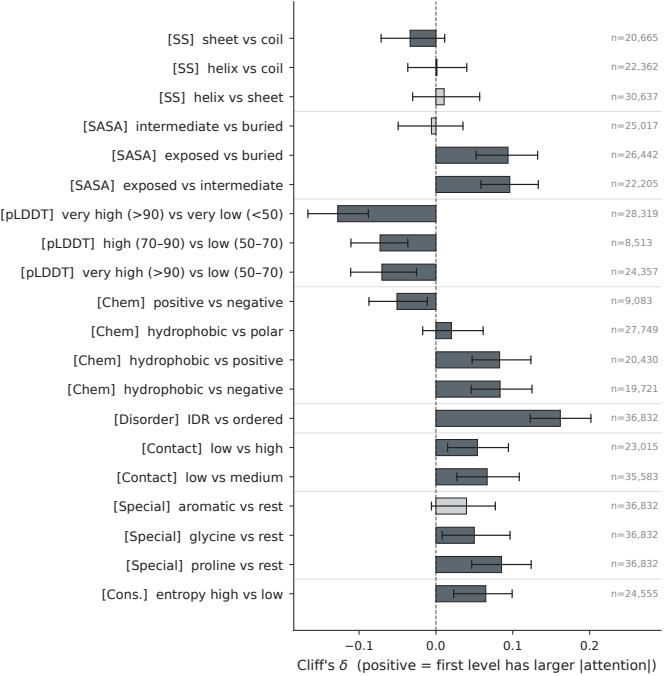

mean Δα of the archetype's genes
0.025
0.000
−0.025
−0.050
−0.075
−0.100
−0.125
0
20
40
60
80
100
epoch
archetype
A0 (n = 3,202)
A1 (n = 3,264)
A2 (n = 2,413)

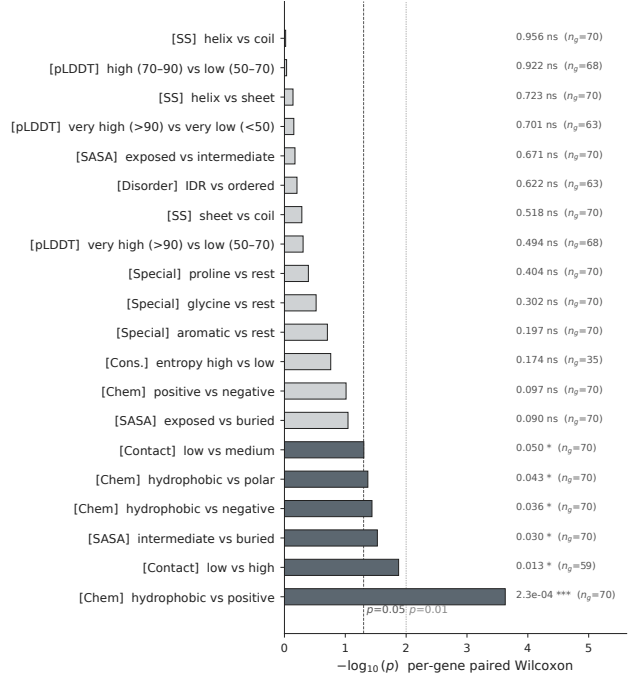

A0
A1
A2
Cytosolic ribosome
Mitochondrial ribosome
Ribosome biogenesis
Translation initiation
OXPHOS, nuclear-encoded
Splicing factors
DNA replication and repair
Cell cycle
Chaperones (HSP)
Proteasome
Autophagy
Actin cytoskeleton
Microtubules
Intermediate filaments
ECM: collagens
ECM: other
Complement
Coagulation and serum
Cytokines and chemokines
Interferon response
Immediate early genes
Signaling kinases
Wnt signaling
Notch signaling
TGF-β/BMP signaling
Hedgehog signaling
Homeobox TFs
Zinc-finger TFs
Forkhead and other TFs
Nuclear receptors
Cytochrome P450
Glutathione and redox
Glycolysis
Lipid metabolism
Apoptosis
archetype
+8
+4
0
−4
−8
Pearson χ² residual

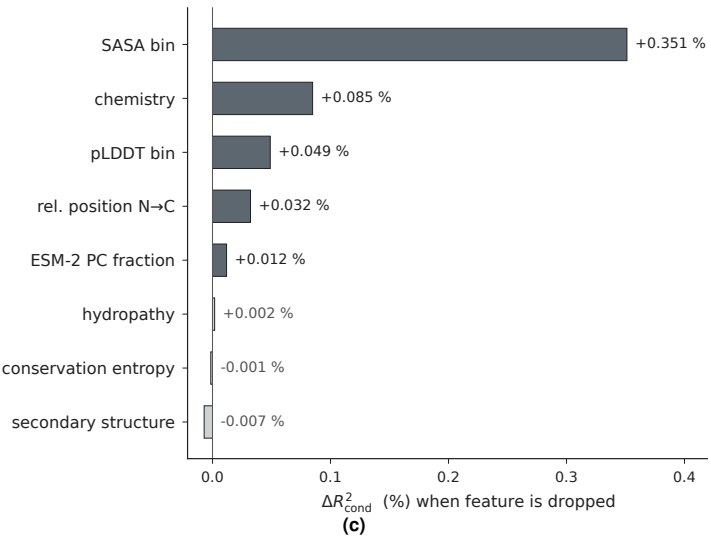
(a)

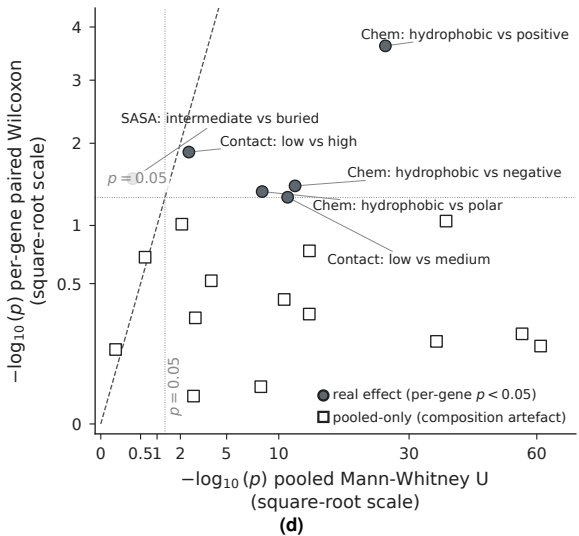

A0
A1
A2
Translation & ribosome
Mitochondria & OXPHOS
Chromatin & DNA processing
Cell cycle
Protein homeostasis
Cytoskeleton & ECM
Immunity & inflammation
Signaling pathways
Transcription factors
Metabolism
archetype
+8
+4
0
−4
−8
Pearson χ² residual

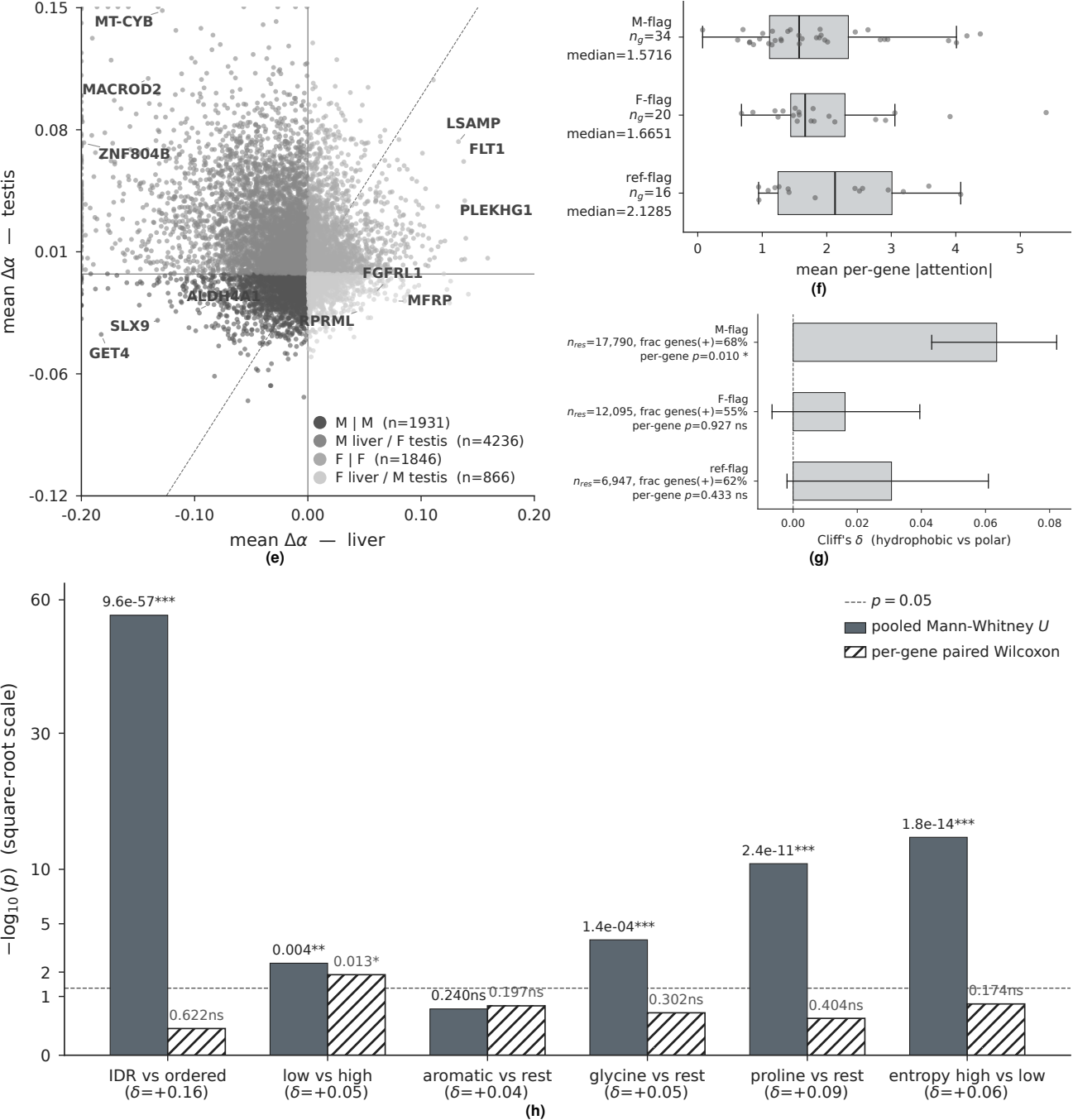
(b)

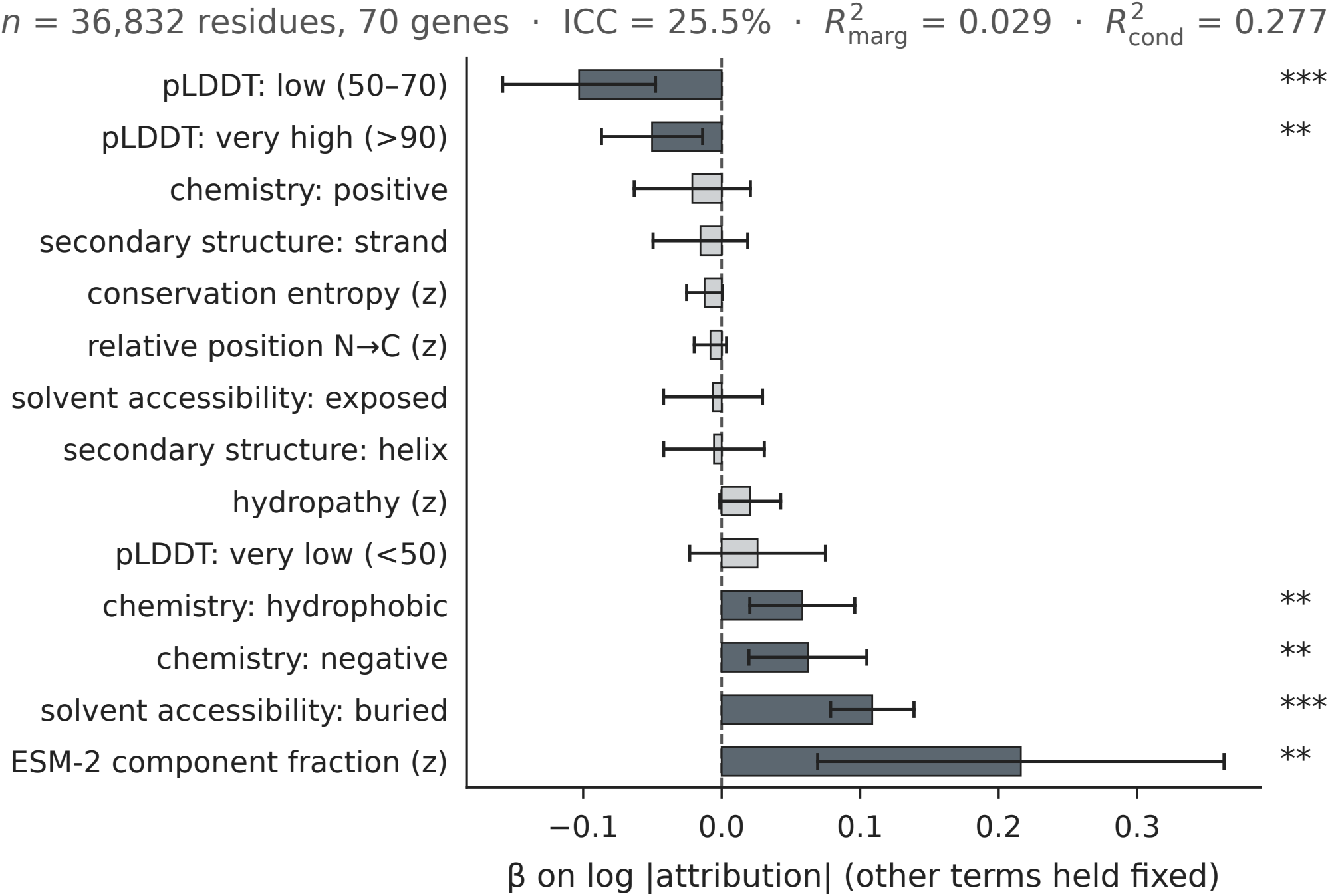

[SS] sheet vs coil
[SS] helix vs coil
[SS] helix vs sheet
[SASA] intermediate vs buried
[SASA] exposed vs buried
[SASA] exposed vs intermediate
[pLDDT] very high (>90) vs very low (<50)
[pLDDT] high (70–90) vs low (50–70)
[pLDDT] very high (>90) vs low (50–70)
[Chem] positive vs negative
[Chem] hydrophobic vs polar
[Chem] hydrophobic vs positive
[Chem] hydrophobic vs negative
[Disorder] IDR vs ordered
[Contact] low vs high
[Contact] low vs medium
[Special] aromatic vs rest
[Special] glycine vs rest
[Special] proline vs rest
[Cons.] entropy high vs low
n=20,665
n=22,362
n=30,637
n=25,017
n=26,442
n=22,205
n=28,319
n=8,513
n=24,357
n=9,083
n=27,749
n=20,430
n=19,721
n=36,832
n=23,015
n=35,583
n=36,832
n=36,832
n=36,832
n=24,555
−0.1
0.0
0.1
0.2
Cliff's δ (positive = first level has larger |attention|)
(a)

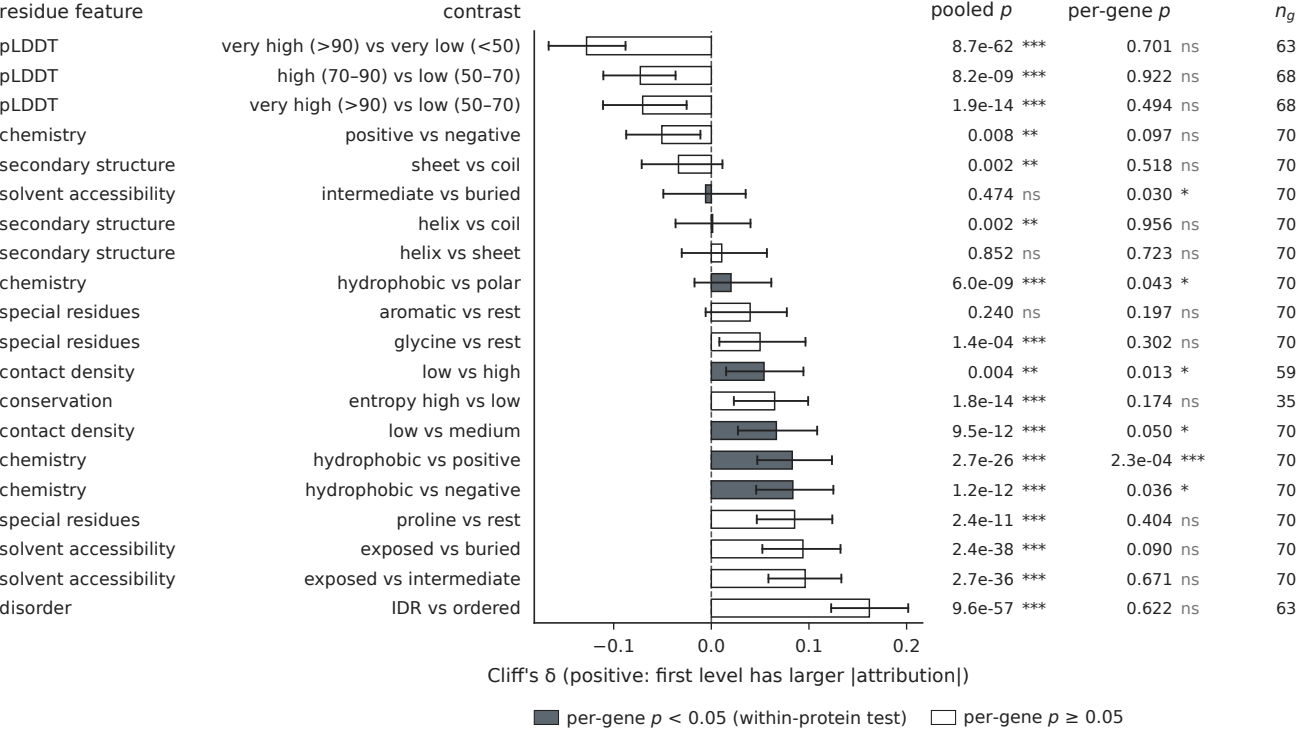

[SS] helix vs coil
[pLDDT] high (70–90) vs low (50–70)
[SS] helix vs sheet
[pLDDT] very high (>90) vs very low (<50)
[SASA] exposed vs intermediate
[Disorder] IDR vs ordered
[SS] sheet vs coil
[pLDDT] very high (>90) vs low (50–70)
[Special] proline vs rest
[Special] glycine vs rest
[Special] aromatic vs rest
[Cons.] entropy high vs low
[Chem] positive vs negative
[SASA] exposed vs buried
[Contact] low vs medium
[Chem] hydrophobic vs polar
[Chem] hydrophobic vs negative
[SASA] intermediate vs buried
[Contact] low vs high
[Chem] hydrophobic vs positive
0.956 ns (n_g=70)
0.922 ns (n_g=68)
0.723 ns (n_g=70)
0.701 ns (n_g=63)
0.671 ns (n_g=70)
0.622 ns (n_g=63)
0.518 ns (n_g=70)
0.494 ns (n_g=68)
0.404 ns (n_g=70)
0.302 ns (n_g=70)
0.197 ns (n_g=70)
0.174 ns (n_g=35)
0.097 ns (n_g=70)
0.090 ns (n_g=70)
0.050 * (n_g=70)
0.043 * (n_g=70)
0.036 * (n_g=70)
0.030 * (n_g=70)
0.013 * (n_g=59)
2.3e-04 *** (n_g=70)
p=0.05 p=0.01
0
1
2
3
4
5
$-\log_{10}(p)$ per-gene paired Wilcoxon
(b)

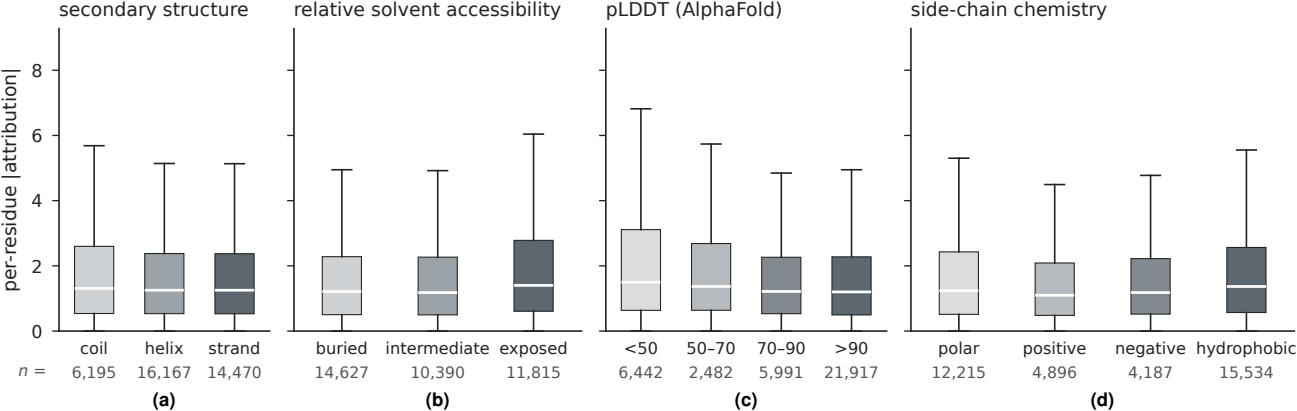

SASA bin
chemistry
pLDDT bin
rel. position N→C
ESM-2 PC fraction
hydropathy
conservation entropy
secondary structure
+0.351 %
+0.085 %
+0.049 %
+0.032 %
+0.012 %
+0.002 %
-0.001 %
-0.007 %
0.0
0.1
0.2
0.3
0.4
ΔR^2_{cond} (%) when feature is dropped
(c)

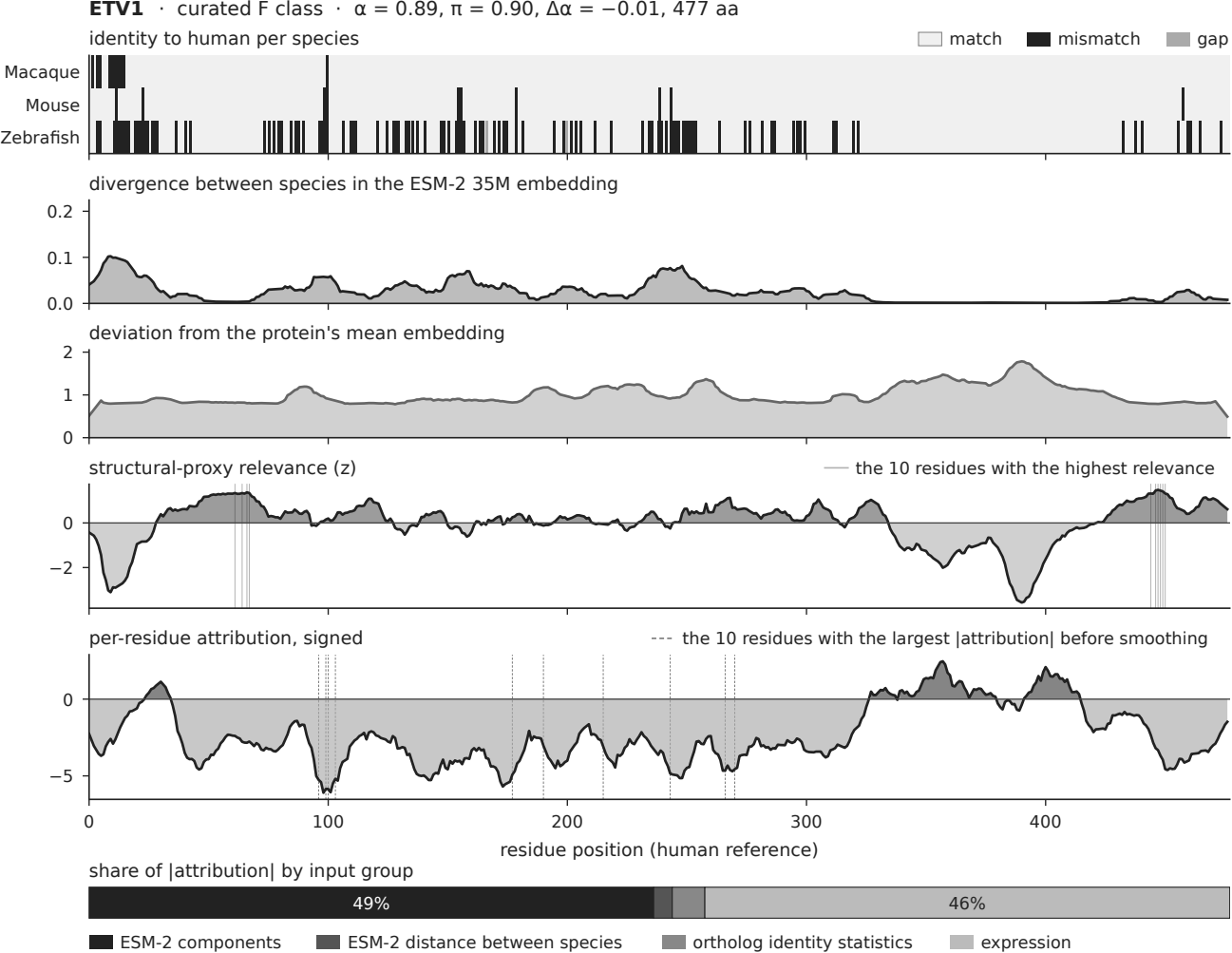

Chem: hydrophobic vs positive
SASA: intermediate vs buried
Contact: low vs high
p = 0.05
Chem: hydrophobic vs negative
Chem: hydrophobic vs polar
Contact: low vs medium
p = 0.05
real effect (per-gene $p < 0.05$)
pooled-only (composition artefact)
$-\log_{10}(p)$ per-gene paired Wilcoxon (square-root scale)
$-\log_{10}(p)$ pooled Mann-Whitney U (square-root scale)
0
0.5
1
2
3
4
0
0.51
2
5
10
30
60
(d)

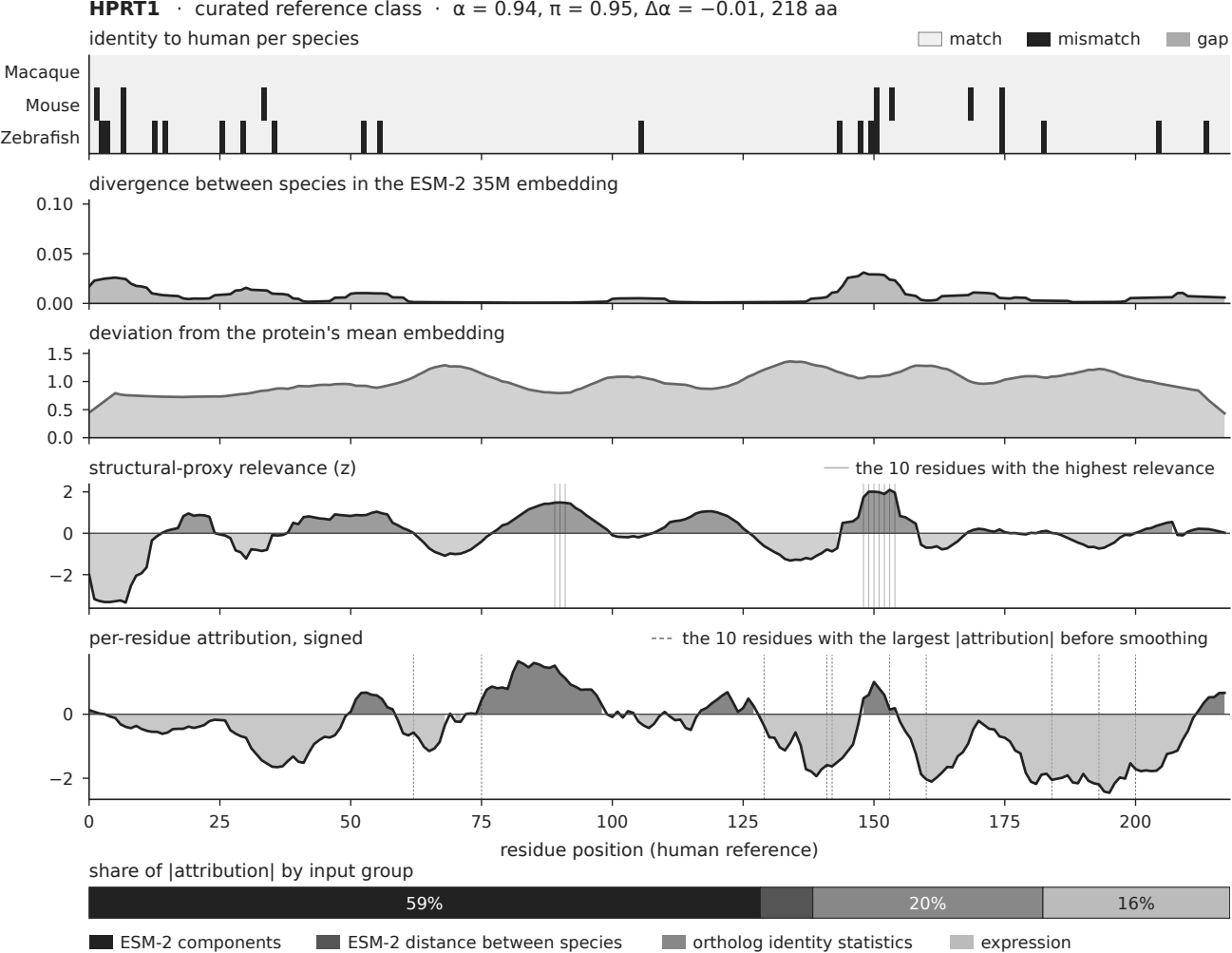

MT-CYB
MACROD2
ZNF804B
LSAMP
FLT1
PLEKHG1
FGFRL1
MFRP
ALDH4A1
SLX9
GET4
M | M (n=1931)
M liver / F testis (n=4236)
F | F (n=1846)
F liver / M testis (n=866)
mean Δα — testis
mean Δα — liver
(e)
M-flag
n_g=34
median=1.5716
F-flag
n_g=20
median=1.6651
ref-flag
n_g=16
median=2.1285
mean per-gene |attention|
(f)
M-flag
n_{res}=17,790, frac genes(+)=68%
per-gene p=0.010 *
F-flag
n_{res}=12,095, frac genes(+)=55%
per-gene p=0.927 ns
ref-flag
n_{res}=6,947, frac genes(+)=62%
per-gene p=0.433 ns
Cliff's δ (hydrophobic vs polar)
(g)
$p = 0.05$
pooled Mann-Whitney U
per-gene paired Wilcoxon
$-\log_{10}(p)$ (square-root scale)
9.6e-57***
0.622ns
0.004**
0.013*
0.240ns
0.197ns
1.4e-04***
0.302ns
2.4e-11***
0.404ns
1.8e-14***
0.174ns
IDR vs ordered
(δ=+0.16)
low vs high
(δ=+0.05)
aromatic vs rest
(δ=+0.04)
glycine vs rest
(δ=+0.05)
proline vs rest
(δ=+0.09)
entropy high vs low
(δ=+0.06)
(h)

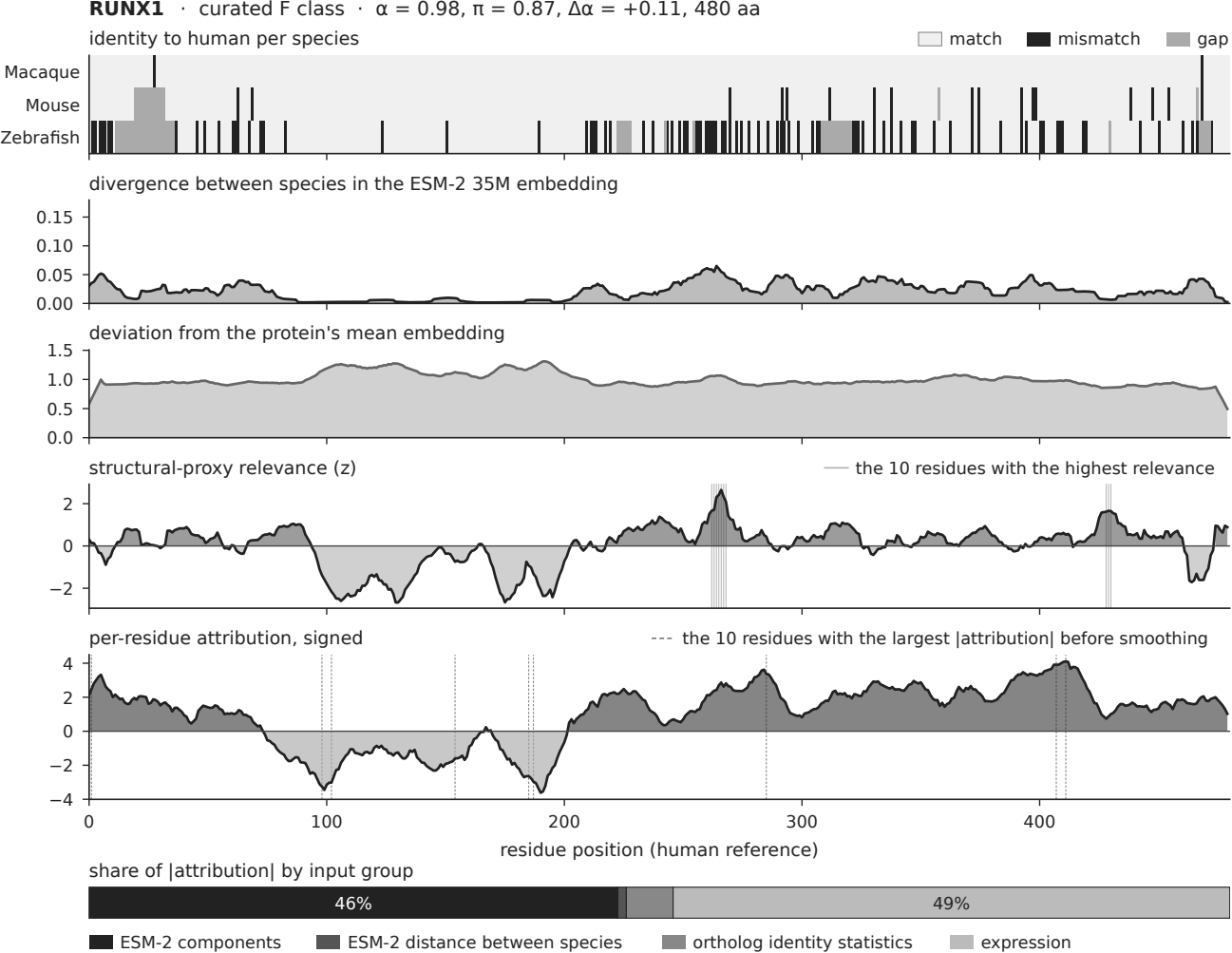

n = 36,832 residues, 70 genes · ICC = 25.5% · R^2_{marg} = 0.029 · R^2_{cond} = 0.277
pLDDT: low (50–70)
pLDDT: very high (>90)
chemistry: positive
secondary structure: strand
conservation entropy (z)
relative position N→C (z)
solvent accessibility: exposed
secondary structure: helix
hydropathy (z)
pLDDT: very low (<50)
chemistry: hydrophobic
chemistry: negative
solvent accessibility: buried
ESM-2 component fraction (z)
***
**
**
**
***
**
−0.1
0.0
0.1
0.2
0.3
β on log |attribution| (other terms held fixed)


residue feature
contrast
pooled p
per-gene p
n_g
pLDDT
very high (>90) vs very low (<50)
8.7e-62 ***
0.701 ns
63
pLDDT
high (70–90) vs low (50–70)
8.2e-09 ***
0.922 ns
68
pLDDT
very high (>90) vs low (50–70)
1.9e-14 ***
0.494 ns
68
chemistry
positive vs negative
0.008 **
0.097 ns
70
secondary structure
sheet vs coil
0.002 **
0.518 ns
70
solvent accessibility
intermediate vs buried
0.474 ns
0.030 *
70
secondary structure
helix vs coil
0.002 **
0.956 ns
70
secondary structure
helix vs sheet
0.852 ns
0.723 ns
70
chemistry
hydrophobic vs polar
6.0e-09 ***
0.043 *
70
special residues
aromatic vs rest
0.240 ns
0.197 ns
70
special residues
glycine vs rest
1.4e-04 ***
0.302 ns
70
contact density
low vs high
0.004 **
0.013 *
59
conservation
entropy high vs low
1.8e-14 ***
0.174 ns
35
contact density
low vs medium
9.5e-12 ***
0.050 *
70
chemistry
hydrophobic vs positive
2.7e-26 ***
2.3e-04 ***
70
chemistry
hydrophobic vs negative
1.2e-12 ***
0.036 *
70
special residues
proline vs rest
2.4e-11 ***
0.404 ns
70
solvent accessibility
exposed vs buried
2.4e-38 ***
0.090 ns
70
solvent accessibility
exposed vs intermediate
2.7e-36 ***
0.671 ns
70
disorder
IDR vs ordered
9.6e-57 ***
0.622 ns
63
−0.1
0.0
0.1
0.2
Cliff's δ (positive: first level has larger |attribution|)
per-gene $p < 0.05$ (within-protein test)
per-gene $p \geq 0.05$


secondary structure
relative solvent accessibility
pLDDT (AlphaFold)
side-chain chemistry
per-residue |attribution|
0
2
4
6
8
coil
helix
strand
buried
intermediate
exposed
<50
50–70
70–90
>90
polar
positive
negative
hydrophobic
n =
6,195
16,167
14,470
14,627
10,390
11,815
6,442
2,482
5,991
21,917
12,215
4,896
4,187
15,534
(a)
(b)
(c)
(d)


ETV1 · curated F class · α = 0.89, π = 0.90, Δα = −0.01, 477 aa
identity to human per species
match
mismatch
gap
Macaque
Mouse
Zebrafish
divergence between species in the ESM-2 35M embedding
0.2
0.1
0.0
deviation from the protein's mean embedding
2
1
0
structural-proxy relevance (z)
the 10 residues with the highest relevance
0
−2
per-residue attribution, signed
the 10 residues with the largest |attribution| before smoothing
0
−5
0
100
200
300
400
residue position (human reference)
share of |attribution| by input group
49%
46%
ESM-2 components
ESM-2 distance between species
ortholog identity statistics
expression


HPRT1 · curated reference class · α = 0.94, π = 0.95, Δα = −0.01, 218 aa
identity to human per species
match
mismatch
gap
Macaque
Mouse
Zebrafish
divergence between species in the ESM-2 35M embedding
0.10
0.05
0.00
deviation from the protein's mean embedding
1.5
1.0
0.5
0.0
structural-proxy relevance (z)
the 10 residues with the highest relevance
2
0
−2
per-residue attribution, signed
the 10 residues with the largest |attribution| before smoothing
0
−2
0
25
50
75
100
125
150
175
200
residue position (human reference)
share of |attribution| by input group
59%
20%
16%
ESM-2 components
ESM-2 distance between species
ortholog identity statistics
expression


RUNX1 · curated F class · α = 0.98, π = 0.87, Δα = +0.11, 480 aa
identity to human per species
match
mismatch
gap
Macaque
Mouse
Zebrafish
divergence between species in the ESM-2 35M embedding
0.15
0.10
0.05
0.00
deviation from the protein's mean embedding
1.5
1.0
0.5
0.0
structural-proxy relevance (z)
the 10 residues with the highest relevance
2
0
−2
per-residue attribution, signed
the 10 residues with the largest |attribution| before smoothing
4
2
0
−2
−4
0
100
200
300
400
residue position (human reference)
share of |attribution| by input group
46%
49%
ESM-2 components
ESM-2 distance between species
ortholog identity statistics
expression